Residual Ternary Algebra and a modular foundation: the s-series, the Bridge Series C, and the 35-component law vector

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Abstract

Residual Ternary Algebra (RTA, also called Resonant Algebra) is built from three kinds of written objects: elementary packaging arithmetic (the ternary map T_3, the residual cardinality 539, the carrier $\mathcal{O}_{\text{res}}$), the residual series $g_s(q) = \sum_{n\geq 0} q^{n^2}/(-q^s;q^s)_n$, and the Bridge Series $C = P_7^{\text{orth}} - H_7$, an orthant-minus-cone sum on the indefinite lattice $\mathbb{Z}^2$ with Gram matrix $\left(\begin{smallmatrix} 2 & 7 \\ 7 & 7 \end{smallmatrix}\right)$. This paper collects, with complete proofs, what can be established in tying these objects to a modular foundation, and what cannot.

Part I treats the s-series g_s (written g_k there): an exact expansion of the coefficients, families of zeros, sign laws for even k, and, for every odd $k \leq 33$, a parity-dominance theorem with complete zero sets (computer-assisted in stated steps). Part II treats the Bridge Series C: a ray-class decomposition over $\mathbb{Q}(\sqrt{35})$, radial asymptotics, a Baker-type obstruction to every scalar modular or mock residual law, and a vector-valued structure from a Siegel theta lift. Part III is new. It proves that the leading radial profile of C lies outside the span $\mathcal{V}_{35}$ of the 35-component law vector, while the profile of the normalized series $q^{-7/20}C$ lies in $\mathcal{V}_{35}$ and obeys its vector-valued S-law; it gives the complete radial expansion of $q^{-7/20}C$ and of 105 companion series through order t^0 at every rational point, with closed formulas, an algebraicity statement at order $t^{-1/2}$, and an exact decomposition of that coefficient by quadratic class. The final part states precisely which links between RTA and this framework are proved, which written RTA objects (T_3, $\mathcal{O}_{\text{res}}$, 539, the packaging identities) have no written link to it, and where the shadow Ξ would have to enter; the last question is open. Part III also determines the shadow structure of the 105 companion series: the exact matrix identities $M^2 = 35P$ and $(MT)^3 = 35\sqrt{35}P$, the resulting parity theorem (a weight-one law forces the trivial multiplier and P-odd vectors), the ξ-images of the cone completions in terms of three unary theta series, the Laplacian defect of the orthant Siegel lift as one product of unary thetas with an exact vanishing criterion, and the line terms of the literal series in closed form. No claim is made about the twin prime conjecture or the Riemann hypothesis.

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Symbol key

symbol	meaning	where defined
General and preliminaries		
$q = \mathbf{e}(\tau)$, $\mathbf{e}(z)$	$\mathbf{e}(z) = e^{2\pi i z}$	§2
Q, B, A	binary quadratic form $n_1^2 + 7n_1n_2 + \frac{7}{2}n_2^2$, its bilinear form, Gram matrix	§2
Q_s	quadratic form of parameter s	Def. 2.1
P_s^{orth}	orthant theta series of Q_s	Def. 2.1
Π_s	product series $(-q^s;q^s)_\infty g_s$	Prop. 2.2
H_s	cone series with walls c_1, c_2 (never redefined)	§10
C	Bridge Series $P_7^{\mathrm{orth}} - H_7$	§34
$w(v)$	cone weight $\frac{1}{2}(\operatorname{sgn} B(c_1,v) - \operatorname{sgn} B(c_2,v))$	§34
I	$\log(6+\sqrt{35})/(2\sqrt{35})$	Thm. 34.6
Part I		
g_k, $g_k(d)$	residual series $\sum q^{n^2}/(-q^k;q^k)_n$, its coefficients	§3
P_n, $\varphi_n(m)$	$\prod_{j\le n}(1+x^j)^{-1}$ and its coefficients	§3
$A_n(m)$, $Q(m)$	$\lvert\varphi_n(m)\rvert$; partitions of m into distinct odd parts	§4
Z_k	zero set of g_k on allowed d	§3
$\mathcal{S}_k$, $\mathrm{OccFilt}_k(\infty)$	square classes mod k; classes eventually nonzero	§3
Ξ_k	a finite set of integers (unrelated to the shadow Ξ)	§6.6
Part II		
g_s	residual series at parameter s	§25.2
V	42-dim. span of the lift functions	§14
E	ray-class partial theta series	§12
C_{axis}, C_{third}	pieces of C	§24
u_L	vector of the theta-lift family	§21
Δ	wall parameter of the cone	§23.1
H_s^{axes}	axis variant of H_s	§23.3
Part III		
$G_{\alpha,\beta}$	Gauss mean	Def. 33.1
D (set), a, h_A	discriminant group; shift; $(\frac{1}{2},0)$	§33
$K1_\mu, \dots, K4_\mu$	$G_{a+\mu,a}, G_{a+\mu,0}, G_{\mu,a}, G_{\mu,0}$	Def. 33.2
$(K2_\mu)$	the 35-component law vector	Def. 33.2
$K2_0$, $K3_p$	components at $\mu=0$, $\mu=p$	Def. 33.2
$\mathcal{V}_{35}$	36-dim. span of $K2, K3, K4$	Def. 33.2, Prop. 33.6
$\mathcal{E}_{36}$, $\mathcal{C}_6$	36- and 6-dim. summands of Part II's 42-dim. lift rep.	Prop. 33.7
$S_{\alpha,\beta}$, wt	companion series and weight $1_{\mathrm{orthant}} - w$	Def. 34.1
$q^{-7/20}C = S_{a,0}$	normalized Bridge Series	§34
u_1, u_2; $K_\pm$	$(21,-1)$, $(0,1)$; cones split into 21 pieces	Prop. 34.3
$\Lambda_{\alpha,\beta}$	leading radial profile	Thm. 34.5
Ar, J_a, J_b, $K(l)$	Gaussian cone integrals	Lem. 35.3
mean(f), W_i, V_i, C	periodic mean, sawtooth, second moment, corner terms	Thm. 35.4
A_0, A_1, A_2	coefficients of $t^{-1}, t^{-1/2}, t^0$	Thm. 35.6
D (term), R, $W[\lambda]$	drift, ray, sawtooth parts of A_1	Prop. 35.11
$R[\frac{7}{2}]$, $R[\frac{595}{2}]$	ray terms on $\pm\mathbb{N}(0,1)$, $\pm\mathbb{N}(21,-1)$	Prop. 35.11
$\lambda \in \{1, \frac{7}{2}, \frac{595}{2}\}$	Q-values $Q(e_1)$, $Q(0,1)$, $Q(21,-1)$	Prop. 35.11
Z_0	zero set of A_0 in X_{216}	Prop. 35.10
Ξ_ε	sawtooth shadow candidate (Category B)	§36
Part IV		
T_3	ternary map $3k \mapsto k$, $3k{+}1 \mapsto 4k{+}2$, $3k{+}2 \mapsto 2k{+}1$	Prop. 38.6

symbol	meaning	where defined
$\mathcal{O}_{\mathrm{res}}$	carrier $\{21+9j : 0 \le j \le 538\}$	Rem. 38.7
539	residual cardinality	Prop. 38.6
Ξ	RTA weight-$\frac{3}{2}$ shadow (not identified)	§38
M, T, P	105×105 S-, T- and parity matrices on the companions	§37
p, E, ρ_O, ρ_H	ν-sign weight, its even part, odd cone weights	Lemma 37.6
$\widehat{H}, \widehat{G}_O, X^c$	cone completions and their ξ_1-images	Prop. 37.7
$\mathcal{B}_{z,z'}, \vartheta^{\pm}$	boundary series; unary thetas in $\frac{7}{2}m^2$, $\frac{5}{2}u^2$	Thm. 37.10
Φ_{orth}	Siegel lift over the orthant arc	Thm. 37.11
ρ_m, ρ_u	reflections $m \mapsto -m$, $u \mapsto -u$	Thm. 37.10

1 Introduction

1.1 Goal

The aim of this work is to tie Residual Ternary Algebra (RTA) to a foundation: to identify which RTA objects are governed by modular, mock-modular or vector-valued modular structure, and to prove each such statement. RTA, as written, has no algebraic axioms. Its objects are (a) elementary arithmetic of the ternary map T_3 and of the residual cardinality 539 (packaging identities, the equitable partition $539 = 3 \cdot 68 + 5 \cdot 67$, the carrier $\mathcal{O}_{\text{res}}$); (b) the residual series g_s; (c) the Bridge Series C and its pieces C_{axis}, C_{third}; and (d) a weight-$\frac{3}{2}$ shadow Ξ and a weight-$\frac{1}{2}$ period-function candidate, described in words but not defined as explicit series. A "foundation" here means a theorem placing an object of (a)–(d) inside a structure with known transformation laws under $\mathrm{SL}_2(\mathbb{Z})$.

1.2 What is achieved

(1) For the s-series (Part I): the coefficient expansion of g_s, its zero families, sign laws for even s, and complete zero sets and residue-class occupancy for odd $s \leq 33$. In particular the residue law of g_7 recorded in RTA follows from the expansion, and its converse holds (Corollary 38.2).

(2) For the Bridge Series (Part II): C and C_{third} share the orthant remainder P_7^{orth}; up to theta and mock pieces this is a partial theta series of one ray class of $\mathbb{Q}(\sqrt{35})$; a Baker-type argument excludes every scalar modular or mock residual law, including every scalar W_7 slash law; the leading radial data carry a vector-valued structure coming from a Siegel theta lift.

(3) For the 35-component law vector (Part III, new): the leading profile of C is $I\,\mathbf{e}(7x/20)K2_0(x)$ and lies outside $\mathcal{V}_{35}$; the normalized series $q^{-7/20}C$ has profile $I\,K2_0 \in \mathcal{V}_{35}$ and satisfies the vector-valued S-law of $\mathcal{V}_{35}$; the radial expansion of $q^{-7/20}C$ and of 105 companion series is computed in closed form through order t^0 at every rational point.

(4) For the foundation question (§38): every written link between RTA and the framework above is listed with its status.

1.3 What is not achieved

(1) No written object links T_3, $\mathcal{O}_{\text{res}}$, the number 539 or the packaging identities to g_s, C or $\mathcal{V}_{35}$. These parts of RTA are not tied to the modular framework by anything proved here.

(2) The shadow Ξ is not identified. The vector-valued law of $\mathcal{V}_{35}$ holds at leading order only; at order $t^{-1/2}$ no law of the same kind is known, and the numerical evidence (Appendix D) is against closure inside the companion span.

(3) No link to the g_s family is proved beyond the shared definition and the residue law.

(4) Nothing here concerns the twin prime conjecture, the Riemann hypothesis, or any security property of a hash construction.

1.4 Status conventions

Every statement in a theorem, proposition, lemma or corollary is **Category A**: proved here; or proved by a finite exact computation (integer, rational, modular or cyclotomic arithmetic, including

ranks modulo a prime together with the argument that transfers them to characteristic 0); or proved by a finite computation in rigorous ball or interval arithmetic (labelled *computer-assisted*); or proved conditional on a cited published theorem, named each time. **Category B** statements (floating-point evidence, fits, singular-value ranks, patterns, interpretations) appear only in remarks labelled "Remark (Category B)", in status boxes marked Category B, or in the appendices. Part II uses status boxes after its results; their wording is part of the record.

1.5 Organisation

Section 2 fixes common notation and resolves notational conflicts between the parts. Part I (Sections 3–8) is the s-series. Part II (Sections 9–32) is the Bridge Series. Part III (Sections 33–36) is the 35-component law vector and the normalized Bridge Series. Section 38 treats the foundation question and Section 39 lists open problems. Appendices A–C contain the computational records of Parts I–II, and Appendix D the Category B record of Part III.

2 Preliminaries and unified notation

2.1 Common conventions

$\mathbb{N} = \{0, 1, 2, \dots\}$; $\mathbf{e}(x) = e^{2\pi i x}$; $q = \mathbf{e}(\tau)$ with $\tau \in \mathbb{H}$. "Radial" means $q = \mathbf{e}(x)e^{-t}$ with $x \in \mathbb{Q}$ and $t \to 0^+$, so that $q^\lambda = \mathbf{e}(\lambda x)e^{-\lambda t}$ for real λ. $(a;q)_n = \prod_{j=0}^{n-1}(1 - aq^j)$. $B_1(y) = y - \frac{1}{2}$, $B_2(y) = y^2 - y + \frac{1}{6}$, $B_3(y) = y^3 - \frac{3}{2}y^2 + \frac{1}{2}y$ are Bernoulli polynomials and $\bar{B}_j(y) = B_j(\{y\})$ their periodic versions.

2.2 The lattice

$A = \left(\begin{smallmatrix} 2 & 7 \\ 7 & 7 \end{smallmatrix}\right)$, $B(x, y) = x^{\mathsf{T}}Ay$, $Q(x) = \frac{1}{2}B(x, x) = x_1^2 + 7x_1x_2 + \frac{7}{2}x_2^2$. The form Q has signature $(1, 1)$ and $\det A = -35$. $D = A^{-1}\mathbb{Z}^2/\mathbb{Z}^2$ has order 35. $a = (\frac{7}{10}, -\frac{1}{5})$, so $Aa = (0, \frac{7}{2})$ and $Q(a) = -\frac{7}{20}$. For general s: $A_s = \left(\begin{smallmatrix} 2 & s \\ s & s \end{smallmatrix}\right)$ and

$$Q_s(n, k) = n^2 + snk + \tfrac{s}{2}k(k + 1).$$

At $s = 7$, $Q_7(v) = Q(v) + \frac{7}{2}v_2 = Q(v + a) + \frac{7}{20}$.

2.3 The residual series and the two series previously both written P_s

Definition 2.1. For $s \geq 1$:

(i) the *s-series* $g_s(q) = \sum_{n\geq 0} q^{n^2}/(-q^s; q^s)_n$ (in Part I written g_k, with k in place of s);

(ii) the *product-normalised s-series* $\Pi_s(q) = (-q^s; q^s)_\infty\, g_s(q)$;

(iii) the *orthant series* $P_s^{\mathrm{orth}}(q) = \sum_{n,k\geq 0} q^{Q_s(n,k)}$.

In the RTA manuscript both Π_s and P_s^{orth} carry the name P_s. They are different series:

Proposition 2.2. *For every $s \geq 1$, in $\mathbb{Z}[[q]]$,*

$$\Pi_s(q) = \sum_{n,k\geq 0} \frac{q^{Q_s(n,k)}}{(q^s; q^s)_k} = \sum_{k\geq 0} \frac{q^{sk(k+1)/2}}{(q^s; q^s)_k} \sum_{n\geq 0} q^{n^2+skn},$$

and $\Pi_s - P_s^{\mathrm{orth}}$ has non-negative coefficients and is non-zero; its coefficient of q^{2s} is at least 1.

Proof. Euler's identity $(-z;p)_\infty = \sum_{k\ge 0} p^{k(k-1)/2} z^k/(p;p)_k$ with $p = q^s$ and $z = q^{s(n+1)}$ gives $(-q^s;q^s)_\infty/(-q^s;q^s)_n = (-q^{s(n+1)};q^s)_\infty = \sum_{k\ge 0} q^{sk(k-1)/2+s(n+1)k}/(q^s;q^s)_k$, and $n^2+sk(k-1)/2+s(n+1)k = Q_s(n,k)$. Multiplying by q^{n^2} and summing over n gives the first identity; both double sums converge in $\mathbb{Z}[[q]]$ because $Q_s(n,k) \ge n^2 + \frac{s}{2}k(k+1)$. Since $1/(q^s;q^s)_k - 1$ has non-negative coefficients, so does $\Pi_s - P_s^{\mathrm{orth}} = \sum_{n,k} q^{Q_s(n,k)}(1/(q^s;q^s)_k - 1)$; the term $(n,k) = (0,1)$ contributes $q^s \cdot q^s$ with coefficient 1. □

Throughout, the Bridge Series is built from P_s^{orth} only. Part II uses P_s^{orth} for P_7^{orth} when $s = 7$ is fixed.

2.4 The Bridge Series

$c_1 = (-1,1)$, $c_2 = (-4,1)$, $w(v) = \frac{1}{2}(\operatorname{sgn} B(c_1,v) - \operatorname{sgn} B(c_2,v))$ with $\operatorname{sgn} 0 = 0$, and

$$H_s = \sum_{v\in\mathbb{Z}^2} w(v) q^{Q_s(v)}, \qquad C = P_7^{\mathrm{orth}} - H_7.$$

H_s is the cone series of Part II, §10, with the fixed walls c_1, c_2 for every s; it is not redefined anywhere in this paper. This is the Bridge Series of RTA (Definition 9.1 there), and it is the object C of Part II.

2.5 Notational conflicts and their resolution

Each part keeps its own local notation where this causes no ambiguity. The following symbols have different meanings in different parts.

symbol	meaning	other meaning, and its name here
P_s	not used	RTA: both Π_s and P_s^{orth} (Proposition 2.2)
$P_n(x)$	Part I: $\prod_{j\le n}(1+x^j)^{-1}$	–
V	Part II: the 42-dimensional span of the lift functions	Part III: $\mathcal{V}_{35}$, the 36-dimensional span of the 35-component law vector
E	Part II: the ray-class partial theta series	Part III: $\mathcal{E}_{36}$, a 36-dimensional summand of a representation
C	the Bridge Series	Part III: $\mathcal{C}_6$, a 6-dimensional summand of a representation
Ξ	§38: the RTA shadow	Part I: Ξ_k, a finite set of integers
Q	the quadratic form above	Part I: $Q(m)$, the number of partitions of m into distinct odd parts
N	Part III: the period $2m$	Part II: the label negation map
k	Part I: the parameter of g_k	Parts II, III: the second lattice coordinate
G	Parts II, III: the Gauss mean of Q_7	Part I, §6.3: an eta-quotient

Part I
The s-series

This part treats $g_k(q) = \sum_{n\geq 0} q^{n^2}/(-q^k;q^k)_n$, the s-series of RTA with $s = k$. Its notation (P_n, φ_n, c_m, $Q(m)$, E_d, O_d, Ξ_k) is local to this part. Results proved with a computer state the computation and what it relies on; the dependence on one cited explicit bound is stated where used.

2.5.1 Summary of Part I

Result	Statement	Nature of proof
Prop. 3.2	$g_k(d) = \sum_{n^2\leq d,\, n^2\equiv d\,(k)} \varphi_n\big(\frac{d-n^2}{k}\big)$	algebraic
Lemmas 4.1–4.6	$\varphi_n(m) = c_m$ for $m \leq n$; $c_m = (-1)^m Q(m)$; $(-1)^m\varphi_n(m) \geq 0$; product form (4.1)	algebraic
Prop. 4.7	odd k: $(-1)^d g_k(d) = E_d - O_d$	algebraic
Thm. 5.1	$k \geq 2$ squarefree $\Rightarrow g_k(k(k+2)) = 0$	algebraic
Thm. 5.3, Cor. 5.4, Thm. 5.5	for odd prime powers k: formula for $g_k(k^2+jk+b^2)$, $b = 1, 2$; zeros at $(k+2)^2-3$ ($k \geq 5$), k^2+2k+4 ($k \equiv 3 \bmod 4$, $k \neq 3, 7$), k^2+3k+4 ($k \equiv 1 \bmod 4$, $k \neq 5$); exact failures; complete zero classes	algebraic
Thm. 6.9	$\omega(k) = 2$, k odd: $g_k((k+2)^2-3) \neq 0$ except $k = 15, 21$	algebraic + exact integer certificate for $s \leq 2500$ + cited bound [20] for $s \geq 2501$
Thm. 7.4, Cor. 7.5, Thm. 7.6	one-sign classes: explicit bound B_c beyond which $g_k \neq 0$ with a definite sign; $k \equiv 2 \pmod 4$: Z_k finite; g_2: sign $(-1)^{\lfloor d/2\rfloor}$, $Z_2 = \{2, 8, 10, 26\}$	algebraic
Thm. 7.7	even $k \geq 8$: $8 \nmid k \Rightarrow g_k((k+2)^2-3) \geq 2$; $k = 2p^a \Rightarrow = 2$; $k = 2^a \Rightarrow < 0$	algebraic
Thm. 8.1	odd $k \leq 33$: $E_d \leq (R_k + 10^{-6})O_d$ with $R_k < 0.906$ for allowed $d \geq D_0(k)$; Z_k complete; $\mathrm{OccFilt}_k(\infty) = \mathcal{S}_k$	analytic reduction + finite computer-assisted verification (interval arithmetic, modular arithmetic)

Scope. Only proved statements are included. For even k no claim is made on classes of d that are not one-sign. For odd $k \geq 35$ no claim is made beyond Sections 4–6.

Dependence on external results. Theorem 6.9 for $s \geq 2501$ and the monotonicity of $Q(n)$ for $n \geq 35200$ (Lemma 6.5) rely on the explicit bound of Chern [20] (Theorem 1.1, error term (3.10), Remark 3), cited in Section 6.3. Computer-assisted steps use the Arb ball arithmetic [21] and exact integer arithmetic; they are listed in Section 8.8 and Appendix A.

3 Definitions

Conventions. $\mathbb{N} = \{0, 1, 2, \dots\}$. All series are formal power series with integer coefficients; $[x^m]F$ is the coefficient of x^m in F. For a positive integer k and an integer $n \geq 0$ put $(-q^k; q^k)_n = \prod_{j=1}^{n}(1 + q^{kj})$.

Definition 3.1. For $k \geq 1$,

$$g_k(q) = \sum_{n \geq 0} \frac{q^{n^2}}{(-q^k; q^k)_n} = \sum_{d \geq 0} g_k(d)\, q^d.$$

For $n \geq 0$ let

$$P_n(x) = \prod_{j=1}^{n} \frac{1}{1 + x^j} = \sum_{m \geq 0} \varphi_n(m)\, x^m, \qquad P_\infty(x) = \prod_{j \geq 1} \frac{1}{1 + x^j} = \sum_{m \geq 0} \varphi_\infty(m)\, x^m,$$

with $P_0 = 1$.

The series for g_k converges in $\mathbb{Z}[[q]]$ because the n-th summand is $q^{n^2} P_n(q^k)$ and has order n^2.

Proposition 3.2 (expansion in d). For all $k \geq 1$ and $d \geq 0$,

$$g_k(d) = \sum_{\substack{n \geq 0,\ n^2 \leq d \\ n^2 \equiv d \pmod{k}}} \varphi_n\!\left(\frac{d - n^2}{k}\right). \tag{3.1}$$

Proof. The n-th summand of $g_k(q)$ is $q^{n^2} P_n(q^k) = \sum_{m \geq 0} \varphi_n(m) q^{n^2 + km}$. The coefficient of q^d collects the pairs (n, m) with $n^2 + km = d$. ∎

The summand of index n in (3.1) is called the *term* n of $g_k(d)$; it is defined when $n^2 \leq d$ and $k \mid d - n^2$, and $m_n = (d - n^2)/k$ is its *index*.

Definition 3.3. Let $\mathcal{S}_k = \{\, n^2 \bmod k : \ n \in \mathbb{Z} \,\} \subseteq \mathbb{Z}/k\mathbb{Z}$. An integer $d \geq 0$ is *allowed* for k if $d \bmod k \in \mathcal{S}_k$.

$$Z_k = \{\, d \geq 0 \text{ allowed for } k : \ g_k(d) = 0 \,\}.$$

$\mathrm{OccFilt}_k(\infty)$ is the set of residue classes $r \in \mathbb{Z}/k\mathbb{Z}$ such that $g_k(d) \neq 0$ for all sufficiently large $d \equiv r \pmod{k}$.

Proposition 3.4. (i) If d is not allowed for k then $g_k(d) = 0$. (ii) $\mathrm{OccFilt}_k(\infty) \subseteq \mathcal{S}_k$, with equality if and only if Z_k is finite.

Proof. (i) If $d \bmod k \notin \mathcal{S}_k$, no n satisfies $n^2 \equiv d \pmod{k}$, so the sum (3.1) is empty. (ii) A class $r \notin \mathcal{S}_k$ has $g_k(d) = 0$ for all $d \equiv r$, hence $r \notin \mathrm{OccFilt}_k(\infty)$. For $r \in \mathcal{S}_k$, $r \in \mathrm{OccFilt}_k(\infty)$ if and only if Z_k contains only finitely many $d \equiv r$. Since there are finitely many classes, equality holds if and only if Z_k is finite. ∎

By Proposition 3.4, the statement $\mathrm{OccFilt}_k(\infty) = \mathcal{S}_k$ ("every square class occurs for all large d") is the statement that Z_k is finite. Section 8 proves it for every odd $k \leq 33$, with Z_k computed exactly.

4 The polynomials φ_n

4.1 Stabilisation, product form, sign rule

Lemma 4.1 (stabilisation). For all $n \geq 0$ and $0 \leq m \leq n$, $\varphi_n(m) = \varphi_\infty(m)$.

Proof. $P_n(x) = P_\infty(x)\prod_{j>n}(1+x^j)$ and $\prod_{j>n}(1+x^j) \equiv 1 \pmod{x^{n+1}}$. ■

Write $c_m = \varphi_\infty(m)$. By Lemma 4.1, $\varphi_n(m) = c_m$ whenever $m \leq n$.

Lemma 4.2 (product form). For $n \geq 0$, as formal power series,

$$\prod_{j=1}^{n} \frac{1}{1+(-x)^j} = \prod_{\substack{j \leq n \\ j \text{ odd}}} (1+x^j) \prod_{\lfloor n/2 \rfloor < l \leq n} \frac{1}{1-x^{2l}}. \tag{4.1}$$

The left side equals $\sum_m (-1)^m \varphi_n(m) x^m$. Consequently, with $a_n(m) = [x^m]$ of the right side,

$$(-1)^m \varphi_n(m) = a_n(m) \geq 0, \qquad a_n(m) = \#\{(\lambda, \mu)\}, \tag{4.2}$$

where (λ, μ) runs over pairs consisting of a set λ of distinct odd integers in $[1, n]$ and a multiset μ of even integers in $(n, 2n]$, with $\sum \lambda + \sum \mu = m$.

Proof. Replacing x by $-x$ in P_n gives the left side and $\sum_m \varphi_n(m)(-x)^m$. For odd j, $1+(-x)^j = 1-x^j$ and $\frac{1}{1-x^j} = \frac{1+x^j}{1-x^{2j}}$. For $j = 2i$, $\frac{1}{1+x^{2i}} = \frac{1-x^{2i}}{1-x^{4i}}$. Multiplying over $j \leq n$, the numerators contribute $\prod_{j \leq n,\, j \text{ odd}}(1+x^j)\prod_{i \leq \lfloor n/2 \rfloor}(1-x^{2i})$ and the denominators contribute $\prod_{j \leq n,\, j \text{ odd}}(1-x^{2j})\prod_{i \leq \lfloor n/2 \rfloor}(1-x^{4i}) = \prod_{j \leq n}(1-x^{2j})$. The factors $1-x^{2i}$, $i \leq \lfloor n/2 \rfloor$, cancel, which leaves (4.1). The right side of (4.1) is a product of series with nonnegative coefficients; its coefficient count is (4.2), because $l \in (n/2, n]$ corresponds to the even integers $2l \in (n, 2n]$. ■

We write $A_n(m) = |\varphi_n(m)| = a_n(m)$ and $A_n(x) = \sum_m a_n(m) x^m$ (the series (4.1)); the meaning is clear from the argument.

Let $Q(m)$ denote the number of partitions of m into distinct odd parts ($Q(0) = 1$).

Lemma 4.3. $P_\infty(x) = \prod_{j \text{ odd}}(1-x^j)$. Hence $c_m = (-1)^m Q(m)$, and $(-1)^m \varphi_n(m) = Q(m)$ for $m \leq n$. The values are

$$(c_0, \ldots, c_{12}) = (1, -1, 0, -1, 1, -1, 1, -1, 2, -2, 2, -2, 3).$$

Proof. $\frac{1}{1+x^j} = \frac{1-x^j}{1-x^{2j}}$ and $\prod_{j \geq 1} \frac{1-x^j}{1-x^{2j}} = \prod_{j \text{ odd}}(1-x^j)$, the factors with j even cancelling against the denominators. Expanding, a partition of m into r distinct odd parts contributes $(-1)^r = (-1)^m$ (each odd part is $\equiv 1 \bmod 2$). Lemma 4.1 gives the rest. The values follow from $Q(0..12) = 1, 1, 0, 1, 1, 1, 1, 1, 2, 2, 2, 2, 3$. ■

Lemma 4.4. $Q(2) = 0$; $Q(m) \geq 1$ for $m \neq 2$; $Q(m) \geq 2$ for $m \geq 8$.

Proof. 2 is not a sum of distinct odd integers. For $m \geq 1$ odd, $\{m\}$ is a partition; for $m \geq 4$ even, $\{1, m-1\}$ is one; $Q(0) = 1$. For $m \geq 9$ odd, $\{m\}$ and $\{1, 3, m-4\}$ are distinct partitions ($m - 4 \geq 5$). For $m \geq 8$ even, $\{1, m-1\}$ and $\{3, m-3\}$ are distinct partitions ($m - 3 \geq 5$, so $m - 3 \neq 3$). ■

Lemma 4.5 (truncation formula). For $n \geq 0$ and $0 \leq m \leq 2n+1$,

$$\varphi_n(m) = c_m + \sum_{x=0}^{m-n-1} (-1)^x Q(x), \qquad A_n(m) = Q(m) + (-1)^m \sum_{x=0}^{m-n-1} (-1)^x Q(x), \tag{4.3}$$

where an empty sum is 0. In particular $\varphi_n(m) = c_m$ for $m \leq n$.

Proof. $P_n = P_\infty \prod_{j>n}(1+x^j)$. A product of two or more distinct factors x^j with $j > n$ has degree $\geq (n+1)+(n+2) = 2n+3$. Hence modulo x^{2n+2}, $\prod_{j>n}(1+x^j) \equiv 1 + \sum_{b=n+1}^{2n+1} x^b$. Therefore

for $m \le 2n+1$, $\varphi_n(m) = c_m + \sum_{b=n+1}^{m} c_{m-b} = c_m + \sum_{x=0}^{m-n-1} (-1)^x Q(x)$. Multiplying by $(-1)^m$ and using (4.2) gives the second formula. ■

Lemma 4.6 (small n). $\varphi_0(m) = \delta_{m,0}$; $\varphi_1(m) = (-1)^m$; and $\varphi_2(m) = 1, -1, 0, 0$ for $m \equiv 0, 1, 2, 3 \pmod 4$.

Proof. $P_1 = 1/(1+x) = \sum (-x)^m$. $(1+x)(1+x^2) = (1-x^4)/(1-x)$, so $P_2 = (1-x)/(1-x^4) = \sum_{i \ge 0} (x^{4i} - x^{4i+1})$. ■

4.2 The case of odd k

Proposition 4.7 (E–O identity). Let k be odd and $d \ge 0$. For each term n of $g_k(d)$ with index m_n, $(-1)^d \varphi_n(m_n) = (-1)^n a_n(m_n)$. Hence

$$(-1)^d g_k(d) = E_d - O_d, \qquad E_d = \sum_{n \text{ even}} a_n(m_n), \quad O_d = \sum_{n \text{ odd}} a_n(m_n), \tag{4.4}$$

the sums running over the terms n of $g_k(d)$; all summands are ≥ 0.

Proof. $d = n^2 + k m_n$ with $n^2 \equiv n$ and $k m_n \equiv m_n \pmod 2$ (k odd), so $(-1)^d = (-1)^{n+m_n}$. By (4.2), $(-1)^{m_n} \varphi_n(m_n) = a_n(m_n)$. Multiply: $(-1)^d \varphi_n(m_n) = (-1)^n a_n(m_n)$. Summing over n and using (3.1) gives (4.4). ■

In particular, for odd k and allowed d: $g_k(d) \ne 0$ if and only if $E_d \ne O_d$, and the term of index n has sign $(-1)^{d+n}$.

5 Zero families

5.1 Squarefree k

Theorem 5.1. If $k \ge 2$ is squarefree, then $g_k\big(k(k+2)\big) = 0$.

Proof. Let $d = k^2 + 2k$. A term n of $g_k(d)$ has $n^2 \le d < (k+1)^2$, so $n \le k$, and $k \mid n^2$. Since k is squarefree, $k \mid n^2$ implies $k \mid n$, so $n \in \{0, k\}$. For $n = 0$ the index is $m = k + 2 \ge 1$ and $\varphi_0(k+2) = 0$ (Lemma 4.6). For $n = k$ the index is $m = 2 \le k$, so $\varphi_k(2) = c_2 = 0$ (Lemma 4.1, Lemma 4.3). Hence $g_k(d) = 0$. ■

Remark 5.2. The hypothesis $k \ge 2$ is needed: $g_1(3) = \varphi_1(2) = 1$. Exact evaluation of $g_k(k(k+2))$ for every $k = 2, \dots, 1001$ gives 0 for all 608 squarefree k and a nonzero value for each of the 392 non-squarefree k (for $k \le 30$ non-squarefree: $g_4 = -1$, $g_8 = 1$, $g_9 = 2$, $g_{12} = -2$, $g_{16} = -2$, $g_{18} = 6$, $g_{20} = -5$, $g_{24} = 6$, $g_{25} = 6$, $g_{27} = 8$, $g_{28} = -8$ at $d = k(k+2)$). This is a computation (Appendix A.1).

5.2 Values at $d = k^2 + jk + b^2$

Theorem 5.3. Let $b \in \{1, 2\}$, $j \in \mathbb{Z}$, and let $k = p^a$ be an odd prime power with

$$k \ge K := \max(j + 3b,\ 2b + 1,\ -j).$$

Then for $d = k^2 + jk + b^2$,

$$g_k(d) = \varphi_b(k + j) + [\, j \ge -2b \,]\, c_{j+2b} + [\, j \ge 2b \,]\, c_{j-2b}, \tag{5.1}$$

where $[\cdot]$ is the Iverson bracket and c_m is as in Lemma 4.3 (with c_m for $m \ge 0$ only occurring when its bracket is 1).

Proof. Let n be a term of $g_k(d)$: $n^2 \equiv d \equiv b^2 \pmod{p^a}$, so $p^a \mid (n-b)(n+b)$. A common prime factor of $n - b$ and $n + b$ divides $2b \in \{2, 4\}$, hence is not p. Thus p^a divides one of the two factors:

$n \equiv \pm b \pmod{k}$. The candidates are $n \in \{b,\ k-b,\ k+b,\ 2k-b,\ 2k+b, \dots\}$ (as $k > 2b$, these are distinct and increasing in this order, and $b \geq 1$).

Large n do not occur. For $n \geq 2k-b$ we would need $(2k-b)^2 \leq d$, i.e. $3k \leq j+4b$. But $k \geq j+3b$ gives $j+4b \leq k+b$, and $3k \leq k+b$ contradicts $k \geq 2b+1$.

The three remaining candidates.

- $n = b$: $b^2 \leq d$ iff $k+j \geq 0$, true since $k \geq -j$. Index $m = (d-b^2)/k = k+j$; the term is $\varphi_b(k+j)$.
- $n = k-b$ (≥ 1): $(k-b)^2 \leq d$ iff $j \geq -2b$. Index $m = (d-(k-b)^2)/k = j+2b$. Since $k \geq j+3b$, $m = j+2b \leq k-b = n$, so the term is c_{j+2b} (Lemma 4.1).
- $n = k+b$: $(k+b)^2 \leq d$ iff $j \geq 2b$. Index $m = j-2b$, and $m \leq n$ since $k \geq j+3b$; the term is c_{j-2b}.

Summing gives (5.1). ■

Corollary 5.4 (three families). Let k be an odd prime power.

1. $g_k\big((k+2)^2-3\big) = 0$ for every $k \geq 5$. For $k=3$: $g_3(22) = -1$.
2. If $k \equiv 3 \pmod 4$: $g_k(k^2+2k+4) = 0$ for every $k \neq 3, 7$. Values: $g_3(19) = -1$, $g_7(67) = +1$.
3. If $k \equiv 1 \pmod 4$: $g_k(k^2+3k+4) = 0$ for every $k \neq 5$. Value: $g_5(44) = -1$.

Proof. (1) $(k+2)^2-3 = k^2+4k+1$: $b=1$, $j=4$, $K=7$. By (5.1), for odd $k \geq 7$: $g = \varphi_1(k+4) + c_6 + c_2 = (-1)^{k+4}+1+0 = 0$. For $k=5$ the terms are $(n, m, \varphi_n(m)) = (1,9,-1), (4,6,+1), (6,2,0)$ (the term $n=4$ has $m=6>4$; by (4.3), $\varphi_4(6) = c_6 + \sum_{x=0}^{1}(-1)^x Q(x) = 1+(1-1) = 1$), so $g_5(46) = 0$. For $k=3$: $d=22$, terms $(1,7,-1), (2,6,0), (4,2,0)$, so $g_3(22) = -1$.

(2) k^2+2k+4: $b=2$, $j=2$, $K=8$. For $k \geq 9$, (5.1) gives $g = \varphi_2(k+2)+c_6 = \varphi_2(k+2)+1$, which is 0 iff $\varphi_2(k+2) = -1$ iff $k+2 \equiv 1 \pmod 4$ (Lemma 4.6) iff $k \equiv 3 \pmod 4$. The odd prime powers $k \equiv 3 \pmod 4$ below 9 are $3, 7$. For $k=7$: $d=67$, terms $(2,9,-1), (5,6,+2)$ [$\varphi_5(6) = c_6 + Q(0) = 2$], so $g_7(67) = +1$. For $k=3$: $d=19$, terms $(1,6,+1), (2,5,-1), (4,1,-1)$, so $g_3(19) = -1$.

(3) k^2+3k+4: $b=2$, $j=3$, $K=9$. For $k \geq 9$, (5.1) gives $g = \varphi_2(k+3)+c_7 = \varphi_2(k+3)-1$, which is 0 iff $\varphi_2(k+3) = 1$ iff $k+3 \equiv 0 \pmod 4$ iff $k \equiv 1 \pmod 4$. The only odd prime power $k \equiv 1 \pmod 4$ below 9 is 5: $d=44$, terms $(2,8,+1), (3,7,-2)$, so $g_5(44) = -1$. ■

(The term lists were computed by exact integer arithmetic. For all odd prime powers $k \leq 1001$ the exact values of g_k at these d were also computed directly: the only nonzero values are those listed. Appendix A.1.)

Theorem 5.5 (zero classes for $b \in \{1,2\}$). Let $b \in \{1,2\}$, $j \in \mathbb{Z}$, and $k \geq K$ an odd prime power as in Theorem 5.3. Then $g_k(k^2+jk+b^2) = 0$ in exactly the following cases.

- $b=1$: $j \in \{-2,-1,1,4\}$ (all k).
- $b=2$, $-4 \leq j \leq 3$: $k \equiv 1,3,1,1,1,3,3,1 \pmod 4$ for $j = -4,-3,-2,-1,0,1,2,3$ respectively.
- $b=2$, $j \leq -5$: $k \equiv 1 \pmod 4$ if $j \equiv 1,2 \pmod 4$ and $k \equiv 3 \pmod 4$ if $j \equiv 0,3 \pmod 4$.
- $b=2$, $j \geq 4$: never.

Proof. Let k be odd. We use (5.1) and the values $c_0, \dots, c_7 = 1,-1,0,-1,1,-1,1,-1$ and $c_m = (-1)^m Q(m)$.

$b = 1$. $\varphi_1(k+j) = (-1)^{k+j} = -(-1)^j$. For $j \le -3$ only this term is present, so $g = \pm 1 \neq 0$. For $-2 \le j \le 1$: $j = -2$: $-1 + c_0 = 0$; $j = -1$: $+1 + c_1 = 0$; $j = 0$: $-1 + c_2 = -1$; $j = 1$: $+1 + c_3 = 0$. For $j \ge 2$: $g = -(-1)^j + (-1)^j\big(Q(j+2) + Q(j-2)\big) = (-1)^j\big(Q(j+2) + Q(j-2) - 1\big)$ (here c_{j+2} and c_{j-2} have the sign $(-1)^j$). This vanishes iff $Q(j+2) + Q(j-2) = 1$. Since $Q(j+2) \ge 1$ and $Q(j-2) \ge 1$ unless $j - 2 = 2$ (Lemma 4.4), this holds iff $j = 4$ ($Q(6) + Q(2) = 1 + 0$). For $j = 2, 3$: $Q(4) + Q(0) = 2$, $Q(5) + Q(1) = 2$, nonzero.

$b = 2$, $j \ge 4$. $\varphi_2(k+j) \in \{1, -1, 0, 0\}$ and $c_{j+4} + c_{j-4} = (-1)^j\big(Q(j+4) + Q(j-4)\big)$ with $Q(j+4) \ge 2$ ($j + 4 \ge 8$) and $Q(j-4) \ge 0$, so $|c_{j+4} + c_{j-4}| \ge 2 > |\varphi_2(k+j)|$: $g \neq 0$.

$b = 2$, $-4 \le j \le 3$. Here $g = \varphi_2(k+j) + c_{j+4}$, which vanishes iff $\varphi_2(k+j) = -c_{j+4}$. For $j + 4 = 0, \dots, 7$ we have $-c_{j+4} = -1, 1, 0, 1, -1, 1, -1, 1$. By Lemma 4.6, $\varphi_2(m) = -1$ iff $m \equiv 1$, $= 1$ iff $m \equiv 0$, $= 0$ iff $m \equiv 2, 3 \pmod 4$. So: $j = -4$: $k - 4 \equiv 1$, $k \equiv 1$. $j = -3$: $k - 3 \equiv 0$, $k \equiv 3$. $j = -2$: $k - 2 \in \{2, 3\} \bmod 4$, k odd gives $k \equiv 1$. $j = -1$: $k - 1 \equiv 0$, $k \equiv 1$. $j = 0$: $k \equiv 1$. $j = 1$: $k + 1 \equiv 0$, $k \equiv 3$. $j = 2$: $k + 2 \equiv 1$, $k \equiv 3$. $j = 3$: $k + 3 \equiv 0$, $k \equiv 1$.

$b = 2$, $j \le -5$. Then $j < -2b$ and $j < 2b$, so $g = \varphi_2(k+j)$, zero iff $k + j \equiv 2, 3 \pmod 4$. With k odd: j even requires $k + j \equiv 3$, i.e. $k \equiv 3 - j$; j odd requires $k + j \equiv 2$, i.e. $k \equiv 2 - j$. These give $k \equiv 1$ for $j \equiv 2, 1$ and $k \equiv 3$ for $j \equiv 0, 3 \pmod 4$. ■

(For $b = 2$ and $j \le -5$, the condition $k \ge K$ reads $k \ge -j$.) The cases $(b, j) = (1, 4)$, $(2, 2)$, $(2, 3)$ are the three families of Corollary 5.4. Appendix A.1 lists the exact values of (5.1) for all (b, j) with $-8 \le j \le 16$ and all odd prime powers $k \le 500$ with $k \ge K$ against the exact coefficients of g_k: 5121 cases, no discrepancy.

6 The value of g_k at $(k+2)^2 - 3$ for odd k with two prime factors

Throughout this section $d_k = (k+2)^2 - 3 = k^2 + 4k + 1$.

6.1 Reduction to one pair of terms

For a positive integer s and an integer $k > 2s$ with $k \mid s^2 - 1$ put

$$t = \frac{s^2 - 1}{k}, \quad m_1 = k + 4 - t, \quad m_2 = 4 + 2s - t, \quad n_2 = k - s, \quad u = k - 2s,$$

$$A_1 = A_s(m_1), \qquad A_2 = A_{n_2}(m_2)$$

(so $m_1 - m_2 = u$).

Lemma 6.1 (pair reduction). Let $k \ge 7$ and $\Xi_k = \{s : 1 < s < k/2,\ s^2 \equiv 1 \pmod k\}$. Then

$$g_k(d_k) = (-1)^k + 1 + \sum_{s \in \Xi_k} \big[\varphi_s(m_1) + \varphi_{k-s}(m_2)\big], \tag{6.1}$$

and $\varphi_s(m_1) = (-1)^{m_1} A_1$, $\varphi_{k-s}(m_2) = (-1)^{m_2} A_2$. For odd k the bracket equals $(-1)^{t+1}(A_1 - A_2)$.

Proof. A term n of $g_k(d_k)$ has $n^2 \le k^2 + 4k + 1 < (k+2)^2$, so $n \le k + 1$, and $n^2 \equiv d_k \equiv 1 \pmod k$. The residues $r \in \{0, \dots, k-1\}$ with $r^2 \equiv 1$ include 1 and $k - 1$ ($r = 0$ is excluded since $k \ge 2$), and the others $1 < r < k - 1$ come in pairs $\{s, k - s\}$ with $s \in \Xi_k$ ($s = k/2$ is not a root for $k \ge 4$: if $k = 2h$ with h odd then $h^2 \equiv 1 \bmod 2h$ forces $h = 1$, and if $4 \mid k$ then $(k/2)^2 \equiv 0 \bmod k$). For $n \le k + 1$ the possible n are therefore $n = 1$, $n = k - 1$, $n = k + 1$, and $n \in \{s, k - s\}$, $s \in \Xi_k$ (note $r + k > k + 1$ for $r \ge 2$).

- $n = 1$: $m = (d_k - 1)/k = k + 4$; term $\varphi_1(k+4) = (-1)^k$.

- $n = k-1$: $m = \big(k^2+4k+1-(k-1)^2\big)/k = 6 \le k-1$, so the term is $c_6 = 1$ (Lemma 4.1, 4.3).
- $n = k+1$: $m = \big(k^2+4k+1-(k+1)^2\big)/k = 2$; the term is $c_2 = 0$.
- $n = s$: $m = (k^2+4k+1-s^2)/k = k+4-t = m_1$; term $\varphi_s(m_1)$.
- $n = k-s$: $m = \big(k^2+4k+1-(k-s)^2\big)/k = 4+2s-t = m_2$; term $\varphi_{k-s}(m_2)$.

This is (6.1). By (4.2), $\varphi_n(m) = (-1)^m A_n(m)$. For odd k: $m_1 \equiv 1+t$ and $m_2 \equiv t \pmod 2$, so the bracket is $(-1)^{t+1}A_1 + (-1)^t A_2$. ■

Lemma 6.2 (root geometry). Let k be odd, $s \ge 3$, $k > 2s$, $k \mid s^2-1$. Then: (i) $1 \le t < s/2$; (ii) $u = k-2s$ is odd and $u \ge 3$, $u^2 \ge k+4$, $m_1 \le u^2-1$ and hence $u \ge \sqrt{m_1+1}$; (iii) $\frac{3}{2}s+4 < m_2 \le 2s+3$ and $m_2 < m_1$; (iv) $n_2 \ge s+3$, and with $y_2 = m_2-n_2-1$: $y_2 \le s-1$ whenever $m_2 > n_2$.

Proof. (i) $t \ge 1$ since $s^2-1 \ge k > 0$ is a positive multiple of k; and $k > 2s$ gives $t < (s^2-1)/(2s) < s/2$. (ii) u is odd since k is odd. $u \equiv -2s \pmod k$ gives $u^2 \equiv 4s^2 \equiv 4 \pmod k$. If $u = 1$ then $k \mid -3$, i.e. $k = 3$, contradicting $k > 2s \ge 6$. So $u \ge 3$, $u^2-4 > 0$ is a positive multiple of k, whence $u^2 \ge k+4$. Then $m_1 = k+4-t \le k+3 \le u^2-1$. (iii) $m_2 = 2s+4-t \le 2s+3$ and $m_2 > 2s+4-s/2$. $m_1 - m_2 = u > 0$. (iv) $k = 2s+u \ge 2s+3$. If $m_2 > n_2$: $y_2 = (2s+4-t)-(k-s)-1 = s+3-t-u \le s+3-1-3$. ■

6.2 Estimates for A_1 and A_2

Lemma 6.3. Let $s \ge 3$, k odd, $k > 2s$, $k \mid s^2-1$, and assume that Q is nondecreasing on $[2,\infty)$ (Lemma 6.5 below).

1. For every $m \ge s+4$: $A_s(m) \;\ge\; Q(m) - \dfrac{m-s+1}{2}\,Q(m-s-1)$.
2. Let $e_0 = s+1$ if s is odd and $e_0 = s+2$ if s is even. If $D \ge 0$ is a sum of integers from the set $\{e \text{ even}: \; s < e \le 2s\}$, then $A_s(m) \ge A_s(m-D)$. A number D is such a sum if $D = 0$ or if D is even, $D \ge e_0$ and: (s odd) always; (s even, $s \ge 4$) $D \ne 2s+2$.
3. $A_2 \le Q(m_2) + Q(s)$, and $A_2 = Q(m_2)$ if $m_2 \le n_2$.
4. If $m_1 \ge 5s+8$, then $n_2 > m_2$ and $A_2 = Q(m_2) \le Q(2s+3)$.

Proof. (1) By (4.2), $A_s(m)$ is at least the number of partitions of m into distinct odd parts all $\le s$ (take $\mu = \emptyset$). The number of partitions of m into distinct odd parts having at least one part $j > s$ is at most $\sum_{j \text{ odd},\ s<j\le m} Q(m-j)$ (remove the part j). The arguments $x = m-j$ have the parity of $m-1$ and satisfy $0 \le x \le y := m-s-1$. For $y \ge 3$, $Q(x) \le Q(y)$ for $2 \le x \le y$ (monotonicity) and for $x \in \{0,1\}$ (as $Q(x) = 1 \le Q(y)$, Lemma 4.4). The number of such x is at most $\lfloor y/2 \rfloor + 1 \le (m-s+1)/2$. Hence $A_s(m) \ge Q(m) - \frac{m-s+1}{2}Q(m-s-1)$.

(2) By (4.2), adjoining a fixed multiset of even parts in $(s, 2s]$ to μ is an injective map from the pairs counted by $A_s(m-D)$ to those counted by $A_s(m)$. For j parts, the attainable sums are exactly all even numbers in $[je_0, 2js]$ (each part ranges over all even integers in $[e_0, 2s]$). For s odd, $e_0 = s+1$, and consecutive intervals overlap or touch: $(j+1)(s+1) \le 2js+2$ iff $s-1 \le j(s-1)$. So every even $D \ge s+1$ is attainable. For $s \ge 4$ even, $e_0 = s+2$: for $j = 1, 2$ the intervals are $[s+2, 2s]$ and $[2s+4, 4s]$, missing only $2s+2$ between them; for $j \ge 2$, $(j+1)(s+2) \le 2js+2$ iff $s(j-1) \ge 2j$, true for $s \ge 4$.

(3) By Lemma 6.2 (iv), $n_2 \ge s+3$ so $m_2 \le 2s+3 \le 2n_2+1$ and (4.3) applies: $A_2 = Q(m_2) + (-1)^{m_2}\sum_{x=0}^{y_2}(-1)^x Q(x)$ for $y_2 = m_2-n_2-1 \ge 0$ (empty sum if $m_2 \le n_2$). Since $Q(0) = Q(1)$, the sum equals $\sum_{x=2}^{y_2}(-1)^x Q(x)$ for $y_2 \ge 1$, an alternating sum of a nondecreasing nonnegative

sequence, which has absolute value at most $Q(y_2)$; for $y_2 = 0$ it equals $1 \le Q(s)$. As $y_2 \le s-1$, $A_2 \le Q(m_2) + Q(s)$.

(4) $k = m_1 + t - 4 \ge 5s+5$, so $n_2 = k - s \ge 4s+5 > 2s+3 \ge m_2$ and $A_2 = Q(m_2) \le Q(2s+3)$ by monotonicity. ■

6.3 Chern's bound for $Q(n)$

For $n \ge 1$ let $v(n) = 24n-1$, $x(n) = \pi\sqrt{v(n)}/12$, and

$$\psi(n) = \frac{\sqrt{6}\, e^{x(n)}}{v(n)^{3/4}}, \qquad M(n) = \frac{\pi}{\sqrt{v(n)}}\, I_1\big(x(n)\big),$$

with I_1 the modified Bessel function.

Cited result. Let $G(q) = \prod_{r=1}^{R} (q^{m_r}; q^{m_r})_\infty^{\delta_r} = \sum g(n) q^n$ with distinct positive integers m_r and nonzero integers δ_r, $\sum \delta_r \ge 0$. Put $\Delta_1 = -\frac{1}{2}\sum \delta_r$, $\Delta_2 = \sum m_r \delta_r$, $\Delta_3(\ell) = -\sum \delta_r \gcd^2(m_r,\ell)/m_r$, $\Delta_4(\ell) = \prod (m_r/\gcd(m_r,\ell))^{-\delta_r/2}$, $L = \operatorname{lcm}(m_r)$, $L_{>0} = \{1 \le \ell \le L : \Delta_3(\ell) > 0\}$. S. Chern [20, Theorem 1.1] states: if $\Delta_1 \le 0$ and $\min_r \gcd^2(m_r,\ell)/m_r \ge \Delta_3(\ell)/24$ for all $1 \le \ell \le L$, then for all integers $n > -\Delta_2/24$,

$$g(n) = E(n) + \sum_{\ell \in L_{>0}} 2\pi \Delta_4(\ell) \Big(\frac{24n+\Delta_2}{\Delta_3(\ell)}\Big)^{-\frac{\Delta_1+1}{2}} \sum_{\substack{1 \le k \le N \\ k \equiv \ell \ (L)}} \frac{I_{-\Delta_1-1}\big(\frac{\pi}{6k}\sqrt{\Delta_3(\ell)(24n+\Delta_2)}\big)}{k} \sum_{\substack{0 \le h < k \\ (h,k)=1}} \omega_{h,k}\, e\Big(-\frac{nh}{k}\Big), \quad (6.2)$$

with $N = \lfloor \sqrt{2\pi(n+\Delta_2/24)} \rfloor$ and $|\omega_{h,k}| = 1$. The explicit bound for $E(n)$ is [20, (3.10)] (announced in [20, Remark 3 after Theorem 1.1]); with $\nu = n + \Delta_2/24$, $\varrho = 1/N^2$, and $\Delta_1 = 0$ (so $\Xi_{\Delta_1}(N) = 1$):

$$|E(n)| \le \frac{N^2}{\pi\nu} e^{2\pi\varrho\nu} \sum_{\ell \in L_{>0}} e^{\pi\Delta_3(\ell)/3} + 2e^{2\pi\varrho\nu}\Big\{ \sum_{\ell=1}^{L} \Delta_4(\ell) \exp\Big(\frac{\pi\Delta_3(\ell)}{24} + \sum_r \frac{|\delta_r| e^{-\pi \gcd^2(m_r,\ell)/m_r}}{(1 - e^{-\pi \gcd^2(m_r,\ell)/m_r})^2}\Big) - \sum_{\ell \in L_{>0}} \Delta_4(\ell) e^{\pi\Delta_3(\ell)/24}\Big\}. \quad (6.3)$$

(Numbering as in arXiv:1805.08759v1.)

Application. $\sum_n Q(n) q^n = (-q;q^2)_\infty = \dfrac{(q^2;q^2)_\infty^2}{(q;q)_\infty (q^4;q^4)_\infty}$ (since $(-q;q^2)_\infty = (q^2;q^4)_\infty/(q;q^2)_\infty$ and $1/(q;q^2)_\infty = (q^2;q^2)_\infty/(q;q)_\infty$). Here $m = (1,2,4)$, $\delta = (-1,2,-1)$, $\sum \delta_r = 0$, $\Delta_1 = 0$, $\Delta_2 = -1$, $L = 4$, and

$$(\Delta_3(\ell))_{\ell=1..4} = (\tfrac{1}{4}, -2, \tfrac{1}{4}, 1), \quad (\Delta_4(\ell)^2)_{\ell=1..4} = (1,2,1,1), \quad L_{>0} = \{1,3,4\}.$$

The inequality $\min_r \gcd^2(m_r,\ell)/m_r \ge \Delta_3(\ell)/24$ holds for $\ell = 1,2,3,4$ (the minima are $\frac{1}{4}, 1, \frac{1}{4}, 1$; exact rational comparison in `chern_margins.py`). So (6.2) holds for all $n \ge 1$. The term $k = 1$ of (6.2) is $2\pi \cdot 1 \cdot (4v)^{-1/2} I_1\big(\frac{\pi}{6}\sqrt{v/4}\big) = M(n)$ ($\omega_{1,1} = 1$), where $v = v(n)$ and $I_{-1} = I_1$.

Lemma 6.4 (Bessel bounds). For $y > 0$:

$$\Big(1 - \frac{3}{4y}\Big)\frac{e^y}{\sqrt{2\pi y}} \;\le\; I_1(y) \;\le\; \min\Big(\frac{ye^y}{2},\; \frac{e^y}{\sqrt{2\pi y}}\Big) \le 0.4302\, e^y.$$

Moreover I_1 is increasing and $I_1'(y) = I_0(y) - I_1(y)/y$, $I_0 \ge I_1$.

Proof. $I_1(y) = \frac{y}{\pi}\int_{-1}^{1}\sqrt{1-t^2}\,e^{yt}\,dt$. Substituting $t = 1 - w$ gives $I_1(y) = \frac{y}{\pi}e^y\int_0^2\sqrt{w(2-w)}\,e^{-yw}dw$. *Upper:* $\sqrt{w(2-w)} \le \sqrt{2w}$ and $\int_0^\infty\sqrt{2w}e^{-yw}dw = \sqrt{2}\,\Gamma(\frac{3}{2})y^{-3/2}$ give $I_1 \le e^y/\sqrt{2\pi y}$; also $e^{yt} \le e^y$ and $\int_{-1}^{1}\sqrt{1-t^2}dt = \pi/2$ give $I_1 \le ye^y/2$. Since $y/2$ is increasing and $(2\pi y)^{-1/2}$ is decreasing, $\min(y/2, (2\pi y)^{-1/2})$ is at most the common value at the crossing point $y = (2/\pi)^{1/3}$, namely $\frac{1}{2}(2/\pi)^{1/3} = 0.43012\ldots < 0.4302$. *Lower:* for $w \in [0,2]$, $\sqrt{w(2-w)} = \sqrt{2w}\sqrt{1-w/2} \ge \sqrt{2w}(1-w/2)$, and $\sqrt{2w}(1-w/2) < 0$ for $w > 2$, so $\int_0^2\sqrt{2w}(1-\frac{w}{2})e^{-yw}dw \ge \int_0^\infty\sqrt{2w}(1-\frac{w}{2})e^{-yw}dw = \sqrt{2}\big[\Gamma(\frac{3}{2})y^{-3/2} - \frac{1}{2}\Gamma(\frac{5}{2})y^{-5/2}\big]$. With $\Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$, $\Gamma(\frac{5}{2}) = \frac{3\sqrt{\pi}}{4}$ this gives the lower bound. The series $I_1(y) = \sum_{i\ge0}(y/2)^{2i+1}/(i!(i+1)!)$ has positive coefficients, so I_1 is increasing; $I_1' = I_0 - I_1/y$ is a standard identity; $I_0(y) - I_1(y) = \frac{1}{\pi}\int_0^\pi e^{y\cos\theta}(1-\cos\theta)d\theta \ge 0$. ■

Lemma 6.5. Let $C_E = 138.1955$ and $K = 1.3828$ (the numbers are upper roundings, see below), and $\bar B(n) = C_E + Ke^{x(n)/2}$, $\bar\beta(n) = \bar B(n)/\psi(n)$.

1. For every integer $n \ge 300$: $|Q(n) - M(n)| \le \bar B(n)$. Consequently, for $n \ge 300$,

$$\Big(1 - \frac{3}{4x(n)} - \bar\beta(n)\Big)\psi(n) \;\le\; Q(n) \;\le\; \big(1 + \bar\beta(n)\big)\psi(n). \tag{6.4}$$

2. $\bar\beta$ is nonincreasing on $[300,\infty)$, and $\bar\beta(300) \le 6.7\cdot10^{-3}$, $\bar\beta(1000) \le 1.7\cdot10^{-6}$, $\bar\beta(1804) \le 2.6\cdot10^{-9}$, $\bar\beta(6000) \le 1.2\cdot10^{-18}$.
3. $Q(n+1) \ge Q(n)$ for all $n \ge 2$.

Proof. (1) Let $n \ge 300$, $\nu = n - \frac{1}{24} \ge 299.9$, $N = \lfloor\sqrt{2\pi\nu}\rfloor \ge 43$. By the definition of N, $N^2 \le 2\pi\nu < (N+1)^2$, hence $N^2/(\pi\nu) \le 2$ and $e^{2\pi\varrho\nu} = e^{2\pi\nu/N^2} < e^{(1+1/N)^2} \le e^{(44/43)^2}$. In (6.3) all other quantities depend only on the data of Q. Evaluating them in interval arithmetic ($\sum_{\ell\in L_{>0}} e^{\pi\Delta_3(\ell)/3} = 5.448\ldots$ and the curly bracket $= 18.803\ldots$) gives $|E(n)| \le C_E$ with $C_E = 2e^{(44/43)^2}(5.4482 + 18.8031) \le 138.1955$ (value of the interval upper end: $138.19547\ldots$).

The terms $k \ge 3$ in (6.2) with $k \equiv 1, 3$ (4) have $\Delta_3 = \frac{1}{4}$, $\Delta_4 = 1$ and size at most $\frac{\pi}{\sqrt{v}}I_1(x/k)\cdot\frac{\#\{h\}}{k} \le \frac{\pi}{\sqrt{v}}I_1(x/k)$ (using $|\omega_{h,k}| = 1$, $\#\{h\} \le k$, I_1 increasing); those with $4 \mid k$ have $\Delta_3 = 1$ and size at most $\frac{2\pi}{\sqrt{v}}I_1(2x/k)$, $2x/k \le x/2$. In both cases the Bessel argument is at most $x/2$, so by Lemma 6.4 each such term is at most $\frac{2\pi}{\sqrt{v}}\cdot 0.4302\,e^{x/2}$. There are at most $N \le \sqrt{2\pi\nu} \le \sqrt{\pi v/12}$ terms (since $2\pi\nu = \pi v/12$). So their sum is at most $2\pi\sqrt{\pi/12}\cdot 0.4302\,e^{x/2} \le Ke^{x/2}$ with $K = 1.3828$ (evaluation: $1.38280\ldots$). Together with (6.2), $|Q(n) - M(n)| \le C_E + Ke^{x/2} = \bar B(n)$. Lemma 6.4 gives $(1 - \frac{3}{4x})\frac{e^x}{\sqrt{2\pi x}} \le I_1(x) \le \frac{e^x}{\sqrt{2\pi x}}$, and $\frac{\pi}{\sqrt{v}}\frac{e^x}{\sqrt{2\pi x}} = \psi(n)$ (because $2\pi x = \pi^2\sqrt{v}/6$). Hence $(1 - \frac{3}{4x})\psi \le M \le \psi$, which with the bound on $|Q - M|$ gives (6.4).

(2) $\frac{d}{dn}\ln\psi = \frac{\pi}{\sqrt{v}} - \frac{18}{v} > 0$ for $\sqrt{v} > 18/\pi$, so C_E/ψ is decreasing; $Ke^{x/2}/\psi = \frac{K}{\sqrt{6}}v^{3/4}e^{-x/2}$ and $\frac{d}{dn}\ln(v^{3/4}e^{-x/2}) = \frac{18}{v} - \frac{\pi}{2\sqrt{v}} < 0$ for $\sqrt{v} > 36/\pi$. The numerical values are interval evaluations (`chern_margins.py`).

(3) For $2 \le n \le 35199$ this is checked on the exact table of Q (Appendix A.2). Let $n \ge 35200$, $x = x(n) \ge 240$, $v = v(n)$. Differentiating, $M'(\xi) = \pi^2 v(\xi)^{-1}\big[I_0(x(\xi)) - 2I_1(x(\xi))/x(\xi)\big]$ (use $\frac{dx}{d\xi} = \pi/\sqrt{v}$, $12/\sqrt{v} = \pi/x$ and $I_1' = I_0 - I_1/x$), which is $\ge \pi^2 v(\xi)^{-1}(1 - 2/x(\xi))I_1(x(\xi))$ by $I_0 \ge I_1$. On $[n, n+1]$: $v(\xi) \le v(n+1)$, and $(1 - 2/x)$, I_1 are increasing, so with Lemma 6.4,

$$M(n+1) - M(n) \;\ge\; \Lambda(n) := \pi\Big(1 - \tfrac{2}{x}\Big)\Big(1 - \tfrac{3}{4x}\Big)\frac{\psi(n)\sqrt{v(n)}}{v(n+1)} \;\ge\; 3\,\frac{\psi(n)}{\sqrt{v}} = 3\sqrt{6}\,e^x v^{-5/4}$$

(using $x \ge 240$, $v(n+1) \le 1.0001\,v$). On the other hand $x(n+1) - x(n) \le \pi/\sqrt{v} < 0.0035$ so $\bar{B}(n) + \bar{B}(n+1) \le 2C_E + 2.004\,K e^{x/2}$. By (1), $Q(n+1) - Q(n) \ge \Lambda(n) - \bar{B}(n) - \bar{B}(n+1)$, which is positive as soon as

$$2C_E\, v^{5/4} e^{-x} + 2.004\,K\, v^{5/4} e^{-x/2} < 3\sqrt{6}.$$

The left side is nonincreasing in n (for $\sqrt{v} > 60/\pi$, $\frac{d}{dn}\ln(v^{5/4}e^{-x/2}) = \frac{30}{v} - \frac{\pi}{2\sqrt{v}} < 0$, and similarly for e^{-x}), and at $n = 35200$ it is $< 10^{-40}$ (interval evaluation). ■

6.4 The inequality $A_1 > A_2$

Theorem 6.6. Let k be odd, $s \ge 3$, $k > 2s$, $k \mid s^2 - 1$. Then $A_1 > A_2$, except for $(s,k) = (4,15)$ and $(8,21)$, where $A_1 = A_2 = 2$ and $A_1 = A_2 = 6$ respectively.

The proof has two parts: a finite exact computation for $3 \le s \le 2500$ (Proposition 6.7) and an analytic argument for $s \ge 2501$ (Proposition 6.8).

Proposition 6.7 (finite range; computer-assisted). For every pair (s,k) with $3 \le s \le 2500$, k odd, $k > 2s$, $k \mid s^2 - 1$ (there are 15 616 pairs), $A_1 > A_2$ except for $(4,15)$ and $(8,21)$.

Proof. All arithmetic is exact integer arithmetic. The table of $Q(0), \dots, Q(35200)$ is computed exactly. For each pair, $A_2 = A_{n_2}(m_2)$ is computed by (4.3) (valid since $m_2 \le 2n_2 + 1$, Lemma 6.2). A lower bound $\mathrm{LB} \le A_1$ is computed as follows: choose $m'' \le m_1$ with $m'' \equiv m_1 \pmod 2$ such that $D = m_1 - m''$ is a sum of even integers in $(s, 2s]$ (this is decided exactly by the criterion of Lemma 6.3(2): $D = 0$, or D even and $\lceil D/2s \rceil e_0 \le D$), and set $\mathrm{LB} = A_s(m'')$ computed by (4.3) if $m'' \le 2s + 1$, and otherwise $\mathrm{LB} = Q(m'') - \sum_{j \text{ odd},\ s < j \le m''} Q(m'' - j)$. By Lemma 6.3(2) and the proof of Lemma 6.3(1), $\mathrm{LB} \le A_s(m'') \le A_1$ (the inequality $Q(x) \le Q(y)$ is not used here: the exact sum is subtracted). The candidates are the integers m'' with $m_2 < m'' \le \min(m_1, 35200)$ satisfying the two conditions above (they are tried in the order m_1, $4s+3, 4s+2, 4s+1, 4s$, then downward from $\min(m_1, 8s, 35200)$; the order only affects speed), and the first with $\mathrm{LB} > A_2$ is accepted. The result: 15 587 pairs are certified ($\mathrm{LB} > A_2$, so $A_1 > A_2$). The remaining 29 pairs all have $s \le 32$; they are listed in the output file and are settled by the exact computation of $A_1 = [x^{m_1}] \prod_{j \text{ odd} \le s}(1 + x^j) \prod_{s/2 < i \le s}(1 - x^{2i})^{-1}$ (Lemma 4.2), which was carried out for **all** pairs with $s \le 300$ (995 pairs): $A_1 > A_2$ for every pair except $(4,15)$ and $(8,21)$; the smallest ratio A_1/A_2 among the others is 1.7811 at $(s,k) = (34,77)$. For the two exceptions the terms of $g_k(d_k)$ are listed in Section 6.6; they give $A_1 = A_2 = 2$ for $(4,15)$ and $A_1 = A_2 = 6$ for $(8,21)$. ■

Proposition 6.8 (analytic range). Under Lemma 6.5, $A_1 > A_2$ for all (s,k) as in Theorem 6.6 with $s \ge 2501$.

Proof. Let $s \ge 2501$. All arguments of Q below are ≥ 300, so (6.4) applies, and $\bar{\beta}$ is nonincreasing. Define

$$\lambda(n) = 1 - \frac{3}{4x(n)} - \bar{\beta}(n),$$

which is increasing in n.

Case 1: $m_1 \le 5s+7$**.** Put $y_1 = m_1 - s - 1$, $n_a = \frac{3}{2}s + 4$. By Lemma 6.2, $m_1 > m_2 > n_a$ and $y_1 > s/2 + 3$. By Lemma 6.3 (1),(3),

$$A_1 - A_2 \;\ge\; Q(m_1) - \tfrac{m_1-s+1}{2} Q(y_1) - Q(m_2) - Q(s).$$

Divide by $\psi(m_1)$ and use (6.4):

$$\frac{A_1 - A_2}{\psi(m_1)} \;\ge\; \lambda(m_1) - \tfrac{m_1-s+1}{2}\big(1+\bar\beta(y_1)\big)\frac{\psi(y_1)}{\psi(m_1)} - \big(1+\bar\beta(m_2)\big)\frac{\psi(m_2)}{\psi(m_1)} - \big(1+\bar\beta(s)\big)\frac{\psi(s)}{\psi(m_1)}.$$

We bound the four terms.

First term. $\lambda(m_1) \ge \lambda(n_a) =: L$.

Third term ($m_2 = m_1 - u$). Write $\psi(m_2)/\psi(m_1) = e^{f(u)}$ with $v_1 = v(m_1)$ and

$$f(u) = -\big(x(m_1) - x(m_1-u)\big) + \tfrac{3}{4}\ln\frac{v_1}{v_1 - 24u}.$$

Then $f'(u) = \dfrac{18 - \pi\sqrt{v_1 - 24u}}{v_1 - 24u} < 0$ (as $\sqrt{v(m_2)} > 18/\pi$), so f is decreasing in u and, with $u_0 = \sqrt{m_1+1} \le u$ (Lemma 6.2), $f(u) \le f(u_0)$. By concavity of $x(\cdot)$, $x(m_1) - x(m_1-u_0) \ge u_0 x'(m_1) = \pi u_0/\sqrt{v_1} > \pi/\sqrt{24}$ (as $(m_1+1)/(24m_1-1) > 1/24$). The logarithmic term is at most $\frac{3}{4}\ln\big(1 - 24u_0/v_1\big)^{-1}$, which is decreasing in m_1, hence at most its value at $m_1 = n_a$. Therefore

$$\frac{\psi(m_2)}{\psi(m_1)} \le r_0 := e^{-\pi/\sqrt{24}}\Big(1 - \frac{24\sqrt{n_a+1}}{24n_a - 1}\Big)^{-3/4}, \qquad \text{third term} \le (1+\bar\beta(n_a))\, r_0 =: T.$$

Second term. $m_1 - s + 1 \le 4s+8$. By concavity, $x(m_1) - x(y_1) \ge (s+1)x'(m_1) = \pi(s+1)/\sqrt{v_1} \ge \pi(s+1)/\sqrt{120s+167}$ (since $m_1 \le 5s+7$). Also $v(m_1)/v(y_1) = \frac{24m_1-1}{24m_1-24s-25}$ is decreasing in m_1, so $\le \frac{36s+95}{12s+71} < 3$ at $m_1 = n_a$. Hence the second term is at most

$$\theta(s) = (2s+4)\big(1+\bar\beta(\tfrac{s}{2}+3)\big)3^{3/4}e^{-\pi(s+1)/\sqrt{120s+167}}.$$

Fourth term. $x(m_1) - x(s) \ge (m_1 - s)x'(m_1) = \pi\frac{m_1-s}{\sqrt{24m_1-1}} \ge \pi\frac{s/2+4}{\sqrt{120s+167}}$ (numerator minimal at $m_1 = n_a$, denominator maximal at $m_1 = 5s+7$), and $v(m_1)/v(s) \le (120s+167)/(24s-1)$. So the fourth term is at most

$$U(s) = \big(1+\bar\beta(s)\big)e^{-\pi(s/2+4)/\sqrt{120s+167}}\Big(\frac{120s+167}{24s-1}\Big)^{3/4}.$$

Monotonicity in s. L is increasing and T is decreasing in s (both $1+\bar\beta(n_a)$ and r_0 decrease as n_a grows). For θ: $\bar\beta$ factor nonincreasing, and $\frac{d}{ds}\ln\big[(s+2)e^{-\pi(s+1)/\sqrt{120s+167}}\big] = \frac{1}{s+2} - \frac{\pi(60s+107)}{(120s+167)^{3/2}} < 0$ for $s \ge 100$ (the second fraction is $\ge 0.139\, s^{-1/2}$, the first is $\le 0.1\, s^{-1/2}$). For U: $\frac{d}{ds}\big[-\pi(\frac{s}{2}+4)(120s+167)^{-1/2}\big] = -\pi\frac{30s-156.5}{(120s+167)^{3/2}} < 0$ and $\frac{d}{ds}\frac{3}{4}\ln\frac{120s+167}{24s-1} < 0$; the $\bar\beta$ factor is nonincreasing. Hence the lower bound $\mathrm{M}_1(s) = L - T - \theta - U$ is nondecreasing in s, and at $s = 2501$ interval evaluation gives $L \ge 0.99046$, $T \le 0.53316$, $\theta \le 6.73\cdot 10^{-3}$, $U \le 2.52\cdot 10^{-3}$, so $\mathrm{M}_1(2501) \ge 0.448 > 0$ (`chern_margins.py`). Hence $A_1 > A_2$ in Case 1 for all $s \ge 2501$.

Case 2: $m_1 \ge 5s+8$**.** By Lemma 6.3 (4), $A_2 \le Q(2s+3)$. Choose $m'' \in \{4s, 4s+1, 4s+2, 4s+3\}$ with $m'' \equiv m_1 \pmod 2$ and $D = m_1 - m'' \ne 2s+2$ (of the two admissible values, the corresponding D differ by 2, so one of them is not $2s+2$). Then $D \ge m_1 - (4s+3) \ge s+5$ is even, so by Lemma

6.3 (2), $A_1 \geq A_s(m'')$. By Lemma 6.3 (1) and monotonicity, $A_s(m'') \geq Q(4s) - (\frac{3}{2}s+2)Q(3s+2)$ (as $m'' \geq 4s$, $\frac{m''-s+1}{2} \leq \frac{3}{2}s+2$, $m''-s-1 \leq 3s+2$). Hence

$$\frac{A_1 - A_2}{\psi(4s)} \;\geq\; \lambda(4s) - c_1(s) - c_2(s),$$

$$c_1 = (\tfrac{3}{2}s+2)\big(1+\bar\beta(3s+2)\big)\frac{\psi(3s+2)}{\psi(4s)}, \qquad c_2 = \big(1+\bar\beta(2s+3)\big)\frac{\psi(2s+3)}{\psi(4s)}.$$

By concavity $x(4s) - x(3s+2) \geq (s-2)\pi/\sqrt{96s-1}$ and $x(4s) - x(2s+3) \geq (2s-3)\pi/\sqrt{96s-1}$, and $v(4s)/v(3s+2) = \frac{96s-1}{72s+47}$, $v(4s)/v(2s+3) = \frac{96s-1}{48s+71}$. Thus $c_1 \leq (\frac{3}{2}s+2)(1+\bar\beta(3s+2))e^{-\pi(s-2)/\sqrt{96s-1}}\big(\frac{96s-1}{72s+47}\big)^{3/4}$ and $c_2 \leq (1+\bar\beta(2s+3))e^{-\pi(2s-3)/\sqrt{96s-1}}\big(\frac{96s-1}{48s+71}\big)^{3/4}$. These bounds are decreasing in s for $s \geq 2501$ (the logarithmic derivatives of the elementary factors are $\frac{1}{s+4/3} - \frac{\pi(48s+95)}{(96s-1)^{3/2}} + O(s^{-2}) < 0$ and $-\frac{\pi(96s+142)}{(96s-1)^{3/2}} + O(s^{-2}) < 0$, and the $\bar\beta$ factors are nonincreasing), while $\lambda(4s)$ is increasing. At $s = 2501$: $\lambda(4s) \geq 0.99415$, $c_1 \leq 5.13\cdot 10^{-4}$, $c_2 \leq 2.1\cdot 10^{-14}$, so the margin is $\geq 0.9936 > 0$. ■

Proof of Theorem 6.6. Propositions 6.7 and 6.8 cover all $s \geq 3$. ■

Dependence. Proposition 6.8 depends on [20] only through (6.2)–(6.3) (Lemma 6.5 (1)). Proposition 6.7 and the exact tables do not depend on [20]. The margins in Proposition 6.8 are large: if every occurrence of $\bar\beta$ is replaced by a constant 0.25, the lower bounds at $s = 2501$ are still positive (0.06 in Case 1), whereas $\bar\beta(1253) \leq 2\cdot 10^{-7}$.

6.5 The theorem

Theorem 6.9. Let k be an odd integer with exactly two distinct prime factors. Then $g_k\big((k+2)^2-3\big) \neq 0$, except for $k = 15$ and $k = 21$, for which $g_k\big((k+2)^2-3\big) = 0$.

For odd prime powers $k \geq 5$, $g_k\big((k+2)^2-3\big) = 0$ (Corollary 5.4).

Proof. Let $k = p^a q^b$ with $p < q$ odd primes; $k \geq 15$. The solutions of $x^2 \equiv 1 \pmod k$ are, by the Chinese remainder theorem, exactly four residues $\pm 1, \pm s_0$ with $s_0 \not\equiv \pm 1$. Hence Ξ_k consists of exactly one element s (one of $s_0, k-s_0$ lies in $(1, k/2)$), and $s \geq 3$ since $s^2 \equiv 1$, $s = 2$ would give $k \mid 3$. By Lemma 6.1 and $(-1)^k + 1 = 0$,

$$g_k\big((k+2)^2-3\big) = (-1)^{t+1}(A_1 - A_2).$$

By Theorem 6.6, $A_1 \neq A_2$ unless $(s,k) \in \{(4,15),(8,21)\}$, and for these two pairs $A_1 = A_2$, so $g_k(d_k) = 0$. ■

6.6 The cases $k = 15$ and $k = 21$

For $k = 15$, $d = 286$: the terms $(n, m, \varphi_n(m))$ are $(1,19,-1)$, $(4,18,+2)$, $(11,11,-2)$, $(14,6,+1)$, $(16,2,0)$ ($s = 4$, $t = 1$, $A_1 = A_4(18) = 2$, $A_2 = A_{11}(11) = 2$). For $k = 21$, $d = 526$: $(1,25,-1)$, $(8,22,+6)$, $(13,17,-6)$, $(20,6,+1)$, $(22,2,0)$ ($s = 8$, $t = 3$, $A_1 = A_8(22) = 6$, $A_2 = A_{13}(17) = 6$). In both cases the sum is 0.

Computer-assisted items in this section: the table of Q up to 35200 and its monotonicity; the interval evaluation of C_E, K, $\bar\beta$ and of the margins at $s = 2501$; the finite certificate of Proposition 6.7 (Appendix A.2).

7 Even k

7.1 Signs of the terms and finiteness of the zero set in "one-sign" classes

Lemma 7.1. Let $k \geq 1$ and $d \geq 0$. The term n of $g_k(d)$ equals $(-1)^{m_n} A_n(m_n)$. Two terms n, n' of $g_k(d)$ have the same sign factor $(-1)^{m_n} = (-1)^{m_{n'}}$ if and only if $n^2 \equiv n'^2 \pmod{2k}$.

Proof. The first statement is (4.2). Also $m_n - m_{n'} = (n'^2 - n^2)/k$, which is even if and only if $2k \mid n'^2 - n^2$. ■

Definition 7.2. A residue class $c \in \mathcal{S}_k$ is a *one-sign class* if $n^2 \equiv n'^2 \pmod{2k}$ whenever $n^2 \equiv n'^2 \equiv c \pmod{k}$. For such a class and every $d \equiv c \pmod{k}$, all terms of $g_k(d)$ have the same sign factor $\varepsilon_d = (-1)^{m_n}$ and

$$g_k(d) = \varepsilon_d \sum_n A_n(m_n), \qquad \text{so } g_k(d) \neq 0 \iff A_n(m_n) \geq 1 \text{ for some term } n. \tag{7.1}$$

Lemma 7.3. If $k \equiv 2 \pmod 4$, every class $c \in \mathcal{S}_k$ is a one-sign class.

Proof. Let $k = 2h$, h odd, $n^2 \equiv n'^2 \pmod k$. Then $n \equiv n' \pmod 2$, so $n^2 - n'^2 = (n-n')(n+n')$ is a product of two even numbers and $4 \mid n^2 - n'^2$; also $h \mid n^2 - n'^2$. As $\gcd(4, h) = 1$, $4h = 2k$ divides $n^2 - n'^2$. ■

Theorem 7.4. Let $k \geq 2$ and let c be a one-sign class. Let n_0 be the least integer $n \geq 3$ with $n^2 \equiv c \pmod k$, $L = \lfloor n_0/2 \rfloor + 1$, and

$$B_c = n_0^2 + k\big(2L(L-1) + 1\big).$$

Then $g_k(d) \neq 0$ for every $d \equiv c \pmod k$ with $d \geq B_c$, and $g_k(d)$ has the sign ε_d.

Proof. n_0 exists since if $n^2 \equiv c$ then also $(n+k)^2 \equiv c$ and one of $n, n+k, n+2k$ is ≥ 3. Let $d \geq B_c$. Then n_0 is a term of $g_k(d)$, with index $m = (d - n_0^2)/k \geq 2L(L-1) + 1$. By (7.1) it suffices to show $A_{n_0}(m) \geq 1$.

The even integers $2L$ and $2L+2$ lie in $(n_0, 2n_0]$: $2L = 2\lfloor n_0/2 \rfloor + 2 > n_0$ and $2L + 2 \leq 2n_0$ iff $\lfloor n_0/2 \rfloor + 2 \leq n_0$, true for $n_0 \geq 3$. Every integer $N \geq L(L-1)$ is of the form $aL + b(L+1)$ with $a, b \geq 0$: write $N = qL + r$, $0 \leq r < L$; then $q \geq L - 1 \geq r$ and $N = (q - r)L + r(L+1)$. Hence every even integer $\geq 2L(L-1)$ is a sum of parts $2L$ and $2L+2$. If m is even, take $\lambda = \emptyset$ and μ such a multiset with sum m ($m \geq 2L(L-1) + 2$). If m is odd, take $\lambda = \{1\}$ ($1 \leq n_0$) and μ with sum $m - 1 \geq 2L(L-1)$. By (4.2), $A_{n_0}(m) \geq 1$. ■

Corollary 7.5. If $k \equiv 2 \pmod 4$ then Z_k is finite, $Z_k \subset [0, \max_c B_c)$, and $\mathrm{OccFilt}_k(\infty) = \mathcal{S}_k$.

The sets Z_k for $k \equiv 2 \pmod 4$, $k \leq 34$, are computed exactly in Appendix B.2 ($d < \max_c B_c$ is a finite range; the program evaluates $g_k(d)$ exactly for all $d \leq \max_c B_c + 3000$). For example:

$$Z_2 = \{2, 8, 10, 26\}, \qquad Z_6 = \{3, 6, 12, 18, 21, 24, 30, 46, 48, 78\},$$

$$Z_{10} = \{5, 6, 10, 15, 20, 24, 29, 30, 34, 40, 45, 50, 60, 70, 80, 90, 120\},$$

$$Z_{14} = \{2, 7, 8, 11, 14, 21, 22, 28, 32, 35, 37, 42, 44, 46, 53, 56, 70,$$
$$77, 84, 86, 88, 98, 102, 112, 126, 140, 154, 168, 182, 224\},$$

with $\max_c B_c = 42, 186, 710, 1778$ and $|Z_k| = 4, 10, 17, 30$ respectively.

7.2 The case $k = 2$

Theorem 7.6. For every $d \geq 0$, $g_2(d) = 0$ or $\operatorname{sign} g_2(d) = (-1)^{\lfloor d/2 \rfloor}$. Moreover $g_2(d) \neq 0$ for all even $d \geq 42$ and all odd $d \geq 19$, and the zero set is

$$Z_2 = \{2, 8, 10, 26\}.$$

Proof. Sign. Let n be a term of $g_2(d)$, $m = (d - n^2)/2$. If d is even, $n = 2j$ and $m = d/2 - 2j^2 \equiv d/2 \pmod 2$. If d is odd, n is odd and $n^2 \equiv 1 \pmod 8$, so $m = \frac{d-1}{2} - \frac{n^2-1}{2} \equiv \frac{d-1}{2} \pmod 2$ (as $4 \mid \frac{n^2-1}{2}$). In both cases $(-1)^m = (-1)^{\lfloor d/2 \rfloor}$, independent of n, and by (4.2) every term $\varphi_n(m) = (-1)^m A_n(m)$ has that sign (or is 0).

Nonvanishing. The two classes $c = 0$ (n even) and $c = 1$ (n odd) are one-sign classes (Lemma 7.3). For $c = 0$: $n_0 = 4$, $L = 3$, $B_0 = 16 + 2(2 \cdot 3 \cdot 2 + 1) = 42$. For $c = 1$: $n_0 = 3$, $L = 2$, $B_1 = 9 + 2(2 \cdot 2 \cdot 1 + 1) = 19$. By Theorem 7.4, $g_2(d) \neq 0$ for even $d \geq 42$ and odd $d \geq 19$.

Finite range. The exact values $g_2(d)$ for $0 \leq d \leq 41$ are:

d	0	1	2	3	4	5	6	7	8	9	10	11	12	13
$g_2(d)$	1	1	0	-1	1	1	-1	-1	0	2	0	-2	1	1

d	14	15	16	17	18	19	20	21	22	23	24	25	26	27
$g_2(d)$	-1	-2	1	3	-1	-2	1	2	-2	-3	1	4	0	-4

d	28	29	30	31	32	33	34	35	36	37	38	39	40	41
$g_2(d)$	2	3	-2	-4	1	5	-2	-5	3	5	-3	-5	2	7

The zeros in $d < 42$ are $2, 8, 10, 26$; all nonzero values have the sign $(-1)^{\lfloor d/2 \rfloor}$. ■

7.3 The value at $(k+2)^2 - 3$ for even k

Theorem 7.7. Let $k \geq 8$ be even and $d_k = (k+2)^2 - 3$.

1. $g_k(d_k) = 2 + \sum_{s \in \Xi_k} (-1)^t (A_1 + A_2)$ with $t = (s^2 - 1)/k$, and every $s \in \Xi_k$ is odd.
2. If $8 \nmid k$, every t is even and $g_k(d_k) \geq 2$.
3. If $k = 2p^a$ with p an odd prime ($k \geq 10$), then $\Xi_k = \emptyset$ and $g_k(d_k) = 2$.
4. If $k = 2^a$ with $a \geq 3$, then $\Xi_k = \{k/2 - 1\}$ and $g_k(d_k) = 2 - A_1 - A_2 < 0$. Values: $g_8(97) = -3$, $g_{16}(321) = -8$, $g_{32}(1153) = -31$, $g_{64}(4353) = -236$, $g_{128}(16897) = -5174$.

Proof. (1) Lemma 6.1 with $(-1)^k + 1 = 2$; for even k, $m_1 \equiv m_2 \equiv t \pmod 2$, so the bracket is $(-1)^t(A_1 + A_2)$. A root s of $s^2 \equiv 1 \pmod k$ with k even is odd.

(2) s odd gives $8 \mid s^2 - 1 = tk$, and $v_2(k) \leq 2$ forces $v_2(t) \geq 1$. All pair terms are ≥ 0.

(3) Modulo p^a the roots are ± 1 and modulo 2 the root is 1, so $s^2 \equiv 1 \pmod{2p^a}$ implies $s \equiv \pm 1 \pmod{2p^a}$: only $s = 1, k - 1$, which are excluded from Ξ_k.

(4) The solutions of $x^2 \equiv 1 \pmod{2^a}$, $a \geq 3$, are ± 1 and $2^{a-1} \pm 1$. So $\Xi_k = \{s\}$ with $s = k/2 - 1$ (> 1), and $t = \frac{(s-1)(s+1)}{k} = \frac{(k/2-2)(k/2)}{k} = \frac{k}{4} - 1$, which is odd since $k/4$ is even. Hence $g_k(d_k) = 2 - (A_1 + A_2)$. Here $n_2 = k - s = \frac{k}{2} + 1$ and $m_2 = 4 + 2s - t = \frac{3k}{4} + 3$, which is odd (as $\frac{3k}{4}$ is even). For $k \geq 16$ and $o \in \{1, 3, 5\}$, the pair $\lambda = \{o\}$, $\mu = \{m_2 - o\}$ is admissible in (4.2) for $A_{n_2}(m_2)$: $o \leq n_2$; $m_2 - o$ is even; $m_2 - o \leq \frac{3k}{4} + 2 \leq k + 2 = 2n_2$; and $m_2 - o > \frac{k}{2} + 1 = n_2$ iff $o < \frac{k}{4} + 2$, true for $o \leq 5$ when $k \geq 16$. These are three different pairs, so $A_2 \geq 3$, $A_1 \geq 0$ and $g_k(d_k) \leq -1$. For $k = 8$ the terms of $g_8(97)$ are $(1, 12, +1)$, $(3, 11, -2)$, $(5, 9, -3)$, $(7, 6, +1)$, $(9, 2, 0)$, with sum -3. The values for $k = 16, \ldots, 128$ are exact evaluations (Appendix A.1). ■

Scope. Lemma 7.1 shows that for $4 \mid k$ there are classes c in which two roots n, n' have $n^2 \not\equiv n'^2 \pmod{2k}$; in such a class the terms of $g_k(d)$ have different signs and Theorem 7.4 does not apply. For $k = 4, 8, 12$ these classes are: $c \equiv 0$ ($k = 4$), $c \equiv 1$ ($k = 8$), $c \equiv 0, 4$ ($k = 12$) (computed by enumeration of n modulo $2k$). Nothing is claimed for such classes.

8 Parity dominance for odd $k \leq 33$

Throughout this section k is odd, $3 \leq k \leq 33$, and d is allowed for k (Definition 3.3). The notation $a_n(m)$, $A_n(x)$, E_d, O_d is that of Section 4: $a_n(m) = |\varphi_n(m)|$, $A_n(x) = \sum_m a_n(m) x^m$ is the product (4.1), and by (4.4)

$$(-1)^d g_k(d) = E_d - O_d, \qquad E_d = \sum_{n \text{ even}} a_n(m_n), \quad O_d = \sum_{n \text{ odd}} a_n(m_n).$$

8.1 Statement

For k in the table of Section 8.7, $R_k < 1$ and $D_0(k)$ are the numbers computed there (Table 8.2). Put $\eta_t = 10^{-6}$.

Theorem 8.1 (parity dominance). Let $k \in \{3, 5, \ldots, 33\}$.

1. For every allowed $d \geq D_0(k)$: $O_d > 0$ and $E_d \leq (R_k + \eta_t)\, O_d$. Consequently $(-1)^d g_k(d) \leq -(1 - R_k - \eta_t) O_d < 0$; in particular $g_k(d) \neq 0$ and $\operatorname{sign} g_k(d) = (-1)^{d+1}$.
2. Z_k is a finite set, $\max Z_k < D_0(k)$, and Z_k is the explicit set of Appendix B.1 (cardinalities and maxima in Table 8.5).
3. $\mathrm{OccFilt}_k(\infty) = \mathcal{S}_k$: the residue classes mod k that occur for all large d are exactly the squares mod k.

Part 3 follows from Part 2 and Proposition 3.4. Part 2 follows from Part 1 and the finite certificate of Section 8.7. Part 1 is proved in Sections 8.2 to 8.5: it combines (a) a local limit estimate for a family of integer-valued random variables attached to A_n (Sections 8.2, 8.3), (b) a comparison of each even term with the two odd terms $n \pm k$ (Section 8.4) and (c) an upper bound for the even terms far from the saddle, and a lower bound for one odd term (Section 8.5).

Computer-assisted steps: the finite computations of Algorithms C (cells), R (block bound), T (tail) and the modular certificate. Each is specified below and reproduced by the code listed in Appendix A.3; they use interval (ball) arithmetic with the Arb library through python-flint [21], at 128-bit working precision, except for the modular certificate, which uses exact integer arithmetic modulo two primes.

8.2 The tilted model

Let $n \geq 1$ and $0 < r < 1$. Put $t = -\ln r$ and

$$p_j = \frac{r^j}{1+r^j} = \frac{1}{1+e^{jt}} \ (j \text{ odd},\ j \leq n), \qquad w_l = r^{2l} = e^{-2lt} \ \big(\tfrac{n}{2} < l \leq n\big).$$

Let B_j (j odd $\leq n$) be independent Bernoulli variables with $P(B_j = 1) = p_j$, and N_l ($n/2 < l \leq n$, l integer) independent geometric variables with $P(N_l = i) = (1-w_l)w_l^i$, $i \geq 0$, all independent, and

$$S = S_n(r) = \sum_{j \text{ odd} \leq n} jB_j + \sum_{n/2 < l \leq n} 2l\, N_l \in \mathbb{N}.$$

Lemma 8.2. $E\, z^S = A_n(rz)/A_n(r)$ for $|z| \leq 1$. Hence $P(S = m) = a_n(m)\, r^m / A_n(r)$ for all integers m (both sides vanish for $m < 0$).

Proof. $E\, z^{jB_j} = 1 - p_j + p_j z^j = \frac{1+(rz)^j}{1+r^j}$ and $E\, z^{2lN_l} = \frac{1-w_l}{1-w_l z^{2l}} = \frac{1-r^{2l}}{1-(rz)^{2l}}$. Multiply and use (4.1). ■

Put $\mu = ES$, $\sigma^2 = \operatorname{Var} S$, κ_3 the third cumulant, and $\varphi(\theta) = Ee^{i\theta S}$. With $v_j = p_j(1-p_j) = \big(4\cosh^2(jt/2)\big)^{-1}$ and $\lambda_l = \frac{4w_l}{(1-w_l)^2} = \sinh^{-2}(lt)$:

$$\varphi(\theta) = \prod_{j \text{ odd} \leq n} \big(1 - p_j + p_j e^{ij\theta}\big) \prod_{n/2 < l \leq n} \frac{1-w_l}{1-w_l e^{2il\theta}}, \tag{8.1}$$

$$\mu = \sum_j jp_j + \sum_l \frac{2lw_l}{1-w_l}, \quad \sigma^2 = \sum_j j^2 v_j + \sum_l l^2 \lambda_l, \quad \kappa_3 = \sum_j j^3 v_j(1-2p_j) + \sum_l (2l)^3 \frac{w_l(1+w_l)}{(1-w_l)^3}. \tag{8.2}$$

(The cumulants of jB_j are jp_j, $j^2p_j(1-p_j)$, $j^3p_j(1-p_j)(1-2p_j)$; those of $2lN_l$ are $\frac{2lw}{1-w}$, $\frac{4l^2w}{(1-w)^2}$, $\frac{8l^3w(1+w)}{(1-w)^3}$.)

In all applications $u > 0$ is given and $r = e^{-u/n}$, i.e. $t = u/n$. Then μ/n^2 and σ^2/n^3 are bounded above and below by the enclosures of Table 8.1 (Lemma 8.5).

Monotonicity of the mean. For fixed n, $\mu_n(r)$ is continuous and strictly increasing in $r \in (0,1)$ ($\frac{d\mu}{dr} = \sigma^2/r > 0$ from Lemma 8.2: $\mu = r\, A_n'(r)/A_n(r)$), with $\mu \to 0$ as $r \to 0$ and $\mu \to \infty$ as $r \to 1$ (the geometric means $2lw_l/(1-w_l) \to \infty$). Equivalently $u \mapsto \mu_n(e^{-u/n})$ is strictly decreasing and onto $(0,\infty)$. So for every real $m > 0$ there is exactly one $u > 0$ with $\mu_n(e^{-u/n}) = m$.

8.3 The local limit estimate

8.3.1 Fourier inversion

Lemma 8.3. Let $0 < \theta_0 \leq \pi$ and put $\tilde\varphi(\theta) = \varphi(\theta)e^{-i\mu\theta}$, $g(\theta) = e^{-\sigma^2\theta^2/2}$. For every integer m,

$$\Big|\sqrt{2\pi}\,\sigma P(S = m) - e^{-(m-\mu)^2/(2\sigma^2)}\Big| \leq M + T + Q,$$

$$M = \frac{\sigma}{\sqrt{2\pi}} \int_{|\theta| \leq \theta_0} |\tilde\varphi - g|\, d\theta, \quad T = \operatorname{erfc}\Big(\frac{\sigma\theta_0}{\sqrt{2}}\Big), \quad Q = \frac{2\sigma}{\sqrt{2\pi}} \int_{\theta_0}^{\pi} |\varphi(\theta)|\, d\theta.$$

Proof. S is integer valued, so $P(S = m) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \varphi(\theta) e^{-im\theta} d\theta$. With $x = m - \mu$, $e^{-im\theta}\varphi = \tilde\varphi e^{-ix\theta}$, and $e^{-x^2/(2\sigma^2)} = \frac{\sigma}{\sqrt{2\pi}} \int_{\mathbb{R}} g\, e^{-ix\theta} d\theta$. Hence the difference is $\frac{\sigma}{\sqrt{2\pi}}\big[\int_{|\theta|\leq\pi} (\tilde\varphi - g) e^{-ix\theta} d\theta -$

$\int_{|\theta|>\pi} g\, e^{-ix\theta} d\theta\big]$. Split $|\theta| \le \theta_0$ and $\theta_0 < |\theta| \le \pi$; use $|\tilde{\varphi}| = |\varphi|$, $|\varphi(-\theta)| = |\varphi(\theta)|$ and $\int_{\theta_0<|\theta|\le\pi} g + \int_{|\theta|>\pi} g = \int_{|\theta|>\theta_0} g = \frac{\sqrt{2\pi}}{\sigma}\operatorname{erfc}(\sigma\theta_0/\sqrt{2})$. ■

We use these elementary facts: (E1) $1 - x^2/2 \le \cos x \le 1 - x^2/2 + x^4/24$ for real x (integrate $\sin x \le x$ repeatedly); (E2) Taylor's formula with integral remainder: for $F \in C^r[0,\theta]$, $|F(\theta) - \sum_{s<r} F^{(s)}(0)\theta^s/s!| \le \sup|F^{(r)}|\,|\theta|^r/r!$; (E3) $|e^z - 1| \le |z| e^{\max(0,\operatorname{Re} z)}$; (E4) $\operatorname{erfc}(z) \le e^{-z^2}/(z\sqrt{\pi})$ for $z > 0$ (since $\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}}\int_z^\infty e^{-s^2} ds \le \frac{2}{\sqrt{\pi}}\int_z^\infty \frac{s}{z} e^{-s^2} ds$).

8.3.2 The logarithm of φ and its third derivative

For $|\theta| \le \pi$ every factor in (8.1) has positive real part after normalisation: $1 - p_j + p_j e^{iy} = (1-p_j)(1 + q_j e^{iy})$ with $q_j = p_j/(1-p_j) = e^{-jt} < 1$, so $\operatorname{Re}(1 + q_j e^{iy}) \ge 1 - q_j > 0$; and $\operatorname{Re}(1 - w_l e^{iy}) \ge 1 - w_l > 0$. Let $f(\theta) = \operatorname{Log}\varphi(\theta)$, the sum of the principal logarithms of the factors; f is real-analytic on $\mathbb{R}$, $e^f = \varphi$, $f(0) = 0$, and $f^{(s)}(0) = i^s \kappa_s$.

Lemma 8.4 (third derivative). (a) Let $0 < p \le \frac{1}{2}$, $v = p(1-p)$, $|y| \le t' < \pi$ and $h = \dfrac{pe^{iy}}{1 - p + pe^{iy}}$. Then

$$|h(1-h)(1-2h)| \le \frac{v\,[(1-2p) + p|y|]}{\cos(y/2)^3} \le \frac{v(1-2p) + t'vp}{\cos(t'/2)^3}.$$

(b) For $0 < w < 1$, $y \in \mathbb{R}$ and $g = \dfrac{we^{iy}}{1 - we^{iy}}$: $|g(1+g)(1+2g)| \le \dfrac{w(1+w)}{(1-w)^3}$. (c) Let $n \ge 1$, $u > 0$, $0 < \tau_0 < \pi$, $\theta_0 = \tau_0/n$. Then $\sup_{|\theta|\le\theta_0} |f'''(\theta)| \le n^4 \kappa(\tau_0)$ with

$$\kappa(\tau_0) = \frac{K_3 + \tau_0 K_4}{\cos(\tau_0/2)^3} + 8K_G,$$

$$K_3 \ge n^{-4} \sum_{j \text{ odd} \le n} j^3 v_j (1 - 2p_j),\ K_4 \ge n^{-5} \sum_j j^4 v_j p_j,\ K_G \ge n^{-4} \sum_l l^3 \frac{w_l(1+w_l)}{(1-w_l)^3}.$$

Proof. (a) With $e = e^{iy}$: $1 - h = \frac{1-p}{1-p+pe}$, $1 - 2h = \frac{1-p-pe}{1-p+pe}$, so $h(1-h)(1-2h) = \dfrac{v\,e\,(1-p-pe)}{(1-p+pe)^3}$. Numerator: $|1 - p - pe| = |(1-2p) + p(1-e)| \le (1-2p) + p|y|$ since $|1 - e^{iy}| = 2|\sin(y/2)| \le |y|$. Denominator: $|1 - p + pe|^2 = (1-p)^2 + p^2 + 2p(1-p)\cos y = 1 - 4v\sin^2(y/2) \ge 1 - \sin^2(y/2) = \cos^2(y/2)$ since $4v \le 1$. Finally $\cos(y/2) \ge \cos(t'/2) > 0$ and $|y| \le t'$. (b) $1 + g = \frac{1}{1-we}$, $1 + 2g = \frac{1+we}{1-we}$, so $|g(1+g)(1+2g)| = \frac{w|1+we|}{|1-we|^3} \le \frac{w(1+w)}{(1-w)^3}$. (c) $f(\theta) = \sum_j [\ln(1-p_j) + B_j(ij\theta)] + \sum_l [\ln(1 - w_l) + G_l(2il\theta)]$ with $B_j(z) = \ln(1 + q_j e^z)$, $G_l(z) = -\ln(1 - w_l e^z)$. $B_j'''(z) = h(1-h)(1-2h)$ with $h = q_j e^z/(1+q_j e^z)$, and $G_l'''(z) = g(1+g)(1+2g)$ with $g = w_l e^z/(1-w_l e^z)$ (differentiate $B' = h$, $h' = h(1-h)$, and $G' = g$, $g' = g(1+g)$). By the chain rule $f''' = \sum_j (ij)^3 B_j'''(ij\theta) + \sum_l (2il)^3 G_l'''(2il\theta)$. For $|\theta| \le \theta_0$ and odd $j \le n$, $|j\theta| \le t_j := j\tau_0/n \le \tau_0 < \pi$, so by (a), $|B_j'''| \le \dfrac{v_j(1-2p_j) + t_j v_j p_j}{\cos(\tau_0/2)^3}$; summing $j^3|B_j'''|$ gives at most $\cos(\tau_0/2)^{-3}\big[\sum j^3 v_j(1-2p_j) + \frac{\tau_0}{n}\sum j^4 v_j p_j\big] \le n^4 \cos(\tau_0/2)^{-3}(K_3 + \tau_0 K_4)$. By (b), $\sum_l (2l)^3 |G_l'''| \le 8\sum_l l^3 \frac{w_l(1+w_l)}{(1-w_l)^3} \le 8n^4 K_G$. ■

8.3.3 Lattice sums

Lemma 8.5 (lattice sums). Let $n_0 \ge 1$, let $f \in C^1[0, 1 + 1/n_0]$ and $n \ge n_0$. Write TV for the total variation and sup for the supremum norm of f on the indicated interval.

- (odd) $\left|\frac{1}{n}\sum_{j\ \mathrm{odd}\le n} f(j/n) - \frac{1}{2}\int_0^1 f\right| \le \frac{TV_{[0,1+1/n_0]}f + \frac{1}{2}\sup|f|}{n}$.
- (even-index geometric) $\left|\frac{1}{n}\sum_{n/2<l\le n} f(l/n) - \int_{1/2}^1 f\right| \le \frac{TV_{[1/2,1+1/n_0]}f + \sup|f|}{n}$.

Proof. (odd) The intervals $[(j-1)/n, (j+1)/n]$, j odd $\le n$, are disjoint with union $[0,b]$, $b = 1$ (n even) or $1+1/n$ (n odd). On each, $f(j/n)$ and the mean value $\frac{n}{2}\int f$ both lie between the minimum and maximum of f, so they differ by at most the variation of f on the interval. Summing and dividing by n: $|\frac{1}{n}\sum f(j/n) - \frac{1}{2}\int_0^b f| \le TV/n$; and $|\frac{1}{2}\int_1^b f| \le \frac{\sup|f|}{2n}$. (geometric) The intervals $[(l-\frac{1}{2})/n, (l+\frac{1}{2})/n]$ are disjoint with union $[a, 1+\frac{1}{2n}]$, $a = \frac{1}{2}+\frac{1}{2n}$ (n even) or $\frac{1}{2}$ (n odd); the same argument gives TV/n, and the endpoint mismatch with $[\frac{1}{2}, 1]$ costs at most $\sup|f|/n$. TV and sup over $[0, 1+1/n]$ are bounded by those over $[0, 1+1/n_0]$. ■

The sums needed are of the form $n^{-(q+1)}\sum_j j^q F(cuj/n) = \frac{1}{n}\sum f(j/n)$ with $f(y) = y^q F(cuy)$. The nine sums are listed in Table 8.1 (with $z = cuy$ and $w = e^{-z}$, $p(z) = \frac{1}{1+e^z}$, $v(z) = p(1-p)$, $\lambda(z) = \frac{4w}{(1-w)^2}$):

name	sum	q	c	$F(z)$	summation
SB	$n^{-3}\sum j^2 v_j$	2	1	$v(z)$	odd $j \le n$
SG	$n^{-3}\sum l^2 \lambda_l$	2	2	$\lambda(z)$	$\frac{n}{2} < l \le n$
$V4$	$n^{-5}\sum j^4 v_j$	4	1	$v(z)$	odd
QG	$n^{-5}\sum l^4 \lambda_l$	4	2	$\lambda(z)$	geometric
$K3a$	$n^{-4}\sum j^3 v_j(1-2p_j)$	3	1	$v(1-2p)$	odd
$K4a$	$n^{-5}\sum j^4 v_j p_j$	4	1	vp	odd
KGr	$n^{-4}\sum l^3 \frac{w_l(1+w_l)}{(1-w_l)^3}$	3	2	$\frac{w(1+w)}{(1-w)^3}$	geometric
MB	$n^{-2}\sum j p_j$	1	1	$p(z)$	odd
MG	$n^{-2}\sum \frac{l w_l}{1-w_l}$	1	2	$\frac{w}{1-w}$	geometric

By (8.2) and the table: $\sigma^2/n^3 = SB + SG$, $\mu/n^2 = MB + 2MG$, and $\kappa_3/n^4 = K3a + 8KGr$.

Algorithm E (enclosure of one lattice sum). Input: F (with its derivative formula), q, c, a ball $U = [u_a, u_b]$ for u, n_0, an integer Q_e (we use $Q_e = 600$), and the flag odd/geometric. Put $[y_a, y_b] = [0,1]$ (odd) or $[\frac{1}{2}, 1]$ (geometric) and $y_e = 1 + 1/n_0$. (1) J: split $[y_a, y_b]$ into Q_e equal sub-intervals Y_i and take the ball sum $\sum_i \mathrm{width}\cdot\big(Y_i^q\, F(cUY_i)\big)$ (ball arithmetic: the product encloses $f(y) = y^q F(cuy)$ for all $y \in Y_i$, $u \in U$). (2) Split $[y_a, y_e]$ into Q_e sub-intervals Y_i'; TV is the sum of $\mathrm{width}\cdot\sup_{Y_i'\times U}|f'|$ with $f'(y) = qy^{q-1}F(cuy) + y^q\, cu\, F'(cuy)$, and sup is the maximum over i of the upper bound of $|f|$ on $Y_i' \times U$. (3) Return the interval $[\,J/2 - e,\ J/2 + e\,]$ with $e = (TV + \sup/2)/n_0$ (odd) or $[\,J - e,\ J + e\,]$ with $e = (TV + \sup)/n_0$ (geometric). The derivative formulas are $v' = -(1-2p)v$, $p' = -v$, $(v(1-2p))' = -v((1-2p)^2 - 2v)$, $(vp)' = -v(p(1-2p) + v)$, $\lambda' = -\frac{4w(1+w)}{(1-w)^3}$, $\big(\frac{w(1+w)}{(1-w)^3}\big)' = -\frac{w(1+4w+w^2)}{(1-w)^4}$, $\big(\frac{w}{1-w}\big)' = -\frac{w}{(1-w)^2}$ (in z, $w = e^{-z}$; verified symbolically).

By Lemma 8.5, the interval returned in Algorithm E contains $n^{-(q+1)}\sum_j j^q F(cuj/n)$ for **every** $n \ge n_0$ and every $u \in U$. We write X_{lo}, X_{hi} for its endpoints, $X \in \{SB, \dots, MG\}$.

8.3.4 The main arc

Proposition 8.6. Let $n \ge n_0$, $u \in U$, $0 < \tau_0 < \pi$, $\theta_0 = \tau_0/n$, $S_{lo} = SB_{lo} + SG_{lo}$ (so $\sigma^2 \ge S_{lo} n^3$), $\kappa = \kappa(\tau_0)$ computed from $K_3 = K3a_{hi}$, $K_4 = K4a_{hi}$, $K_G = KGr_{hi}$, and suppose $c := \tau_0\kappa/S_{lo} < 3$.

Then

$$M \le D_1 := \frac{\kappa}{6\sqrt{2\pi}\, S_{lo}^{3/2}\sqrt{n_0}\,(\frac12-\frac{c}{6})^2}, \qquad T \le D_2 := \frac{e^{-z^2}}{z\sqrt{\pi}},\ z = \tau_0\sqrt{S_{lo}n_0/2}.$$

Proof. By (E2) with $r=3$ and Lemma 8.4(c), $f(\theta) = i\mu\theta - \frac{\sigma^2\theta^2}{2} + R(\theta)$ with $|R| \le K_3'|\theta|^3/6$, $K_3' = n^4\kappa$. Then $\tilde\varphi = ge^R$, and by (E3), $|\tilde\varphi - g| \le g|R|e^{|R|}$. For $|\theta| \le \theta_0$: $|R| \le \frac{K_3'\theta_0}{6}\theta^2$ and $K_3'\theta_0 = n^3\kappa\tau_0 \le c\,\sigma^2$ (as $\sigma^2 \ge S_{lo}n^3$). Hence $ge^{|R|} \le e^{-a\theta^2}$ with $a = \sigma^2(\frac12 - \frac{c}{6}) > 0$, and

$$\int_{|\theta|\le\theta_0} |\tilde\varphi - g| \le \frac{K_3'}{6}\cdot 2\int_0^\infty \theta^3 e^{-a\theta^2}d\theta = \frac{K_3'}{6a^2}.$$

Multiplying by $\sigma/\sqrt{2\pi}$: $M \le \frac{K_3'}{6\sqrt{2\pi}\sigma^3(1/2-c/6)^2} \le \frac{n^4\kappa}{6\sqrt{2\pi}S_{lo}^{3/2}n^{9/2}(\dots)^2} = \frac{\kappa}{6\sqrt{2\pi}S_{lo}^{3/2}\sqrt{n}(1/2-c/6)^2} \le D_1$. For T: $\sigma\theta_0/\sqrt{2} = \sigma\tau_0/(n\sqrt{2}) \ge \tau_0\sqrt{S_{lo}n/2} \ge z$ and erfc is decreasing, $e^{-z^2}/(z\sqrt{\pi})$ is decreasing in z; (E4). ■

8.3.5 Minor arcs

Lemma 8.7 (modulus). For $\theta \in \mathbb{R}$,

$$|\varphi(\theta)| \le \exp\Big(-\sum_{j\ \text{odd}\le n} v_j(1-\cos j\theta)\Big)\prod_{n/2<l\le n}\big(1+\lambda_l\sin^2(l\theta)\big)^{-1/2}.$$

Proof. $|1-p+pe^{iy}|^2 = 1-2v(1-\cos y) \le e^{-2v(1-\cos y)}$. Also $\big|\frac{1-w}{1-we^{iy}}\big|^2 = \frac{(1-w)^2}{(1-w)^2+4w\sin^2(y/2)} = \big(1+\lambda\sin^2(y/2)\big)^{-1}$; put $y = 2l\theta$. ■

Fix $n \ge n_0$, a cell $U = [u_a, u_b] \ni u$, the enclosures of Algorithm E, and τ_0. Put $\tau = n\theta$ and write $\mathrm{pref} = \sqrt{2\pi S_{hi}}/\pi$, where $S_{hi} = SB_{hi} + SG_{hi}$, so that $\sigma \le \sqrt{S_{hi}}\,n^{3/2}$ and $\frac{2\sigma}{\sqrt{2\pi}} = \sigma\sqrt{2/\pi} \le \mathrm{pref}\cdot n^{3/2}$. Split $[\theta_0,\pi]$ into $A = [\tau_0/n,\ 1.5/n]$, $B = [1.5/n,\ \pi - 1.5/n]$, $C = [\pi - 1.5/n,\ \pi]$ (these are non-empty since $n \ge n_0 > 3$).

Lemma 8.8 (region A). Let $[\tau_l,\tau_h] \subset [\tau_0, 1.5]$, $\tau_l \ge 0$, $Z = \lambda(u_a)\min(1,\tau_h^2)$ and $\ell = \ln(1+Z)/Z$ (here $\lambda(u) = 4e^{-u}/(1-e^{-u})^2$). For $\tau = n\theta \in [\tau_l,\tau_h]$, $|\varphi(\theta)| \le e^{-nb}$ with $b = b_o^+ + b_g^+$ ($x^+ = \max(x,0)$),

$$b_o = \tfrac12\tau_l^2 SB_{lo} - \tfrac{1}{24}\tau_h^4 V4_{hi}, \qquad b_g = \tfrac{\ell}{2}\big(\tau_l^2 SG_{lo} - \tfrac13\tau_h^4 QG_{hi}\big).$$

Proof. By Lemma 8.7 it suffices to bound the two sums. By (E1), $\sum_j v_j(1-\cos j\theta) \ge \sum_j v_j\big(\frac{(j\theta)^2}{2} - \frac{(j\theta)^4}{24}\big) \ge n\big(\frac{\tau^2}{2}SB - \frac{\tau^4}{24}V4\big)$ (as $\sum j^2 v_j = n^3 SB$, $\sum j^4 v_j = n^5 V4$, $\theta = \tau/n$), and this is $\ge nb_o$. Since $z_l = \lambda_l\sin^2(l\theta) \le \lambda_l\min(1,(l\theta)^2) \le Z$ (here $l\theta \le \tau \le \tau_h$, and $\lambda_l = \lambda(2lu/n) \le \lambda(u) \le \lambda(u_a)$ because $2l/n > 1$ and λ is decreasing), concavity of $\ln(1+z)$ gives $\ln(1+z_l) \ge \ell z_l$. And $\sin^2 x \ge x^2 - x^4/3$ ($\sin^2 x = \frac{1-\cos 2x}{2}$, (E1)). Hence $\frac12\sum_l \ln(1+z_l) \ge \frac{\ell}{2}\sum_l \lambda_l\big((l\theta)^2 - \frac{(l\theta)^4}{3}\big) \ge nb_g$, using $\sum l^2\lambda_l = n^3 SG$, $\sum l^4\lambda_l = n^5 QG$. Both exponents are nonnegative, so each may be replaced by its positive part. ■

Lemma 8.9 (Abel step and region B). (a) Let $\sin\theta \ne 0$, $X = 1/|\sin\theta|$, and let $v_1 \ge v_3 \ge \cdots \ge v_J \ge 0$ (J odd). Then $\sum_{j\ \text{odd}\le J} v_j(1-\cos j\theta) \ge \sum_{j\ \text{odd},\,X+1\le j\le J} v_j$. (b) Let $Y_0 = \frac{3}{n_0} + \big[1.5\,(1-\frac{1.5^2}{6n_0^2})\big]^{-1} < 1$ and $b_B = \frac12\cdot\frac{1-Y_0}{M}\sum_{i=1}^M v\big(u_b(Y_0 + i\frac{1-Y_0}{M})\big)$ with $v(z) = (2\cosh(z/2))^{-2}$, $M = 300$. For $\theta \in [1.5/n, \pi - 1.5/n]$, $n \ge n_0$, $u \in U$: $|\varphi(\theta)| \le e^{-nb_B}$.

Proof. (a) Let $N(j) = \frac{j+1}{2}$ (the number of odd integers in $[1,j]$), $c_i = 1-\cos i\theta$ and $C_j = \sum_{i\ \text{odd}\le j} c_i = N(j) - s_j$ with $s_j = \sum_{i\ \text{odd}\le j}\cos i\theta = \frac{\sin((j+1)\theta)}{2\sin\theta}$ (telescoping $2\sin\theta\cos i\theta = \sin((i+$

$1)\theta)-\sin((i-1)\theta))$. Since $|s_j|\le\min(N(j),X/2)$, $C_j\ge h(j):=(N(j)-X/2)^+$. Put $v_{J+2}=0$. By Abel summation, $\sum_j v_jc_j=\sum_j(v_j-v_{j+2})C_j\ge\sum_j(v_j-v_{j+2})h(j)=\sum_j v_j\big(h(j)-h(j-2)\big)$ with $h(-1)=0$. Here h is non-decreasing, so every difference is ≥ 0, and $h(j)-h(j-2)=1$ whenever $N(j-2)\ge X/2$, i.e. $j\ge X+1$. This gives (a). (b) For $\theta\in[1.5/n,\pi-1.5/n]$, $|\sin\theta|\ge\sin(1.5/n)\ge\frac{1.5}{n}\big(1-\frac{1.5^2}{6n^2}\big)$, so $(X+3)/n\le Y_0$ for $n\ge n_0$. Drop the second product in Lemma 8.7 (each factor is ≤ 1). By (a) with J the largest odd integer $\le n$, $\sum_j v_j(1-\cos j\theta)\ge\sum_{j\ \mathrm{odd}\in[X+1,J]}v_j$. As $v(\cdot)$ is even and decreasing on $[0,\infty)$, $v_j\ge\frac12\int_j^{j+2}v(yu/n)dy$; the smallest odd $j\ge X+1$ is $<X+3$, and $J+2>n$, so the sum is $\ge\frac12\int_{X+3}^{n}v(yu/n)dy=\frac{n}{2}\int_{(X+3)/n}^{1}v(ux)dx\ge\frac{n}{2}\int_{Y_0}^{1}v(u_bx)dx\ge nb_B$, because a right Riemann sum underestimates the integral of a decreasing function. ■

Lemma 8.10 (region C). Let $b_C=\frac12\cdot\frac{1/2-1/n_0}{M}\sum_{i=1}^{M}v\big(u_b(\frac{1}{n_0}+i\frac{1/2-1/n_0}{M})\big)$. For $\theta=\pi-\phi$, $0\le\phi\le 1.5/n$, $n\ge n_0$, $u\in U$: $|\varphi(\theta)|\le e^{-nb_C}$.

Proof. For odd $j\le n/2$: $j\phi\le 0.75<\pi/2$, so $\cos j\theta=-\cos j\phi<0$ and $1-\cos j\theta\ge 1$. By Lemma 8.7, $|\varphi|\le\exp(-\sum_{j\ \mathrm{odd}\le n/2}v_j)$. With J the largest odd integer $\le n/2$ we have $J+2>n/2$, and $v_j\ge\frac12\int_j^{j+2}v(yu/n)dy$, so $\sum\ge\frac12\int_1^{n/2}v(yu/n)dy=\frac{n}{2}\int_{1/n}^{1/2}v(ux)dx\ge\frac{n}{2}\int_{1/n_0}^{1/2}v(u_bx)dx\ge nb_C$. ■

Proposition 8.11 (local limit theorem on a cell). Let $U=[u_a,u_b]$, $n_0\ge 4$ and $\tau_0\in(0,1.5]$ be such that: $c=\tau_0\kappa(\tau_0)/S_{lo}<2.9$; $n_0b\ge\frac12$ for each sub-cell of Lemma 8.8 when $[\tau_0,1.5]$ is cut into 240 equal parts; $n_0b_B\ge\frac32$ and $n_0b_C\ge\frac32$. Put

$$D_3=\operatorname{pref}\Big[\sqrt{n_0}\sum_i(\tau_{i+1}-\tau_i)e^{-n_0b_i}+\pi n_0^{3/2}e^{-n_0b_B}+1.5\sqrt{n_0}\,e^{-n_0b_C}\Big],\quad \delta=D_1+D_2+D_3.$$

Then for all $n\ge n_0$, $u\in U$, integers m:

$$\Big|\sqrt{2\pi}\,\sigma P(S_n(e^{-u/n})=m)-e^{-(m-\mu)^2/(2\sigma^2)}\Big|\le\delta.$$

Moreover $S_{lo}n^3\le\sigma^2\le S_{hi}n^3$ and $M_{lo}n^2\le\mu\le M_{hi}n^2$, where $M=MB+2MG$.

Proof. Lemma 8.3 with $\theta_0=\tau_0/n$ gives the bound $M+T+Q$. $M\le D_1$ and $T\le D_2$ by Proposition 8.6 (all quantities there were enclosed uniformly in $n\ge n_0$, $u\in U$). For $Q=\frac{2\sigma}{\sqrt{2\pi}}\int_{\theta_0}^{\pi}|\varphi|$ we use $\frac{2\sigma}{\sqrt{2\pi}}\le\operatorname{pref}\,n^{3/2}$, $\operatorname{pref}=\sqrt{2\pi S_{hi}}/\pi=\sqrt{2S_{hi}/\pi}$. Region A ($\theta=\tau/n$, $d\theta=d\tau/n$): $\int_A|\varphi|\le\frac1n\sum_i(\tau_{i+1}-\tau_i)e^{-nb_i}$, so its contribution to Q is $\le\operatorname{pref}\sqrt{n}\sum_i(\tau_{i+1}-\tau_i)e^{-nb_i}$. Region B: length $\le\pi$ and Lemma 8.9: contribution $\le\operatorname{pref}\,\pi n^{3/2}e^{-nb_B}$. Region C: length $1.5/n$ and Lemma 8.10: $\le\operatorname{pref}\,1.5\sqrt{n}\,e^{-nb_C}$. Each of the three terms has the form $n^ae^{-n\beta}$ with $a\in\{\frac12,\frac32\}$, which is non-increasing for $n\ge a/\beta$; the hypotheses give $n_0\beta\ge a$. So the three contributions at $n\ge n_0$ are bounded by their values at $n=n_0$. Finally $D_1\propto n^{-1/2}$ and D_2 is decreasing in n (only $S_{lo}n_0$ enters, as a lower bound for n). The bounds on σ^2,μ are (8.2) and Table 8.1. ■

Algorithm C (cells). Fix $n_0=1467$. For $i=19,\dots,331$ let u_i be the double-precision number nearest $i/100$. The 312 cells are $U_i=[u_i,u_{i+1}]$, $i=19,\dots,330$; they are contiguous and cover $[u_{19},u_{331}]$. For each cell: compute the nine enclosures by Algorithm E ($Q_e=600$); for each $\tau_0\in\{0.05i:\ i=2,\dots,24\}$ test the hypotheses of Proposition 8.11 and compute $\delta(\tau_0)$; keep the admissible τ_0 with least δ. All operations are in ball arithmetic at 128 bits; δ is rounded up to double precision. All later uses of the cells take u-intervals whose endpoints are double-precision numbers inside $[u_{19},u_{331}]$.

Theorem 8.12 (result of Algorithm C). All 312 cells admit an admissible τ_0. The following hold: $\max\delta=0.08727$ (on $[3.30,3.31]$), $\min\delta=0.02847$; $\tau_0\in[0.10,0.55]$; $\max D_1=0.06539$, $\max D_2=2.21\cdot10^{-3}$, $\max D_3=1.99\cdot10^{-2}$; $\max c=1.1870$. Consequently the conclusion of Proposition 8.11 holds with $n_0=1467$ on every cell. The cell-wise values are stored in

`data/pd/cells56_sharp.json` (Appendix A.3), and Table 8.4 of Section 8.7 lists the first and last cells.

The proof is the computation. (The code is `code/pd/clt56.py`, `make_cells56.py`; a full regeneration produced byte-identical data, see Appendix A.3.)

8.4 The block inequality

Fix odd $k \in [3, 33]$, $N_1 = 1500$ and the window parameters $U_A < U_B$ of Table 8.3 (the same numbers for all k are listed there). Call an even $n \geq N_1$ **admissible for** d if $m_n = (d - n^2)/k$ is a positive integer. For admissible n put $e_n = a_n(m_n)$; $n_\pm = n \pm k$ are odd, $n_\pm^2 \equiv n^2 \pmod k$, and $m_\pm = m_n - \frac{n_\pm^2 - n^2}{k}$, i.e. $m_+ = m_n - 2n - k$, $m_- = m_n + 2n - k$; $o_\pm = a_{n_\pm}(m_\pm)$ whenever $m_\pm \geq 0$ (otherwise there is no such term). By Section 8.2 let $u = u_n$ be the unique number with $\mu_n(e^{-u/n}) = m_n$ and put $r = e^{-u/n}$.

The finite differences. For even n and $j = n + t$, $(-1)^j = (-1)^t$. By (4.1), $A_N(r) = \prod_{j \leq N}(1 + (-r)^j)^{-1}$. Put $s = 1/n$, $\xi_t = 1 + ts$, $z_t = (-1)^t r^{\,n+t} = (-1)^t e^{-u\xi_t}$, and for $\varepsilon \in \{+1, -1\}$ let $T_+ = \{1, \dots, k\}$, $T_- = \{-k + 1, \dots, 0\}$. Define

$$\Lambda_\varepsilon = \varepsilon \sum_{t \in T_\varepsilon} \big(-\log(1 + z_t)\big), \quad \Delta\mu_\varepsilon = \varepsilon \sum_{t \in T_\varepsilon} \Big(-\xi_t \frac{z_t}{1 + z_t}\Big), \quad \Delta\sigma_\varepsilon = \varepsilon \sum_{t \in T_\varepsilon} \Big(-\xi_t^2 \frac{z_t}{(1 + z_t)^2}\Big). \tag{8.3}$$

Then

$$\log \frac{A_{n+\varepsilon k}}{A_n}(r) = \Lambda_\varepsilon, \qquad \mu_{n+\varepsilon k}(r) - \mu_n(r) = n\,\Delta\mu_\varepsilon, \qquad \sigma^2_{n+\varepsilon k}(r) - \sigma^2_n(r) = n^2 \Delta\sigma_\varepsilon. \tag{8.4}$$

Indeed $\log A_N = -\sum_{j \leq N} \log(1 + (-r)^j)$; with $z = (-r)^j$ one has $r\,\frac{dz}{dr} = jz$, so $r\frac{d}{dr}[-\log(1 + z)] = -\frac{jz}{1+z}$ and $r\frac{d}{dr}\big[-\frac{jz}{1+z}\big] = -\frac{j^2 z}{(1+z)^2}$; summing these over $j \leq N$ gives $\log A_N$, μ_N, σ_N^2 (Lemma 8.2: $\mu_N = r\frac{d}{dr}\log A_N$, $\sigma_N^2 = r\frac{d}{dr}\mu_N$), and passing from n to $n + k$ adds the terms $j = n + 1, \dots, n + k$, while passing to $n - k$ removes $j = n - k + 1, \dots, n$, with $j = n\xi_t$.

Lemma 8.13 (block inequality). Let $n \geq N_1$ be even and admissible, $U = [u_a, u_b] \ni u_n$ a box with $[u_a(1 - k/N_1),\, u_b(1 + k/N_1)] \subset [u_{19}, u_{331}]$, δ_U the maximum of δ over the cells meeting that interval, S_b a lower bound of S_{lo} over the cells meeting U, and S_ε a lower bound of S_{lo} over the cells meeting the u-range of the partner $n + \varepsilon k$ (namely $[u_a(1 - k/N_1), u_b]$ for $\varepsilon = -1$ and $[u_a, u_b(1 + k/N_1)]$ for $\varepsilon = +1$). Let $s_b = 1/N_1$, $\gamma \geq (1 - k s_b)^{-3}$. Let $\Lambda_\varepsilon, \Delta\mu_\varepsilon, \Delta\sigma_\varepsilon$ be enclosed for $u \in U$, $s \in [0, s_b]$, and put

$$\rho_\varepsilon = \exp\big(\Lambda_\varepsilon - u(2\varepsilon + ks)\big)\big(1 + q_\varepsilon\big)^{-1/2}, \quad q_\varepsilon = \frac{s_b \sup|\Delta\sigma_\varepsilon|}{S_b}, \quad X_\varepsilon = -(2\varepsilon + ks) - \Delta\mu_\varepsilon,$$

$$\beta_\varepsilon = 1 - \frac{\sup X_\varepsilon^2 \; s_b\,\gamma}{2S_\varepsilon} - \delta_U, \qquad R_U = \frac{1 + \delta_U}{\frac{1}{2}\sum_{\varepsilon = \pm 1} \inf \rho_\varepsilon \cdot \beta_\varepsilon}$$

(the suprema and infima are over $u \in U$, $s \in [0, s_b]$). If $\beta_\pm > 0$ then $m_\pm \geq 0$ and $e_n \leq R_U \cdot \frac{1}{2}(o_{n-k} + o_{n+k})$.

(Here $(n_\varepsilon^2 - n^2)/(kn) = 2\varepsilon + ks$ for $n_\varepsilon = n + \varepsilon k$.)

Proof. Write $a_N(m) = A_N(r) r^{-m} P_N(S_N(r) = m)$ (Lemma 8.2) for $N \in \{n, n_\pm\}$ with the same r; S_{n_ε} is the model of Section 8.2 at level $u_\varepsilon = u n_\varepsilon / n \in \{u(1 + \varepsilon k s)\}$, which lies in the range of the partner. Then

$$\frac{o_\varepsilon}{e_n} = \frac{A_{n_\varepsilon}}{A_n}(r)\, r^{m_n - m_\varepsilon}\, \frac{P_{n_\varepsilon}(m_\varepsilon)}{P_n(m_n)}, \qquad r^{m_n - m_\varepsilon} = r^{(n_\varepsilon^2 - n^2)/k} = e^{-u(2\varepsilon + ks)},$$

because $(n_\varepsilon^2-n^2)/k=(2\varepsilon kn+k^2)/k=n(2\varepsilon+ks)$ and $r=e^{-us}$. Apply Proposition 8.11 at (n,u) with $m=m_n=\mu_n$ and at $(n_\varepsilon,u_\varepsilon)$ with $m=m_\varepsilon$ ($n_\varepsilon\ge N_1-k\ge 1467=n_0$; the u-values lie in cells by the hypothesis):

$$P_n(m_n)\le\frac{1+\delta_U}{\sqrt{2\pi}\sigma_n},\qquad P_{n_\varepsilon}(m_\varepsilon)\ge\frac{e^{-x_\varepsilon^2/(2\sigma_\varepsilon^2)}-\delta_U}{\sqrt{2\pi}\sigma_\varepsilon}\ge\frac{1-x_\varepsilon^2/(2\sigma_\varepsilon^2)-\delta_U}{\sqrt{2\pi}\sigma_\varepsilon},$$

where $\sigma_\varepsilon=\sigma_{n_\varepsilon}(r)$ and $x_\varepsilon=m_\varepsilon-\mu_{n_\varepsilon}(r)=-\frac{n_\varepsilon^2-n^2}{k}-(\mu_{n_\varepsilon}-\mu_n)=nX_\varepsilon$ by (8.4) (as $\mu_n=m_n$). Since $\sigma_\varepsilon^2=\sigma_n^2+n^2\Delta\sigma_\varepsilon\le\sigma_n^2(1+s\sup|\Delta\sigma_\varepsilon|/S_b)\le\sigma_n^2(1+q_\varepsilon)$ (using $\sigma_n^2\ge S_bn^3$), $\sigma_n/\sigma_\varepsilon\ge(1+q_\varepsilon)^{-1/2}$. Also $x_\varepsilon^2/(2\sigma_\varepsilon^2)\le\frac{X_\varepsilon^2n^2}{2S_\varepsilon n_\varepsilon^3}=\frac{X_\varepsilon^2 s}{2S_\varepsilon}\big(\frac{n}{n_\varepsilon}\big)^3\le\frac{\sup X_\varepsilon^2 s_b\gamma}{2S_\varepsilon}$ since $n_\varepsilon\ge n(1-ks)$. Combining: if $\beta_\varepsilon>0$ then $P_{n_\varepsilon}(m_\varepsilon)>0$, so m_ε is in the support of S_{n_ε}, hence $m_\varepsilon\ge 0$ and o_ε is a term; and $o_\varepsilon\ge e_n\,\rho_\varepsilon\beta_\varepsilon/(1+\delta_U)$. Adding the two inequalities: $\frac12(o_++o_-)\ge e_n\frac{\frac12\sum\rho_\varepsilon\beta_\varepsilon}{1+\delta_U}$, i.e. $e_n\le R_U\frac12(o_++o_-)$. ■

Algorithm R. Input k, U_A,U_B (Table 8.3), $H=0.0025$, the cells of Theorem 8.12. Let s_b be the upper end of the ball $1/N_1$, γ the upper end of $(1-ks_b)^{-3}$, $n_{box}=\mathrm{round}((U_B-U_A)/H)$, and for $i<n_{box}$ the box $U_i=[U_A+iH,\ U_A+(i+1)H]$ (the last one ends at U_B). For each box compute δ_U, S_b, S_ε from the cells (an exact comparison of double-precision endpoints; the program asserts that the cells meeting the range are contiguous and cover it), enclose $\Lambda_\varepsilon,\Delta\mu_\varepsilon,\Delta\sigma_\varepsilon$ by interval arithmetic in $u\in U_i$, $s\in[0,s_b]$ (the $2k$ terms $t\in T_\pm$ are summed with interval arithmetic in the variables (u,s)), check $\beta_\varepsilon>0$, and compute R_{U_i} (rounded up). Return $R_k=\max_i R_{U_i}$ (rounded up).

Proposition 8.14 (block bound). For each odd $k\in[3,33]$ Algorithm R succeeds (all $\beta_\varepsilon>0$, all required cells exist) and returns the value R_k of Table 8.2; in all cases $R_k<1-10^{-6}$. Consequently, for d allowed and every even $n\ge N_1$ admissible for d with $u_n\in[U_A,U_B]$: $e_n\le R_k\cdot\frac12(o_{n-k}+o_{n+k})$, and

$$\sum_{n\text{ even admissible},\ u_n\in[U_A,U_B]} e_n\le R_k\,O_d.$$

Proof. The computation is the content of the first assertion (code `code/pd/blocks56.py`). For the second, each $u_n\in[U_A,U_B]$ lies in a box U_i, and Lemma 8.13 applies with $R_{U_i}\le R_k$ ($n\ge N_1$ gives $s\le s_b$). Every odd N occurs at most twice among $\{n\pm k\}$ ($n=N-k$ as n_+ and $n=N+k$ as n_-) with weight $\frac12$ each, so $\sum_n\frac12(o_{n-k}+o_{n+k})\le\sum_{N\text{ odd}}a_N(m_N)=O_d$. ■

8.5 The tails

Let $\Phi_1(u)=\frac12\int_0^1\log(1+e^{-uy})dy+\frac12\int_1^2\big[-\log(1-e^{-uy})\big]dy$, $g(u)=-\log(1-e^{-u})$, and for $\alpha=n/\sqrt{d}$

$$F_k(\alpha,u)=\alpha\Phi_1(u)+\frac{(1-\alpha^2)u}{k\alpha},\qquad \rho(\alpha)=\frac{1/\alpha^2-1}{k}=\frac{m_n}{n^2}.$$

Lemma 8.15. (a) $\Phi_1(u)=\big[\mathrm{Li}_2(-e^{-u})+\frac{\pi^2}{12}+\mathrm{Li}_2(e^{-u})-\mathrm{Li}_2(e^{-2u})\big]/(2u)$ for $u>0$. (b) For even $n\ge2$ and $u>0$: $\log A_n(e^{-u/n})\le n\Phi_1(u)+\log 2$. (c) For $n\ge1$, $u>0$: $\log A_n(e^{-u/n})\ge n\Phi_1(u)-\frac12\log 2-g(u)$. (d) For n even with $n^2\le d$, $m=(d-n^2)/k\in\mathbb{Z}$ and any $u_c>0$: $a_n(m)\le A_n(e^{-u_c/n})e^{u_cm/n}\le 2\exp\big(\sqrt{d}\,F_k(n/\sqrt{d},u_c)\big)$. (e) $\partial_\alpha^2F_k=2u/(k\alpha^3)>0$; hence for fixed u, $F_k(\cdot,u)$ on an interval $[x,y]$ is at most $\max(F_k(x,u),F_k(y,u))$, and for α in an interval with midpoint c and half-width h, $F_k(\alpha,u)\ge F_k(c,u)-h\,|\partial_\alpha F_k(c,u)|$.

Proof. (a) $\frac{d}{dy}\mathrm{Li}_2(-e^{-uy})=u\log(1+e^{-uy})$ and $\frac{d}{dy}\mathrm{Li}_2(e^{-uy})=u\log(1-e^{-uy})$ (from $\mathrm{Li}_2'(z)=-\log(1-z)/z$), and $\mathrm{Li}_2(-1)=-\pi^2/12$. Integrating, $\int_0^1\log(1+e^{-uy})dy=[\mathrm{Li}_2(-e^{-u})+\pi^2/12]/u$ and $\int_1^2[-\log(1-e^{-uy})]dy=[\mathrm{Li}_2(e^{-u})-\mathrm{Li}_2(e^{-2u})]/u$. By (4.1), with $r=e^{-u/n}$, $\log A_n=\sum_{j\text{ odd}\le n}f_B(j/n)+\sum_{n/2<l\le n}f_G(l/n)$, $f_B(x)=\log(1+e^{-ux})$, $f_G(x)=-\log(1-e^{-2ux})$, both

decreasing, and $\Phi_1 = \frac{1}{2}\int_0^1 f_B + \int_{1/2}^1 f_G$. (b) For odd $j \ge 3$, $f_B(j/n) \le \frac{n}{2}\int_{(j-2)/n}^{j/n} f_B$; these intervals are disjoint in $[1/n, 1]$, and $f_B(1/n) \le \log 2$. So $\sum_{\text{odd}} f_B \le \log 2 + \frac{n}{2}\int_0^1 f_B$. For n even, the integers $l \in (n/2, n]$ give the intervals $[(l-1)/n, l/n]$ with union $[1/2, 1]$, and $f_G(l/n) \le n\int_{(l-1)/n}^{l/n} f_G$. Add. (c) $f_B(j/n) \ge \frac{n}{2}\int_{j/n}^{(j+2)/n} f_B$; the intervals for odd $j \le n$ cover $[1/n, 1]$, so $\sum f_B \ge \frac{n}{2}\int_{1/n}^1 f_B \ge \frac{n}{2}\int_0^1 f_B - \frac{1}{2}\log 2$ ($f_B \le \log 2$). $f_G(l/n) \ge n\int_{l/n}^{(l+1)/n} f_G$ for $l = \lfloor n/2 \rfloor + 1, \dots, n$; the union contains $[\frac{1}{2} + \frac{1}{n}, 1]$ and $f_G \le f_G(\frac{1}{2}) = g(u)$ on $[\frac{1}{2}, 1]$, so $\sum f_G \ge n\int_{1/2}^1 f_G - g(u)$. (d) All $a_n(m') \ge 0$ (Lemma 4.2), so $a_n(m) r^m \le A_n(r)$; with $r = e^{-u_c/n}$ and (b): $\log a_n(m) \le n\Phi_1(u_c) + \log 2 + u_c m/n$, and $u_c m/n = \sqrt{d}\, u_c(1-\alpha^2)/(k\alpha)$, $n = \alpha\sqrt{d}$. (e) $F_k = \alpha\Phi_1 + \frac{u}{k}(\frac{1}{\alpha} - \alpha)$ and the first term is linear in α; $\partial_\alpha^2 = \frac{u}{k}\cdot\frac{2}{\alpha^3}$. Convex functions lie below the chord's maximum and above their tangents. ■

Lemma 8.16 (small α). Let $3 \le k \le 33$, $n \ge 2$ even, $\alpha = n/\sqrt{d} \le 0.005$ and $u_c = k\alpha^2/2$. Then

$$F_k(\alpha, u_c) \le F_{\text{tiny}}(\alpha) := \frac{\alpha}{2}\Big(2 - \log k - 2\log\alpha + \frac{3k\alpha^2}{8}\Big), \quad g(u_c) \le \log\frac{2d}{k},$$

F_{tiny} is increasing on $(0, 1/\sqrt{k})$. Hence $\log a_n(m) \le \sqrt{d}\, F_{\text{tiny}}(0.005) + \log 2 + \log(2d/k)$.

Proof. $1 - e^{-x} = 2e^{-x/2}\sinh(x/2) \ge xe^{-x/2}$, so $g(u) \le -\log u + u/2$. Bound $\int_0^1 \log(1 + e^{-uy})dy \le \log 2$ and $-\log(1 - e^{-uy}) \le -\log(uy) + uy/2$ on $[1, 2]$, using $\int_1^2 \log y\, dy = 2\log 2 - 1$: $\Phi_1(u) \le \frac{1}{2}(1 - \log 2 - \log u + \frac{3u}{4})$. With $u = u_c = k\alpha^2/2$: $-\log 2 - \log u_c = -\log k - 2\log\alpha$, $\frac{3u_c}{4} = \frac{3k\alpha^2}{8}$, and $\frac{(1-\alpha^2)u_c}{k\alpha} \le \frac{\alpha}{2}$; add. Monotonicity: $F'_{\text{tiny}}(\alpha) = -\log(\sqrt{k}\alpha) + \frac{9k\alpha^2}{16} > 0$ for $\sqrt{k}\alpha < 1$. Finally $g(u_c) \le -\log u_c + \frac{u_c}{2} = \log\frac{2d}{kn^2} + \frac{k\alpha^2}{4} \le \log\frac{2d}{k} - \log 4 + 0.001$. ■

Lemma 8.17 (central odd term). Let $A_0 \in (0, 1)$, d allowed, $d \ge D$ where $A_0\sqrt{D} \ge N_1$, $h = k/\sqrt{d}$. There is an odd integer $n^* \in [A_0\sqrt{d},\, A_0\sqrt{d} + 2k]$ with $k \mid d - n^{*2}$; $m^* = (d - n^{*2})/k \ge 0$ if $A_0 + 2h < 1$. Suppose $\underline{M}(U_{LO}) \ge (1/A_0^2 - 1)/k$ and $\overline{M}(U_{HI}) \le (1/(A_0 + 2h)^2 - 1)/k$, where $\underline{M}(u) \le \mu_n(e^{-u/n})/n^2 \le \overline{M}(u)$ are the enclosures of Table 8.1 valid for all $n \ge N_1$. Then the saddle u^* of n^* satisfies $U_{LO} \le u^* \le U_{HI}$. Let $\delta_c = \max\delta$ over the cells meeting $[U_{LO}, U_{HI}]$ and $S_{hi} \ge \sigma^2/n^3$ for $n \ge N_1$, $u \in [U_{LO}, U_{HI}]$. Then $o^* := a_{n^*}(m^*)$ satisfies

$$\log o^* \ge \sqrt{d}\,\Psi_{\text{low}} - C^*, \quad C^* = \tfrac{1}{2}\log 2 + g(U_{LO}) + \tfrac{1}{2}\log 2\pi + \tfrac{3}{2}\log(A_0\sqrt{d} + 2k) + \tfrac{1}{2}\log S_{hi} - \log(1 - \delta_c),$$

where $\Psi_{\text{low}} \le \min\{F_k(\alpha, u) : \alpha \in [A_0, A_0 + 2h],\, u \in [U_{LO}, U_{HI}]\}$.

Proof. The interval contains $2k$ consecutive integers, so one in every class mod $2k$; as k is odd, CRT gives an odd n^* in the class $n_0 \pmod k$ where $n_0^2 \equiv d \pmod k$ (d allowed). $\alpha^* = n^*/\sqrt{d} \in [A_0, A_0 + 2h]$ and $n^* \ge A_0\sqrt{D} \ge N_1$. Let $\rho^* = m^*/n^{*2} = \rho(\alpha^*) \in [(1/(A_0 + 2h)^2 - 1)/k, (1/A_0^2 - 1)/k]$. Then $\mu_{n^*}(e^{-U_{LO}/n^*})/n^{*2} \ge \rho^* \ge \mu_{n^*}(e^{-U_{HI}/n^*})/n^{*2}$ and since $u \mapsto \mu_{n^*}(e^{-u/n^*})$ is strictly decreasing, $u^* \in [U_{LO}, U_{HI}]$. At $m = \mu$, Proposition 8.11 gives $P(S = m^*) \ge (1 - \delta_c)/(\sqrt{2\pi}\sigma)$, with $\sigma \le \sqrt{S_{hi}}\, n^{*3/2}$. Lemma 8.15(c) with $g(u^*) \le g(U_{LO})$: $a_{n^*}(m^*) = A(r^*)r^{*-m^*}P \ge \exp(n^*\Phi_1(u^*) + u^*m^*/n^* - \frac{1}{2}\log 2 - g(U_{LO}))\cdot\frac{1-\delta_c}{\sqrt{2\pi}\sqrt{S_{hi}}\, n^{*3/2}}$, and $n^*\Phi_1(u^*) + u^*m^*/n^* = \sqrt{d}\, F_k(\alpha^*, u^*)$, $n^* \le A_0\sqrt{d} + 2k$. ■

Lemma 8.18 (window). Let $\alpha_1 \ge (1 + k\underline{M}(U_A))^{-1/2}$, $\alpha_2 \le (1 + k\overline{M}(U_B))^{-1/2}$, $D_0 \ge (N_1/\alpha_1)^2$. For $d \ge D_0$ and every even admissible n with $\alpha = n/\sqrt{d} \in [\alpha_1, \alpha_2]$: $n \ge N_1$ and $u_n \in [U_A, U_B]$.

Proof. $n = \alpha\sqrt{d} \ge \alpha_1 N_1/\alpha_1 = N_1$. $\rho(\alpha)$ is decreasing, $\rho(\alpha_1) \le \underline{M}(U_A)$ and $\rho(\alpha_2) \ge \overline{M}(U_B)$. The saddle satisfies $\mu_n/n^2 = \rho(\alpha)$; so $\mu_n(e^{-u_n/n})/n^2 \le \underline{M}(U_A) \le \mu_n(e^{-U_A/n})/n^2$ gives $u_n \ge U_A$ (monotonicity), and similarly $u_n \le U_B$. ■

Algorithm T. Input k and the numbers of Table 8.3 ($U_A, U_B, U_{LO}, U_{HI}, A_0$), the constants $\underline{M}(U_A)$, $\overline{M}(U_B)$, $\underline{M}(U_{LO})$, $\overline{M}(U_{HI})$, S_{hi} (from Algorithm E on $[U_{LO}, U_{HI}]$ cut into 35 equal pieces, $n_0 = 1500$) and δ_c. Set $\eta_t = 10^{-6}$, α_1, α_2, D_0 as in Lemma 8.18 (rounded in the safe direction),

$h = k/\sqrt{D_0}$, $\alpha_m = A_0 + h$. 1. (Window condition) verify $A_0\sqrt{D_0} \ge N_1$ and the two hypotheses of Lemma 8.17 for $d = D_0$. 2. Ψ_{low}: cut $[U_{LO}, U_{HI}]$ into boxes V_i of width 10^{-4} and take the minimum over i of the lower end of $F_k(\alpha_m, V_i) - h|\partial_\alpha F_k(\alpha_m, V_i)|$ (Lemma 8.15(e)). 3. α-boxes: $[0.005 + 0.0025i,\ 0.005 + 0.0025(i+1)]$ for $i \ge 0$, the last one truncated at α_1 (left tail), and $[\alpha_2 + 0.0025i, \alpha_2 + 0.0025(i+1)]$ truncated at 1 (right tail). For each box $[x, y]$ choose u_c (a double near the minimiser of $F_k(\frac{x+y}{2}, \cdot)$; any positive value is valid) and put $\text{Fup} = \max(F_k(x, u_c), F_k(y, u_c))^{\uparrow}$, $\text{gap} = \text{Fup} - \Psi_{\text{low}}$, $w_{box} = \sqrt{D_0}\,\text{gap} + g(u_c)$. 4. Tiny part: $\text{tiny_gap} = F_{\text{tiny}}(0.005) - \Psi_{\text{low}}$, $w_{\text{tiny}} = \sqrt{D_0}\,\text{tiny_gap} + \log(2D_0/k) + 1$. 5. $C^*(D_0)$ as in Lemma 8.17 at $d = D_0$, and $T(D_0) = \log\sqrt{D_0} + \log 2 + \max(\max_{box} w_{box}, w_{\text{tiny}}) + C^*(D_0) - \log\eta_t$. 6. Check $\max \text{gap} \le -2.5/\sqrt{D_0}$ and $\text{tiny_gap} \le -4.5/\sqrt{D_0}$ and $T(D_0) \le 0$.

Proposition 8.19 (tail bound). Suppose Algorithm T succeeds. Then for every allowed $d \ge D_0$,

$$\sum_{\substack{n \text{ even}, 2 \le n \le \sqrt{d} \\ n/\sqrt{d} \notin [\alpha_1, \alpha_2],\ k \mid d - n^2}} a_n\Big(\frac{d - n^2}{k}\Big) \le \eta_t\, o^* \le \eta_t\, O_d.$$

Proof. Fix $d \ge D_0$, put $x = \sqrt{d}$. By Lemma 8.17 $\log o^* \ge x\Psi_{\text{low}} - C^*(d)$: the window $[A_0, A_0 + 2k/x]$ is contained in the one at D_0, the u^*-hypotheses at d follow from those at D_0 (the right side of the second inequality increases with d), and Ψ_{low} is a lower bound on the larger window. For an even n with $\alpha = n/x \in [0.005, \alpha_1] \cup [\alpha_2, 1]$, α lies in a box $[x_b, y_b]$ and Lemma 8.15(d),(e) give $\log a_n \le x \max(F_k(x_b, u_c), F_k(y_b, u_c)) + \log 2 \le x\text{Fup} + \log 2$ (and $+g(u_c)$ may be added, harmlessly). For $\alpha < 0.005$, Lemma 8.16 gives $\log a_n \le xF_{\text{tiny}}(0.005) + \log 2 + \log(2d/k)$. There are at most $x/2 \le x$ even n. Hence the tail sum divided by o^* is at most $\exp(\tau(x))$ with

$$\tau(x) = \log x + \log 2 + \max\Big(\max_{box}[x(\text{Fup} - \Psi_{\text{low}}) + g(u_c)],\ x(F_{\text{tiny}}(0.005) - \Psi_{\text{low}}) + \log(2x^2/k)\Big) + C^*(x).$$

It suffices to show $\tau(x) \le \log\eta_t$ for $x \ge \sqrt{D_0}$; by hypothesis $\tau(\sqrt{D_0}) = T(D_0) + \log\eta_t \le \log\eta_t$. The function of x in the box part is $\log x + x\,\text{gap} + \frac{3}{2}\log(A_0 x + 2k) + \text{const}$ with derivative $\le \text{gap} + \frac{1}{x} + \frac{3A_0}{2(A_0 x + 2k)} \le \text{gap} + \frac{2.5}{x} \le 0$ for $x \ge \sqrt{D_0}$ (as $\text{gap} \le -2.5/\sqrt{D_0} \le -2.5/x$). The tiny part is $x\,\text{tiny_gap} + \log x + \log(2x^2/k) + \frac{3}{2}\log(A_0 x + 2k) + \text{const}$, with derivative $\le \text{tiny_gap} + \frac{1+2+1.5}{x} \le 0$. A maximum of non-increasing functions is non-increasing. Finally $o^* \le O_d$. ■

Proposition 8.20 (results of Algorithms R and T). For every odd $k \in \{3, \dots, 33\}$ both algorithms succeed with the parameters of Table 8.3 and give the values of Table 8.2 (in particular $R_k < 1 - 2\eta_t$, $T(D_0) < 0$, and $\max \text{gap} \cdot \sqrt{D_0}$ as listed).

This is a computation (Appendix A.3). Its inputs are the cells (Theorem 8.12) and the constants of Table 8.3.

8.6 Proof of Theorem 8.1

Part 1. Let k be odd, $3 \le k \le 33$, $d \ge D_0(k)$ allowed. The even n with $n^2 \le d$, $k \mid d - n^2$ are split by $\alpha = n/\sqrt{d}$. The terms with $\alpha \in [\alpha_1, \alpha_2]$ satisfy $n \ge N_1$ and $u_n \in [U_A, U_B]$ (Lemma 8.18), so by Proposition 8.14 their sum is $\le R_k O_d$. The remaining terms are bounded by Proposition 8.19 by $\eta_t O_d$; the term $n = 0$ vanishes for $d > 0$. Therefore $E_d \le (R_k + \eta_t)O_d$. By Lemma 8.17, $O_d \ge o^* > 0$. With (4.4), $(-1)^d g_k(d) = E_d - O_d \le -(1 - R_k - \eta_t)O_d < 0$ (Table 8.2: $R_k < 0.906$).

Part 2 (finite range). Section 8.7 describes the modular certificate for $0 \le d \le 6 \cdot 10^7$ and the exact evaluation of the candidates, giving Table 8.5 and Appendix B.1: the set of allowed $d \le 6 \cdot 10^7$ with $g_k(d) = 0$. Since $6 \cdot 10^7 > D_0(k)$ for each k, Part 1 and the finite computation together give Z_k in full; and $\max Z_k < D_0(k)$ (Table 8.5).

Part 3. By Proposition 3.4.

8.7 The finite certificate, the tables, and the zero sets

8.7.1 The modular certificate

Lemma 8.21. Let k be odd and $D \ge 0$. Put $a_0[m] = \delta_{m,0}$ and, for $n \ge 1$ and $0 \le m \le (D-n^2)/k$, $a_n[m] = a_{n-1}[m] - (-1)^n a_n[m-n]$ (with $a_n[m'] = 0$ for $m' < 0$). Then $a_n[m] = a_n(m)$ and for $0 \le d \le D$

$$G[d] := \sum_{n \ge 0,\ n^2 \le d,\ k \mid d-n^2} (-1)^n a_n\Big(\frac{d-n^2}{k}\Big) = (-1)^d g_k(d).$$

If $G[d] \not\equiv 0 \pmod p$ for some integer modulus $p \ge 2$, then $g_k(d) \ne 0$.

Proof. $A_n(x)(1+(-x)^n) = A_{n-1}(x)$ by (4.1); compare coefficients of x^m. The identity for G is (4.4). The recursion for $a_n[m]$ needs only $m \le (D-n^2)/k$, a range that decreases with n, so the entries computed at step $n-1$ suffice. Reduction mod p commutes with the integer recursion. ■

Algorithm M. Input k, $D = 6 \cdot 10^7$ and a modulus $p < 2^{63}$; compute $G[d] \bmod p$ for all $0 \le d \le D$ by Lemma 8.21 in 64-bit unsigned arithmetic (sums of two residues $< 2p < 2^{64}$ do not overflow), and print the allowed d with $G[d] \equiv 0$. (`cert/modcert_copy.c`; $O(D^{3/2}/k)$ operations.) Run with the two primes $p_1 = 2^{61} - 1 = 2305843009213693951$ and $p_2 = 4611686018427387847$ for each odd $k \in [3, 33]$ (the moduli need not be prime for the argument; both are).

Theorem 8.22 (zero sets). For each odd $k \in \{3, \dots, 33\}$ let C_k be the set of allowed $d \le 6 \cdot 10^7$ with $G[d] \equiv 0$ modulo both p_1 and p_2. Then every $d \in C_k$ has $g_k(d) = 0$ (exact evaluation in integer arithmetic), so $Z_k \cap [0, 6 \cdot 10^7] = C_k$. The cardinalities and maxima are in Table 8.5, the elements in Appendix B.1. In every case $\max Z_k < D_0(k) \le 6 \cdot 10^7$.

Proof. If d is allowed, $d \le D$ and $d \notin C_k$ then $G[d] \not\equiv 0$ for p_1 or p_2, so $g_k(d) \ne 0$ by Lemma 8.21. For $d \in C_k$, $g_k(d)$ is computed exactly by `code/zero_sets.py` and vanishes (the program also evaluates $g_k(d)$ exactly for all allowed $d \le \max C_k$ and asserts that the zeros are precisely C_k). ■

8.7.2 Tables

Table 8.2. Results of Algorithms R and T ($N_1 = 1500$, $\eta_t = 10^{-6}$, $H = 0.0025$). $\eta_k = 1 - R_k - 10^{-6}$ truncated to five decimals. Entries are printed to the digits shown; the stored values (`data/pd/pd56_results.json`) are directed-rounded doubles, and D_0 is exact.

k	R_k	$\max\delta$ on block cells	η_k	α_1	α_2	D_0	$T(D_0)$	$\max\text{gap}\cdot\sqrt{D_0}$
3	0.5077783	0.05933	0.49222	0.36739	0.74689	16,669,607	-69.25	-108.45
5	0.5601190	0.04374	0.43988	0.33225	0.72095	20,382,618	-54.18	-92.85
7	0.5989585	0.03885	0.40104	0.30672	0.70327	23,916,479	-49.60	-88.00
9	0.6311271	0.04069	0.36887	0.28726	0.68958	27,265,747	-47.21	-85.45
11	0.6601314	0.04215	0.33986	0.27013	0.67591	30,835,058	-48.67	-86.81
13	0.6867128	0.04462	0.31328	0.26126	0.66711	32,963,377	-40.85	-78.86
15	0.7098506	0.04621	0.29014	0.25133	0.65684	35,620,105	-38.58	-76.51
17	0.7316365	0.04789	0.26836	0.24350	0.64926	37,947,145	-35.41	-73.27
19	0.7529213	0.04973	0.24707	0.23414	0.64067	41,043,238	-36.87	-74.70
21	0.7740846	0.05103	0.22591	0.22915	0.63381	42,849,645	-32.55	-70.32
23	0.7949990	0.05225	0.20499	0.22225	0.62687	45,550,651	-32.75	-70.50
25	0.8160826	0.05402	0.18391	0.21627	0.62111	48,102,974	-32.56	-70.28

k	R_k	$\max \delta$ on block cells	η_k	α_1	α_2	D_0	$T(D_0)$	$\max \text{gap} \cdot \sqrt{D_0}$
27	0.8379773	0.05526	0.16202	0.21361	0.61493	49,312,289	-27.62	-65.28
29	0.8596991	0.05641	0.14029	0.20890	0.60963	51,557,554	-26.95	-64.58
31	0.8819648	0.05761	0.11803	0.20472	0.60507	53,684,094	-26.10	-63.72
33	0.9052174	0.05905	0.09478	0.20099	0.59981	55,698,896	-25.05	-62.64

Table 8.3. Window parameters (inputs; chosen from the critical point of F_k, any values passing the checks would do). In the continuation, $\underline{M}, \overline{M}$ are enclosures of μ_n/n^2 at the indicated points and S_{hi} is the maximum of $SB_{hi} + SG_{hi}$ over 35 equal pieces of $[U_{LO}, U_{HI}]$.

k	UA	UB	ULO	UHI	A0
3	0.21	1.1	0.346	0.519	0.508
5	0.27	1.38	0.435	0.653	0.466
7	0.31	1.59	0.501	0.751	0.438
9	0.34	1.76	0.554	0.83	0.417
11	0.36	1.89	0.598	0.896	0.401
13	0.39	2.02	0.636	0.953	0.387
15	0.41	2.12	0.669	1.004	0.375
17	0.43	2.22	0.699	1.049	0.365
19	0.44	2.3	0.726	1.089	0.356
21	0.46	2.38	0.751	1.127	0.348
23	0.47	2.45	0.774	1.161	0.341
25	0.48	2.52	0.795	1.193	0.334
27	0.5	2.58	0.815	1.223	0.329
29	0.51	2.64	0.834	1.251	0.323
31	0.52	2.7	0.851	1.277	0.318
33	0.53	2.75	0.868	1.302	0.313

Table 8.3 (continued). Lattice-sum constants at $n \geq 1500$.

k	$\underline{M}(U_A)$	$\overline{M}(U_B)$	$\underline{M}(U_{LO})$	$\overline{M}(U_{HI})$	S_{hi}
3	2.136238	0.264198	1.210238	0.744639	4.13818
5	1.611788	0.184786	0.920200	0.553556	2.59855
7	1.375649	0.145980	0.772751	0.458497	1.94612
9	1.235346	0.122548	0.680379	0.399027	1.58216
11	1.154952	0.108081	0.616503	0.357888	1.35068
13	1.050030	0.095924	0.568702	0.327282	1.18826
15	0.988744	0.087856	0.531774	0.303090	1.06913
17	0.933259	0.080721	0.501364	0.283879	0.97518
19	0.907444	0.075594	0.476250	0.268270	0.90042
21	0.859252	0.070921	0.454692	0.254579	0.83830
23	0.836728	0.067164	0.436159	0.243176	0.78640
25	0.815164	0.063686	0.420230	0.233110	0.74291

k	$\underline{M}(U_A)$	$\overline{M}(U_B)$	$\underline{M}(U_{LO})$	$\overline{M}(U_{HI})$	S_{hi}
27	0.774688	0.060908	0.405871	0.224213	0.70460
29	0.755671	0.058301	0.392911	0.216345	0.67074
31	0.737406	0.055853	0.381839	0.209391	0.64236
33	0.719850	0.053924	0.371233	0.203005	0.61565

Table 8.4. Sample of the 312 cells of Theorem 8.12 (all cells: `data/pd/cells56_sharp.json`). Over all cells: $\max\delta = 0.08727$ (cell $[3.30, 3.31]$, $\tau_0 = 0.55$), $\min\delta = 0.02847$, $\tau_0 \in [0.10, 0.55]$, $\max D_1 = 0.06539$, $\max D_2 = 2.21\cdot10^{-3}$, $\max D_3 = 1.99\cdot10^{-2}$, $\max c = 1.1870$; every monotonicity flag of Proposition 8.11 holds.

u-cell	τ_0	c	D_1	D_2	D_3	δ	S_{lo}	S_{hi}	M_{lo}	M_{hi}
[0.19,0.20]	0.10	1.187	0.0641	2.00e-41	3.32e-21	0.06412	12.3906	13.8664	2.2538	2.3995
[0.49,0.50]	0.10	0.441	0.0302	1.01e-07	7.19e-07	0.03021	1.9378	2.0371	0.7746	0.8003
[0.79,0.80]	0.15	0.419	0.0308	1.02e-06	3.80e-06	0.03080	0.7263	0.7530	0.4165	0.4276
[1.19,1.20]	0.20	0.393	0.0331	3.02e-05	6.32e-05	0.03319	0.2983	0.3073	0.2286	0.2341
[1.69,1.70]	0.25	0.375	0.0376	5.66e-04	8.21e-04	0.03903	0.1310	0.1347	0.1286	0.1315
[2.19,2.20]	0.35	0.438	0.0457	5.11e-04	7.28e-04	0.04697	0.0679	0.0698	0.0810	0.0828
[2.69,2.70]	0.45	0.490	0.0547	7.12e-04	9.96e-04	0.05641	0.0390	0.0401	0.0550	0.0563
[3.19,3.20]	0.55	0.536	0.0646	1.15e-03	7.47e-03	0.07320	0.0242	0.0249	0.0395	0.0405
[3.30,3.31]	0.55	0.523	0.0654	1.98e-03	1.99e-02	0.08727	0.0219	0.0226	0.0370	0.0379

Table 8.5. Zero sets of Theorem 8.22.

k	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33
$\#Z_k$	9	17	24	20	43	54	55	80	89	88	105	76	68	121	127	88
$\max Z_k$	43	176	247	118	273	545	565	1137	766	2055	837	936	1375	1934	1712	1587

8.8 Computer-assisted steps and what each relies on

The following steps are carried out by programs; everything else in Section 8 is proved by hand above.

1. **Lattice-sum enclosures** (Algorithm E) for the 312 cells and for the window points: Arb ball arithmetic, 128 bits, through python-flint [21].
2. **Cells** (Algorithm C, Theorem 8.12): the same arithmetic. Output: δ for 312 cells.

3. **Block bound** (Algorithm R, Proposition 8.14), for 16 values of k and about $(U_B - U_A)/0.0025$ boxes each: interval arithmetic over Arb balls.
4. **Tail bound** (Algorithm T, Propositions 8.19, 8.20): Arb balls, with the dilogarithm of Arb.
5. **Modular certificate** (Algorithm M) and the exact evaluation of C_k (Theorem 8.22): integer arithmetic only (a C program with 64-bit residues; exact Python integers).

The only places where floating-point numbers enter as inputs are: the double-precision endpoints of cells and boxes (each is an exactly representable binary number, converted without rounding to a ball), the choice of τ_0, u_c, $U_A, \dots, A_0$ (any value is valid; correctness never depends on how they were chosen), and output roundings, which are all directed (upward for upper bounds, downward for lower bounds). The programs were re-run from their inputs for this paper: every output file (window parameters, cells, constants, results) is identical to the stored one (Appendix A.3). This checks reproducibility, not independence of the method.

8.9 Remarks

An independent derivation of the block step (separate code, a different decomposition of the circle into five regimes, cells of width 0.005 on $[0.40, 2.90]$, a fourth-derivative remainder) reports, for $k = 27, 29, 31, 33$, block ratios 0.79055, 0.80677, 0.82294, 0.83906, below the values R_k of Table 8.2. These numbers play no role in the proofs above and were not recomputed for this paper.

What is proved and what is not. Theorem 8.1 is proved for the sixteen values $k = 3, \dots, 33$ through the finite data above. It is a statement about each fixed k; no bound uniform in k is claimed, and no extension to $k \geq 35$ or to even k is asserted.

Part II
The Bridge Series C

9 Introduction to Part II

9.1 Setting and question

The series studied here arise as the "Bridge Series C" of the author's S^2-11DM^2ET-X programme. The question posed there was whether C satisfies a *residual law*: a modular part plus a controlled remainder, or an exact weight-$\frac{1}{2}$ slash relation. This part is self-contained. All objects are defined from scratch in §10, and nothing in it depends on, or is claimed to bear on, other parts of that programme.

9.2 Summary

In one sentence: C and C_{third} share the remainder P_s^{orth}; up to theta and mock pieces, P_s^{orth} is one ray class partial theta series of $\mathbb{Q}(\sqrt{35})$; its regulator-type singularity rules out every scalar modular or mock residual law (Baker); and the obstruction is resolved not by a scalar law but by a vector-valued weight-1 quantum modular structure, coming from a Siegel theta lift to weight-0 Maass forms, whose cocycle is computed in closed form. The precise statements are collected in §11.

9.3 What is not claimed

No claim is made about the twin prime conjecture or the Riemann hypothesis, and nothing here concerns any termination claim of the wider programme. We do not claim that C, C_{third} or P_s^{orth} is modular, mock modular or quantum modular as a scalar q-series. The quantum-modular statements concern the fundamental-domain cone-sum vector of §15. The K3 statements of §26 concern a shared representation, not a shared modular form.

9.4 Organisation

Section 10 fixes conventions and notation, and §11 states the main theorems. Sections 12–18 prove them: the ray class decomposition, the radial asymptotics and the Baker obstruction, the profile law and the theta lift, the Lewis–Zagier period formula, the Eisenstein/cusp split, the Herglotz closed form, and the $L(0)$ data for the cusp rows. Sections 19–21 treat the lift family for general s (tensor structure, dimensions, the cusp content of u_L). Sections 22–25 treat H_s, the cone, C_{third} and the dependence on s. Section 26 gives the K3 realisation, §27 the uniqueness of the vector completion, and §28 the obstructions for coefficient-emitting maps. Sections 29–32 collect consequences, numerical checks, dependencies and open problems. Appendix C lists the scripts that carry out the exact and numerical verifications.

10 Conventions and notation

10.1 Conventions on status

Every result carries one of two labels.

- **Category A**: proved in this paper; or *exact*, meaning a finite verification in rational, integer, modular or cyclotomic arithmetic over a stated finite set (including exact ranks, and ranks modulo a prime ℓ together with a stated argument that transfers them to characteristic 0); or proved *conditional on* a named published theorem, stated each time. Where a written proof is long and only its structure is reproduced here, it is marked "Proof (outline)".
- **Category B**: statements whose only support is floating-point or multiprecision numerics (including singular-value ranks, FFT character counts and fits), PSLQ recognition, identification by citation only, interpretation, analogy, patterns observed on finitely many cases, and conjectures.

Numerical checks reported inside a Category A status box are supporting evidence for a statement that is proved or exact. The scripts behind every exact and numerical claim are listed in Appendix C. Section 31 lists the cited results on which proofs depend. *Computer-assisted proofs.* A finite verification carried out in rigorous ball (interval) arithmetic (Arb), in which every quantity entering the final inequality is a proven enclosure of the exact value, counts as Category A and is labelled "Category A (computer-assisted)". *Naming.* Results are numbered by section, continuously with the rest of the paper.

10.2 The lattice and the quadratic polynomial

Write $\mathbf{e}(x) = e^{2\pi i x}$ and $q = \mathbf{e}(\tau)$, $\tau \in \mathbb{H}$. Let

$$A = \begin{pmatrix} 2 & 7 \\ 7 & 7 \end{pmatrix}, \qquad B(x,y) = x^{\mathsf{T}} A y, \qquad Q(x) = \tfrac{1}{2} B(x,x) = x_1^2 + 7x_1x_2 + \tfrac{7}{2}x_2^2.$$

Then $\det A = -35$, Q has signature $(1,1)$, $L = \mathbb{Z}^2$ is an *odd* lattice ($Q(L) \not\subset \mathbb{Z}$), $L' = A^{-1}\mathbb{Z}^2$ with $A^{-1} = \begin{pmatrix} -1/5 & 1/5 \\ 1/5 & -2/35 \end{pmatrix}$, $|L'/L| = 35$, and the level is 70 (the least N with NA^{-1} even integral). Put

$$Q_7(n,k) = Q(n,k) + \tfrac{7}{2}k = n^2 + 7nk + \tfrac{7}{2}k(k+1), \qquad Q_-(n,k) = n^2 + 7nk + \tfrac{7}{2}k(k-1) = Q_7(-n,-k).$$

With $a = (\frac{7}{10}, -\frac{1}{5})$ one has $Aa = (0, \frac{7}{2})$, $Q(a) = -\frac{7}{20}$, hence

$$Q_7(v) = Q(v+a) + \tfrac{7}{20} \qquad (v \in \mathbb{Z}^2). \tag{10.1}$$

The characteristic vector of the odd lattice is $w_c = A^{-1}(0, \frac{1}{2}) = (\frac{1}{10}, -\frac{1}{35})$; it satisfies $\mathbf{e}(Q(n)) = \mathbf{e}(B(n, w_c))$ for $n \in \mathbb{Z}^2$ and $w_c \equiv a \pmod{L'}$. Let $\varepsilon = 6 + \sqrt{35}$, the fundamental unit of $K = \mathbb{Q}(\sqrt{35})$ ($\mathrm{N}\varepsilon = +1$), and

$$I = \frac{\log \varepsilon}{2\sqrt{35}} = 0.209419820314714631473\ldots$$

10.3 The series

Let $c_1 = (-1, 1)$, $c_2 = (-4, 1)$; then $Q(c_1) = -\frac{5}{2}$, $Q(c_2) = -\frac{17}{2}$ (negative norm), and for $v = (n,k)$, $B(c_1, v) = 5n$, $B(c_2, v) = -n - 21k$. With $\operatorname{sgn}(0) = 0$ put $w(v) = \frac{1}{2}\big(\operatorname{sgn} B(c_1, v) - \operatorname{sgn} B(c_2, v)\big)$ (so $w = \frac{1}{2}$ on the walls). Define

$$P_s^{\mathrm{orth}} = \sum_{n,k \ge 0} q^{Q_7(n,k)}, \qquad H_s = \sum_{v \in \mathbb{Z}^2} w(v)\, q^{Q_7(v)}, \qquad C := P_s^{\mathrm{orth}} - H_s,$$

$$C_{\mathrm{axis}} = \tfrac{3}{2} + \sum_{n \ge 1} q^{n^2} + \sum_{k \ge 1} q^{7k(k+1)/2}, \quad C_{\mathrm{third}} = \sum_{n,k \ge 1} q^{Q_-(n,k)}, \qquad C_{\mathrm{mixed}} := C - C_{\mathrm{axis}} - C_{\mathrm{third}}.$$

Further, $\theta_3 = \sum_{n \in \mathbb{Z}} q^{n^2}$, $R := \sum_{n \ge 0,\, k \ge 1} q^{Q_7(n,k)}$,

$$\Theta_{\mathrm{hol}} := \sum_{n \ge 1,\, k \ge 0} q^{Q_-(n,k)} - \sum_{n \le 0,\, k \le -1} q^{Q_-(n,k)},$$

the holomorphic part of a Zwegers indefinite theta function [18], and $f_{a,b,c}$ is the Hecke-type double sum of Hickerson–Mortenson [8, Def. 0.2], $f_{a,b,c}(x,y,q) = \sum_{\operatorname{sgn}(r) = \operatorname{sgn}(s)} \operatorname{sgn}(r)(-1)^{r+s} x^r y^s q^{a\binom{r}{2} + brs + c\binom{s}{2}}$, with $\operatorname{sgn}(r) = 1$ for $r \ge 0$ and -1 for $r < 0$ in this formula.

Finally, for a rational $x = h/m$ ($\gcd(h,m) = 1$) define the Gauss mean

$$G(x) = (2m)^{-2} \sum_{(n,k) \bmod 2m} \mathbf{e}\big(x Q_7(n,k)\big).$$

Throughout, "radial" refers to $q = \mathbf{e}(x)e^{-t}$ with x rational and $t \to 0^+$, and t^r coefficients are those of the resulting asymptotic expansion.

10.4 Notation for the lift family

For an integer $s \ge 5$ let $A_s = \left(\begin{smallmatrix} 2 & s \\ s & s \end{smallmatrix}\right)$, $Q_s(n,k) = n^2 + snk + \frac{s}{2}k(k+1)$, $a = A_s^{-1}(0, \frac{s}{2})$, $c_0 = s/(4(s-2))$ and $\gamma = \left(\begin{smallmatrix} -1 & -s \\ 2 & 2s-1 \end{smallmatrix}\right)$ (so that $Q_s(v) = Q(v+a) + c_0$; at $s = 7$ these are the objects above).

- $n_s := s(s-2)$ (the radicand: $35, 48, 99$ at $s = 7, 8, 11$); D_s is the discriminant of the order of the form $(140, 48, 396)$; $\varepsilon_s = (s-1) + \sqrt{n_s}$.
- N_s is the number of labels of the lift family; $N_s = 4n_s$ for odd s (so $N_7 = 140$).
- S_{lab}, T are the label matrices of the S- and T-laws (Theorem 11.6); $P = S_{\mathrm{lab}}^2$ is the label involution and $W_P := \mathrm{Fix}(P)$ ($\dim W_P = 72$ at $s = 7$). The representation of $\mathrm{SL}_2(\mathbb{Z})$ in the standard presentation is $\rho(S) := S_{\mathrm{lab}}^{-1}$, $\rho(T) := T$ (Remark 19.1); on W_P, $S_{\mathrm{lab}} = S_{\mathrm{lab}}^{-1}$.
- V is the span of the values of the lift vector (the function span); V_{Eis} is the $\mathrm{SL}_2(\mathbb{Z})$-cyclic span of the constant-term vector, $d_{\mathrm{Eis}} := \dim V_{\mathrm{Eis}}$, $d_{\mathrm{cusp}} := \dim V - d_{\mathrm{Eis}}$; $V^{\perp}$ is the complement of V in W_P.
- N denotes the negation map $u_{\alpha,\beta} \mapsto u_{-\alpha,-\beta}$ on labels (not to be confused with N_s); Z_p is the local reflection $x \mapsto -x$ on a p-primary factor, and $Z_{(m)} = \bigotimes_{p|m} Z_p$.
- $W_p^{\pm}$ are the even and odd parts of a local Weil representation; A_3, B_1 are the 3- and 1-dimensional summands of the 2-adic factor ρ_2 (Theorem 19.3); $X^{\pm} = W_5^{\pm} \otimes W_7^{\pm}$ at $s = 7$.
- H_s always denotes the cone series of §10 with the fixed walls $c_1 = (-1, 1)$, $c_2 = (-4, 1)$ (for every s). The axes-cone series H_s^{axes} of §23 is a separate object. We write $g := \gcd(24, s-8)$ and $\Delta := (s-2)(7s-32)$.

11 Main results

Each theorem is followed by its status; proofs are in the sections indicated.

Theorem 11.1 (Shared remainder)**.** *As formal q-series,*

(i) $C_{\mathrm{third}} = q^8 f_{2,7,7}(-q^{10}, -q^{14}, q) + P_s^{\mathrm{orth}}$;

(ii) $C - C_{\mathrm{third}} = -H_s - q^8 f_{2,7,7}(-q^{10}, -q^{14}, q)$;

(iii) $\Theta_{\mathrm{hol}} = q^8 f_{2,7,7}(-q^{10}, -q^{14}, q) + \theta_3 = C_{\mathrm{third}} + \sum_{n \geq 1} q^{n^2} - R$, *and* $P_s^{\mathrm{orth}} = R + \frac{1}{2}(1 + \theta_3)$;

(iv) $C_{\mathrm{mixed}} = -H_s - q^8 f_{2,7,7}(-q^{10}, -q^{14}, q) - C_{\mathrm{axis}}$ *vanishes through* q^{293} *but not identically:* $C_{\mathrm{mixed}} = -\frac{1}{2}q^{294} + \frac{1}{2}q^{301} - q^{330} + q^{337} - q^{368} + q^{375} - \cdots$.

Status (Category A). (i) is proved by reindexing, and (ii)–(iv) follow from (i) and the definitions (§12.1). All identities are also checked with exact rational coefficients through q^{3000}; the first nonzero terms in (iv) are an exact computation.

Theorem 11.2 (Real-quadratic decomposition and ray class)**.** *Let $\mathcal{K}$ be the narrow ray class modulo $\mathfrak{p}_2\mathfrak{p}_5\infty_1\infty_2$ of $K = \mathbb{Q}(\sqrt{35})$ containing $\mathfrak{p}_7(\alpha_1)$ for any $\alpha_1 \equiv 1 \bmod {}^{\times}\mathfrak{p}_2\mathfrak{p}_5$ with signs $(-,+)$. Then*

$$2P_s^{\mathrm{orth}} = E + 1 + \theta_3 - \Theta_{\mathrm{hol}}, \qquad E(q) = \sum_{\mathfrak{a} \in \mathcal{K}} q^{(\mathrm{N}\mathfrak{a}+7)/20},$$

and E is the sum of $q^{Q_7(v)}$ over $v + a$ in one fundamental cone for the action of the automorph $\gamma = \begin{pmatrix} -1 & -7 \\ 2 & 13 \end{pmatrix}$ (eigenvalues $6 \pm \sqrt{35}$) on the positive sector.

Status (Category A). Proved in §§12.2–12.3; the proof uses that $\mathbb{Z}[\sqrt{35}]$ is the maximal order and that its unit group is $\pm\varepsilon^{\mathbb{Z}}$. The cone identity is also checked exactly to q^{3000} for four fundamental cones.

Theorem 11.3 (Radial asymptotics and partial zeta values)**.** *As $t \to 0^+$, with $q = e^{-t}$, $P_s^{\mathrm{orth}} = I/t + a_P\, t^{-1/2} + \frac{11-7I}{20} + O(t^{1/2})$ with $a_P = \sqrt{\pi}(49-2\sqrt{14})/140 = 0.5256172043\ldots$; $E = 2I/t + \frac{1-7I}{10} + O(t)$ with no half-integral powers; $\zeta(0,\mathcal{K}) = \frac{1}{10}$, $\zeta(-1,\mathcal{K}) = \frac{29}{60}$, $\operatorname{Res}_{s=1}\zeta(s,\mathcal{K}) = \log\varepsilon/(20\sqrt{35})$; and $C = I/t + a_C\, t^{-1/2} + \frac{11-7I}{20} + O(t^{1/2})$ with*

$$a_C = \sqrt{\pi}\left(833 + 204\sqrt{14} - 14\sqrt{1190}\right)/2380 = 0.8291426171\ldots$$

The full table of coefficients is Table 13.

Status (Category A). The rows for P_s^{orth}, R, E and a_P are exact, by symbolic two-dimensional Euler–Maclaurin (Proposition 13.1). $\zeta(0,\mathcal{K})$ and $\zeta(-1,\mathcal{K})$ are exact rational arithmetic, confirmed by five independent Shintani cone decompositions. $a_H = H_{-1/2}(7)$ and $a_C = C_{-1/2}(7)$ are the $s = 7$ values of the closed forms of Theorem 22.1; they agree symbolically with the radical forms above. The constant term 0 of H_s is part of Theorem 22.1.

Theorem 11.4 (Gauss-mean pattern and zero set)**.** *For $\gcd(h,m) = 1$: $G(h/m) = 0$ if $2 \mid m$ or $5 \mid m$; $|G(h/m)| = 1/m$ if $\gcd(m,70) = 1$; $|G(h/m)| = \sqrt{7}/m$ if $7 \mid m$ and $\gcd(m,10) = 1$. Moreover $\lim_{t\to 0^+} t\, S(\mathbf{e}(x)e^{-t}) = I \cdot G(x)$ for $S \in \{P_s^{\mathrm{orth}}, C, C_{\mathrm{third}}\}$ and every rational x.*

Status (Category A). The pattern is proved in §13.5 and checked on all 3374 coprime pairs with $m \le 105$. The limit follows from leading-order Euler–Maclaurin (Proposition 13.1); it is checked numerically down to $t = 6.25 \cdot 10^{-4}$.

Theorem 11.5 (Baker obstruction: scalar no-go)**.** *(i) (Additive.) There is no identity $S = F + \mathcal{E}$ with $S \in \{C, C_{\mathrm{third}}\}$, F in the $\overline{\mathbb{Q}}$-span $\mathcal{M}$ of §13 (holomorphic modular forms with algebraic coefficients, signature-$(1,1)$ Zwegers holomorphic parts with rational characteristics, unary/partial/false theta series), and $\mathcal{E}(e^{-t}) = o(1/t)$.*

(ii) (Multiplicative.) For holomorphic modular forms $g \not\equiv 0$ and F of any weights, levels and multipliers, with complex coefficients, and any q^r factor: $C_{\mathrm{third}} \cdot g \neq F$ and $C \cdot g \neq F$.

(iii) (Scalar W_7.) For no k and no constant λ is $C(\tau) - \lambda(7\tau)^{-k}C(-1/(7\tau))$ radially bounded at all rationals.

Status (Category A). (i) is proved *conditional on* Baker's theorem [1] and on the q-expansion principle (algebraicity of expansions at all cusps), both cited. (ii) is proved; for C the nonvanishing input is the closed form $a_C = \sqrt{\pi}(833 + 204\sqrt{14} - 14\sqrt{1190})/2380 \neq 0$ (Theorem 11.3). (iii) is an exact finite computation: 178 of 270 test rationals (denominator < 30) have exactly one of $G(x)$, $G(-1/(7x))$ zero, including $x = 5/7$; the zero pattern is that of Theorem 11.4.

Theorem 11.6 (Theta lift and vector-valued S/T law)**.** *Let $u_{\alpha,\beta}$ ($\alpha \in (\{0,a\}+L')/\mathbb{Z}^2$, $\beta \in \{0,-a\}$; 140 labels) be the Siegel-theta lifts of §14. Each $u_{\alpha,\beta}$ is a weight-0 Laplace eigenfunction with $\lambda = \frac{1}{4}$, and*

$$u_{\alpha,\beta}(-1/w) = \frac{\mathbf{e}(B(\alpha,\beta))}{\sqrt{35}} \sum_{p \in L'/\mathbb{Z}^2} u_{p-\beta,\alpha}(w), \qquad u_{\alpha,\beta}(w+1) = \mathbf{e}\big(-Q(\alpha) - B(\alpha,w_c)\big)\, u_{\alpha,\alpha+\beta+w_c}(w).$$

The resulting S, T matrices are unitary, $S^2 = P$ (the label involution $(\alpha,\beta) \mapsto (-\alpha,-\beta)$), $(ST)^3 = I$, and on the 72-dimensional P-invariant subspace $S^2 = (ST)^3 = I$: a 72-dimensional

representation of $\mathrm{PSL}_2(\mathbb{Z})$ *on label space; we write* $W_P := \mathrm{Fix}(P)$. *The functions* $u_{\alpha,\beta}$ *span a* 42*-dimensional subspace* $V \subset W_P$, *invariant under* S *and* T *(Theorem 11.8, §16.3). Here* S *is the label matrix* S_{lab}; *the representation of* $\mathrm{SL}_2(\mathbb{Z})$ *in the standard presentation is* $\rho(S) := S_{\mathrm{lab}}^{-1}$, $\rho(T) := T$ *(Remark 19.1).*

Status (Category A). The S-law is proved by Poisson summation (§14.3), independently of [3]; the T-law is elementary. The relations $S^2 = P$, $P^2 = I$, $[P,T] = 0$ and $(ST)^3 = I$ on all of $\mathbb{C}^{140}$ are exact (verified at all 48 Galois conjugates over $\mathbb{Q}(\zeta_{140})$). $\dim V = 42$ is exact (Proposition 19.4). The label representation is exactly $\rho_{2x^2/5} \otimes \rho_{x^2/7} \otimes \rho_2$ (Theorem 19.3) and is isomorphic to the Weil representation of the even lattice N_2 (Theorem 26.2). The S/T laws are also verified numerically on all 140 rows ($\le 1.2 \cdot 10^{-15}$; supporting).

Theorem 11.7 (Lewis–Zagier period formula). *For* $z \in \mathbb{H}$ *let* $\Psi(z) = E_L(-1/z) - (z/\sqrt{35})E_R(z)$ *and* $\Psi^-(z) = E_L(-1/z) + (z/\sqrt{35})E_R(z)$ *(§15.1), and for* $z \in \mathbb{C} \setminus i\mathbb{R}$ *put*

$$\Omega(z) = \frac{z}{\pi\sqrt{35}}\Big(\int_0^{i\infty} [u_R, R_z^{1/2}] - \frac{\log \varepsilon}{2}\Big),$$

the integral taken along $i\mathbb{R}_{>0}$ *with the principal square root.*

(i) Ω *is holomorphic on* $\mathbb{C} \setminus i\mathbb{R}$; *on* $\mathbb{H} \cap \{\mathrm{Re}\, z > 0\}$ *it equals* Ψ, *and on* $\mathbb{H} \cap \{\mathrm{Re}\, z < 0\}$ *it equals* $-\Psi^-$. *In particular* Ψ *continues holomorphically to* $U = \mathbb{H} \cup \{\mathrm{Re}\, z > 0\}$.

(ii) Let ψ *be the order-*t^0 *radial* S*-cocycle of §15.1. For rational* $x > 0$, $\psi(x) = \Omega(x)$; *for rational* $x < 0$, $\psi(x) = -\Omega(x) = \frac{|x|}{\pi\sqrt{35}}\big(\int_0^{i\infty}[u_R, R_x^{1/2}] - \frac{\log\varepsilon}{2}\big)$.

(iii) The analogous closed form holds on every row of the 140*-label vector cocycle* Φ, *which is real-analytic on* $\mathbb{R} \setminus \{0\}$ *and satisfies the three-term relation* (15.2).

Status (Category A). Proved in §15.4, given Lemma 15.1, the branch lemma (Lemma 15.2), the S-law of Theorem 11.6, the sector-uniform Euler–Maclaurin expansion (Proposition 13.1) and the Fourier interchange R2 (§15.6). Numerically: the period formula agrees with exact Bernoulli sums at 84 rationals (denominators ≤ 60, both signs) to $1.1 \cdot 10^{-13}$; $|\Omega - \Psi| \approx 10^{-16}$ at three points with $\mathrm{Re}\, z > 0$ and $|\Omega + \Psi^-| \approx 10^{-16}$ at three points with $\mathrm{Re}\, z < 0$ (where $|\Omega - \Psi| = 0.076, 0.069, 0.539$); all 140 rows to $2.3 \cdot 10^{-14}$; the three-term relation to $7 \cdot 10^{-15}$. The phrase "quantum modular form in Zagier's sense" for the real factor $|cx + d|$ is terminology (Category B).

Theorem 11.8 (Eisenstein/cusp split). $u_L = u_{a,0}$ *and* u_R *are pure Eisenstein (combinations of the* 8 *genus series attached to* $\psi \circ \mathrm{N}$, $\psi \bmod 20$*). The* 42*-dimensional function span* V *splits* $\mathrm{SL}_2(\mathbb{Z})$*-equivariantly as* $V_{\mathrm{Eis}} \oplus V_{\mathrm{cusp}}$ *with* $\dim V_{\mathrm{Eis}} = 36$ *(from the* 116 *Eisenstein components) and* $\dim V_{\mathrm{cusp}} = 6$ *(from the* 24 *cusp components, §16). The cusp components are exact combinations of* 12 *dihedral Maass forms* ($\lambda = \frac{1}{4}$) *from the ray class characters of conductor* $(2\sqrt{35})$ *of pure infinity type; these* 12 *forms occur only in* 6 *fixed linear combinations. Consequently* $\psi_{\mathrm{cusp}} \equiv 0$ *for the scalar cocycle.*

Status (Category A). u_L is Eisenstein by a hand proof independent of PARI (§16.1), with an exact coefficient match for all $n \le 60\,000$. The dimensions 36 and 6 and the S,T-invariance of the split are exact: $V_{\mathrm{Eis}} = A_3 \otimes X^+$ and $V_{\mathrm{cusp}} = B_1 \otimes X^-$ (Proposition 19.4). $\psi_{\mathrm{cusp}} \equiv 0$ follows because ψ involves only u_L and u_R.

Category B. The expression of the cusp components through the 12 named dihedral forms rests on floating-point fits (§16.2); the rank 6 of the 12 forms inside the cusp components is floating, and is explained exactly by $\dim(B_1 \otimes X^-) = 6$. Cuspidality $\Leftrightarrow \chi \ne \chi \circ \sigma$, the level 19600, the nebentypus and newform status are standard theory [11], cited.

Theorem 11.9 (Herglotz closed form)**.** *On* $U = \mathbb{H} \cup \{\operatorname{Re} z > 0\}$,

$$\Psi(z) = \frac{1}{8} \sum_{j\in\{0,2,4,6\}} \sum_{l\in\{0,1\}} \mathbf{e}(-j/8)\, R_{j,l}(z), \qquad R_\chi(z) = \sum_{m\geq 1} a_\chi(m) \sum_{s=1}^{140} \rho_s^\chi\, \psi_\Gamma\Big(\frac{s}{140} + \frac{m}{20z}\Big),$$

with $\psi_\Gamma = \Gamma'/\Gamma$ *and the data of §17. In particular (given R1)* $\psi(x)$ *for rational* $x > 0$ *equals this expression.*

Theorem 11.10 (Closed form of E_R)**.**

$$E_R(z) = \frac{\sqrt{35}}{140} \sum_\chi w_\chi \sum_{n,k\geq 1} \widehat{b}_\chi(n)\, \check{a}_\chi(k)\, \mathbf{e}(nkz/140), \qquad w_\chi = \mathbf{e}(-j/8)/16.$$

Status (Category A). Theorems 11.9 and 11.10 are proved in §17 from the Eisenstein expansion of E_L (§16.1), the S-law (Theorem 11.6), Lemma 15.1 and Theorem 11.7(i), which is used on $\operatorname{Re} z > 0$ only. The proof is given in outline; the full estimates are in the companion report listed in Appendix C. Numerically, Theorem 11.9 holds to $5.5\cdot 10^{-35}$ on $\mathbb{H}$, to $\leq 9.2\cdot 10^{-16}$ off $\mathbb{H}$ and to $5.7\cdot 10^{-35}$ at rationals; Theorem 11.10 holds for all $N \leq 14\,000$ (627 nonzero coefficients identical as elements of $\mathbb{Z}[\phi]$). No arithmetic (Gauss-sum or ideal-theoretic) proof of Theorem 11.10 is claimed.

Remark (Category B) 11.11 (Cusp rows and $L(0)$)**.** For every cusp row c and each of the seven rationals $x \in \{1, \frac{1}{3}, 2, \frac{3}{5}, \frac{7}{4}, -1, -\frac{2}{3}\}$, the radial constants $\kappa_c(h/k)$ entering $\Phi_c(x)$ are explicit finite combinations of ray-class Hecke L-values $L_{\mathfrak{m}_k}(0, \chi')$ of K; only totally odd characters contribute, and the $L(0)$ values lie in $\mathbb{Z}[\zeta_{12}]$. Replacing $L(0,\cdot)$ by $L(\frac{1}{2},\cdot)$ fails by up to 0.32.

Category B. Verified numerically at the seven points for all 24 cusp rows, to $4\cdot 10^{-14}$ with PARI `lfun` values and to $1.2\cdot 10^{-14}$ with PARI-free inputs (§18). Not proved.

Status (Category A). The 12 values $L(0,\chi) \in \mathbb{Z}[\zeta_{12}]$ are exact (PARI-free Shintani sums), and the structure $\mathrm{Cl}^+_{(2\sqrt{35})\infty_1\infty_2} \cong \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/24$ with its σ-action, types and conductors is certified by hand (§18).

12 The real-quadratic decomposition and the ray class

12.1 Elementary identities (Theorem 11.1)

Proof of Theorem 11.1. (i). In $q^8 f_{2,7,7}(-q^{10}, -q^{14}, q)$ put $n = r+1$, $k = s+1$. The signs $(-1)^{r+s}$ from $x = -q^{10}$, $y = -q^{14}$ cancel those in f, and on $r, s \geq 0$ the exponent is $8 + 10r + 14s + r(r-1) + 7rs + \frac{7}{2}s(s-1) = Q_-(n,k)$; this orthant gives C_{third}. On $r, s < 0$, after $(r,s) = (-1-n, -1-k)$ the exponent is $Q_7(n,k)$ with sign -1; this gives $-P_s^{\text{orth}}$.

(ii) is (i) combined with $C = P_s^{\text{orth}} - H_s$. For (iii): in the second sum defining Θ_{hol} substitute $(n,k) = (-n', -k')$ with $n' \geq 0$, $k' \geq 1$; since $Q_-(-v) = Q_7(v)$ it equals R. The first sum is $C_{\text{third}} + \sum_{n\geq 1} q^{n^2}$ (split $k \geq 1$ and $k = 0$). Hence $\Theta_{\text{hol}} = C_{\text{third}} + \sum_{n\geq 1} q^{n^2} - R$. The $k = 0$ row of P_s^{orth} gives $P_s^{\text{orth}} = R + \sum_{n\geq 0} q^{n^2} = R + \frac{1}{2}(1 + \theta_3)$, and then (i) gives $\Theta_{\text{hol}} = q^8 f + \theta_3$. (iv) is the definition combined with (ii). □

Status (Category A). All identities in Theorem 11.1 were in addition checked with exact rational coefficients through q^{3000}; the H_s summation box was enlarged by 30 with identical output.

Remark 12.1 (Origin of C_{mixed})**.** At q^{294} the point $(21,-1)$ lies on the c_2-wall $n+21k=0$, has $w=\frac{1}{2}$ and lies outside the orthant, contributing $-\frac{1}{2}$. Points such as $(22,-1)$ (q^{330}) and $(-22,1)$ (q^{337}) lie in the H_s cones but outside the orthant. Thus "$C = C_{\text{axis}} + C_{\text{third}}$" holds only through q^{293}. The coefficient set $\{0,1,\frac{3}{2}\}$ holds through q^{101} only (first coefficient 2 at q^{102}, first 3 at q^{144}, first $\frac{1}{2}$ at q^{294}); through q^{3000} the set is $\{0,\frac{1}{2},1,\frac{3}{2},2,3\}$. The support $\{0,1,2,4\}$ mod 7 holds for all terms since $Q_7 \equiv n^2 \pmod 7$. (Category A, exact.)

Consequently, every part of C and of C_{third} other than P_s^{orth} is a signature-$(1,1)$ mock/indefinite theta object or a unary theta series, and the residual-law question is a question about P_s^{orth} alone. Note also that R is a cone sum of the *indefinite* form Q_7 (positive only on the closed orthant), so it cannot be split by inclusion–exclusion into full lattice theta series: $\sum_{\mathbb{Z}^2} q^{Q_7}$ diverges.

12.2 Automorphs and the fundamental cone

In the coordinates $\nu = v + a$, let $\rho_1 = \begin{pmatrix} 1 & 7 \\ 0 & -1 \end{pmatrix}$, $\rho_2 = \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix}$. Both are integral reflections preserving Q; ρ_1 fixes the ray $\nu_2 = 0$, ρ_2 fixes $\nu_1 = 0$, the two walls of $C_0 = \{\nu_1 > 0, \nu_2 > 0\}$. Their product $\gamma = \rho_2\rho_1$ has eigenvalues $6 \pm \sqrt{35}$. The reflections swap $a + \mathbb{Z}^2$ and $-a + \mathbb{Z}^2$; γ preserves $a + \mathbb{Z}^2$. Coset points never lie on the walls ($\nu_1 \equiv 0.7$, $\nu_2 \equiv 0.8 \bmod 1$). Hence $C_0 \cup \rho_2 C_0$ is a fundamental domain F for $\langle\gamma\rangle$ on the positive sector S^+, and

$$E := R + C_{\text{third}} + \sum_{n\geq 1} q^{n^2} = \sum_{v:\ v+a\in F} q^{Q_7(v)} \tag{12.1}$$

for any γ-fundamental cone $F = [r, \gamma r) \subset S^+$. With Theorem 11.1(iii) this gives $R - (C_{\text{third}} + \sum_{n\geq 1} q^{n^2}) = -\Theta_{\text{hol}}$ and, adding, $2P_s^{\text{orth}} = E + 1 + \theta_3 - \Theta_{\text{hol}}$.

Status (Category A). Proved; reflections and coset behaviour checked symbolically; (12.1) checked exactly to q^{3000} for $r = (1,0), (1,1), (7,-1), (1,2)$. Also $E_{-a} = E_a$, so the sign-twisted partner $E_a - E_{-a}$ (a weight-1 Hecke form in Zwegers' $c_2 = \gamma c_1$ set-up) vanishes for this coset; this holds for every $s \geq 5$ (Theorem 24.1(ii)).

12.3 Ray-class identification (Theorem 11.2)

Put $(X,Y) = 10\nu \in (7,-2) + 10\mathbb{Z}^2$ and $\beta = (2X+7Y) + Y\sqrt{35} \in \mathbb{Z}[\sqrt{35}]$. Then $Q(\nu) = \mathrm{N}(\beta)/400$, $\beta(\gamma\nu) = \varepsilon\beta(\nu)$, β runs over $\beta_0 + 10\mathfrak{p}_2$ with $\beta_0 = -2\sqrt{35}$ and $\mathfrak{p}_2 = (2, 1+\sqrt{35})$, and $\nu \in S^+ \Leftrightarrow \beta \gg 0$. With $\mathfrak{d}' = \gcd((\beta_0), 10\mathfrak{p}_2) = \mathfrak{p}_2^2\mathfrak{p}_5$ and $\mathfrak{f} = \mathfrak{p}_2\mathfrak{p}_5$, the ideals $\mathfrak{a} = (\beta)\mathfrak{d}'^{-1}$ are exactly $\alpha\mathfrak{p}_7$ with $\alpha \equiv 1 \bmod {}^\times\mathfrak{f}$ and $\operatorname{sgn}\alpha = (-,+)$: one narrow ray class $\mathcal{K}$ modulo $\mathfrak{f}\infty_1\infty_2$. Totally positive units $\equiv 1 \bmod {}^\times\mathfrak{f}$ are $\varepsilon^{\mathbb{Z}}$ ($\varepsilon \equiv 1 \bmod \mathfrak{p}_2$ and $\bmod \mathfrak{p}_5$), so γ-orbits correspond bijectively to ideals in $\mathcal{K}$, and $\mathrm{N}\mathfrak{a} = 20Q_7(v) - 7$. This yields Theorem 11.2. Sanity check: the first norms are $13, 73, 133 = 7\cdot 19, 153 = 9\cdot 17, 173, \ldots$, all $\equiv 13 \bmod 20$.

Status (Category A). Proved: hand algebra with symbolic checks of $Q = \mathrm{N}(\beta)/400$ and $\beta(\gamma\nu) = \varepsilon\beta$. The region-to-class bijection was also checked exactly against PARI's independent ideal enumeration up to norm 60 000.

Remark 12.2 (Group data)**.** $\mathrm{Cl}(K) = \mathbb{Z}/2$, $\mathrm{Cl}^+(K) = (\mathbb{Z}/2)^2$; $|\mathrm{Cl}^+_{\mathfrak{f}\infty_1\infty_2}| = 16$, with structure $\mathbb{Z}/8 \times \mathbb{Z}/2$. Category A: $\mathrm{Cl}(K)$ and the order 16 are hand-certified (Minkowski bound $\sqrt{140}/2 \approx 5.92$; $h\varphi(\mathfrak{f})2^2/[U : U_{\mathfrak{f},1}] = 2\cdot 4\cdot 4/2$); the group structure $\mathbb{Z}/8 \times \mathbb{Z}/2$ and the narrow class group are from PARI (`bnfcertify`$= 1$).

12.4 Analytic consequences

By Hecke, $\zeta(s,\mathcal{K}) = \sum_{\mathfrak{a}\in\mathcal{K}} \mathrm{N}\mathfrak{a}^{-s}$ continues meromorphically with a single simple pole at $s=1$, so

$$E(e^{-t}) = e^{-7t/20}\Big[\frac{20\,\mathrm{Res}_{s=1}\,\zeta(s,\mathcal{K})}{t} + \sum_{j\ge 0}\zeta(-j,\mathcal{K})\frac{(-t/20)^j}{j!}\Big]$$

has an expansion in integral powers of t; by Siegel–Klingen the $\zeta(-j,\mathcal{K})$ are rational. The exact computation gives $\mathrm{Res}_{s=1}\,\zeta(s,\mathcal{K}) = \log\varepsilon/(20\sqrt{35})$, $\zeta(0,\mathcal{K}) = \frac{1}{10}$ $\zeta(-1,\mathcal{K}) = 20\,\zeta_E(-1) = \frac{29}{60}$, where $\zeta_E(s) = \sum_{\nu\in F\cap(a+\mathbb{Z}^2)} Q(\nu)^{-s}$, $\zeta_E(0) = \frac{1}{10}$, $\zeta_E(-1) = \frac{29}{1200}$.

Status (Category A). The exact values are Euler–Maclaurin computations in rational arithmetic; the half-integral coefficients vanish identically (symbolic); independently confirmed by five Shintani cones $[r,\gamma r)$, $r\in\{(1,0),(1,1),(1,2),(2,1),(3,2)\}$ ($|\det| = 2,23,58,43,130$), each giving $\frac{1}{10}$ and $\frac{29}{1200}$. The appeal to Hecke and Siegel–Klingen [7, 15] is to standard theorems cited by name; rationality is used only as a sanity check.

Category B. The residue matches the narrow ray-class residue formula $R_{\mathfrak{f}}/(w\sqrt{d_K}\,\mathrm{N}\mathfrak{f})$ with $d_K = 140$, $\mathrm{N}\mathfrak{f} = 10$, $w = 1$; this was not checked against a published statement. The Mellin transform of P_s^{orth} is a Shintani-type cone zeta whose residue is "class-number-like".

13 Asymptotics and the Baker obstruction

13.1 Two-dimensional Euler–Maclaurin

All radial expansions in this paper rest on the following expansion for cone sums.

Proposition 13.1 (Cone Euler–Maclaurin)**.** *For every rational x, the shifted-lattice cone sums of e^{-tQ} that make up $H_s(\mathbf{e}(x)e^{-t})$ and $E_L(x+h)$ have a full asymptotic expansion in powers of $t^{1/2}$. The coefficients are area terms, edge Bernoulli sums with the half-weight rule $\tilde{B}_r$ at the edges, and corner terms. The remainders are explicit integrals of periodic Bernoulli functions and are bounded uniformly in every sector $|\arg t| \le \pi/2 - \delta$.*

Status (Category A). Proved, by a one-dimensional lemma (Zagier's asymptotic lemma with the Euler–Maclaurin remainder $\frac{(-1)^{p+1}}{p!}\int_0^\infty \tilde{B}_p(y-u)f^{(p)}(y)\,dy$), a quadrant lemma, and a decomposition of the closed cone into translated quadrants. The sector-uniform bound is order-3 two-variable Euler–Maclaurin with every remainder bounded; only $|\tilde{B}_p|$ enters the bounds. The written proof is in the companion report (Appendix C); its structure is reproduced here.

In particular we use the following consequence, which we call **R1**: for rational x, any cone sum f of this type and $|\arg t| \le \pi/2-\delta$, $|t|\le 1$,

$$f(\mathbf{e}(x)e^{-t}) = A/t + h\,t^{-1/2} + \kappa + O(|t|^{1/2}),$$

with the coefficients of the real-t expansion. Thus radial expansions hold non-tangentially.

13.2 Radial expansions

Table 13 lists the coefficients of $t^{-1}, t^{-1/2}, t^0$ at $q = e^{-t}$. The method is two-dimensional Euler–Maclaurin on shifted quadrants; for H_s the closed cone $\langle(0,1),(21,-1)\rangle$ is decomposed into 21 residue classes $(s,0)+\mathbb{N}v_1+\mathbb{N}v_2$ minus half-walls (decomposition exact to q^{2000}). Consistency: $2P_s^{\mathrm{orth}} = E+1+\theta_3-\Theta_{\mathrm{hol}}$ forces the $t^{-1/2}$ coefficient of Θ_{hol} to be $\sqrt{\pi}-2a_P = \sqrt{\pi}(21+2\sqrt{14})/70$. For $Z(s) = \sum_{(n,k)\ge 0,\ne 0} Q_7^{-s}$: simple poles at $s=1$ (residue I) and $s=\frac{1}{2}$ (residue $(49-2\sqrt{14})/140$), and $Z(0) = -(9+7I)/20$; the $\log\varepsilon$ in $Z(0)$ comes entirely from the shift $\frac{7}{20}$.

series	t^{-1}	$t^{-1/2}$	t^0
P_s^{orth}	I	$\sqrt{\pi}(49-2\sqrt{14})/140 = 0.5256172043$	$(11-7I)/20 = 0.4767030629$
C_{third}	I	$-\sqrt{\pi}(49-2\sqrt{14})/140$	$(11-7I)/20$
R	I	-0.3606097211	$(1-7I)/20$
E	$2I$	0	$(1-7I)/10$
H_s	0	$a_H = \sqrt{\pi}(\sqrt{1190}-17\sqrt{14})/170 = -0.3035254127$	0
C	I	$a_C = \sqrt{\pi}(833+204\sqrt{14}-14\sqrt{1190})/2380 = 0.8291426171$	$(11-7I)/20$

Table 13: Radial coefficients at $q = e^{-t}$. Status as in Theorem 11.3; the H_s and C entries are the closed forms of Theorem 22.1.

Lemma 13.2 (Zwegers parts carry no $1/t$ term)**.** *The holomorphic part of any signature-*$(1,1)$ *indefinite theta* $\vartheta_{a,b}^{c_1;c_2}$ *with* c_1, c_2 *of negative norm and rational* a, b *has* $1/t$ *coefficient* 0 *at* $q = e^{-t}$*. This covers* H_s*,* Θ_{hol} *and* $q^8 f_{2,7,7}$*.*

Sketch. The weight is $+1$ on a cone $\mathcal{C}$ and -1 on $-\mathcal{C}$ (walls contribute $O(t^{-1/2})$). The mean of $\mathbf{e}(B(\nu, b))$ over the coset is the same on $\mathcal{C}$ and $-\mathcal{C}$, and Q is even, so the leading integrals cancel. □

Status (Category A). Proved (symmetry argument; null cones excluded). Numerically $t\,H_s \to 0$ (fit -0.0002).

13.3 The additive obstruction (Theorem 11.5(i))

Let $\mathcal{M}$ be the $\overline{\mathbb{Q}}$-span of: (i) $q^r f$, $r \in \mathbb{Q}$, f a holomorphic weight-1 form on some $\Gamma(N)$ with $\overline{\mathbb{Q}}$ coefficients; (ii) $q^r f$ with f of half-integral weight, and f of integral weight $k \geq 2$ without q^r shift; (iii) q^r times Zwegers holomorphic parts as in Lemma 13.2; (iv) unary, partial and false theta series $\sum_{n \geq n_0} \psi(n) n^\epsilon q^{an^2+bn}$ with ψ periodic and $\overline{\mathbb{Q}}$-valued, $\epsilon \in \{0, 1\}$, $a > 0$.

Proof of Theorem 11.5(i). Multiply by t and let $t \to 0^+$. Class (i) contributes $2\pi\beta$ with $\beta \in \overline{\mathbb{Q}}$ (algebraic expansion of $f|S$); (ii) contributes 0; (iii) contributes 0 by Lemma 13.2; (iv) contributes 0 or $\mathrm{mean}(\psi)/(2a) \in \overline{\mathbb{Q}}$. So $I = 2\pi\beta + \gamma'$ with $\beta, \gamma' \in \overline{\mathbb{Q}}$, i.e. $\log(6+\sqrt{35}) + 4\sqrt{35}\, i\beta \log(-1) - 2\sqrt{35}\gamma' = 0$, a nontrivial $\overline{\mathbb{Q}}$-linear relation among 1, $\log(6+\sqrt{35})$, $\log(-1) = i\pi$. Since $\log(6+\sqrt{35})$ (real, nonzero) and $i\pi$ are $\mathbb{Q}$-linearly independent, Baker's theorem [1] says $1, \log\alpha_1, \log\alpha_2$ are $\overline{\mathbb{Q}}$-linearly independent: contradiction. □

Status (Category A). Proved, conditional on Baker's theorem [1] and on the algebraicity of cusp expansions (Shimura's $\mathbb{Q}(\zeta_N)$-rationality, i.e. the q-expansion principle), both cited. The radial expansion used is R1 (§13.1).

Category B. Reading: the remainder of C is "weight-1 sized" at $q \to 1$ with a regulator as coefficient; weight-$\frac{1}{2}$ false thetas are $O(t^{-1/2})$ and cannot carry it. An Appell–Lerch formula for E or P_s^{orth} with algebraic data would likewise be excluded, but this needs the $q \to 1$ asymptotics of Appell–Lerch sums, which were not verified; that obstruction is *conditional*.

13.4 The multiplicative obstruction (Theorem 11.5(ii))

Proof. Along $q = e^{-t}$, $F/g = c\,t^{-j} e^{-\alpha/t}(1 + O(e^{-\beta/t}))$ times a power series in t, so only integer-step powers of t occur; but C_{third} has the nonzero term $-a_P t^{-1/2}$ and C the nonzero term $a_C t^{-1/2}$ (Table 13). □

Status (Category A). Proved. For C the nonvanishing of a_C rests on its closed form (Theorem 11.3). This covers every space with $k \in \{\frac{1}{2}, 1, \frac{3}{2}, 2\}$ and $N \mid 140$ or 280; no Sturm-bound search is needed.

13.5 Gauss means and the scalar W_7 obstruction

Proof of Theorem 11.4 (pattern). If $2 \mid m$, shift $k \mapsto k+m$: $(Q_7(n,k+m) - Q_7(n,k))/m = 7n + 7k + \frac{7}{2}(m+1)$ and the phase $\mathbf{e}(7h(m+1)/2) = -1$, so $G = 0$. If m is odd, Q_7 is m-periodic and 2 is invertible mod m; with $n' = n + 7 \cdot 2^{-1}k$, $hQ_7 \equiv h[n'^2 - \frac{35}{4}k^2 + \frac{7}{2}k]$ and $G = m^{-2}g_n g_k$ with $|g_n| = \sqrt{m}$. If $5 \mid m$, the shift $(n,k) \mapsto (n - m/5, k + m/5)$ changes the constant part by $j/5$, $j \equiv 1 \bmod 5$, so $G = 0$. If $\gcd(m, 35) = 1$, completing the square gives $|g_k| = \sqrt{m}$. If $7^e \| m$, $5 \nmid m$, by CRT the 7-part of g_k is $7\sum_{k \bmod 7^{e-1}} \mathbf{e}\big(h(-\frac{5}{4}k^2 + \frac{k}{2})/7^{e-1}\big)$, of modulus $7^{(e+1)/2}$, so $|G| = \sqrt{7}/m$. □

Status (Category A). Proved; verified on all 3374 coprime pairs with $m \le 105$, zero mismatches. For $x = h/7$, $G = \big(\frac{h}{7}\big) i/\sqrt{7}$.

For the scalar W_7 question: a singularity $\sim G(x)I/(2\pi\varepsilon')$ at x (with $\tau = x + i\varepsilon'$) maps under W_7 to $\sim |7x|^2 G(-1/(7x))I/(2\pi\varepsilon')$, so boundedness of $C - \lambda(7\tau)^{-k}C(-1/(7\tau))$ would force $G(x) = \lambda(7x)^{-k}|7x|^2 G(-1/(7x))$ for all x. This fails at the 178 zero-mismatch points; on the 60 pairs of common support the moduli obey an exact *weight-1* relation with $|\lambda| = 7^{-1/2}$ but the phase varies ($\pi/2$ at $1/7$ and 1, $-\pi/6$ at $1/3$ and $3/7$); for W_{35}, W_{70}, W_{140} there are no pairs with both values nonzero. The map $\tau \mapsto -1/(7\tau)$ does not normalise $\Gamma_0(70)$.

Status (Category A). Zero/nonzero pattern and moduli: exact, by Theorem 11.4 (the floating-point exponential sums, nonzero values ≥ 0.034 and zeros $< 10^{-9}$, agree).

Category B. The phase values quoted above are floating-point (with a wide margin) and are not separately proved.

Category B. The scalar failure is *vector mixing*: the leading profiles obey a vector-valued law (§14.1), under which W_7-type maps mix components; this is not a contradiction with modularity of the vector.

14 The theta lift and the vector-valued S/T law

14.1 Leading order: radial profiles

For $\alpha, \beta \in \mathbb{Q}^2$ let $G_{\alpha,\beta}(x)$ be the mean over $\nu \in \alpha + \mathbb{Z}^2$ of $\mathbf{e}(xQ(\nu) + B(\nu,\beta))$; this is the $1/t$ radial profile (divided by the cone measure) of any cone sum over that coset and twist (leading-order Euler–Maclaurin, as in Theorem 11.4 and R1). For rational α, β and rational x the summand is periodic on $\alpha + \mathbb{Z}^2$, and the mean is the average over one period. Write $A^* = (A_{11}, A_{22}) = (2, 7)$, so $\frac{1}{2}A^{-1}A^* = (\frac{1}{2}, 0)$.

Theorem 14.1 (Profile S/T law)**.** *For all $\alpha, \beta \in \mathbb{Q}^2$ and all $x \in \mathbb{Q} \setminus \{0\}$,*

$$G_{\alpha,\beta}(-1/x) = \frac{\mathbf{e}(B(\alpha,\beta))}{\sqrt{35}\,|x|} \sum_{p \in L'/\mathbb{Z}^2} G_{\beta+p,-\alpha}(x),$$

$$G_{\alpha,\beta}(x+1) = \mathbf{e}\big(-Q(\alpha) - B(\alpha, w_c)\big)\, G_{\alpha,\,\alpha+\beta+w_c}(x).$$

The T-law may equivalently be written with $\frac{1}{2}A^{-1}A^* = (\frac{1}{2}, 0)$ in place of w_c (both in the phase and in the label), since $(\frac{1}{2}, 0) - w_c \in L'$ and $G_{\alpha,\beta+\mu} = \mathbf{e}(B(\alpha,\mu))G_{\alpha,\beta}$ for $\mu \in L'$.

Proof. T-law. For $\nu=\alpha+n$, $Q(\nu)=Q(\alpha)+B(\alpha,n)+Q(n)$ and $\mathbf{e}(Q(n))=\mathbf{e}(B(n,w_c))$ (indeed $B(n,w_c)=n_2/2$ and $Q(n)\equiv\frac{7}{2}n_2^2\equiv n_2/2 \bmod 1$). Hence $\mathbf{e}(Q(\nu))=\mathbf{e}(-Q(\alpha)-B(\alpha,w_c))\,\mathbf{e}(B(\nu,\alpha+w_c))$; average over the coset.

S-law. (1) *Gaussian-weighted sums.* Let $M\subset\mathbb{R}^2$ be a lattice, $c\in\mathbb{R}^2$, Π a function on $c+M$ periodic modulo a sublattice $\Lambda\subset M$ of finite index, and N complex symmetric with $\operatorname{Re}N>0$. Splitting $c+M$ into Λ-cosets and applying Poisson summation on each,

$$\sum_{\eta\in c+M}\Pi(\eta)e^{-\pi\delta\,\eta^{\mathsf T}N\eta}=\frac{\operatorname{mean}_{c+M}(\Pi)}{\operatorname{vol}(M)\,\delta\,(\det N)^{1/2}}+O\big(\delta^{-1}e^{-\kappa/\delta}\big)\qquad(\delta\to0^+),$$

since $\operatorname{Re}N^{-1}>0$; here $(\det N)^{1/2}$ is the branch on the (convex) Siegel domain that is positive on real N, and $\kappa>0$ and the O-constant are uniform for N in a compact neighbourhood of I. (2) *Poisson.* Fix $x\neq0$, $\epsilon>0$, put $Z_\epsilon=\epsilon I+\frac{i}{x}A$ ($\operatorname{Re}Z_\epsilon=\epsilon I>0$) and $g(\nu)=\mathbf{e}(B(\nu,\beta))e^{-\pi\nu^{\mathsf T}Z_\epsilon\nu}=\mathbf{e}(-Q(\nu)/x+B(\nu,\beta))e^{-\pi\epsilon|\nu|^2}$, a Schwartz function. By (1) with $M=\mathbb{Z}^2$, $N=I$, $\delta=\epsilon$: $\sum_{\nu\in\alpha+\mathbb{Z}^2}g(\nu)=\epsilon^{-1}G_{\alpha,\beta}(-1/x)+O(\epsilon^{-1}e^{-\kappa/\epsilon})$. On the other hand, $\sum_n g(\alpha+n)=\sum_{k\in\mathbb{Z}^2}\widehat g(k)\mathbf{e}(k\cdot\alpha)$ with $\widehat g(k)=(\det Z_\epsilon)^{-1/2}\exp(-\pi(k-A\beta)^{\mathsf T}Z_\epsilon^{-1}(k-A\beta))$. Write $k=A\mu$, $\mu\in L'$, $\eta=\mu-\beta\in-\beta+L'$, so $\mathbf{e}(k\cdot\alpha)=\mathbf{e}(B(\alpha,\beta))\mathbf{e}(B(\eta,\alpha))$. With $N_\epsilon=(I-i\epsilon xA^{-1})^{-1}$ one has $Z_\epsilon=\frac{i}{x}A\,N_\epsilon^{-1}$ and $AZ_\epsilon^{-1}A=-ixA+\epsilon x^2N_\epsilon$, so

$$\sum_{\nu\in\alpha+\mathbb{Z}^2}g(\nu)=\frac{\mathbf{e}(B(\alpha,\beta))}{(\det Z_\epsilon)^{1/2}}\sum_{\eta\in-\beta+L'}\mathbf{e}\big(xQ(\eta)+B(\eta,\alpha)\big)\,e^{-\pi\epsilon x^2\eta^{\mathsf T}N_\epsilon\eta}.$$

By (1) with $M=L'$ ($\operatorname{vol}=1/35$), $N=N_\epsilon\to I$, $\delta=\epsilon x^2$, and $\operatorname{mean}_{-\beta+L'}=\frac{1}{35}\sum_{p\in L'/\mathbb{Z}^2}G_{p-\beta,\alpha}(x)$, the right side is $\mathbf{e}(B(\alpha,\beta))\,[(\det Z_\epsilon)(\det N_\epsilon)]^{-1/2}(\epsilon x^2)^{-1}\sum_p G_{p-\beta,\alpha}(x)$ up to $O(\epsilon^{-1}e^{-\kappa'/\epsilon})$. (3) *The factor $|x|$.* $(\det Z_\epsilon)(\det N_\epsilon)=\det(\frac{i}{x}A)=35/x^2$ for all ϵ. Along $Z_s=\epsilon I+s\frac{i}{x}A$, $s\in[0,1]$, the eigenvalues $\epsilon+is\lambda_j/x$ ($\lambda_1>0>\lambda_2$ the eigenvalues of A, $\lambda_1\lambda_2=-35$) stay in the right half-plane, so $(\det Z_\epsilon)^{1/2}=\prod_j(\epsilon+i\lambda_j/x)^{1/2}$ (principal roots) $\to(i\lambda_1/x)^{1/2}(i\lambda_2/x)^{1/2}=\sqrt{35}/|x|$, the arguments $\pm\pi/4$ cancelling for either sign of x (signature $(1,1)$); and $(\det N_\epsilon)^{1/2}\to1$. The continuous product, whose square is constant, therefore equals $\sqrt{35}/|x|$. Comparing the coefficients of ϵ^{-1} gives $G_{\alpha,\beta}(-1/x)=\mathbf{e}(B(\alpha,\beta))(\sqrt{35}|x|)^{-1}\sum_p G_{p-\beta,\alpha}(x)$. Finally $G_{-\gamma,-\delta}=G_{\gamma,\delta}$ ($\nu\mapsto-\nu$, Q even) and $p\mapsto-p$ permutes $L'/\mathbb{Z}^2$, so $\sum_p G_{p-\beta,\alpha}=\sum_p G_{\beta+p,-\alpha}$. □

Status (Category A). Proved above; the analytic inputs are Poisson summation for Schwartz functions and the complex Gaussian Fourier transform. Numerically verified independently of the proof: the S-law on all 140 labels at $x\in\{1,-1,\frac13,-\frac23\}$ (560 tests, 280 with nonzero value), on 13 labels at 12 values of x (both signs), and on $40+60$ random rational pairs (α,β) outside the family ($3+60$ with nonzero value), with maximal deviation $3.3\cdot10^{-15}$; the T-law on all 140 labels to $5.6\cdot10^{-16}$. With the signed factor x in place of $|x|$ the ratio is exactly -1 at $x<0$.

Corollary 14.2 (Consequences of Theorem 14.1)**.** *(i) The* 140*-vector* $\mathbf{G}(x)=(G_{\alpha,\beta}(x))$ *satisfies* $P\mathbf{G}=\mathbf{G}$, $\mathbf{G}(-1/x)=|x|^{-1}S\mathbf{G}(x)$ *and* $\mathbf{G}(x+1)=T\mathbf{G}(x)$ *with the matrices of Theorem 11.6; hence* $\mathbf{G}(\gamma x)=|cx+d|^{-1}\rho(\gamma)\mathbf{G}(x)$ *for all* $\gamma\in\mathrm{SL}_2(\mathbb{Z})$ *and rational* $x\neq-d/c$. *(ii)* $G_{a,0}\big(x/(140x+1)\big)\,|140x+1|=G_{a,0}(x)$ *for all rational* $x\neq0,-\frac{1}{140}$. *(iii) For* $\gcd(m,70)=1$, $m\,\mathbf{e}(-7/(20m))\,G(1/m)=m\,G_{a,0}(1/m)$ *depends only on* $m \bmod 70$. *(iv) For the fundamental-domain vector (all components are cone sums over the same cone measure), the* $1/t$ *coefficients* $A(x)$ *satisfy* $|x|\,A(-1/x)=S\,A(x)$ *at every rational* $x\neq0$.

Proof. (i) The reductions $G_{\alpha+n,\beta} = G_{\alpha,\beta}$, $G_{\alpha,\beta+\mu} = \mathbf{e}(B(\alpha,\mu))G_{\alpha,\beta}$ ($\mu \in L'$), $G_{-\alpha,-\beta} = G_{\alpha,\beta}$ are those of the $u_{\alpha,\beta}$, so the laws of Theorem 14.1 have exactly the matrices S, T; $|cx+d|$ is a cocycle and ρ is well defined on $\mathrm{Fix}(P)$. (ii) Apply the S-law at $x' = -1/x-140$, then $G_{p,-a}(y-140) = G_{p,-a}(y)$ (since $70Q(\nu) \in \mathbb{Z}$ for $\nu \in L'$), then the S-law at $-1/x$. (iii) The S-law gives $m\,G_{a,0}(1/m) = 35^{-1/2}\sum_p G_{p,-a}(-m)$, and $mQ(\nu) \bmod 1$ depends only on $m \bmod 70$ for $\nu \in L'$. (iv) Each $1/t$ coefficient is (cone measure)$\times G$, by leading-order Euler–Maclaurin (R1); apply (i). □

Status (Category A). Proved given Theorem 14.1 ((iv) also uses R1). Numerically: (i) for random words in S, T to $6.6 \cdot 10^{-16}$, (ii) at 6 rationals to $3 \cdot 10^{-15}$, (iii) for all $m < 1200$ (spread $1.2 \cdot 10^{-13}$). Part (ii) is the Weil-kernel invariance used in Theorem 24.6(ii), and (iii) is the Gauss-sum input of Theorem 24.4.

At the cusp pair $(0,\infty)$ no exact vector-valued *modular* law with algebraic data can hold: at $x = 0$ each cone sum has singular part $IG/t \neq 0$, while at ∞ there is no q^0 term (Q has no nonzero rational zeros since 35 is not a square); an exact law would force $I \in \pi\overline{\mathbb{Q}} + \overline{\mathbb{Q}}$, excluded by Baker (Category A, conditional on Baker). At order $t^{-1/2}$ the *quadrant*-cone vector fails the weight-1 S-law ($|\mathrm{LHS}|/|\mathrm{RHS}| \in [0.49, 2.97]$ over 16 tests, either wall convention), while the *fundamental-domain* vector has vanishing $t^{-1/2}$ coefficients ($\leq 5 \cdot 10^{-15}$) (Category B: floating-point ratios and residuals). For the scalar rows the vanishing that is actually needed is proved in the course of the proof of Theorem 11.7 (§15.4).

14.2 Siegel theta and Poisson summation

Fix a majorant (a point of the γ-geodesic) with $Q = Q_+ + Q_-$ and put $G_\tau(\nu) = \mathbf{e}(\tau Q_+(\nu) + \bar\tau Q_-(\nu))$, $\Theta_{\alpha,\beta}(\tau) = \sum_{\nu\in\alpha+\mathbb{Z}^2} \mathbf{e}(B(\nu,\beta))G_\tau(\nu)$. Poisson summation, with $\widehat{G}_{-1/\tau}(A\eta) = (|\tau|/\sqrt{35})G_\tau(\eta)$ (the two square roots of $\det(M/i) = 35|\tau|^2$ are complex conjugate), gives

$$\Theta_{\alpha,\beta}(-1/\tau) = |\tau|\,\frac{\mathbf{e}(B(\alpha,\beta))}{\sqrt{35}} \sum_{p\in L'/\mathbb{Z}^2} \Theta_{p-\beta,\alpha}(\tau). \tag{14.1}$$

14.3 The lift

Integrating over the closed geodesic of $\langle\gamma^k\rangle$ ($k = 2$ is the common stabiliser order on all labels), unfolding with $\int_{\mathbb{R}} e^{-a\cosh 2w}\,dw = K_0(a)$ and geodesic length $\log\varepsilon$, and multiplying by $\sqrt{y}$ ($w = x + iy$) gives

$$u_{\alpha,\beta}(w) = \sqrt{y}\Big[\log\varepsilon\cdot\delta_{\alpha\in\mathbb{Z}^2} + \tfrac{1}{2}\sum_{\nu \bmod \gamma^2} \mathbf{e}(B(\nu,\beta))\,K_0(2\pi|Q(\nu)|y)\,\mathbf{e}(Q(\nu)x)\Big],$$

the sum over $\nu \in \alpha + \mathbb{Z}^2$, $\nu \neq 0$, in all four sign regions $\pm S^+$, $\pm T^+$ (negative-norm vectors give negative frequencies; the fundamental cones are $[e_1, \gamma e_1)$, $[f, \gamma f)$, $f = (1,-1)$, and their negatives). Each term $\sqrt{y}K_0(2\pi|n|y)\mathbf{e}(nx)$, and $\sqrt{y}$, is a Δ-eigenfunction with $\lambda = \frac{1}{4}$. The scalar rows are $u_L = u_{a,0}$ and $u_R = \sum_p u_{p,-a}$; for these the formula reduces to $u_L = \sqrt{y}\sum_{\nu\in(a+\mathbb{Z}^2)/\langle\gamma\rangle} K_0(\cdot)\mathbf{e}(\cdot)$ and $u_R = \sqrt{y}[\log\varepsilon + \sum_{0\neq\nu\in L'/\langle\gamma\rangle} \mathbf{e}(B(\nu,-a))K_0(\cdot)\mathbf{e}(\cdot)]$. Reductions: $u_{\alpha+n,\beta} = u_{\alpha,\beta}$ ($n \in \mathbb{Z}^2$), $u_{\alpha,\beta+\mu} = \mathbf{e}(B(\alpha,\mu))u_{\alpha,\beta}$ ($\mu \in L'$), $u_{-\alpha,-\beta} = u_{\alpha,\beta}$.

Proof of Theorem 11.6. Integrate (14.1) over the geodesic and use $\sqrt{\mathrm{Im}(-1/\tau)} = \sqrt{y}/|\tau|$; the factor $|\tau|$ cancels and gives the weight-0 S-law; the constant term $\sqrt{y}\log\varepsilon$ comes from $\nu = 0$ and appears exactly when $\alpha \in \mathbb{Z}^2$. For $(a,0)$ the right side is $\frac{1}{\sqrt{35}}\sum_p u_{p,a} = u_R/\sqrt{35}$ using $u_{p,a} = u_{-p,-a}$, i.e. $u_L(-1/w) = u_R(w)/\sqrt{35}$. The T-law follows from $\mathbf{e}(Q(n)) = \mathbf{e}(B(n,w_c))$; the naive even-lattice

T-law fails by $O(1)$. With labels $\alpha \in (\{0,a\}+L')/\mathbb{Z}^2$ (70 classes) and $\beta \in \{0,-a\}$ the family is closed; each row of S has exactly 35 entries of modulus $1/\sqrt{35}$, T is monomial. □

Status (Category A). S-law proved by Poisson summation; T-law derived. Numerically on all 140 rows at three points with $y \approx 1$: $|u(-1/w) - Su(w)| \le 1.2 \cdot 10^{-15}$, $|u(w+1) - Tu(w)| \le 7 \cdot 10^{-16}$, $|Pu - u| \le 6 \cdot 10^{-16}$; on the 72-dimensional P-invariant subspace $S^2 = I$, $(ST)^3 = I$ to $1.3 \cdot 10^{-15}$. Borcherds [3, Thm. 4.1] gives the same automorphy factor $(c\tau+d)^{b^+/2}(c\bar\tau+d)^{b^-/2}$; it is not needed. The S-law $u_L(-1/w) = u_R(w)/\sqrt{35}$ was also checked to $6 \cdot 10^{-11}$, and $u_L(w+1) = \mathbf{e}(13/20)u_L(w)$ exactly.

Remark 14.3. The cyclic S,T-span of the row $e_{(a,0)}$ has dimension 60 in label space $\mathbb{C}^{140}$; after projection to $\mathrm{Fix}(P)$ (rows multiplied by $\frac{1}{2}(I+P)$, which is what acts on the P-invariant vector u) its rank is 36. The positive Fourier coefficients satisfy $f_{u_L} = 2E_L$, $f_{u_R} = 2E_R$ (the automorph $-\rho_1$ maps S^+ to S^- preserving the coset and phase), checked for all norms ≤ 25 (Category A for the automorph argument; the numerical check is supporting).

Remark 14.4 (Normalisation of the lift)**.** The lift can be written as a sum over $\langle\gamma\rangle$-orbits or as $\frac{1}{2}\sum_{\bmod \gamma^2}$. For the scalar rows these agree exactly: γ maps $a+\mathbb{Z}^2$ to itself ($\gamma a - a = (0,-1)$) and preserves Q; it fixes no nonzero vector (hyperbolic), so every $\langle\gamma\rangle$-orbit is the disjoint union of exactly two $\langle\gamma^2\rangle$-orbits with equal Q; and the phase $\mathbf{e}(B(\nu,\beta))$ is γ-invariant for $\beta = 0$, and for $\beta = -a$ on L' because $\gamma^{-1}a - a \in \mathbb{Z}^2$ and $B(L',\mathbb{Z}^2) \subset \mathbb{Z}$. Hence $\frac{1}{2}\sum_{\nu \bmod \gamma^2} = \sum_{\nu \bmod \gamma}$ termwise for u_L and u_R (Category A, elementary). For labels whose coset is not γ-stable the $\frac{1}{2}\sum_{\bmod \gamma^2}$ form is the definition.

15 The Lewis–Zagier cocycle and quantum modularity

15.1 Definitions

Let $F = [e_1, \gamma e_1)$ (half-open, vertex excluded). For labels (α,β) set $E_{\alpha,\beta}(\tau) = \sum_{\nu \in F \cap (\alpha+\mathbb{Z}^2)} \mathbf{e}(B(\nu,\beta))\mathbf{e}(\tau Q(\nu))$, $E_L = E_{a,0}$ and $E_R = \sum_p E_{p,-a}$. Thus $E_L(\tau) = \sum_{\mathfrak{a}\in\mathcal{K}} \mathbf{e}(\tau \mathrm{N}\mathfrak{a}/20)$ and $E = q^{7/20}E_L$. At rational x, $E(x+iy) = 2I\,G(x)/t + \kappa(x) + O(t)$ with no $t^{-1/2}$ term. Write $\kappa_L(x)$ and $\kappa_{p,-a}(x)$ for the radial constant terms of E_L and $E_{p,-a}$ at x. The *scalar cocycle* is

$$\psi(x) = \kappa_L(-1/x) - \frac{|x|}{\sqrt{35}} \sum_p \kappa_{p,-a}(x) \qquad (x \in \mathbb{Q}\setminus\{0\}),$$

a finite Shintani-type sum of B_2, B_1B_1 and roots of unity at each rational; hence $\psi(h/m) \in \mathbb{Q}(\zeta_N, \sqrt{35})$. The real factor $|x|$ is the one dictated by the profile law (Theorem 14.1). On $\mathbb{H}$ we use

$$\Psi(z) = E_L(-1/z) - \frac{z}{\sqrt{35}}E_R(z), \qquad \Psi^-(z) = E_L(-1/z) + \frac{z}{\sqrt{35}}E_R(z),$$

whose radial constant terms at rational $x > 0$, respectively $x < 0$, are $\psi(x)$ (Theorem 11.7(ii)). For the vector, let $f_{\alpha,\beta}$ be the γ-averaged positive-frequency parts of $u_{\alpha,\beta}$, κ the vector of their radial constant terms, and $\Phi(x) = \kappa(-1/x) - |x|\,S\kappa(x)$. Normalisation: $\Phi_{(a,0)} = 2\psi$ (since $f_{u_L} = 2E_L$).

15.2 The Green-form lemma

Let $R_z(w) = \mathrm{Im}\, w/((w-z)(\bar w - z))$, $[u,v] = u_w v\,dw + uv_{\bar w}\,d\bar w$ and $T_n(w) = \sqrt{y}K_0(2\pi|n|y)\mathbf{e}(nx)$.

Lemma 15.1 (Green-form lemma)**.** *For $z \in \mathbb{H}$, $F_n(z) := \int_z^{z+i\infty}[T_n, R_z^{1/2}]$ (vertical ray) equals $-\frac{\pi}{2}\mathbf{e}(nz)$ for $n > 0$ and 0 for $n < 0$, and $\int_z^{z+i\infty}[\sqrt{y}, R_z^{1/2}] = \frac{1}{2}$.*

Proof. $\Delta_w R_z^{1/2} = \frac{1}{4} R_z^{1/2}$ and the form is closed (a direct computation). R_z is holomorphic in z and the endpoint term vanishes ($R^{-1/2} \to 0$ like $|w-z|^{1/2}$), so F_n is holomorphic (in detail: the integrand is $O(s^{-1/2})$ at the lower end, so one works with translation covariance gives $F_n = C_n \mathbf{e}(nz)$. For $n < 0$ boundedness forces $C_n = 0$. For $n > 0$, Laplace's method as $y \to \infty$ with $K_\nu(X) = \sqrt{\pi/(2X)} e^{-X}(1 + O(1/X))$ (DLMF 10.40.2) gives the limit $(\sqrt{\pi a}/2) \int_0^\infty s^{-1/2} e^{-as} ds = \pi/2$ with $a = 2\pi n$, and an overall minus sign. For $n = 0$ the integral is $\frac{1}{4} B(\frac{1}{2}, 1) = \frac{1}{2}$. □

$F_{n,\eta}(z) = \int_{z+i\eta}^{z+i\infty}$; by closedness and the z-holomorphy of R_z, $\partial_{\bar z} F_{n,\eta}(z) = -T_n \, \partial_{\bar w} R_z^{1/2}$ at $w = z + i\eta$, which tends to 0 locally uniformly, while $F_{n,\eta} \to F_n$ locally uniformly);

Status (Category A). Proved; the symbolic ingredients are checked by computer algebra. Independent 30-digit quadrature at $z \in \{0.3 + 0.8i, -0.17 + 0.45i, 1.1 + 1.3i\}$, $n \in \{\pm 1, \pm 2, 3\}$ and $n = 0$ agrees to $\le 7 \cdot 10^{-22}$. This is the $s = \frac{1}{2}$ case of the Lewis–Zagier construction [10, Ch. II §2]; the proof does not depend on the citation. By translation covariance the lemma holds for every real $n \neq 0$.

15.3 The branch lemma

For $z \in \mathbb{C} \setminus i\mathbb{R}$ write $\varsigma(z) = \operatorname{sgn} \operatorname{Re} z$.

Lemma 15.2 (Branch lemma)**.** *Let $z \in \mathbb{H} \setminus i\mathbb{R}$ and let C_z be the image of the vertical ray $-1/z + i\mathbb{R}_{\ge 0}$ under $w' \mapsto -1/w'$, an arc from z to 0. On the ray take the positive root of $R_{-1/z}$; on C_z take the root of R_z continued from the principal root on $i\mathbb{R}_{>0}$ near 0. Then, for $w \in C_z$,*

$$R_{-1/z}(-1/w)^{1/2} = \varsigma(z)\, z\, R_z(w)^{1/2}.$$

Proof. Since $-1/w + 1/z = (w - z)/(wz)$ and $\operatorname{Im}(-1/w) = \operatorname{Im} w/|w|^2$, one has $R_{-1/z}(-1/w) = z^2 R_z(w)$. Both roots are continuous and nonvanishing along C_z, so the two sides agree up to a sign that is constant on C_z. Compare at $w \to 0$ along C_z, where C_z is tangent to $i\mathbb{R}$. There $R_z(w) = \operatorname{Im} w \, z^{-2}(1 + o(1))$, and the principal root of z^{-2} is $\varsigma(z)/z$ (the root with positive real part), so $\varsigma(z)\, z\, R_z(w)^{1/2} = (\operatorname{Im} w)^{1/2}(1 + o(1)) > 0$. On the other side $-1/w \to i\infty$ on the ray, where $R_{-1/z}(-1/w) > 0$ and its positive root is taken. Hence the sign is $+$. □

Status (Category A). Elementary. Checked numerically at points of both half-planes (the sign is $+1$ for $\operatorname{Re} z > 0$ and -1 for $\operatorname{Re} z < 0$).

15.4 Proof of the period formula

Proof of Theorem 11.7. (i) Holomorphy. On the path $w = it$, $R_z(it) = t/(t^2 + z^2)$. For $z \notin i\mathbb{R}$ and $t > 0$ we have $t^2 + z^2 \notin (-\infty, 0]$, so the principal roots of $R_z(it)$ and of its $\bar w$-derivative are holomorphic in z. The integrand decays exponentially as $t \to 0$ (since $u_R(it) = \sqrt{35}\, u_L(i/t)$ and u_L has no constant term) and is $O(t^{-2})$ as $t \to \infty$ (the $1/t$ terms of the constant-term integrand cancel). The convergence is locally uniform in z, so Ω is holomorphic on $\mathbb{C} \setminus i\mathbb{R}$.

Identification on $\mathbb{H}$. Fix $z \in \mathbb{H} \setminus i\mathbb{R}$ and put $\varsigma = \varsigma(z)$. Lemma 15.1 at the point $-1/z$, applied to u_L (no constant term, $f_{u_L} = 2E_L$), gives

$$\int_{-1/z}^{-1/z + i\infty} [u_L, R_{-1/z}^{1/2}] = -\pi E_L(-1/z).$$

Pull back by the involution $g : w \mapsto -1/w$, which maps the vertical ray from $-1/z$ onto the arc C_z from z to 0. The Green form is natural: $g^*[u, v] = [u \circ g, v \circ g]$. By Theorem 11.6, $u_L \circ g = u_R/\sqrt{35}$, and by Lemma 15.2, $R^{1/2}_{-1/z} \circ g = \varsigma\, z\, R^{1/2}_z$ on C_z. By bilinearity,

$$-\pi E_L(-1/z) = \frac{\varsigma z}{\sqrt{35}} \int_{C_z} [u_R, R^{1/2}_z].$$

Lemma 15.1 at z, applied to u_R (constant term $\sqrt{y}\log\varepsilon$, $f_{u_R} = 2E_R$), gives $\int_z^{z+i\infty}[u_R, R^{1/2}_z] = -\pi E_R(z) + \frac{1}{2}\log\varepsilon$. The form $[u_R, R^{1/2}_z]$ is closed on $\mathbb{H}\setminus\{z\}$ (both entries are Δ-eigenfunctions with eigenvalue $\frac{1}{4}$), R_z has no zeros on $\mathbb{H}$, and the continuation of $R^{1/2}_z$ around the loop formed by the vertical ray from z, the imaginary axis and C_z is single-valued (it is the positive root on the vertical ray and near $i\infty$, and the principal root on $i\mathbb{R}_{>0}$). At $w = z$ the integrand is $O(|w - z|^{-1/2})$, so small arcs around z contribute $o(1)$; the contribution at height $Y \to \infty$ is $O(1/Y)$; at 0 the arc C_z is tangent to $i\mathbb{R}$ and u_R decays exponentially. Hence

$$\int_{C_z} [u_R, R^{1/2}_z] = \int_z^{z+i\infty} [u_R, R^{1/2}_z] - \int_0^{i\infty} [u_R, R^{1/2}_z] = -\pi E_R(z) + \tfrac{1}{2}\log\varepsilon - \int_0^{i\infty} [u_R, R^{1/2}_z].$$

Combining the last two displays and multiplying by ς/π (note $\varsigma^2 = 1$),

$$\Omega(z) = \varsigma\, E_L(-1/z) - \frac{z}{\sqrt{35}} E_R(z),$$

which is $\Psi(z)$ for $\operatorname{Re} z > 0$ and $-\Psi^-(z)$ for $\operatorname{Re} z < 0$. In particular Ψ and Ω agree on $\mathbb{H}\cap\{\operatorname{Re} z > 0\}$, so Ψ continues holomorphically to U.

(ii) Boundary values. Let $x > 0$ be rational and let $z = x + iy \to x$ non-tangentially in $\mathbb{H}$. With $t = 2\pi y$ and $t'' = -2\pi i(-1/z + 1/x) = t/(xz)$ (so that $\mathbf{e}(-1/z) = \mathbf{e}(-1/x)e^{-t''}$, and $t'' \to 0$ non-tangentially), R1 (§13.1) gives

$$E_L(-1/z) = \frac{A_L(-1/x)}{t''} + \kappa_L(-1/x) + O(|t|^{1/2}), \qquad E_{p,-a}(z) = \frac{A_p(x)}{t} + \frac{h_p(x)}{t^{1/2}} + \kappa_{p,-a}(x) + O(|t|^{1/2}),$$

where E_L has no $t''^{-1/2}$ term (Lemma 24.3). Since $1/t'' = xz/t$, the $1/t$ part of $\Psi(z)$ is

$$\frac{z}{t}\Big(x\, A_L(-1/x) - \frac{1}{\sqrt{35}}\sum_p A_p(x)\Big) = 0$$

exactly, by Corollary 14.2(iv) on the row $(a, 0)$, whose S-row is $\frac{1}{\sqrt{35}}\sum_p$ over the labels $(p, -a)$. Therefore

$$\Psi(z) = -\frac{z}{\sqrt{35}}\Big(\sum_p h_p(x)\Big)t^{-1/2} + \kappa_L(-1/x) - \frac{x}{\sqrt{35}}\sum_p \kappa_{p,-a}(x) + o(1).$$

By (i), $\Psi = \Omega$ near x and Ω is holomorphic at x, so Ψ is bounded as $z \to x$. Hence $\sum_p h_p(x) = 0$, and the limit is $\Omega(x) = \kappa_L(-1/x) - \frac{x}{\sqrt{35}}\sum_p \kappa_{p,-a}(x) = \psi(x)$.

For rational $x < 0$ the same computation applies to Ψ^-: its $1/t$ part is $\frac{z}{t}\big(xA_L(-1/x) + \frac{1}{\sqrt{35}}\sum_p A_p(x)\big) = \frac{z}{t}\big(-|x|A_L(-1/x) + \frac{1}{\sqrt{35}}\sum_p A_p(x)\big) = 0$ by Corollary 14.2(iv); $\Psi^- = -\Omega$ is bounded near x, so the $t^{-1/2}$ coefficient vanishes, and $-\Omega(x) = \kappa_L(-1/x) + \frac{x}{\sqrt{35}}\sum_p \kappa_{p,-a}(x) = \psi(x)$.

(iii) Rows. For a row $u = u_{\alpha,\beta}$ with S-row combination U, the same argument (Lemma 15.1 with the constant terms $a_0(u), a_0(U)$, the S-law of Theorem 11.6, Lemma 15.2, Corollary 14.2(iv) and R2 for the termwise integration) gives the split-contour formula (15.1) below. The functions $J_U(x)$ and $J_u(-1/x)$ there are holomorphic in x off $i\mathbb{R}$, so Φ is real-analytic on $\mathbb{R}\setminus\{0\}$. The three-term relation follows from $f(z + 1) = Tf(z)$ and $(ST)^3 = I$. □

The order of the argument in (ii) matters: the profile law removes the $1/t$ singularity exactly; boundedness, which comes from the holomorphic continuation (i), then removes the $t^{-1/2}$ term and identifies the limit.

The split-contour form used numerically is $\psi(x) = -\frac{1}{\pi}\big[\frac{x}{\sqrt{35}}(\frac{1}{2}\log\varepsilon - J_R(x)) + J_L(-1/x)\big]$ for $x > 0$, $J(z) = \int_i^{i\infty}[u, R_z^{1/2}]$. For every row $u = u_{\alpha,\beta}$, with U the corresponding S-row combination and a_0 the constant term,

$$\Phi_u(x) = -\frac{2}{\pi}\Big[\,|x|\frac{a_0(U)}{2} - \frac{a_0(u)}{2} - |x|J_U(x) + J_u(-1/x)\Big]. \tag{15.1}$$

From $f(z+1) = Tf(z)$ and $(ST)^3 = I$,

$$\Phi(z) = T\,\Phi\big(z/(z+1)\big) + \frac{z}{z+1}\,TST\,\Phi(z+1). \tag{15.2}$$

Category B. Supporting numerics for (ii)–(iii): a holomorphic fit from $\operatorname{Im} z \geq 0.01$ reproduces the real-axis values to $2\cdot 10^{-12}$ at 169 rationals; negative side with the factor $|x|$ to $1.6\cdot 10^{-14}$. All 140 rows at $x \in \{1, \frac{1}{3}, 2, \frac{3}{5}, \frac{7}{4}, -1, -\frac{2}{3}\}$: singular cancellation $9.0\cdot 10^{-16}$, no $t^{-1/2}$ terms $5.7\cdot 10^{-16}$, $P\kappa = \kappa$ to $5.3\cdot 10^{-15}$, closed form to $2.3\cdot 10^{-14}$; (15.2) at $z \in \{1, \frac{1}{2}, \frac{2}{3}, \frac{3}{2}\}$ to $7\cdot 10^{-15}$.

The automorphy factor. With the holomorphic factor x in place of $|x|$ the cocycle at $x < 0$ is unbounded (it grows like $1/y$); the bounded object on the negative side is Ψ^-, i.e. the real factor $|x|$. Numerically $\psi(-x) = \overline{\psi(x)}$ to $8\cdot 10^{-14}$ (Category B; this symmetry is not proved here and is not used). Near 0, $\psi(x) \approx -0.0549i\,x^2 + 0.0206\,x^3$ (from $x = 0.01, 0.05$); ψ is not analytic at 0. Since T acts by phases and S, T generate, every h_γ is real-analytic off $\gamma^{-1}\infty$. Example: $\psi(1) = 0.00688399376041986919032158 7 - 0.03825971221697307224258855\,i$.

Category B. Interpretation: the 72-dimensional fundamental-domain vector is a vector-valued quantum modular form of weight 1 in Zagier's sense for the real automorphy factor $|cx+d|$ ("weight $(\frac{1}{2}, \frac{1}{2})$"), i.e. the f-function of a $\lambda = \frac{1}{4}$ Maass vector, with exceptional set $\{0\}$ for S; E itself is then a "renormalised" weight-1 quantum modular object with the singular parts $I \cdot G(x)$ removed. The Lewis–Zagier period-function reading [10] and the broader context of [4] are cited for context only.

15.5 The cocycle of the E-part

Theorem 15.3 (Cocycle of the E-part)**.** *Let $s = 7$, and let u_L and E_L be as in §15.*

(i) (Constant) For $u = \sqrt{y}\sum A_n K_0(2\pi|n|y)\mathbf{e}(nx) + b\sqrt{y}$, $\int_z^{i\infty}[u, R_z^{1/2}] = -\frac{\pi}{2}f_u(z) + \frac{b}{2}$, so $C = -\pi/2$ and $E_L(z) = -\frac{1}{\pi}\int_z^{i\infty}[u_L, R_z^{1/2}]$.

(ii) (Boundary step, both signs of $cx+d$) If $u_L \circ \gamma' = u_L$, then for rational $x \neq \gamma'^{-1}\infty$,

$$\kappa_L(x) - |cx+d|^{-1}\kappa_L(\gamma' x) = -\frac{1}{\pi}\int_{\gamma'^{-1}\infty}^{i\infty}[u_L, R_x^{1/2}],$$

with branch sign $\operatorname{sgn}(cx+d)$, and the singular profile obeys $A(x) = |cx+d|\,A(\gamma' x)$.

(iii) (Vector law) For all 140 labels $\lambda = (\alpha, \beta)$, $u \circ \gamma' = \rho(\gamma')u$ for every $\gamma' \in \mathrm{SL}_2(\mathbb{Z})$, with ρ generated by the S- and T-matrices of Theorem 11.6. The constant terms $b_\lambda = \log\varepsilon \cdot \delta_{\alpha\in\mathbb{Z}^2}$ (here $\varepsilon = 6 + \sqrt{35}$) are carried inside the theta identity. The vector cocycle with b holds for general γ', and on the L-row at $\gamma' = S$ it reduces exactly to the S-formula for ψ (§15.1).

Status (Category A). (i): the integral is holomorphic and translation-covariant, so it equals $C\,\mathbf{e}(nz)$; the asymptotics as $y \to \infty$ give $C = -\pi/2$ (as in Lemma 15.1; the constant term contributes $b/2$). (ii): closedness of the Green form, the non-tangential expansion R1 and the positive-root branch comparison (Lemma 15.2 is the case $\gamma' = S$). (iii): Poisson summation at a fixed majorant, the T-law, the reductions, exact γ^2-periodicity and cycle integration, label by label.

Category B. The vector identity at $\gamma' = STS$, ST^2S, ST^3S agrees to $\le 3.3 \cdot 10^{-12}$; the scalar cocycle at 45 rationals to $7 \cdot 10^{-13}$; the quadrature check of C to 10^{-18}; non-tangential κ fits in 4 directions.

15.6 Interchange of sum and integral

R2. The Fourier expansion of u may be integrated termwise against $R_z^{1/2}$ along the paths used above: the frequencies satisfy $|Q| \ge n_0 > 0$ ($n_0 = 4/140$ for u_R, 0.35 for u_L), there are $O(X)$ γ-orbits with $|Q| \le X$, and dominated convergence applies.

Status (Category A). Proved (elementary estimates).

16 The Eisenstein/cusp split

16.1 u_L is Eisenstein

Modulo $\mathfrak{f}\infty_1\infty_2$ ($\mathfrak{f} = \mathfrak{p}_2\mathfrak{p}_5$; $\mathrm{Cl}_{\mathfrak{f}}^{+} = \mathbb{Z}/8 \times \mathbb{Z}/2$) each sign region of u_L is one ray class: $S^+ \mapsto (5,0)$, $S^- \mapsto (1,0)$, $T^+ \mapsto (1,1)$, $T^- \mapsto (5,1)$, with lattice counts equal to ideal counts for every $n \le 60\,000$. With $d = (4,0)$ the class of a totally negative element $\equiv 1 \bmod \mathfrak{f}$, $S^- = S^+ + d$ and $T^- = T^+ + d$, so each character coefficient carries $(1 + \bar\chi(d))$ and the characters $\chi_{j,l}$ with j odd (exactly the non-σ-invariant ones) drop out:

$$u_L(w) = \tfrac{1}{4} \sum_{j \in \{0,2,4,6\}} \mathbf{e}(-j/8)\big[U^{\cos}_{\chi_{j,0}}(w) + i\,U^{\sin}_{\chi_{j,1}}(w)\big],$$

$$U_\chi^{\cos\,/\,\sin} = \sqrt{y} \sum_{(\mathfrak{a},10)=1} \chi(\mathfrak{a}) K_0\Big(\frac{2\pi \mathrm{N}\mathfrak{a}\, y}{20}\Big)\{2\cos, 2\sin\}\Big(\frac{2\pi \mathrm{N}\mathfrak{a}\, x}{20}\Big).$$

Each of the 8 characters is a genus character $\psi \circ \mathrm{N}$, ψ of conductor dividing 20, normalised by $\psi(7) = \chi(\mathfrak{p}_7)$ (Table 14), with coefficients $\lambda_\chi(n) = 1_{(n,10)=1} \sum_{d|n} \psi(d)(\psi\chi_{140})(n/d)$.

χ	ψ	$\psi\chi_{140}$	type
$\chi_{0,0}$	1	χ_{140}	$(0,0)$
$\chi_{0,1}$	χ_{-4}	χ_{-35}	$(1,1)$
$\chi_{4,0}$	$\chi_5^2 = (\frac{5}{\cdot})$	χ_{28}	$(0,0)$
$\chi_{4,1}$	χ_{-20}	χ_{-7}	$(1,1)$
$\chi_{2,0}$	$\chi_5^3\chi_{-4}$	$\chi_5\chi_{-7}$	$(0,0)$
$\chi_{2,1}$	χ_5^3	$\chi_5\chi_{-4}\chi_{-7}$	$(1,1)$
$\chi_{6,0}$	$\chi_5\chi_{-4}$	$\chi_5^3\chi_{-7}$	$(0,0)$
$\chi_{6,1}$	χ_5	$\chi_5^3\chi_{-4}\chi_{-7}$	$(1,1)$

Table 14: The genus characters $\chi_{j,l}(c) = \mathbf{e}(jc_1/8 + lc_2/2)$; χ_5 is the quartic character mod 5 with $\chi_5(2) = i$.

Status (Category A). Hand proof, independent of PARI: $\mathrm{Cl}(K) = \mathbb{Z}/2$; $|\mathrm{Cl}_{\mathfrak{f}}^{+}| = 16$; $S^- = S^+ + d$ by $\beta(-\nu) = -\beta(\nu)$, $d \neq 0$ since $-\varepsilon^k \equiv -1 \bmod \mathfrak{p}_5$; the 8 characters $\psi \circ \mathrm{N}$ ($\psi \bmod 20$) are distinct, satisfy $\psi \circ \mathrm{N}(d) = 1$, and hence are exactly the characters with $\chi(d) = 1$. Exact coefficient match $A(n) = \#(S^+ \cup S^-)$, $B(n) = \#(T^+ \cup T^-)$ for every $n \leq 60\,000$ (error 10^{-15}). Calling the pieces Maass–Eisenstein series $E_{\psi,\psi\chi_{140}}(\cdot, \frac{1}{2})$ is standard (Category B).

16.2 The Eisenstein/cusp split $(36 + 6)$ and the dihedral forms

Every one of the 140 components is exactly a ray class theta series modulo $\mathfrak{m} = \mathfrak{p}_2^3\mathfrak{p}_5\mathfrak{p}_7\infty_1\infty_2$ (fit over n coprime to 70 up to 54 880; max residual $4.4 \cdot 10^{-14}$ over 280 fits). Expanding in characters: mixed-type content $\leq 10^{-29}$; 116 components have cusp content $\leq 3 \cdot 10^{-30}$ (pure Eisenstein); 24 components have Eisenstein content $\leq 8 \cdot 10^{-31}$ (pure cusp). *Function-space dimensions.* Since distinct frequencies give independent mode functions, $\dim \mathrm{span}\{u_{\alpha,\beta}\}$ is the rank of the label×frequency coefficient matrix. This rank is 42; the Eisenstein components span 36 and the cusp components 6 dimensions, and both subspaces are S- and T-invariant. These statements are exact (Proposition 19.4); the floating-point rank computations agree (invariance residuals $\leq 8 \cdot 10^{-16}$; with independent code at $|Q| \leq 12$ and $|Q| \leq 25$, gap 6.4 versus $6 \cdot 10^{-15}$). The 24 cusp components are the $\beta = -a$ labels with $70\alpha_1 \in \{7, 21, 49, 63\}$ and $7 \mid$ denominator of α_2, excluding the four labels $70\alpha = (49, 56), (63, 42), (7, 28), (21, 14)$, which give identically zero functions. In all, 11 labels (all with $\beta = -a$) give identically vanishing $u_{\alpha,\beta}$. This vanishing is exact: for the first five ($\alpha \in \{(\frac{7}{10}, \frac{4}{5}), (\frac{9}{10}, \frac{3}{5}), (\frac{1}{10}, \frac{2}{5}), (\frac{3}{10}, \frac{1}{5}), (\frac{1}{2}, 0)\}$) the automorph γ, for the remaining six ($\alpha = (\frac{1}{2}, \frac{j}{7})$, $1 \leq j \leq 6$; also $j = 0$) the automorph $-\gamma$, maps $\alpha + \mathbb{Z}^2$ onto itself, commutes with γ^2, leaves the kernel of the lift unchanged (it depends on ν evenly and γ-covariantly, and the closed geodesic is γ-stable), and multiplies the phase $\mathbf{e}(B(\nu, \beta))$ by -1 (because, with $g = \pm\gamma$ the relevant automorph, $g^{-1}\beta - \beta = \mu \in L'$ with $B(\alpha, \mu) \equiv \frac{1}{2}$); so the sum over ν modulo γ^2 equals its own negative. Checked exactly (rational arithmetic) over all 140 labels: these are exactly 11 labels. (Category A.)

Modulo $\mathfrak{p}_2\mathfrak{p}_5\infty_1\infty_2$ the non-σ-invariant characters all have *mixed* type (weight-1 holomorphic, cancelling in u); pure-type (Maass) cusp characters first appear at modulus $(2\sqrt{35})\infty_1\infty_2 = \mathfrak{p}_2^2\mathfrak{p}_5\mathfrak{p}_7\infty_1\infty_2$. There, of 96 characters, 24 are non-σ-invariant of pure type, all of conductor $(2\sqrt{35})$ (12 of type $(0, 0)$, 12 of type $(1, 1)$), 48 are mixed and 24 are genus. With $\chi = \chi_{j,l,1}$, j odd mod 24, $\sigma : j \mapsto 13j$, one obtains one form per σ-orbit and each l:

$$u_\chi(z) = \sqrt{y} \sum_{n \geq 1} \lambda_\chi(n) K_0(2\pi n y) \cdot \{2\cos(2\pi n x)\ (l = 0);\ 2i \sin(2\pi n x)\ (l = 1)\}, \qquad \lambda_\chi(n) = \sum_{\mathrm{N}\mathfrak{a} = n} \chi(\mathfrak{a}),$$

6 even and 6 odd, of level $140 \cdot \mathrm{N}(\mathfrak{f}_\chi) = 19600$, nebentypus $\chi_{140}\omega_0$ with $\omega_0(n) = \chi((n))$ even mod 35 (consistent on 27 primes). Each cusp component is $u_c(w) = \sum_\chi 2\widehat{F}_c(\chi) u_\chi(w/140)$ with $\widehat{F}_c(\chi) = \frac{1}{48}\mathbf{e}(k_{c,\chi}/60)$; e.g. for $c = 73$, $k = 7, 57, 47, 37, 27, 17$ for $j = 1, 3, 5, 7, 9, 11$ ($l = 0$), shifted by 30 for $l = 1$.

Category B. The component fits and the Eisenstein/cusp classification of components are floating-point (PARI ray-class data for $\mathfrak{m}$; base field certified with `bnfcertify`$= 1$); phases recognised to $6 \cdot 10^{-9}$ from 10^{-14}-accurate fits.

Status (Category A). The structure of $\mathrm{Cl}^+_{(2\sqrt{35})\infty_1\infty_2}$ is hand-certified (§18); the structure at the fitting modulus $\mathfrak{p}_2^3\mathfrak{p}_5\mathfrak{p}_7\infty_1\infty_2$ is PARI-only. The dimensions 36 and 6, the S, T-invariance of the split, and the fact that the 24 cusp components are exactly the nonzero labels of odd theta characteristic are exact (Proposition 19.4).

Category B. Maass's construction [11] of dihedral $\lambda = \frac{1}{4}$ forms from pure-type Hecke characters; cuspidality $\Leftrightarrow \chi \neq \chi \circ \sigma$; the level formula $d_K \mathrm{N}(\mathfrak{f}_\chi)$; newform status; Hecke's theory [7] cited generically.

16.3 Function span, S and T spectra

Let $V = \mathrm{span}\{u_{\alpha,\beta}(w)\} \subset \mathbb{C}^{140}$ be the span of the values of the vector u (equivalently the column span of the coefficient matrix). Since $u(-1/w) = Su(w)$ and $u(w+1) = Tu(w)$, V is S,T-invariant, and $V \subset \mathrm{Fix}(P)$.

Proposition 16.1. *(i)* $\dim V = 42 = 36 + 6$, $V = V_{\mathrm{Eis}} \oplus V_{\mathrm{cusp}}$, *both summands* S,T*-invariant. (ii) On* $\mathrm{Fix}(P)$ *(*$\dim 72$*),* S *has eigenvalues* ± 1 *with multiplicities* $36/36$ *(*$\mathrm{tr}\, S|_{\mathrm{Fix}(P)} = 0$*); on* V *they are* $21/21$*, namely* $18/18$ *on* V_{Eis} *and* $3/3$ *on* V_{cusp}*. (iii)* T *has order* 140*. On* $\mathrm{Fix}(P)$ *its eigenvalues are* $\mathbf{e}(r)$ *with* 36 *distinct* $r \in \frac{1}{140}\mathbb{Z}/\mathbb{Z}$*, of multiplicities* 1 *(*12 *values),* 2 *(*18 *values) and* 4 *(*6 *values:* $r = \frac{19}{140}, \frac{31}{140}, \frac{59}{140}, \frac{111}{140}, \frac{131}{140}, \frac{139}{140}$*). On* V_{Eis} *each of the* 36 *values occurs once; on* V_{cusp} *the eigenvalues are the* 6 *values of multiplicity* 4*. Hence* $\mathrm{Fix}(P) = V \oplus V^\perp$ *with a* 30*-dimensional* T*-stable complement* $V^\perp$ *(multiplicities* $0, 1, 2$*), which the lift functions do not realise. (iv) The* 24 *cusp characters form* 12 σ*-orbits of size* 2 *and give* 12 *distinct dihedral forms, all with nonzero content; the* 24×48 *content matrix has rank* 6*, so the* 12 *forms occur only in* 6 *fixed linear combinations.*

Status (Category A). (i) by Proposition 19.4; (ii) by the exact Gauss-sum traces of the local factors (Theorem 19.3), which give $n_\pm = d/2$ on each of the four sectors; (iii) order 140 exact (monomial T, rational phases), multiplicities from the exact phases and the sector decomposition. Floating-point computations agree (singular-value gap $17.1 \to 1.8 \cdot 10^{-13}$, invariance residuals $\leq 8 \cdot 10^{-16}$).

Category B. (iv): the rank 6 of the content matrix is floating-point. Its value is explained exactly by $V_{\mathrm{cusp}} = B_1 \otimes X^-$ (dim = 6). The complement $V^\perp = B_1 \otimes X^+ \oplus A_3 \otimes X^-$ is killed by the lift exactly (Proposition 19.4); the positive-frequency partners $f_{\alpha,\beta}$ span the same V. Weight-1 vectors lie in $\mathrm{Fix}(-P)$ (Remark 25.3(e)), so they cannot realise $V^\perp$ either.

Consequence for ψ. Both representations of the scalar ψ involve only u_L and u_R, which are Eisenstein; hence $\psi_{\mathrm{cusp}} \equiv 0$, $\psi = \psi_{\mathrm{Eis}}$ (Category A given the above). The 24 cusp rows of Φ are nonzero (e.g. $\Phi_{73}(1) = 0.120446470003 + 0.176437422094\, i$); their identification as Lewis–Zagier period functions of the cusp vector is Category B.

17 The Herglotz closed form (Theorems 11.9 and 11.10)

The proof below is given in outline; the full estimates are in the companion report listed in Appendix C.

17.1 Data

For each genus character $\chi = \chi_{j,l}$ (Table 14) with its $\psi \bmod 20$: $a(m) = \psi(m)1_{(m,10)=1}$ (period 20, $a(0) = 0$); $b(k) = \psi(k)\chi_{140}(k)$ (period 140, $b(0) = b(140) = 0$); parities $\eta_a = \psi(-1)$, $\eta_b = \psi(-1)\chi_{140}(-1) = \eta_a$, so $\eta_a\eta_b = 1$ (χ_{140} is even, K being real). Gauss sums $\widehat{b}(n) = \sum_{r \bmod 140} b(r)\mathbf{e}(rn/140)$, $\check{a}(k) = \sum_{r \bmod 20} a(r)\mathbf{e}(kr/20)$; $\rho_n = -\widehat{b}(n)/(2\pi i \cdot 140)$, which is 140-periodic with $\rho_0 = 0$, $\rho_{-n} = \eta_b\rho_n$, $\sum_{s=1}^{140} \rho_s = 0$. Put $G_\chi(\tau) = \sum_{m,k\geq 1} a(m)b(k)\mathbf{e}(mk\tau)$, $\varphi(X) = \sum_s \rho_s \psi_\Gamma(s/140 + X)$, $R_\chi(z) = \sum_{m\geq 1} a(m)\varphi(m/(20z))$, $w_\chi = \mathbf{e}(-j/8)/16$. Equivalently $R_\chi(z) = -140z \sum_{m,n\geq 1} a(m)\rho_n/(7m + nz)$. When $\psi' = \psi\chi_{140}$ is primitive mod 140, $\rho_s = -\tau(\psi')\overline{\psi'(s)}/(2\pi i \cdot 140)$.

Inputs. (I1) $E_L = \sum_\chi w_\chi G_\chi(\cdot/20)$ (§16; $w_\chi = \bar\chi(S^+)/16$). (I2) The Maass S-law (Theorem 11.6). (I3) Lemma 15.1 and Theorem 11.7(i) *on* $\operatorname{Re} z > 0$ *only*: there $\Omega = \Psi$, so Ψ continues holomorphically to $U = \mathbb{H} \cup \{\operatorname{Re} z > 0\}$. Input (I1) was also checked exactly for $20Q \le 2000$.

Remark 17.1 (Use of Theorem 11.7 in this section)**.** Every step below takes place in U. The functions R_χ are holomorphic on U; the difference D of the Liouville step is used on $\{\operatorname{Re} z > 0\}$; the growth bound is on $S = \{1 \le \operatorname{Re} z \le 141,\ \operatorname{Im} z \le -1\} \subset \{\operatorname{Re} z > 0\}$; and the 140-translates of U cover $\mathbb{C}$, which is how $P = D/z$ becomes entire. No step uses Theorem 11.7 on $\operatorname{Re} z < 0$, where $\Omega = -\Psi^- \ne \Psi$.

17.2 Proof outline

Lemma 17.2 (Mittag-Leffler)**.** $B(u) := \sum_{k\ge1} b(k)\mathbf{e}(ku)$ *continues to* $P(\mathbf{e}(u))/(1-\mathbf{e}(140u))$, $\deg P \le 139$, *and for* $u \notin \frac{1}{140}\mathbb{Z}$, $B(u) = \sum_n \rho_n/(u - n/140)$ *(symmetric summation)* $= \sum_{s=1}^{140} \rho_s \pi \cot \pi(u - s/140)$*; the one-sided series converge.*

(Same poles and residues, both vanish at $\pm i\infty$, Liouville in $\mathbf{e}(u)$; one-sided convergence by Abel summation.)

Lemma 17.3. *For* $X \notin (-\infty, -\frac{1}{140}]$*:* $\varphi(X) = -140\sum_{n\ge1} \rho_n/(n + 140X)$. *With* $\beta_1 = \sum_s \rho_s B_1(s/140)$ *(*$= 0$ *if* $\eta_b = +1$*), uniformly in* $|\arg X| \le \pi - \delta$, $\varphi(X) = \beta_1/X + O_\delta(|X|^{-2})$ *for* $|X| \ge 1$*. Consequently* R *is holomorphic on* $U = \mathbb{H} \cup \{\operatorname{Re} z > 0\}$ *(since* $\arg X_m = -\arg z \in (-\pi, \pi/2)$*), and* $|R(z)| \le C(1 + |z|)$ *on* $S = \{1 \le \operatorname{Re} z \le 141,\ \operatorname{Im} z \le -1\}$ *(odd tail by Abel summation).*

Proposition 17.4. *For* $z \in \mathbb{H}$*:* $G_\chi(-1/(20z)) = 2R_\chi(z) + T_\chi(z)$ *with* $T_\chi(z) = -\lim_M \sum_{0<|m|\le M} a(m)\varphi(m/(20z))$*. Hence Theorem 11.9 is equivalent to*

$$\sum_\chi w_\chi T_\chi(z) = (z/\sqrt{35})\, E_R(z) \qquad (z \in \mathbb{H}). \tag{17.1}$$

Proposition 17.5. *For every genus* χ *and* $z \in \mathbb{H}$, $T_\chi(z) = \frac{z}{140}\sum_{n,k\ge1} \widehat{b}(n)\check{a}(k)\mathbf{e}(nkz/140)$*, absolutely convergent.*

For nontrivial ψ this is two-variable summation by parts (kernel $F(m,n) = 1/(nz + 7m)$ with uniform $O(|w|^{-3})$ bounds on $\Delta_m\Delta_n F$, $D_n^2 F$) followed by the cotangent evaluation $\lim_M \sum_{|m|\le M} a(m)/(7m + nz) = -\frac{2\pi i}{140}\sum_k \check{a}(k)\mathbf{e}(nkz/140)$. For trivial ψ the n-outer sum diverges and a second-order argument is needed; the constants $\pm\frac{2}{5}\sum_n \rho_n/(nz)$ cancel. They are genuinely nonzero: $\sum \rho_n/n = i\log\varepsilon/(70\pi)$ to 15 digits (Category B context via the class number formula; not used).

Liouville step. Let $D = \Psi - 2\sum_\chi w_\chi R_\chi$, holomorphic on $\{\operatorname{Re} z > 0\}$ by (I3) and the lemma. On $\mathbb{H}$, $D = zP$ with $P(z) = \sum_{N\ge1} p_N \mathbf{e}(Nz/140)$ (E_R has no constant term since $\nu = 0 \notin F$; its frequencies lie in $\frac{1}{70}\mathbb{Z}$). P extends to an entire 140-periodic function, holomorphic in $q' = \mathbf{e}(z/140)$ on $\mathbb{C}^*$ with Laurent coefficients p_N, $N \ge 1$. A growth bound on S (proved in the companion report: $|\Psi| \le C(1 + |z|)^2$) gives $|P| \le C'(1 + \log|q'|)$ for $|q'| \ge e^{2\pi/140}$; Cauchy's estimates force $p_N = 0$. Hence $P \equiv 0$, which is (17.1) (Theorem 11.10) and $D \equiv 0$ (Theorem 11.9). Periodicity and absence of a constant term alone would not suffice ($\mathbf{e}(z/140)$); the $o(|q'|)$ growth is essential.

Arithmetic form of Theorem 11.10. With $\nu \in L'$, $(u,v) = A\nu$, $\theta = (7u-2v) + u\sqrt{35}$: $\nu \mapsto \theta$ is a bijection $L' \to \mathfrak{p}_2$, $Q(\nu) = -\mathrm{N}(\theta)/140$, $B(\nu,a) = \mathrm{Re}\,\theta/10$, and the coefficient of $\mathbf{e}(Nz/140)$ in E_R is $c_R(N/140) = \frac{\sqrt{35}}{140}\sum_\chi w_\chi \sum_{nk=N} \widehat{b}(n)\check{a}(k)$. For rational $x > 0$ it follows, by Theorem 11.7(ii), that $\psi(x) = \kappa_L(-1/x) - (x/\sqrt{35})\sum_p \kappa_{p,-a}(x)$ is the boundary value of the right-hand side of Theorem 11.9.

Category B. Reading of R_χ as the Lewis–Zagier Eisenstein period function at $s = \frac{1}{2}$ for (ψ, ψ') [10, Ch. IV]; "twisted Herglotz/Kronecker-limit" terminology. Contrast with Radchenko–Zagier [14], whose untwisted $F(x) = \sum(\psi_\Gamma(nx) - \log nx)/n$ has dilogarithmic rational values; here no log-regularisation is needed and the combination takes algebraic values at rationals. Radchenko–Zagier also record Herglotz's evaluation at the same unit $6+\sqrt{35}$ (suggestive only). Zagier's Kronecker limit formula [17] is context only. Individual $R_\chi(x)$ were not tested for algebraicity.

18 $L(0)$ identities for the cusp rows

For a cusp row c, $\kappa_c(x)$ is the t^0 coefficient of $f_c(\mathbf{e}(x)e^{-t})$; by Mellin, $\kappa_c(x) = Z_c(0;x)$ with $Z_c(w;x) = \sum A_Q \mathbf{e}(Qx)Q^{-w}$ entire. For $x = h/k$ with k composed of $2, 5, 7$, $\mathbf{e}(h\mathrm{N}\mathfrak{a}/(140k))$ is a ray class function mod $\mathfrak{m}_k$ (checked on all classes to norm 60 000); for $3 \mid k$ the ideals divisible by the inert prime 3 give one extra ray-class sum. This yields

$$\kappa_c(h/k) = \sum_{\chi'} \widehat{G}_{c,h/k}(\chi')\, L_{\mathfrak{m}_k}(0,\chi'),$$

and, for $3 \nmid k$,

$$\kappa_c(h/k) = \frac{1}{\varphi(140k)} \sum_{\psi \bmod 140k} \tau(\bar{\psi})\psi(h) \sum_\chi \widehat{F}_c(\chi)\, L(0, u_\chi \otimes \psi).$$

Only totally odd characters contribute; for those $L(0,\chi) = W(\chi)\cdot 140 \cdot L(1,\bar{\chi})/\pi^2$ (checked to $2\cdot 10^{-15}$). Type-$(0,0)$ and mixed-type characters have $L(0) = 0$ (Category A from the exact PARI-free Shintani sums of Appendix 18.1; the PARI values agree).

Category B. Numerically verified, conditional on PARI `lfun` (38 digits; primitive characters via `bnrmap`, imprimitive Euler factors restored): all 24 cusp rows at all 7 points to $4\cdot 10^{-14}$ (Table 21). Not proved. Central-value hypothesis rejected: replacing $L(0)$ by $L(\frac{1}{2})$ gives $\Phi_{73}(1) = 0.0349 + 0.0748i$, max deviation 0.32 (0.31 with only the 12 totally odd central values). At $x = 1$ the 24 values lie in a 6-dimensional space.

Hand certification of $\mathrm{Cl}^+_{(2\sqrt{35})\infty_1\infty_2}$. From $1 \to U/U_{\mathfrak{m},1} \to (\mathcal{O}/\mathfrak{m})^\times \times \{\pm\}^2 \to \mathrm{Cl}^+_{\mathfrak{m}} \to \mathrm{Cl} \to 1$ with $(\mathcal{O}/(2))^\times = \langle\sqrt{35}\rangle \cong \mathbb{Z}/2$, $\mathbb{F}_5^\times = \langle 2\rangle$, $\mathbb{F}_7^\times = \langle 3\rangle$, $U = \langle -1, \varepsilon\rangle$, $\mathrm{Cl} = \mathbb{Z}/2$ generated by $\mathfrak{p}_{13}$ ($\mathfrak{p}_{13}^2 = (22+3\sqrt{35})$), the Smith normal form gives invariants $[2,2,24]$, order $96 = 2\cdot 48\cdot 4/4$. The σ-action, infinity types and conductors are read off and match PARI. (Category A, PARI-free.)

18.1 PARI-free certification

Status (Category A). An exact, finite computation, or a standard theorem applied to exact data. Hand inputs: $\varepsilon = 6+\sqrt{35}$ is the fundamental unit and $\mathrm{Cl} = \mathbb{Z}/2$ (generated by $\mathfrak{p}_{13}$, $\mathfrak{p}_{13}^2 = (22+3\sqrt{35})$).

- The ray class group mod $\mathfrak{m} = \mathfrak{p}_2^3\mathfrak{p}_5\mathfrak{p}_7\infty_1\infty_2$, constructed by hand from $U \to (\mathcal{O}/\mathfrak{m})^\times \times \{\pm\}^2 \to \mathrm{Cl}^+_{\mathfrak{m}} \to \mathrm{Cl} \to 1$, has order 96 and structure $\mathbb{Z}/24 \times \mathbb{Z}/2 \times \mathbb{Z}/2$; the same holds for $\mathfrak{f} = (2\sqrt{35})\infty_1\infty_2$.

χ (σ-pair)	type	$L(\frac{1}{2}, u_\chi)$	$L(0)$
$(1,\cdot,1) \sim (13,\cdot,1)$	$(0,0)$	$1.109240202417 - 0.205788403698i$	0
	$(1,1)$	$0.131513747745 + 0.708885114750i$	$-(6+4\sqrt{3}) + (6\sqrt{3}-4)i$
$(3,\cdot,1) \sim (15,\cdot,1)$	$(0,0)$	$1.057777176883 - 0.300492329005i$	0
	$(1,1)$	$-0.123002454086 - 0.432986722370i$	$-12 + 4i$
$(5,\cdot,1) \sim (17,\cdot,1)$	$(0,0)$	$0.514021161380 + 0.466537408032i$	0
	$(1,1)$	$1.122786592523 - 1.237062791397i$	$(4\sqrt{3}-6) - (4+6\sqrt{3})i$

Table 15: Central values and $L(0)$ for six of the twelve dihedral forms; the other six ($j = 7, 9, 11 \sim 19, 21, 23$) are complex conjugates. All 12 central values are nonzero. PARI `lfun`, 38 digits; $L(0)$ values recognised to 10^{-10}.

- The independent class assignment of all 140 components reproduces the PARI fit exactly: the 22 empty (label, sign) slots, every content g, the Eisenstein/cusp/mixed norms (differences $\leq 6.7 \cdot 10^{-16}$), the same 24 cusp components with coefficients $\frac{1}{48}\mathbf{e}(k/60)$, and the label-row ranks 60 and 12 (in function space 36 and 6, §16.3). The theta identity holds for all n with $|Q| \leq 400$ (residual $\leq 8.9 \cdot 10^{-16}$); the statement for all n is Category B.
- Exact Shintani cone sums give the 12 values $L(0, \chi)$ of the totally odd, non-σ-invariant characters of conductor $(2\sqrt{35})$ in $\mathbb{Z}[\zeta_{12}]$: $-(6+4\sqrt{3}) \pm (6\sqrt{3}-4)i$, $-12 \pm 4i$, $(4\sqrt{3}-6) \mp (4+6\sqrt{3})i$, each twice (one per σ-pair). Validations: $\sum_c \zeta(0, c) = 0$; $L(0, \chi) = 0$ for all 72 non-totally-odd characters; $L(\chi) = L(\chi \circ \sigma)$; a genus test exact for all 8 characters mod 20; two independent moduli agree exactly.
- All 7×24 cusp-row identities of Theorem 11.11 hold to $1.2 \cdot 10^{-14}$ with PARI-free inputs. No discrepancy was found.

Not redone without PARI: $L(\frac{1}{2}, \chi)$, $L(1, \chi)$, root numbers. Characters are determined only up to $\chi \leftrightarrow \chi \circ \sigma$. Note that the group mod $\mathfrak{m}$ (order 96) is not the group mod $\mathfrak{p}_2\mathfrak{p}_5\infty_1\infty_2$ (order 16, $\mathbb{Z}/8 \times \mathbb{Z}/2$) used in §12.

19 The theta-lift family: tensor structure

For $s \geq 5$ let A_s, Q_s, a, c_0 and γ be as in §10.4, and let $u_{\alpha,\beta}$ be the theta lifts of §14 built from A_s (for $s = 7$ these are the 140 functions of Theorem 11.6).

Remark 19.1 (The S versus S^{-1} label convention)**.** Let S_{lab} be the label matrix of Theorem 11.6, i.e. $u(-1/w) = S_{\mathrm{lab}} u(w)$. For $s = 7, 8, 11$ one checks exactly: $S_{\mathrm{lab}}^2 = P$, $(S_{\mathrm{lab}}T)^3 = I$ on all labels, $(S_{\mathrm{lab}}T)^3 \neq S_{\mathrm{lab}}^2$, and $(S_{\mathrm{lab}}^{-1}T)^3 = S_{\mathrm{lab}}^{-2}$. So the representation of $\mathrm{SL}_2(\mathbb{Z})$ in the standard presentation $S^4 = 1$, $(ST)^3 = S^2$ is $\rho(S) := S_{\mathrm{lab}}^{-1}$, $\rho(T) := T$. On W_P one has $S_{\mathrm{lab}} = S_{\mathrm{lab}}^{-1}$. (Category A, exact.)

Remark 19.2 (Label operators)**.** Label maps (negation, γ, the local reflections Z_p) are realised as monomial matrices Y with $Y[i, j(i)] = \mathbf{e}(\mathrm{ph}_i)$, the convention of the permutation matrix of the family. With this convention neg $= \rho(-I) = S^2$, and neg, γ, Z_p ($p \mid s$) and Z_q ($q \mid s-2$) commute exactly with S and T (exact exponent arithmetic modulo $2D^2$ at $s = 5, 7, 9, 11, 13, 15, 25$). (Category A.)

19.1 Exact tensor factorisation at $s = 7$

Let $g = (14, 6)/70$, a generator of $L'/\mathbb{Z}^2 \cong \mathbb{Z}/35$, and $a_2 = (\frac{1}{2}, 0)$, so that $a = a_2 + a_{\mathrm{odd}}$ with $a_{\mathrm{odd}} = (\frac{1}{5}, -\frac{1}{5}) \in L'$. In the adapted basis $u'_{x,\epsilon,\delta} := u_{xg+\epsilon a_2,\, -\delta a_2}$ ($x \in \mathbb{Z}/35 \cong \mathbb{Z}/5 \times \mathbb{Z}/7$ by CRT, $\epsilon, \delta \in \{0, 1\}$) put $F_p[x, y] = \mathbf{e}(4xy/p)$, $T_p = \operatorname{diag} \mathbf{e}(2x^2/p)$, and let S_2, T_2 be the 4×4 monomial matrices on the theta characteristics (ϵ, δ): $S_2 : 00 \mapsto 00$, $01 \leftrightarrow 10$, $11 \mapsto -11$; $T_2 : 00 \leftrightarrow 01$, $10 \mapsto i \cdot 10$, $11 \mapsto i \cdot 11$.

Theorem 19.3 (Tensor factorisation, $s = 7$)**.** *(i) In the adapted basis $u'_{x,\epsilon,\delta}$ built on the generator $g = (14, 6)/70 = (7, 3)/35$, exactly $S_{\mathrm{lab}} = (F_5/\sqrt{5}) \otimes (F_7/\sqrt{7}) \otimes S_2$ and $T = T_5 \otimes T_7 \otimes T_2$. (ii) $\rho = \rho_5 \otimes \rho_7 \otimes \rho_2$, where ρ_p is the Weil representation of $(\mathbb{Z}/p, q_p)$ with $q_5 = 2x^2/5$ and $q_7 = x^2/7$, and $\rho_2 = (\mathbf{e}(3/4)S_2^{-1}, T_2)$ acting on the 4-dimensional space of theta characteristics, is a faithful representation of $\mathrm{SL}_2(\mathbb{Z}/4)$ (image of order 48). Hence ρ factors through $\mathrm{SL}_2(\mathbb{Z}/140) = \mathrm{SL}_2(\mathbb{Z}/4) \times \mathrm{SL}_2(\mathbb{F}_5) \times \mathrm{SL}_2(\mathbb{F}_7)$. (iii) $\rho_2 = A_3 \oplus B_1$, where A_3 is the span of the three even characteristics $00, 01, 10$ (irreducible, T-spectrum $\{1, -1, i\}$) and B_1 is the odd characteristic 11 ($S, T \mapsto \mathbf{e}(1/4)$). (iv) $P = \rho(-I) = Z_5 \otimes Z_7 \otimes I_4$, where Z_p is the reflection $x \mapsto -x$. Hence $W_P = (W_5^+ \otimes W_7^+ \oplus W_5^- \otimes W_7^-) \otimes \mathbb{C}^4$; the mixed local parities are absent. The restrictions are $16W_5^+ + 12W_5^-$, $12W_7^+ + 8W_7^-$ and $18(A_3 + B_1)$. (v) $\operatorname{tr} \rho_5(S)|_{W_5^\pm} = (-1, 0)$, $\operatorname{tr} \rho_7(S)|_{W_7^\pm} = (0, -1)$, $\operatorname{tr} \rho_2(S)|_{A_3} = -i$, $\operatorname{tr} \rho_2(S)|_{B_1} = i$, so $\operatorname{tr} S = 0$ on each of the four sectors below and $n_\pm = 36/36$ on W_P.*

Proof outline. Every entry of S_{lab} is one term $\mathbf{e}(r)/\sqrt{35}$ with $r \in \mathbb{Q}$, and T is monomial, so (i) is a finite identity over $\mathbb{Q}(\zeta_{140})$, checked exactly. For (ii), $(\overline{F_p} T_p)^3 = c\, Z_p$ exactly in $\mathbb{Z}[\zeta_p]$ with $c^2 = 125$ ($p = 5$) and $c^2 = -343$ ($p = 7$); with Gauss's sign of the quadratic Gauss sum this gives the scalars $\gamma_5 = -1$, $\gamma_7 = \mathbf{e}(3/4)$, so ρ_p is exactly the Weil representation of q_p; by the congruence theorem for Weil representations of finite quadratic modules of odd level [16], ρ_p factors through $\mathrm{SL}_2(\mathbb{Z}/p)$. ρ_2 satisfies $S^4 = I$, $(ST)^3 = S^2 = -I$, $T^4 = I$ exactly, and the pair group $(\rho_2(g), g \bmod 4)$ has exactly 48 elements. The scalars cancel in the product, which equals $(S_{\mathrm{lab}}^{-1}, T)$. (iv) and (v) follow from the local Gauss sums. □

Status (Category A). Exact (rational and cyclotomic arithmetic), conditional on the cited congruence theorem for the odd local factors [16] (hypotheses: exact match with the Weil formulas, odd level). The local forms are determined as follows: a 140th-root exponent a decomposes as $a = 35x_4 + 28x_5 + 20x_7$, i.e. $x_4 \equiv 3a \bmod 4$, $x_5 \equiv 2a \bmod 5$, $x_7 \equiv 6a \bmod 7$, which gives $q_5 = 2x^2/5$ and $q_7 = x^2/7$. (Since 2 is a non-square mod 5 and 3 is a non-square mod 7, these are not isometric to $x^2/5$ and $3x^2/7$.)

Proposition 19.4 (V is the matched-sector space; local reflections)**.** *Let $X^\pm = W_5^\pm \otimes W_7^\pm$. At $s = 7$: (i) negation acts on u' as $Z_5 \otimes Z_7 \otimes I_4$, and the unit γ ($\varepsilon = 6 + \sqrt{35}$) acts as $I_5 \otimes Z_7 \otimes \operatorname{diag}(1, 1, 1, -1)$; (ii) $V = \mathrm{Fix}(N) \cap \mathrm{Fix}(\gamma) = A_3 \otimes X^+ \oplus B_1 \otimes X^-$, of dimension $36 + 6 = 42$, and the lift kills $V^\perp = B_1 \otimes X^+\,(12) \oplus A_3 \otimes X^-\,(18)$; (iii) $V_{\mathrm{Eis}} = A_3 \otimes X^+$ and the cuspidal part of V is $V_{\mathrm{cusp}} = B_1 \otimes X^-$. The 24 cusp components of §16 are exactly the nonzero labels with $(\epsilon, \delta) = (1, 1)$, and the 11 identically vanishing labels are exactly the 6 inconsistent orbits ($x_5 x_7 = 0$, $\epsilon = \delta = 1$); (iv) each of the 42 classes is a free $\langle N, \gamma \rangle$-orbit (sizes $1, 2, 4$: $3, 15, 24$ classes), with proportionality constants ± 1 in the u' basis; (v) the local reflections Z_5, Z_7 and $Z_2 := \operatorname{diag}(1, 1, 1, -1)$ commute with ρ. Their signs (Z_5, Z_7, Z_2) are $(+, +, +)$ on V_{Eis}, $(-, -, -)$ on V_{cusp}, $(+, +, -)$ on $B_1 \otimes X^+$ and $(-, -, +)$ on $A_3 \otimes X^-$. Thus $Z_5 Z_7 = +1$ on W_P (this is P) and $Z_7 Z_2 = +1$ exactly on V (this is γ). S commutes with all three, and $n_\pm = d/2$ on every joint eigenspace.*

Proof outline. (i) The identities $u_{-\alpha,-\beta} = u_{\alpha,\beta}$ and $u_{\gamma\alpha,\gamma\beta} = u_{\alpha,\beta}$ hold as identities of functions, and their actions on u' are computed exactly. (ii) Hence $V \subseteq \mathrm{Fix}(N) \cap \mathrm{Fix}(\gamma)$, which has dimension 42. The exact Fourier matrix (sums of 4900th roots of unity, $|Q| \le 25$) has rank 42 modulo a prime

above $\ell = 29401 \equiv 1 \bmod 4900$; rank cannot increase under reduction, so $\dim V \geq 42$. (iii) The constant-term vector is $\delta_{x=0} \otimes (e_{00} + e_{01})$, a pure tensor with 2-adic part in A_3; its cyclic span is $A_3 \otimes X^+$ (the argument is that of Theorem 20.1(iii)). The cuspidal functions of the span are those orthogonal to V_{Eis} (the $\lambda = \frac{1}{4}$ constant-term criterion); by unitarity this is $B_1 \otimes X^-$. □

Status (Category A). Exact: (i), (ii), (iv), (v) and the label statements of (iii). The mechanism that kills $V^\perp$ is the γ phase flip: the unit acts by -1 at 7 and by -1 on the odd characteristic, the analogue of the exact vanishing of the 11 labels (§16.2). Part (iii) uses the cited congruence theorem and the standard $\lambda = \frac{1}{4}$ constant-term criterion.

Category B. Calling Z_5, Z_7, Z_2 the Atkin–Lehner operators W_5, W_7, W_4 on the lifted forms (not checked on the forms). The Hecke-character reading: a character contributes only if $\psi(\varepsilon) = \psi(-1) = 1$; genus characters are even at the ramified primes and land in $A_3 \otimes X^+$; characters odd at 5 and 7 are not genus characters, hence not σ-invariant, and give Maass cusp forms in $B_1 \otimes X^-$. The identification of V_{cusp} with specific dihedral Hecke eigenforms rests on the floating fits of §16.

19.2 General odd s

Theorem 19.5 (Tensor factorisation for odd s)**.** *Let $s \geq 5$ be odd and $n = n_s$. Then $L'/\mathbb{Z}^2 \cong \mathbb{Z}/n$ is generated by $g = A_s^{-1} e_2$, $a = a_2 + a_{\mathrm{odd}}$ with $a_2 = (\frac{1}{2}, 0)$ and $a_{\mathrm{odd}} = (1, -1)/(s-2) \in L'$, and in the basis $u'_{x,\epsilon,\delta} := u_{xg + \epsilon a_2, -\delta a_2}$ ($N_s = 4n$ labels):*

$$T = \mathrm{diag}_x\, \mathbf{e}(-x^2/n) \otimes T_2, \qquad S_{\mathrm{lab}} = \left[\mathbf{e}(-2xy/n)/\sqrt{n}\right]_{x,y} \otimes S_2,$$

with $S_2 : (\epsilon, \delta) \mapsto (\delta, \epsilon)$ with phase $(-1)^{\epsilon\delta}$, and T_2 as above; S_2, T_2 do not depend on s. (i) With $\rho(S) := S_{\mathrm{lab}}^{-1}$, $\rho = \bigotimes_{p^e \| n} \rho_{q_p} \otimes \rho_2$, where ρ_{q_p} is the Weil representation of $(\mathbb{Z}/p^e, q_p)$, $q_p(x) = -v_p x^2/p^e$, $v_p = (n/p^e)^{-1} \bmod p^e$, and $\rho_2 = (\mathbf{e}(3/4) S_2^{-1}, T_2)$ is the 4-dimensional $s = 7$ factor. The Weil scalar is $\mathbf{e}(1/4)$ for every odd s, since $n \equiv 3 \bmod 4$. ρ factors through $\mathrm{SL}_2(\mathbb{Z}/4n) = \mathrm{SL}_2(\mathbb{Z}/4) \times \prod \mathrm{SL}_2(\mathbb{Z}/p^e)$, so $N_s = 4s(s-2)$ is the number of labels and the order of T. (ii) Negation acts as $\bigotimes_p Z_p \otimes I_4$; γ acts as $x \mapsto (s-1)x$ with phase $(-1)^{\epsilon\delta}$, i.e. as $\bigotimes_{p|s} Z_p \otimes \bigotimes_{p|s-2} I \otimes Z_2$. In fact $\gamma \equiv (s-1) \cdot I$ on all of $L'/\mathbb{Z}^2$, because $(\gamma - (s-1)I) A_s^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$; and $s - 1 \equiv -1 \bmod p^e$ for $p \mid s$, $\equiv +1$ for $p \mid s-2$ (the reduction of $\varepsilon_s = (s-1) + \sqrt{n_s}$ at the ramified primes). (iii) $P = S_{\mathrm{lab}}^2 = Z_n \otimes I_4$, so $\dim W_P = 4 \cdot \frac{n+1}{2} = 2(s-1)^2$ (this is 72 at $s = 7$). (iv) $\mathrm{Fix}(N) \cap \mathrm{Fix}(\gamma) = \{Z_{(s)} Z_{(s-2)} = +1,\ Z_{(s)} Z_2 = +1\}$ has dimension $3E(s)E(s-2) + O(s)O(s-2) = s(s-1)$, with $E(m) = \frac{m+1}{2}$, $O(m) = \frac{m-1}{2}$ and $Z_{(m)} = \bigotimes_{p|m} Z_p$. Hence $\dim V \leq s(s-1)$. There are exactly $2s-3$ forced-zero labels, and the odd-characteristic sector $B_1 \otimes (\mathbb{Z}/s)^- \otimes (\mathbb{Z}/(s-2))^-$ has dimension $(s-1)(s-3)/4$.

Status (Category A). Symbolic proof for all odd s (the T- and S-entries from $B(g,g) = -2/n$, $B(g, a_2) = 0$, $w_c = g/2$, and the CRT split of $-x^2/n$); exact verification entry by entry at $s = 5, 7, 9, 11, 13, 15, 21, 25$ (including the prime powers 9 and 25). The congruence statement in (i) is conditional on the cited theorem [16]; its convention was checked against the exact $s = 7$ relations. Note $\dim W_P = 2(s-1)^2$, while $3\prod(p^e + 1)/2$ is the bound for d_{Eis}.

Remark 19.6 (The two generators at $s = 7$)**.** Theorem 19.3 uses $g = (7, 3)/35$, Theorem 19.5 uses $A_7^{-1} e_2 = (7, -2)/35 = 11 \cdot (7, 3)/35$ in $L'/\mathbb{Z}^2$. By CRT the local factors of Theorem 19.3(i) combine to the $\mathbb{Z}/35$-phases $\mathbf{e}(13xy/35)/\sqrt{35}$ for S_{lab} ($\frac{4}{5} + \frac{4}{7} = \frac{48}{35}$) and $\mathbf{e}(24x^2/35)$ for T ($\frac{2}{5} + \frac{2}{7} = \frac{24}{35}$). Under $x \mapsto 11x$ these become $\mathbf{e}(-2xy/35)$ and $\mathbf{e}(-x^2/35)$, since $121 \cdot 13 \equiv -2$ and $121 \cdot 24 \equiv -1 \pmod{35}$; these are the phases of Theorem 19.5 at $n = 35$. The two descriptions are the same representation in two bases.

19.3 Sectors: parity and the orbit lemma

Proposition 19.7 (Odd-parity sectors are cusp)**.** *If $\rho(T)$ has no eigenvalue 1 on a ρ-invariant subspace σ, the projection of the lift onto σ is cuspidal. Consequently $d_{\mathrm{cusp}} \geq (s-1)(s-3)/4$ wherever the B_1 sector lies in V, and $d_{\mathrm{Eis}} \leq 3\prod(p^e+1)/2$ whenever no odd parity is carried only by primes with $e \geq 2$ (every tested odd $s \leq 99$; the first undecided cases are $s = 225, 1225$).*

Status (Category A). Proved: no constant term at any cusp. *Category B:* equality $d_{\mathrm{Eis}} = 3\prod(p^e+1)/2$, i.e. that the all-even sector carries no cusp content.

Proposition 19.8 (Parity fact)**.** *For every odd s, every Δ label has even second coordinate x_2. Hence Z_{all} acts as negation on Δ, fixes every $\langle\gamma,-1\rangle$-orbit, and every sector with $\varepsilon_{\mathrm{all}} = -1$ vanishes.*

Status (Category A). Symbolic: $D\cdot A^{-1} = -2\,\mathrm{adj}A$ gives even second coordinates on L'; $a = (s^2, -2s)$; D is even.

Proposition 19.9 (Orbit lemma)**.** *For every odd s, $c_\varepsilon(O_0, q_{\min}) = (1+\varepsilon_{\mathrm{all}})N_{O_0}(q_{\min})$, where $O_0 = \{\pm\nu_0, \pm\gamma\nu_0\}$ and $\nu_0 = (s,-2)$. This is nonzero on every sector with $\varepsilon_{\mathrm{all}} = +1$. Extra reduced classes do occur, at $s = 27, 245, 363, 847, 2025, \ldots$ (16 cases with $s \leq 30001$), but they never land in $Z_S O_0$.*

Status (Category A). Symbolic (Nagell bound, $\gamma^2\nu_0 \equiv \nu_0$, stabiliser $\{1, Z_{\mathrm{all}}\}$, exclusion of extra classes). Also checked exactly for odd $s \leq 401$ and at all 16 exceptions. At $s = 27$ the sector run certifies $\dim V = 702$ (exact), in agreement with Theorem 20.1. Combined with Proposition 19.8, the matched sectors with $\varepsilon_{\mathrm{all}} = +1$ have a nonzero lift. Injectivity therefore follows wherever the sector is irreducible (squarefree $s(s-2)$). *Open:* $\dim V = s(s-1)$ when $e \geq 2$.

20 The theta-lift family: dimensions

20.1 Dimensions of V, V_{Eis} and the cusp part

Theorem 20.1 (Dimensions)**.** *(i)* Orbit bound. $\dim V \leq \#\{$*consistent $\langle\gamma,-1\rangle$-orbits*$\}$*, and the number of orbits is $s(s-1)$ for odd s and $(s/2)(s/2+1)$ for even s (Burnside, with $|\mathrm{Fix}\,\gamma| = 2(s-2)$, $|\mathrm{Fix}(-\gamma)| = 2s$ and $|\Delta[2]| = 4$, using (ii) of Theorem 19.5). (ii)* Exact ranks. *For every $s = 5, \ldots, 51$ and for $s = 72$, the exact rational rank of the integer orbit-count matrix equals the upper bound (for $s = 8, 18, 32, 50, 72$ the half-cone bound of (iv)). Hence $\dim V$ is known exactly there; for odd $s \leq 51$, $\dim V = s(s-1)$. (iii)* Eisenstein dimension. *For odd s, $d_{\mathrm{Eis}} = 3\prod_{p^e \| n_s}(p^e+1)/2$, at every odd s whose local factors with $e \geq 2$ are certified ($p^e \in \{9, 25, 27, 49\}$ suffices for odd $s \leq 51$). For even $s = 2t$, with $2^k \| 2t(t-1)$ and v the odd part of $2t(t-1)$, $d_{\mathrm{Eis}} = d_2(k,v)\prod_{p^e\|v}(p^e+1)/2$, where $d_2(k,v) = 2^{k-1}+4$ (k even), $2^{k-1}+6$ (k odd, $v \equiv 1 \bmod 4$), $2^{k-1}+5$ (k odd, $v \equiv 3 \bmod 4$), certified for $k \leq 5$ (every even $s \leq 62$). (iv)* The $s = 2m^2$ mechanism. *γ has a square root $g \in \mathrm{SO}(A_s)(\mathbb{Q})$ with $\det g = 1$ iff $\mathrm{tr}\,\gamma + 2 = 2s$ is a perfect square, i.e. iff $s = 2m^2$; then $g = (\gamma + I)/(2m) = \begin{pmatrix} 0 & -m \\ 1/m & 2m \end{pmatrix}$ (in unit language $\varepsilon_s = \eta^2$, $\eta = m + \sqrt{m^2-1}$). Splitting each cone $[r, \gamma r)$ as $[r, gr) \sqcup g[r, gr)$ gives an exact upper bound with gap exactly $m(m-1)$ below the orbit bound for $m = 2, \ldots, 7$. It equals the exact lower bound for $m = 2, \ldots, 6$: $\dim V = 18, 84, 260, 630, 1302$ at $s = 8, 18, 32, 50, 72$. At $s = 98$ the half-cone bound 2408 is not sharp: there are two further relations and $\dim V = 2406$ (Theorem 20.5). (v)* Table *($d_{\mathrm{cusp}} = \dim V - d_{\mathrm{Eis}}$ is the exact dimension of the cuspidal functions in the span):*

s	$\dim V$	d_{Eis}	d_{cusp}	s	$\dim V$	d_{Eis}	d_{cusp}
5	20	18	2	6	12	12	0
7	42	36	6	8	18	18	0
9	72	60	12	10	30	30	0
11	110	90	20	12	42	36	6
13	156	126	30	14	56	48	8
15	210	126	84	16	72	48	24
17	272	162	110	18	84	60	24
19	342	270	72	20	110	90	20
21	420	240	180	22	132	108	24
23	506	288	218	24	156	120	36
25	600	468	132	26	182	126	56
27	702	546	156	28	210	168	42
29	812	630	182	30	240	144	96
31	930	720	210	32	260	126	134
33	1056	576	480	34	306	198	108
35	1190	432	758	36	342	270	72
37	1332	684	648	38	380	300	80
39	1482	798	684	40	420	270	150
41	1640	882	758	42	462	240	222
43	1806	1386	420	44	506	288	218
45	1980	990	990	46	552	432	120
47	2162	1080	1082	48	600	288	312
49	2352	1800	552	50	630	312	318
51	2550	1350	1200	72	1302	540	762

In particular $d_{\mathrm{cusp}} > 0$ *for every* s *in the table except* $s = 6, 8, 10$*, and at* $s = 11$*:* $\dim V = 110$*,* $d_{\mathrm{Eis}} = 90$*,* $d_{\mathrm{cusp}} = 20$*.*

Proof notes. (i) is. (ii): one counts the points of each orbit at each norm in the four cones and takes the exact rank over $\mathbb{Q}$ (FLINT `fmpz_mat`); a rank of evaluations is a lower bound for $\dim V$. (iii), odd s: by Theorem 19.5 the constant-term vector (labels with $\alpha \in \mathbb{Z}^2$, i.e. $x = 0$, $\epsilon = 0$) is the pure tensor $\delta_0 \otimes \cdots \otimes \delta_0 \otimes (e_{00} + e_{01})$, with 2-adic part in A_3. Since $\rho(\mathrm{SL}_2(\mathbb{Z}))$ is the full product of the local images, the cyclic span of a pure tensor is the tensor product of the local cyclic spans. These are A_3 (irreducible, dimension 3) and the cyclic span of δ_0 in ρ_{q_p}, of dimension $(p^e + 1)/2$: by hand for $e = 1$, and by exact certification (Krylov rank mod ℓ equal to the $O(\Delta)$-orbit upper bound) for the listed p^e. Even s: the family is literally the Weil representation of $L'/\mathbb{Z}^2$ (an exact identity of the S, T matrices), with local factors certified as stated. (iv): the square root and the half-cone bound are exact; the $s = 8$ relations $N_{O12} = N_{O7} + N_{O9}$ and $N_{O2} = N_{O15} + N_{O19}$ are instances. □

Status (Category A). (i); (ii) and the table entries for $\dim V$, exact per s; (iii) for the stated s, conditional on the cited congruence theorem (odd s) and on the exact local certifications; (iv) the square-root criterion and the upper bounds for $m \leq 7$, with equality for $m \leq 6$; $d_{\mathrm{cusp}} = \dim V - d_{\mathrm{Eis}}$ as the dimension of cuspidal functions, given the $\lambda = \frac{1}{4}$ constant-term criterion.

Category B. $\dim V = s(s-1)$ for *all* odd s (the lower bound is open); $d_2(k, v)$ for $k \geq 7$ ($k = 6$ is Proposition 20.9). Floating-point singular-value ranks and FFT counts agree with the exact values above, except for visible truncation effects at low Q (e.g. $s = 51$).

Remark 20.2 (Sector reading of the cusp dimension, odd s)**.** By Theorem 19.5(iv), $V \subseteq A_3 \otimes (\text{even} \otimes \text{even}) \oplus B_1 \otimes (\text{odd} \otimes \text{odd})$, where even/odd refer to $Z_{(s)}$ and $Z_{(s-2)}$. The all-even sector $A_3 \otimes \bigotimes_p W_p^+$ has dimension d_{Eis}. Wherever $\dim V = s(s-1)$, the cusp part is $d_{\text{cusp}} = O(s)O(s-2) + 3\big(E(s)E(s-2) - \prod(p^e+1)/2\big) \geq (s-1)(s-3)/4 > 0$, with equality iff s and $s-2$ are both prime powers. This reproduces $2, 6, 12, 20, 30, 84, 110, 72, 180, 218, 132$ at $s = 5, \ldots, 25$ and 758 at $s = 41$. Category A at the certified s; for general odd s it is Category B, through the open lower bound. The family cusp part is therefore a local (Weil-parity) phenomenon, not a class-group one.

20.2 The case $s = 2m^2$

Let $s = 2m^2$ with m odd and $g = (\gamma + I)/(2m)$, as in Theorem 20.1(iv).

Proposition 20.3 (Complement trace)**.** *For every odd m: $\#\mathrm{Fix}(g) = 2(m-1)$ and $\#\mathrm{Fix}(-g) = 2(m+1)$ on the complement, $\mathrm{tr}(g \mid \mathrm{Fix}(N, \gamma)) = 2m$, and the $g = -1$ eigenspace X of $\mathrm{Fix}(N)$ has dimension $m(m-1)$.*

Status (Category A). $A(g \mp 1)^{-1}$ is integral, and $\gamma = 1$ on the complement C. Exact at $m = 3, 5, 7$ as well.

Proposition 20.4 (Half-cone relations)**.** *If g acts on M_7-classes (more generally on the classes of $\mathbb{Z}^2 + \mathbb{Z}t$) and m is prime, the span of the half-cone g-relations $e_c - e_{gc}$ is exactly $\delta_{\text{iso}} \otimes X$, of dimension $m(m-1)$.*

Status (Category A). Proved for prime m, given that g is consistent on the classes. That consistency is checked exactly only at $m = 3, 5, 7$. At $m = 3, 5$ the exact half-cone kernel equals $\delta_{\text{iso}} \otimes X$ vector by vector. *Open:* g-consistency for all odd m, and composite m (the m-part is not cyclic). In general the kernel can be larger than the g-relations; see Theorem 20.5.

Theorem 20.5 ($\dim V(98) = 2406$)**.** *At $s = 98$ ($m = 7$), $\dim V = 2406$. The kernel of the lift on the 98-dimensional 7-adic old space has dimension 44. It is spanned by the $42 = m(m-1)$ g-relations and two h-relations, R_1 (at $q = 11/12$) and R_2 (at $q = 1/4$). Here $h = (g+I)/4 \in \mathrm{SO}(A)(\mathbb{Q})$, with $h^2 = g$ and $h^{\mathsf{T}}Ah = A$. The half-cone upper bound 2408 is not sharp.*

Status (Category A). *Lower bound:* the exact rank is 2406 (flint, $Q_{\max} = 40$; old-piece ranks are stable to $Q_{\max} = 130$). The new pieces are injective, and all 44 missing dimensions lie in the old space. *Upper bound:* take the overlattice $M = \bigcap_{k=0}^{7} h^k M_7$, $M_7 = \mathbb{Z}^2 + \mathbb{Z}t_0$, with basis $(4, 0), (1, 1/7)$, $[M_7 : M] = 4$ and Gram matrix $(2Q)\begin{pmatrix} 32 & 64 \\ 64 & 32 \end{pmatrix}$ (even, $\det -3072$). Exact HNF gives $hM = M$ and $hM^\sharp = M^\sharp$. Moreover h maps the $+$ set of R_i bijectively onto its $-$ set as M-cosets ($32 + 32$ for R_1, $16 + 16$ for R_2). The sets are γ-invariant ($\gamma = h^4$). Since h has positive eigenvalues $2 \pm \sqrt{3}$, it maps fundamental cones for $\langle\gamma\rangle$ to fundamental cones, so $N_{S^+}(q) = N_{S^-}(q)$ for all q, and the theta identities hold. The same argument with g on M_7 (eigenvalues $7 \pm \sqrt{48}$) proves the 42 g-relations. The 44 relations are independent (exact rank 44), so $\dim V \leq 2450 - 44 = 2406$.

Category B. Extra h-relations beyond $m(m-1)$ are expected exactly when $2m+2$ is a square, i.e. $m = 2k^2 - 1$ ($m = 7, 17, 31, \ldots$); their number is given by Proposition K1 below. Beyond $m = 7$ the actual lift rank is not computed.

The two h-relations at $m = 7$ are the case $k = 2$ of the following count. Fix $k \geq 2$, $m = 2k^2 - 1$, $s = 2m^2$, and let $g = (\gamma + I)/(2m)$ and $h = (g + I)/(2k)$. Let rank_g and rank_h be the ranks of the orbit-sum maps $F \to \mathrm{Fun}(\langle X, -1\rangle\text{-orbits on } M^\sharp/M)$, $X \in \{g, h\}$, with $F = \mathrm{Fun}(M_t^\sharp/M_t)^{\langle\gamma, -1\rangle}$, computed by exact orbit sums.

Proposition 20.6 (K1)**.** $\operatorname{rank}_g = m(m+1)$ *and* $\operatorname{rank}_h = m(m+1) - k(k-1)$*. Hence the number of extra h-relations is exactly $k(k-1)$ for every $k \geq 2$. Moreover* $\dim F = 2m^2 = s$*.*

Proof outline. Order model. $M_t = \mathbb{Z}e_1 \oplus \mathbb{Z}f$ with Gram matrix $\left(\begin{smallmatrix} 2 & 2m \\ 2m & 2 \end{smallmatrix}\right)$ and $gf = -e_1 + 2mf$, so $(M_t, B, g) \cong (R_m, \operatorname{Tr}(x\bar{y}), \cdot\varepsilon_m)$ with $R_n = \mathbb{Z}[\sqrt{n^2-1}]$, $\varepsilon_n = n + \sqrt{n^2-1}$; hence $C := M_t^\sharp / M_t \cong R_m / 2\sqrt{d_m} R_m$, $d_n = n^2 - 1$. *Pell self-similarity.* With $u = (e_1 + f)/(2k)$ one has $hu = f$, $hf = -u + 2kf$ and $\operatorname{Gram}(u, f) = \left(\begin{smallmatrix} 2 & 2k \\ 2k & 2 \end{smallmatrix}\right)$, so $N_1 = M_t + hM_t = \mathbb{Z}u \oplus \mathbb{Z}f$ is even, h-stable, and $(N_1, h) \cong (R_k, \varepsilon_k)$: h is "g at level k". $I = N_1/M_t$ is isotropic of order $2k$ and $C_1 = I^\perp / I \cong R_k / 2\sqrt{d_k} R_k$. *Hecke reduction.* On $V_0 = \operatorname{Fun}(C)^{\langle g, -1 \rangle}$ (so $\operatorname{rank}_g = \dim V_0$), rank_h is the rank of $1 + \tau$ with $\tau = (2k)^{-1} \uparrow \circ \bar{h} \circ \downarrow$, where $\downarrow$ sums over I-cosets in $I^\perp$ and $\uparrow$ is its adjoint. *Spectrum.* $\downarrow\uparrow = 2k$, so τ^2 is an orthogonal projection, $\tau^3 = \tau$, and the (-1)-eigenspace of τ has dimension $a - b$, with a, b the numbers of $\langle h^2, -1 \rangle$- and $\langle h, -1 \rangle$-orbits on C_1. *Burnside.* On $R_n / 2\sqrt{d} R_n$, $\langle \varepsilon, -1 \rangle$ acts through $\{\pm 1, \pm\varepsilon\}$ with $|\operatorname{Fix}(-1)| = 4$, $|\operatorname{Fix}(\mp\varepsilon)| = 2(n \mp 1)$; so there are $n(n+1)$ orbits of $\langle \varepsilon, -1 \rangle$ and $2n^2$ of $\langle \varepsilon^2, -1 \rangle$. Level m gives $\operatorname{rank}_g = m(m+1)$ and $\dim F = 2m^2$; level k gives $a = 2k^2$, $b = k(k+1)$, so the extra count is $2k^2 - k(k+1) = k(k-1)$. □

Status (Category A). Proved for all $k \geq 2$ (the Burnside counts are also brute-forced for $n = 2, \ldots, 40$). *Exact checks of the reformulation:* Exact computation (over $\mathbb{Q}$, $k = 2, 3, 4$) gives $\operatorname{rank}(I + \tau) = 54, 300, 980$, equal to rank_h, with minimal polynomial $x^3 - x$; the same computation at $k = 5$ ($m = 49$) and $k = 6$ ($m = 71$) gives extra $= 20, 30$ and $\operatorname{rank} \tau = 50, 72$, with ranks computed mod 1000003 and mod 998244353 in agreement. Those mod-p ranks are consistency checks of a proved statement, not independent proofs.

Corollary 20.7. *On V_0, τ has eigenvalues $+1$, -1, 0 with multiplicities $k(k+1)$, $k(k-1)$, $m(m+1) - 2k^2$; so $\operatorname{tr} \tau = 2k$ and $\operatorname{rank} \tau = 2k^2$. The extra relations are explicit: $\uparrow$ of the $\bar{h}$-odd $\langle \bar{g}, -1 \rangle$-invariant functions on $C_1 = N_1^\sharp / N_1$. In particular the number of closed orbits is $m(m+1) = \operatorname{rank}_g$.*

Status (Category A). Consequence of Proposition K1.

Proposition 20.8 (Sharpness of Proposition K1 on the old space)**.** *For $k = 2, 3, 4, 5$ ($m = 7, 17, 31, 49$), the theta lift on $F = \operatorname{Fun}(M_t^\sharp / M_t)^{\langle \gamma, -1 \rangle}$ has rank exactly $m(m+1) - k(k-1) =$ 54, 300, 980, 2430. So at these m the old space has no relations beyond the g- and h-relations.*

Status (Category A). The upper bound is Proposition K1. The lower bound is the rank over $\mathbb{F}_{2^{31}-1}$ of the exact (orbit × norm) count matrix of the $x \in M_t^\sharp$ with $0 < |Q(x)| \leq Q_{\max}$, in an exact fundamental domain for $\langle \gamma \rangle$; a rank modulo a prime is a lower bound for the rank over $\mathbb{Q}$. At $k = 2$ this reproduces the old-piece ranks $42 + 12$ of Theorem 20.5. *Not claimed:* $\dim V(578) = 83532$, which would also need injectivity and independence of the new pieces at $s = 578$.

Status (Category A). Proposition K1: proved for all $k \geq 2$. The symbolic-in-k ingredients and the Burnside counts (brute-forced for $n = 2, \ldots, 40$) are checked by computer. Consistency checks of the reformulation: exact over $\mathbb{Q}$ for $k = 2, 3, 4$, $\operatorname{rank}(I + \tau) = 54, 300, 980$ with minimal polynomial $x^3 - x$; modulo two primes at $k = 5, 6$ ($m = 49, 71$), extra $= 20, 30$ and $\operatorname{rank} \tau = 50, 72$.

Dimension consequences (Category A as upper bounds; sharpness open). Proposition K1 is a statement about the lattice-level rank rank_h, which is a proved upper bound for the old lift rank. Every h-relation is a proved theta relation (the argument of Theorem 20.5), so for $s = 2m^2$, $m = 2k^2 - 1$, subtracting the $k(k-1)$ extra relations gives *proved upper bounds*: $\dim V(98) \leq 2406$ (equality, Theorem 20.5) and $\dim V(578) \leq 83532$, and likewise at every $m = 2k^2 - 1$. Whether these bounds are sharp at $m = 17, 31, \ldots$ is *not* proved.

20.3 The case $s = 64$

Proposition 20.9 ($s = 64$)**.** $\dim W_P(64) = 1986$. *The 2-adic cyclic span d_2 at $k = 6$ equals 36, so the 2-adic factor of the function span is $36 \cdot 16 = 576$ at $s = 64$.*

Status (Category A). W_P is an exact count. For the upper bound, $d_2 \le \#O(\Delta_2)$-orbits $= 42$, minus a 6-dimensional S, T-invariant subspace $Y \perp \delta_0$ that is verified exactly in $\mathbb{Z}[\zeta_{64}]$, leaving 36. The lower bound 36 is exact. The value $2^{k-1} + 4$ for general k remains Category B.

20.4 Prime-square sectors

For $s = p^2$ the isotropic projector Π splits each sector into an *old* piece, the p-local invariants, and a *new* piece, their complement; both are subrepresentations.

Proposition 20.10 (Prime-square sectors: ranks)**.** *Exactly:*

s	$(+,+)$ *old*	$(+,+)$ *new*	$(-,-)$ *new*	*total*
9	*12*	*48*	*12*	*72*
11	*18*	*72*	*20*	*110*
25	*36*	*432*	*132*	*600*

Every piece is injective. The ν_0 coefficient is nonzero on every new piece and zero on every old piece.

Status (Category A). Exact integer ranks. The irreducible decompositions are Proposition 20.12; a uniform proof is open.

Let $s = p^2$ with p an odd prime. Write $W_p^{\pm}$ for the even and odd parts of the Weil representation of $\mathbb{Z}/p^2$ (or of $\mathbb{Z}/q$), and A_3 for the irreducible 3-dimensional 2-adic factor.

Theorem 20.11 (Prime-square structure)**.** *Assume (C1) Theorem 19.5 and (C2) CRT surjectivity, both proved above, and (C3) the cited Nobs–Wolfart / Cliff–McNeilly–Szechtman decomposition $W_p^+ = \mathbf{1} \oplus R$, with R irreducible of dimension $(p^2-1)/2$, and $W_q^{\pm}$ irreducible for primes q. Let $\varepsilon = (\varepsilon_q)_{q \mid s-2}$, with $Z_{(s)} = +$. Then:*

(i) The sector is $\Sigma_\varepsilon = W_p^+ \otimes (\bigotimes_q W_q^{\varepsilon_q}) \otimes A_3$. Its p-local invariants are $\mathbf{1} \otimes (\bigotimes_q W_q^{\varepsilon_q}) \otimes A_3$. Summed over the ε with $Z_{(s-2)} = +$, these have total dimension $3(s-1)/2$.

(ii) If $s-2$ is squarefree, then $\Sigma_\varepsilon = (p\text{-local invariants}) \oplus R \otimes (\bigotimes_q W_q^{\varepsilon_q}) \otimes A_3$ is a sum of two non-isomorphic irreducibles.

Status (Category A). Conditional on (C3), which is cited and not re-proved; (C1) and (C2) are proved above; (C3) is Category A for the primes covered by Propositions 20.14 and 20.15. Irreducibility of the complement follows from the outer-tensor argument. *Caveats:* if $s-2$ is composite, the sector splits by prime sign (see $s = 121$ below). If $q^2 \mid s-2$, the complement is reducible.

Proposition 20.12 (Exact splits)**.** *The "$(+,+)$" sectors split as follows: $s = 9$: $12 + 48$; $s = 11$: $18 + 72$; $s = 25$: $36 + 432$; $s = 49$: $72 + 1728$. In each case the commutant is 2 and the smaller piece is exactly the p-local invariant subspace. The $(-,-)$ sectors at $s = 9, 11, 25$ (dimensions $12, 20, 132$) are irreducible, and the mixed sectors vanish.*

Status (Category A). The sectors are built with an exact invariance certificate (0 failures). The commutant is computed mod ℓ on an exactly verified basis. It is lifted to $\mathbb{C}$ using $\ell \nmid |\mathrm{Mp}_2(\mathbb{Z}/N)|$; this divisibility is machine-checked (at $s = 49$ with the two primes $\ell = 1272914161$ and $\ell = 2036662657$, both giving commutant 2 and pieces of dimensions 72 and 1728). *Standard input, cited:* a Weil representation factors through $\mathrm{Mp}_2(\mathbb{Z}/\mathrm{ord}\, T)$. For $\gamma \in \Gamma(N)$ the cited Borcherds/Scheithauer $\Gamma_0(N)$ formula gives $\rho_D(\gamma) = \chi_D(\gamma)I$. After Proposition 20.14, the splits at $s = 9, 25$ need neither this nor (C3). The $s = 9, 25, 49$ splits agree with Theorem 20.11(ii), where $s - 2 = 7, 23, 47$ is prime.

Proposition 20.13 (Label-space counts at $s = 121$ and $s = 169$)**.** *Let the p-local invariants of a sector be the vectors fixed by $\langle T^a, ST^aS^{-1}\rangle$, where $a \equiv 1 \pmod{p^2}$ and $a \equiv 0 \pmod{n_T/p^2}$; this definition does not use the tensor factorisation.*

(i) $s = 121$ *($p = 11$, $s-2 = 7\cdot 17$; signs ordered Z_7, Z_{11}, Z_{17}). The nonzero sectors are $(+,+,+)$, $(-,+,-)$, $(+,-,-)$ and $(-,-,+)$, of dimensions* 6588, 4392, 1920 *and* 1620. *They contain at most* 108, 72, 0 *and* 0 *independent p-local invariants respectively, for a total of at most $180 = 3(s-1)/2$.*

(ii) $s = 169$ *($p = 13$). The $(+,+)$ sector has dimension* 21420 *and contains at most $252 = 3(s-1)/2$ independent p-local invariants; the $(-,-)$ sector has dimension* 6972 *and contains none.*

Given the cited factorisation of a Weil representation through $\mathrm{Mp}_2(\mathbb{Z}/n_T)$, these upper bounds are equalities.

Status (Category A). Sector dimensions: exact orbit counts. Invariant counts: $2k - \mathrm{rank}[X \mid SX]$ computed modulo ℓ, which bounds the characteristic-0 count from above. At $s = 121$ two primes ($\ell = 19903795297$, 99518976481) agree exactly; at $s = 169$ one prime was used. Equality holds if $\ell \nmid |H|$, which is guaranteed when ρ factors through $\mathrm{Mp}_2(\mathbb{Z}/n_T)$ (cited), since ℓ exceeds every prime dividing $|\mathrm{SL}_2(\mathbb{Z}/n_T)|$. The counts agree with Theorem 20.11(i) and do not use (C1) or (C3). Irreducibility of the complements is not claimed.

Proposition 20.14 ((C3) for small p)**.** *For $p \in \{3, 5, 7, 11, 13, 59\}$, the even Weil representation of $\mathbb{Z}/p^2$ is $\mathbf{1} \oplus R$ with R irreducible, and the odd part is irreducible. Together with (C1), this gives the $(+,+)$ decompositions at $s = 9, 25, 49$ in characteristic 0 directly, without the $\ell \nmid |\mathrm{Mp}_2|$ lift.*

Status (Category A). The mod-ℓ commutant is 2 (even part) or 1 (odd part), on exactly verified bases, for both square classes. Semicontinuity bounds the characteristic-0 commutant above by the mod-ℓ one. The explicit invariant line $\delta_{p\mathbb{Z}/p^2}$ bounds it below by 2. The general-p case remains cited (Nobs–Wolfart / Cliff–McNeilly–Szechtman). The irreducibility of 1728 at $s = 49$ also uses W_{47}^+ irreducible (cited; not run).

Category B. *Factor dimensions only:* $s = 59^2 = 3481$, $s - 2 = 7^2 \cdot 71$. The prediction is: in $(+,+,+)$, p-invariants $108 + 2592$ and complement $187920 + 4510080$ (reducible); in $(+,-,-)$, 2520 and 4384800; total p-invariants $5220 = 3(s-1)/2$. No label-space run.

Proposition 20.15 (Weil representations of $\mathbb{Z}/M$)**.** *The even and odd parts of the Weil representation of $\mathbb{Z}/M$ have the following (dimension, commutant): $M = 47$: $(24,1)$, $(23,1)$; $M = 59$: $(30,1)$, $(29,1)$; $M = 71$: $(36,1)$, $(35,1)$; $M = 17^2$: $(145,2)$, $(144,1)$; $M = 19^2$: $(181,2)$, $(180,1)$; $M = 23^2$: $(265,2)$, $(264,1)$. Hence for $q = 47, 59, 71$ both parts are irreducible, and for $p = 17, 19, 23$ the even part of $\mathbb{Z}/p^2$ is $\mathbf{1} \oplus R$ with R irreducible and the odd part is irreducible.*

Status (Category A). Commutants computed mod $\ell \equiv 1 \pmod{M}$ on exactly verified bases, both square classes; the characteristic-0 commutant is bounded above by the mod-ℓ one, and for $\mathbb{Z}/p^2$ below by 2 via the invariant line $\delta_{p\mathbb{Z}/p^2}$. With Proposition 20.14, (C3) is Category A for every prime $q \mid s-2$ and every p^2 occurring at $s = 9, 25, 49$ (in particular $W_{47}^{\pm}$ irreducible, used for the 1728 at $s = 49$), and for $\mathbb{Z}/17^2, \mathbb{Z}/19^2, \mathbb{Z}/23^2$. The general-$p$ statement remains cited (Scheithauer, IMRN 2009, Prop. 4.5 and Thm. 4.7; Cliff–McNeilly–Szechtman, main theorem; Nobs–Wolfart, theorem number not verified).

Category B. At $s = 3481$ the decomposition remains a factor-dimension prediction: the factors are Category A, but the assembled decomposition is conditional on the tensor factorisation and was not computed in label space.

21 Cusp content of the vector u_L

The family statements of §20 concern the whole span V. Whether the single vector $u_L = u_{a,0}$ has cusp content is a different question; the two must be kept apart.

Proposition 21.1 (Cusp criterion for u_L)**.** *Let $G = \mathrm{Cl}^+_{M\infty_1\infty_2}$ with $M = 2dc$, $d = 2(s-2)$, $n_s = c^2 D_0$; let $T \subset G$ be the class set of the coprime part of u_L, $S = \mathrm{Stab}(T)$, $j = [-1]$, $\bar{G} = G/(S + \langle j \rangle)$, $H = (1-\sigma)G$ and $\bar{H}$ its image in $\bar{G}$. (i) The coprime part of u_L is Eisenstein (genus-pure) iff $\bar{H} = 1$, i.e. iff σ acts trivially on $\bar{G}$; the number of non-genus pure characters is then $|\bar{G}| - |\bar{G}^\sigma|$. (ii) For odd s with $c = 1$ (n_s squarefree), only the coprime part occurs and Theorem 21.3 decides: u_L is Eisenstein iff $h(\mathbb{Q}(\sqrt{n_s})) = 2^{\omega(s)}$. (iii) The verdicts for $5 \le s \le 44$ are those of Table 17. (iv) The rule "u_L has cusp content iff $h^+ \neq$ #genera" is false: $s = 14, 17, 18, 22, 23$ have $h^+ =$ #genera and cusp content (all with $\bar{H}$ of order 2). (v) The two notions differ: $d_{\mathrm{cusp}} > 0$ at $s = 5, 7, 13, 15$, where u_L is Eisenstein. The u_L verdict is class-group data ($\bar{H}$) on top of the local structure.*

Status (Category A). (i): reduction (Tate's $\ker \mathrm{N} = C_K^{1-\sigma}$ for cyclic $K/\mathbb{Q}$, plus the dictionary "Maass content of u_L = pure-type character content of its class sets"). (iv): exact finite group computations per s on complete coprime parts (the $s = 18$ coprime part is complete at $B = 150\,000$).

Category B. Sufficiency of the modulus $M = 2dc$ for all norms (stable under $M \to 2M$ wherever tested).

Here $\mathbf{W} \subset (\mathcal{O}/M)^\times$ is the unit stabiliser of $\beta_0 + L$ (unrelated to W_P), S_0 is the coprime-part stabiliser and $\psi(\lambda) = [(\lambda)]$.

Theorem 21.2 (Coset property)**.** *For every s and every c-part $\mathfrak{c}$, $T_\mathfrak{c} + \psi(\mathbf{W}) = T_\mathfrak{c}$, and $S_0 = \psi(\mathbf{W})$. Hence every $T_\mathfrak{c}$ is a union of S_0-cosets.*

Status (Category A). The lemma $T_\mathfrak{c} + \psi(\mathbf{W}) = T_\mathfrak{c}$ is elementary. $S_0 = \psi(\mathbf{W})$ follows from the local condition $R_p = \mathbf{W}_p$, proved for all s and p: it reduces to the fact that ratios of exact-valuation elements of $\beta_0 + L$ lie in the multiplier order of L. The cases are $p \nmid f$; odd $p \mid c$ with $p \mid s$ or $p \mid s-2$; and $p = 2$ in each 2-adic case. Also checked exactly: global $S_0 = \psi(\mathbf{W})$ at $s = 5, \ldots, 44$, and $R_p = \mathbf{W}_p$ at $s \le 150$.

Theorem 21.3 (Squarefree criterion)**.** *Let s be odd with $s(s-2)$ squarefree, and $h = h(\mathbb{Q}(\sqrt{s(s-2)}))$. Then only the coprime c-part occurs, $|\bar{G}^\sigma| = 2^{\omega(s)}\varphi(s-2)$, and u_L is Eisenstein $\iff$ $h = 2^{\omega(s)}$. The number of non-genus characters is $\varphi(s-2)(h - 2^{\omega(s)})$.*

Status (Category A). The proof uses Chevalley's ambiguous class number formula (ray form, unit-norm index 1) and, independently, Gauss genus theory; these are classical theorems applied to exact data. The exact cross-check covers 47 values $s \le 151$: Eisenstein only at $s = 5, 7, 13, 15$.

verdict ($5 \le s \le 44$)	s
Eisenstein	$5, 6, 7, 8, 9, 10, 12, 13, 15, 16, 20, 24, 28$
cusp content	$11, 14, 17, 18, 19, 21, 22, 23, 25, 26, 27$ and every s with $29 \le s \le 44$

Table 17: Verdicts for u_L.

Status (Category A). Verdicts are per s over all (infinitely many) c-parts, using Theorem 21.2 and Proposition 21.4. Coverage at $s = 32, 34, 36, 40$ was completed by enumeration to $B = 200000$. At $s = 24$ there are no mixed-prime parts for any exponents (a local proof), and the parts $(1, 12)$, $(12, 1)$, $(1, 16)$, $(16, 1)$ are genus-pure. Beyond the table, cusp content holds at $s = 53, 55, 57$ (incomplete coverage, matching the prediction of (ii); Category B).

Proposition 21.4 (Deep bad-prime parts)**.** *At $s = 9$ (the 29 parts (e_1, e_2) with $e_1 \le e_2 \le 7$ at the split prime 3, plus $\mathfrak{c} = 1$), $s = 16$ ($\mathfrak{c} = 1, \mathfrak{p}_2$; and 7) and $s = 24$ (15 parts at 2, up to exponent 8; and 11), every tested $\mathfrak{c}$-part is complete and genus-pure. At $s = 18$ the 3-parts are pure, but the coprime part is not, so u_L has cusp content. If $\bar{H} = 1$ and every $T_{\mathfrak{c}}$ is S-invariant, then every $\mathfrak{c}$-part is genus-pure and u_L is Eisenstein; conversely, if $\bar{H} \neq 1$, the coprime part already gives cusp content. By Theorem 21.2 every $T_{\mathfrak{c}}$ is a union of S-cosets, hence S-invariant, so the coprime criterion decides.*

Status (Category A). Per tested part: exact (`matsolvemod` lattice solve, completeness of every class, exact purity test). The reduction lemma, given S-invariance: $(1 + j)1_{T_{\mathfrak{c}}}$ is $(S + \langle j \rangle)$-invariant, hence H-invariant, and $\sigma H = H$.

21.1 Local root numbers at $s = 7$

The 24 cusp characters at $s = 7$ have conductor of norm 140 (ramified at $\mathfrak{p}_2, \mathfrak{p}_5, \mathfrak{p}_7$). The vector-valued components are $\Gamma(140)$-modular and the scalar newforms have level $140^2 = 19600$; the two levels are consistent (Category A).

Proposition 21.5 (Local root numbers)**.** *For the 24 cusp characters, the local root numbers ε_v ($\mathfrak{p}_2, \mathfrak{p}_5, \mathfrak{p}_7, \infty$) are exact elements of $\mathbb{Q}(\zeta_{840})$ of modulus 1. $\varepsilon_{\mathfrak{p}_2}$ is a 6th root of unity; $\varepsilon_{\mathfrak{p}_5}$ is never a root of unity. The global root numbers have non-monic minimal polynomials (e.g. $5t^8 + 6t^4 + 5$), so they are not roots of unity. The sector signs are* products *of local root numbers: $Z_p = \psi_{\mathfrak{p}}(-1) = \varepsilon(\psi_{\mathfrak{p}})\varepsilon(\psi_{\mathfrak{p}}^{-1}) = -1$ ($p = 5, 7$), and $S_p^2 = \omega_{\pi_p}(-1)$. They are not individual local root numbers.*

Status (Category A). The local factors (Tate's formulas) and the character-side identities are exact.

Category B. $\prod_v \varepsilon_v = W$ against PARI's numerical root number ($\le 2.7 \cdot 10^{-63}$). $W_p \leftrightarrow S_p$ holds only up to a fixed unit (the Atkin–Li normalisation, cited and not re-derived).

22 H_s in closed form

Throughout, $s \ge 5$ is an integer, $c_1 = (-1, 1)$, $c_2 = (-4, 1)$, $g := \gcd(24, s - 8)$ and $\Delta := (s - 2)(7s - 32)$ (§10.4). All radial expansions rest on Proposition 13.1.

Theorem 22.1 (H_s in closed form)**.** *(i)* Structure. *$Q(c_1) = -(s - 2)/2$, $Q(c_2) = -(7s - 32)/2$, $B(c_1, c_2) = -4(s - 2)$, so the pair is admissible exactly for $s \ge 5$, and $4Q(c_1)Q(c_2) = \Delta$. $H_s = \frac{1}{2} q^{c_0} \vartheta_{a,0,\mathrm{hol}}^{c_1,c_2} + \Delta_{\mathrm{lines}}(s)$, where Δ_{lines} is a finite sum of unary theta series with explicit s-dependence,*

and H_s has an explicit cone double-sum q-expansion for every $s \geq 5$; $H_5 \equiv 0$. (ii) Radial coefficients at $x = 0$. *Along $q = e^{-t}$, H_s has no integer-order terms at all, and for $r \in \{-\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}\}$ its coefficients are $H_r = \sqrt{2\pi s}\,[R_r(s) + S_r(s,g)\sqrt{\Delta}]$ with*

$$
\begin{aligned}
R_{-1/2} &= -\frac{1}{2(s-2)},\\
S_{-1/2} &= \frac{1}{2(s-2)(7s-32)},\\
R_{1/2} &= -\frac{(s-4)(3s-2)}{48(s-2)^2},\\
S_{1/2} &= \frac{s^3-44s^2+192s+2g^2(s-2)}{48(s-2)^2(7s-32)^2},\\
R_{3/2} &= -\frac{15s^4-76s^3+108s^2+96s-128}{3840(s-2)^3},\\
S_{3/2} &= \frac{\Sigma_{3/2}}{3840(s-2)^3(7s-32)^3},\\
R_{5/2} &= -\frac{(s-4)(105s^5-730s^4+2732s^3-5208s^2+5696s-2560)}{645120(s-2)^4},\\
S_{5/2} &= \frac{\Sigma_{5/2}}{645120(s-2)^4(7s-32)^4},
\end{aligned}
$$

where $\Sigma_{3/2} = 3s^6-152s^5+3872s^4-28160s^3+61440s^2+20g^2s(s^3-18s^2+96s-128)-8g^4(s-2)^2$ and $\Sigma_{5/2} = 15s^9-972s^8+28320s^7-537216s^6+4773888s^5-18923520s^4+27525120s^3+210g^2s^2(s^5-34s^4+448s^3-2816s^2+8192s-8192)-8g^4s(35s^4-504s^3+2940s^2-6832s+5376)+160g^6(s-2)^3$. The denominators are $(s-2)^{r+3/2}(7s-32)^{r+3/2}$, g enters only through g^{2i} with $i \leq r+\frac{1}{2}$, and all four vanish at $s = 5$. (iii) $H_{-1}(x) = 0$ for every rational x. (iv) $s = 7$. $a_H = H_{-1/2}(7)$ and $a_C = C_{-1/2}(7) = P_{-1/2}(7) - H_{-1/2}(7) = 0.829142617\ldots$ symbolically; these are the radical forms of Table 13. (v) $s = 8$, the $\sqrt{\pi}$-rational scale. $\Delta(8) = 144$, and $H_{-1/2}(8) = -\sqrt{\pi}/6$, $H_{1/2}(8) = 5\sqrt{\pi}/54$, $H_{3/2}(8) = -175\sqrt{\pi}/324$, $H_{5/2}(8) = 995\sqrt{\pi}/972$, $C_{-1/2}(8) = 11\sqrt{\pi}/24$. Among $s < 10^6$, Δ is a perfect square only at $s = 5, 8, 26, 77, 365, 1178, 5768, 18725, 91877, 298376$. (vi) Assembly. *The integer orders of $C = P_s^{\mathrm{orth}} - H_s$ at $x = 0$ are those of $(1+E_s)/2$ to all orders, and the half-integer orders are those of $(\theta_3 - \Theta_{\mathrm{hol},s})/2 - H_s$; $C_{-1} = \mathrm{Res}\,/2$, $C_0 = (1+E_0)/2$, $C_1 = E_1/2$.*

Status (Category A). (i): the case analysis of the weight and the characteristic correction are exact identities, checked to q^{400} at 27 values of s. (ii): the integer orders vanish by the parity argument (the order-j Euler–Maclaurin coefficients of the $\pm a$ cone sums differ by $(-1)^j$; each unary theta contributes only half-integer orders); the closed forms come from symbolic Euler–Maclaurin on the expansion of Proposition 13.1, with exact Bernoulli-sum validation for $s = 5..80, 99, 100, 151, 223, 300$. (iii): cancellation of the residue-class averages of $\mathbf{e}(xQ)$ under $\nu \mapsto -\nu$. (iv), (v): evaluation of the closed forms; the Δ-square list is an exact search. (vi): uses in addition Hecke's continuation theorem.

Category B. Supporting numerics: $H_{3/2}$ against small-t fits at 9 values of s (3–4 digits); $H_{5/2}$ only at $s = 6, 8$. The modular completion of $\frac{1}{2}q^{c_0}\vartheta$ is by citation (Zwegers [18], Cor. 2.9).

22.1 H_s at $x = \frac{1}{2}$, $x = \frac{1}{3}$ and higher orders at $x = 0$

Let c_e be the coefficient of $t^{e/2}$ in $H_s(\mathbf{e}(x)e^{-t})$.

Proposition 22.2 (Exact reductions)**.** *For every rational x: (i) the corner terms cancel between Q_+ and Q_-; (ii) every integer order vanishes; (iii) $c_e = \gamma_0 A_e(s) + \sum_r eb_r\, Y_{e,r}(s) + \sum_r ea_r\, X_{e,r}(s,g)$, with universal functions A_e, $Y_{e,r}$, $X_{e,r}$ and finite Gauss data γ_0, eb_r, ea_r given by distribution relations.*

Proposition 22.3 ($x = \frac{1}{2}$)**.** *(i) If s is odd, $\gamma_0 = eb = ea = 0$, so $H_s(-e^{-t}) = O(t^N)$ for every N. (ii) If s is even, $\gamma_0 = 0$, $eb_r = 4 \cdot 2^{1-r}(2 - 2^{1-r})B_r$ for even $r \geq 2$ (and 0 otherwise), and $ea_r = [16 \mid s]\, eb_r$. For $16 \nmid s$:*

$$\frac{H_s(-e^{-t})}{\sqrt{2\pi s}} \sim -\tfrac{1}{8}t^{1/2} + \tfrac{s-4}{64}t^{3/2} - \tfrac{31s^2-168s+192}{3072}t^{5/2} + \tfrac{(s-4)(173s^2-632s+544)}{24576}t^{7/2} - \tfrac{25261s^4-245680s^3+845440s^2-1226240s+634880}{3932160}t^{9/2} + \cdots .$$

For $16 \mid s$ there is an extra $\sqrt{\Delta}$ term at each order; for example $g^2/(8(s-2)(7s-32)^2)$ at $t^{1/2}$

Status (Category A). Exact symbolic computation on the proved expansion (Proposition 13.1). Consistency with an independent exact-term engine to $\leq 7 \cdot 10^{-38}$ through $t^{5/2}$ and to $\leq 5 \cdot 10^{-34}$ through $t^{9/2}$ (these comparisons are support only). Odd s: vanishing to all orders means there is no power-law term. The direct sums are not identically zero: outside the window $t \ll 1/(4m)^2$ the terms $e^{-c/(m^2 t)}$ appear (e.g. $H_7(-e^{-0.01}) \approx -7.95$, floating).

Proposition 22.4 ($x = \frac{1}{3}$)**.** $\gamma_0 = i/\sqrt{3}$, $\mathbf{e}(1/6)/3$, 0 *for $s \equiv 0, 1, 2 \bmod 3$; eb_r depends only on $s \bmod 3$; ea_r depends only on $(s \bmod 9, \varepsilon)$ with $\varepsilon = (-1)^{v_2(\gcd(s,8))}$ (Theorem 22.5). For $s \equiv 2 \bmod 3$ the (γ_0, eb) part of $c_{-1}/\sqrt{2\pi s}$ is $-(1 + i\sqrt{3})/(12s)$, and the higher orders through $t^{5/2}$ are explicit.*

Status (Category A). The dependence on $s \bmod 3$ is proved, and that of ea is Theorem 22.5. The closed form is exact computer algebra; consistency with an independent engine to $\leq 2.4 \cdot 10^{-35}$.

Theorem 22.5 (Period 72 at $x = \frac{1}{3}$)**.** *Let $\varepsilon = (-1)^{v_2(\gcd(s,8))}$. Then ea depends only on $(s \bmod 9, \varepsilon)$, so its period is 72. Explicitly:*

- *for $s \not\equiv 2 \bmod 3$: $ea = eb$ or $ea_r = (-1)^r eb_r$, according to $b \equiv 2$ or $1 \bmod 3$;*
- *for $s \equiv 2 \bmod 9$: $ea = 0$;*
- *for $s \equiv 5 \bmod 9$: $ea = eb$ ($\varepsilon = -1$) or $(-1)^r eb_r$ ($\varepsilon = +1$);*
- *for $s \equiv 8 \bmod 9$: $ea_r = 2^{2-r} i\sqrt{3} \sum_{c<3} \omega^{-\varepsilon c} \tilde{B}_r(c/3)$, which is V for $\varepsilon = -1$ and $(-1)^r V_r$ for $\varepsilon = +1$.*

Status (Category A). Elementary proof: reduction mod 3, a reflection lemma, and the mod-3 Gauss sum. Exact check for $s = 5, \ldots, 1000$ and $r \leq 10$. The split is by ε, not by the parity of s (e.g. $s = 28, 52$ have $ea = eb$).

Proposition 22.6 ($t^{7/2}$, $t^{9/2}$ at $x = 0$)**.** *$H_{9/2} = \sqrt{2\pi s}\,[R_9(s) + S_9(s,g)\sqrt{\Delta}]$, where R_9 is an explicit rational function with denominator $5449973760(s-2)^6$ and S_9 is of degree 10 in g; $H_{9/2}(5) = 0$. The $t^{7/2}$ closed form is likewise explicit.*

Status (Category A). Two independent symbolic routes agree to 170 digits at 8 values of s; exact computer algebra on the proved expansion.

Proposition 22.7 ($t^{11/2}$, $t^{13/2}$)**.** *At $x = 0$, $c_e = \sqrt{2\pi s}\,[R_e + S_e\sqrt{\Delta}]$ for $e = 11, 13$, with R_{11} of denominator $5667972710400(s-2)^7$ and R_{13} of denominator $136031345049600(s-2)^8$; both vanish at $s = 5$. At $x = \frac{1}{2}$ (even s):*

$$Rh_{11} = \frac{(s-4)(675691s^4 - 5253968s^3 + 14999488s^2 - 18628608s + 8491008)}{94371840},$$

and Rh_{13} is likewise explicit, with a $[16 \mid s]\sqrt{\Delta}$ part. At $x = \frac{1}{3}$ the coefficients are given by the universal form with the Gauss data of Theorem 22.5. The full expressions are in the data files listed in Appendix C.

Status (Category A). Exact symbolic coefficients at $K = 13$ (identical to the $K = 9$ ones through $t^{9/2}$). The $R + S\sqrt{\Delta}$ split is interpolated and re-verified at 10 held-out values of s. *Category B support:* agreement with the term-by-term engine (to $\le 3 \cdot 10^{-41}$ where the quadrature allows), and 60-digit direct sums (relative error 10^{-14}–10^{-6}).

Proposition 22.8 ($t^{15/2}$ and $t^{17/2}$ at $x = 0$)**.** *For $e = 15, 17$, $c_e = \sqrt{2\pi s}\,[R_e(s) + S_e(s, g)\sqrt{\Delta}]$: the $R + S\sqrt{\Delta}$ split persists. Explicitly*

$$R_{15} = -\frac{109395s^{16} + 101686679952s^{15} - 3187115618544s^{14} + \cdots - 3332235344216064}{148002103413964800\,(s-2)^9},$$

$$R_{17} = -\frac{(s-4)\,(4849845s^{17} - 139775079348190s^{16} + \cdots - 4580182201382993920)}{472422714097375641600\,(s-2)^{10}},$$

with all coefficients in the data files of Appendix C; S_e is a polynomial in g^2 over $(s-2)^{(e+3)/2}(7s-32)^{(e+3)/2}$. Both vanish at $s = 5$. The factor $(s-4)$ occurs in R_e for $e = 9, 13, 17$.

Status (Category A). Exact symbolic runs ($K = 15$, $K = 17$; the $K = 17$ run reuses the exact $e \le 15$ entries). The split is unique at each sample s and both interpolants pass exact checks at 10 held-out s. $K = 15$ reproduces every $K = 13$ coefficient exactly; the $e \le 13$ entries of $K = 17$ agree with the independent $K = 13$ run; $c_{15}(5) = c_{17}(5) = 0$.

Category B. Independent (floating-point) engine: c_{15} equal at double precision for $s = 6, 7, 9$; c_{17} equal for $s = 6$ and to $\le 9 \cdot 10^{-16}$ for $s = 7, 9$. 60-digit direct sums at $s = 20, 32, 8$: relative errors $2.6 \cdot 10^{-10}$, $4.5 \cdot 10^{-9}$, $8.9 \cdot 10^{-6}$ for c_{15} and $3.5 \cdot 10^{-10}$, $6.1 \cdot 10^{-9}$, $1.2 \cdot 10^{-5}$ for c_{17}.

Structure of S_e. Write $S_e(s-2)^k(7s-32)^k = \sum_j \sigma_{e,2j}(s)g^{2j}$, $k = (e+3)/2$.

Status (Category A). Exact factorisations for each of $e = 11, 13, 15, 17$: only even powers $g^0, \ldots, g^{e+1}$ occur; $\sigma_{e,e+1} = \text{const} \cdot (s-2)^{(e+1)/2}$; for $1 \le j \le (e+1)/2$, $\sigma_{e,2j} = \text{const} \cdot s^{(e+1)/2-j}(s-2)^j(s-8)^{\max(0,e+3-4j)}\varphi_{e,j}(s)$; the cofactors pair up ($\varphi_{e,j} = \varphi_{e,j'}$ up to constants when $2j + 2j' = e + 3$); and $\sigma_{e,0} = s^{(e+1)/2}\cdot$(an irreducible polynomial of degree $e+1$).

Category B. *Observation, not proved:* that this pattern holds for all odd e. Likewise an *observation*: the constant in the top coefficient $\sigma_{e,e+1}$ carries the numerator of the Bernoulli number B_{e+1} (691 for $e = 11$, 3617 for $e = 15$, 43867 for $e = 17$), consistent with the $r = e + 1$ Bernoulli term of the u_2-edge.

22.2 General $x = a/N$: the CRT Factorization Proposition

Proposition 22.9 (General $x = a/N$)**.** *(i) (Periodicity) γ_0, G, eb depend only on $s \bmod N'$, and H, ea only on $s \bmod 24N'$, where $N' = N$ (N odd) or $2N$ (N even).*

*(ii) (**CRT Factorization Proposition**, CFP) $H_{a/N}(c) = [2 \text{ if } N \text{ odd}]\,T_2 \prod_p T_{p^e}$, with local quadratic sums T. The Bernoulli-weighted ea is linear in H but does not factor.*

(iii) *(Prime $p \geq 5$, $p \nmid a\, s(s-2)(7s-32)$)* $H(c) = 2\big(\frac{aQ_2(u_2)}{p}\big) g_p\, \mathbf{e}\big(\frac{a}{p}[-g^2(c+\delta)^2/(2(7s-32)) + c_0]\big)$.

(iv) *(Degenerate $p \geq 5$)* *If $p \mid s$,* $H(c) = 2p\, \mathbf{e}(an_w^2 c^2/p)$. *If $p \mid s-2$,* $ea = 0$. *If $p \mid 7s-32$, H is a delta at one class c^*, and* $ea_r = 2^{2-r} p\, \mathbf{e}(aQ_+(c^* w)/p)\tilde{B}_r(c^*/p)$.

(v) *(Even N)* *The minimal periods of $s \mapsto H_{a/N}$ are* $16, 16, 144, 96$ *for* $N = 2, 4, 6, 8$. *At* $x = \frac{1}{4}$, $ea = 0$ *unless* $4 \mid s$; $ea = eb = (0, \frac{1}{3} - \frac{i}{6}, 0, -\frac{1}{240} + \frac{7i}{1920}, \dots)$ *for* $s \equiv 0 \bmod 16$; *and* $ea = (0, \frac{1+i}{12}, 0, -\frac{1+i}{3840}, \dots)$ *for* $s \equiv 4, 8, 12 \bmod 16$.

Lemma 22.10 (q-adic reduction)**.** *For any prime q, a quadratic sum $\sum_{y \bmod q^n} \mathbf{e}((\alpha y^2 + \beta y + \gamma)/q^n)$ reduces by $q^{v_q(\alpha)}$ to a unit-leading sum. The unit-leading sum is then evaluated by the odd-q Gauss sum, or by the explicit 2-adic formula. This gives a closed form for $H(c)$ for every N.*

Status (Category A). (i), (ii), (iii), (iv) and Lemma 22.10 have elementary proofs. They were also checked exactly in $\mathbb{Z}[\zeta_L]$ against brute force: about $7 \cdot 10^4$, $6.2 \cdot 10^5$ and $7 \cdot 10^5$ tests, with 0 mismatches. For (v), $144 = \mathrm{lcm}(16, 72)$ is explained by (ii). The periods 16 and 96 are proved conceptually in Propositions 22.12 and 22.13 (and also by exhaustive exact check across the proved bound $48N$). The coefficients at $x = \frac{1}{4}, \frac{1}{6}$ are Proposition 22.14.

Proposition 22.11 (Checks of the general formula)**.** *The closed form of $H(c)$ given by the CFP together with Lemma 22.10 agrees exactly with brute force for every $N \in \{2, \dots, 13, 15, 16, 21, 25, 27, 32, 35, 45, 49\}$, every a coprime to N, every $s < 200$ and every c. The runs at $N = 32, 35, 45, 49$ have* 199680, 327600, 421200, 802620 *tests and* 0 *mismatches.*

Status (Category A). Exact arithmetic in $\mathbb{Z}[\zeta_L]$, $L = \mathrm{lcm}(8, 2N)$.

Proposition 22.12 (Even N from the 2-adic reduction)**.** *(i)* $H_{1/2}(c) = 4(-1)^c[16 \mid s]$; *period* 16.

(ii) $H_{a/4}(c) = 0$ *if* $4 \nmid s$; $= 4(1 + i^a)$ *if* $s \equiv 4 \bmod 8$ *or* $s \equiv 8 \bmod 16$; $= 8\, \mathbf{e}(ac^2/4)$ *if* $16 \mid s$; *period* 16.

(iii) $N = 8$: *there are exactly six classes,* $4 \nmid s$ *(zero),* $s \equiv 0 \bmod 32$, $s \equiv 16 \bmod 32$, $s \equiv 8 \bmod 16$, *and* $s \equiv 4 \bmod 8$ *split by* $b \equiv 3$ *or* $1 \bmod 4$; *the period is* $96 = \mathrm{lcm}(32, 48)$, *the* 48 *coming from* $b \bmod 4$ *through* g_3^{-1}.

Status (Category A). (i), (ii): conceptual, from the CFP and Lemma 22.10 with the 2-adic valuations of $A2 = s\Delta/g^2$ class by class; (ii) also checked exactly in $\mathbb{Z}[\zeta_8]$ for $s < 2000$ (31920 tests). (iii): the invariants (and the factors 32 and 3) are derived by hand; pairwise distinctness is proved in Proposition 22.13.

Proposition 22.13 (The six $N = 8$ classes are pairwise distinct)**.** *With* $I_1 = \max_c |H(c)|$, $I_2 = \operatorname{supp} H$, $I_3 =$ *phase of* $H(0)$, $I_4 = H(1)/H(0)$ *(for $a = 1$): class $s \equiv 0 \bmod 32$ has* $(16, \textit{full})$; $s \equiv 16 \bmod 32$ *has* $(16, \textit{odd } c)$; $s \equiv 8 \bmod 16$ *has* $(8, \textit{full})$ *with* $I_3 = \mathbf{e}(1/8)$; *the two $s \equiv 4 \bmod 8$ classes have* $(8, \textit{full})$ *with* $H(0) = +8$ *real, and are separated by* $I_4 = +1$ *($b \equiv 3 \bmod 4$) versus* $I_4 = -1$ *($b \equiv 1 \bmod 4$), for all* $a \in \{1, 3, 5, 7\}$.

Status (Category A). Residue reduction ($H_{a/8}$ depends only on $s \bmod 16$ and on $u_2, w \bmod 16$), hand derivation of I_1–I_4 from Lemma 22.10, and an exact finite computation over all residue tuples satisfying each class's constraints; no scan over s.

ea **tables at $x = a/8$ (Category A).** For $r = 1, \dots, 4$ and $a = 1$: $s \equiv 0 \bmod 32$: $(0, 1 - \sqrt{2}(1+i)/12, 0, -1/64 + 7\sqrt{2}(1+i)/15360)$; $s \equiv 16 \bmod 32$: $(0, -\sqrt{2}(1+i)/12, 0, 7\sqrt{2}(1+i)/15360)$; $s \equiv 8 \bmod 16$: $(0, \sqrt{2}(1+i)/24, 0, -\sqrt{2}(1+i)/30720)$; $s \equiv 4 \bmod 8$: $(\mp 4, \frac{1}{2}, \pm\frac{3}{64}, -\frac{1}{128})$ for $b \equiv 3, 1 \bmod 4$ (independent of a); for $a = 3, 5, 7$ the factor $\sqrt{2}(1+i)$ rotates to $\sqrt{2}(-1+i)$, $\sqrt{2}(-1-i)$, $\sqrt{2}(1-i)$.

Proposition 22.14 ($t^{11/2}$, $t^{13/2}$ at $x = \frac{1}{4}, \frac{1}{6}$, and vanishing)**.** *For $e = 11, 13$ and $x \in \{\frac{1}{4}, \frac{1}{6}\}$, $c_e(s) = \gamma_0 A_e(s) + \sum_r eb_r Y_{e,r}(s) + \sum_r ea_r X_{e,r}(s, g)$ with the universal functions of the $P = 8$, $P = 12$, $K = 13$ runs and the exact class constants (for $x = \frac{1}{4}$, $4 \mid s$: $\gamma_0 = (1+i)/2$, $eb = (0, \frac{1}{3} - \frac{i}{6}, 0, \dots)$; for $x = \frac{1}{6}$: $\gamma_0 = 0$). Moreover:*

(i) *$x = \pm\frac{1}{4}$, $4 \nmid s$: $H_s(\pm i\, e^{-t}) = O(t^M)$ for every M;*

(ii) *$x = \frac{1}{6}$, s odd: $H_s(\mathbf{e}(1/6)e^{-t}) = O(t^M)$ for every M.*

Status (Category A). Universal form: exact symbolic runs with the exact class data. (i): a sign-reversing shift kills $G(j)$, so $\gamma_0 = eb = 0$, and $ea = 0$ by Proposition 22.12(ii). (ii): by the CFP (which applies equally to G and γ_0) the Gauss data contain the $x = \frac{1}{2}$ factor, which vanishes for odd s (Proposition 22.3). Exact zeros at $s = 5, 7, 9, \dots$ confirm both.

Category B. The independent (floating-point) engine agrees to double precision at $s = 8, 12, 16$ ($x = \frac{1}{4}$) and $s = 8, 16$ ($x = \frac{1}{6}$). 60-digit direct sums at $s = 8$: relative errors $2.0 \cdot 10^{-13}$, $1.7 \cdot 10^{-9}$ ($x = \frac{1}{4}$, also $\frac{3}{4}$) and $8.0 \cdot 10^{-10}$, $8.3 \cdot 10^{-7}$ ($x = \frac{1}{6}$) for c_{11}, c_{13}. Larger s were not run (they need $\gtrsim 10^7$ terms).

23 The cone: origin of Δ, the choice of walls, and H_s^{axes}

23.1 The lattice origin of $\Delta = (s-2)(7s-32)$

Let $A = \left(\begin{smallmatrix} 2 & s \\ s & s \end{smallmatrix}\right)$ be the Gram matrix of $Q_2(n, k) = n^2 + snk + \frac{s}{2}k^2$, so $\det A = -s(s-2)$, and let $c_1 = (-1, 1)$, $c_2 = (-4, 1)$ (the choice) and $e_1 = (1, 0)$.

Proposition 23.1 (Lattice origin of Δ)**.** *(i) $7s - 32 = -2Q(c_2)$. Indeed $c_2 = c_1 - 3e_1$, $B(c_1, e_1) = s - 2$, $Q(e_1) = 1$, and $-2Q(c_2) = (s-2) + 6(s-2) - 18 = 7s - 32$.*

(ii) $v := (3s, s-8) = JAc_2$ (J the rotation by $90°$) spans $c_2^{\perp}$, $B(c_2, x) = \det(x, v)$, and $u_2 = v/g$ with $g = \gcd(s-8, 3s) = \gcd(24, s-8)$.

(iii) $2Q(v) = \det A \cdot 2Q(c_2) = s(s-2)(7s-32)$, so $Q_2(u_2) = s\Delta/(2g^2)$.

(iv) For any w with $\det(w, u_2) = 1$, the residual coefficient in the Gauss phase of Proposition 22.9(iii) is $\det A/(4Q(u_2)) = -g^2/(2(7s-32)) = Q(\pi w)$, where π is the B-orthogonal projection onto $\mathbb{R}c_2$ along u_2.

(v) $\Delta = 4Q(c_1)Q(c_2) = B(c_1, c_2)^2 + \det \mathrm{Gram}(c_1, c_2) = 16(s-2)^2 - 9s(s-2)$, with $B(c_1, c_2) = -4(s-2)$ and $\det(c_1, c_2)^2 = 9$.

Status (Category A). All identities are symbolic in s. The degenerate prime cases of Proposition 22.9(iv) are read off: $p \mid 7s - 32$ means that c_2 is isotropic mod p (the residual sum collapses to a delta), and $p \mid s - 2$ is the c_1 analogue. No physical interpretation is offered.

23.2 The cone choice: what is intrinsic and what is tied to the fixed wall c_2

Write $c_a = (-a, 1)$ and $H_s^{(c_1,c)}$ for the cone series with Zwegers-type weight $w = \frac{1}{2}(\operatorname{sgn} B(c_1, v) - \operatorname{sgn} B(c, v))$, so that H_s is $H_s^{(c_1,c_4)}$. The c_1-wall is $n = 0$ (edge $u_1 = (0, 1)$) and the c_2-wall is the edge $u_2 \propto (3s, s - 8)$.

Proposition 23.2 (Edge-class invariance)**.** *$M(n, k) = (-n - sk, k)$ is an integral involution of determinant -1 with $Q_+(Mv) = Q_+(v)$ (including the linear term), and $Mc_a = c_{s-a}$. For admissible a, $H_s^{(c_1,c_a)} = H_s^{(c_1,c_{s-a})}$. Also $-2Q(c_a) = (2a - 1)s - 2a^2$ is invariant under $a \leftrightarrow s - a$, and $4Q(c_1)Q(c_a) = (s - 2)\big((2a - 1)s - 2a^2\big)$.*

Status (Category A). Proof: the difference is $\frac{1}{2}\sum f(v)q^{Q_+(v)}$ with $f = \operatorname{sgn} B(c_{s-a}, \cdot) - \operatorname{sgn} B(c_a, \cdot)$ supported in a sector between two positive lines (finitely many terms per degree); $f \circ M = -f$ and M is a Q_+-preserving bijection, so the sum vanishes. Symbolic identities are checked by computer algebra, with an exact q-series check at $s = 7, 8, 9, 11, 13$ to q^{800} or q^{600}.

Proposition 23.3 (c_1 is forced; the axes cone)**.** *(i) If one wall of the cone is the orthant edge $n = 0$, then $c_1 = \pm(-1, 1)$ (the sign being fixed by orientation), up to its M-image c_{s-1}, which gives the same series. $Q(c_1) = -(s - 2)/2 < 0$ for all $s > 2$.*

(ii) (Axes cone.) Let c_2^{ax} be the primitive vector on the ray of $(-s, 2)$, so that $B(c_2^{\mathrm{ax}}, v) \propto -k$ (the $k = 0$ wall); it is M-fixed, with $Q(-s, 2) = -s(s - 2) < 0$. Then $w = \frac{1}{2}(\operatorname{sgn} n + \operatorname{sgn} k)$ and $C_{\mathrm{mixed}} \equiv 0$ identically, i.e. $C = C_{\mathrm{axis}} + C_{\mathrm{third}}$.

(iii) For any admissible c_2, $C_{\mathrm{mixed}}^{(c_2)} = H_s^{(c_1,c_2^{\mathrm{ax}})} - H_s^{(c_1,c_2)}$, a sector sum with finitely many terms per degree. For $c_2 = (-4, 1)$ at $s = 7$ its first term is the wall point $(21, -1)$, $Q_+ = 294$, weight $-\frac{1}{2}$; this is Theorem 11.1(iv).

Status (Category A). (i): $Ac_1 \propto (1, 0)$. (ii): $P - H = C_{\mathrm{axis}} + C_{\mathrm{third}}$ by direct regrouping (the negative quadrant gives C_{third} since $Q_+(-n, -k) = Q_-(n, k)$); exact check at $s = 5, \ldots, 13$ to q^{600}. (iii): first nonzero degree of C_{mixed} is $27, 294, 375, 242, 189, 168$ for $s = 6, 7, 10, 11, 12, 13$; $\equiv 0$ at $s = 8$ (where $(-4, 1)$ *is* the axes vector); none through q^{600} at $s = 9$. At $s = 7$, any $c_1 \notin \{(-1, 1), (-6, 1)\}$ tested destroys the q^0 normalisation of C.

Proposition 23.4 (Status of $c_2 = (-4, 1)$)**.** *$c_2 = (-4, 1)$ is part of the* definition *of H_s (§10.4): it is fixed at $s = 7$ and kept for all s. Within the class $c \in c_1 + \mathbb{Z}e_1$ (second coordinate $b = 1$) at $s = 7$: $Q(c_a) < 0$ iff $a \in \{1, \ldots, 6\}$; the closed orthant lies strictly inside the $w = 1$ region iff $a \geq 4$; and the C_{mixed} vanishing order is maximal (first nonzero term q^{294}) exactly for the M-class $\{3, 4\}$. Together these give $a = 4$ uniquely among $b = 1$. Among all primitive $(-a, b)$ with $b \leq 6$ it is* not *unique, even up to M: the vanishing order grows as $a/b \to s/2$, and the optimum under the C_{mixed} criterion is the axes vector $(-7, 2)$. For $s \neq 7$ this choice is not forced (the deepest $b = 1$ class is $a \approx s/2$).*

Status (Category A). Exact computations, to q^{1200} at $s = 7$.

Category B. Reading the three conditions as the "selection rule" behind the original choice is an interpretation.

Remark 23.5 (What is intrinsic in Δ)**.** Combining Propositions 23.1–23.4 (all Category A):

- the factor $s-2=-2Q(c_1)$ of Δ is *intrinsic given* c_1, and c_1 is itself forced once one wall is the orthant edge $n=0$;
- the factor $7s-32=-2Q(c_2)$, and with it every $\sqrt{\Delta}$ term, every $(7s-32)^r$ denominator and every g-dependence in the H_s coefficients of §22, is *tied to the choice* $c_2=(-4,1)$; it is invariant only under $a\leftrightarrow s-a$ (Proposition 23.2), and other admissible c_2 give $\Delta(a)=(s-2)((2a-1)s-2a^2)$.

These statements concern H_s as defined; they do not change it.

23.3 The parallel object H_s^{axes}: exact closed forms through $t^{13/2}$

Definition 23.6. $H_s^{\mathrm{axes}}:=H_s^{(c_1,c_2^{\mathrm{ax}})}$, the cone series with $c_1=(-1,1)$ and $c_2^{\mathrm{ax}}=\mathrm{prim}(-s,2)$ (the $k=0$ wall). It is a separate object from H_s; by Proposition 23.3(ii), $H_s^{\mathrm{axes}}=P_s^{\mathrm{orth}}-C_{\mathrm{axis}}-C_{\mathrm{third}}$ exactly.

Theorem 23.7 (Closed forms for H_s^{axes})**.** *Let c_r be the coefficient of t^r in $H_s^{\mathrm{axes}}(e^{-t})$ ($x=0$), $s\geq 3$. All integer orders vanish, and for $r\in\{-\frac12,\frac12,\dots,\frac{13}{2}\}$,*

$$c_r=\sqrt{\pi}\left[R_r^{\mathrm{ax}}(s)+S_r^{\mathrm{ax}}(s)\sqrt{2s}\right],\qquad R_r^{\mathrm{ax}},S_r^{\mathrm{ax}}\in\mathbb{Q}(s),$$

with denominators $(s-2)^{r+3/2}$ times small integers, and no dependence on g. The first rows are

r	R_r^{ax}	S_r^{ax}
$-\frac12$	$\dfrac{1}{s-2}$	$-\dfrac{1}{2(s-2)}$
$\frac12$	$\dfrac{s^2(s-4)}{24(s-2)^2}$	$-\dfrac{(s-4)(3s-2)}{48(s-2)^2}$
$\frac32$	$-\dfrac{s^3(s-4)(s^2-4s+8)}{960(s-2)^3}$	$-\dfrac{15s^4-76s^3+108s^2+96s-128}{3840(s-2)^3}$
$\frac52$	$\dfrac{s^4(s-4)(5s^4-40s^3+126s^2-184s+128)}{80640(s-2)^4}$	$-\dfrac{(s-4)(105s^5-730s^4+2732s^3-5208s^2+5696s-2560)}{645120(s-2)^4}$

and the rows $r=\frac72,\dots,\frac{13}{2}$ are listed in the data files of Appendix C. In particular $c_{-1/2}=\sqrt{\pi}(2-\sqrt{2s})/(2(s-2))=-\sqrt{\pi}/(\sqrt{2s}+2)$, and R_r^{ax} carries the factors $(s-4)$ and $s^{r+3/2}$ for every $r\geq\frac12$.

Status (Category A). Proof: $H^{\mathrm{axes}}=P-C_{\mathrm{axis}}-C_{\mathrm{third}}$; P and C_{third} are expanded by the proved two-dimensional quadrant Euler–Maclaurin lemma (Proposition 13.1), C_{axis} has an exact unary expansion, and the area integrals are evaluated exactly (even moments cancel between P and C_{third}). Everything is symbolic in $\sqrt{s/2}$.

Category B. *Numerical support only.* 70-digit engine numerics with PSLQ at $s=7,\dots,13$ recognise the same field $\mathbb{Q}(\sqrt{2s})$ and the leading term; held-out direct summation at $s=20$ converges towards $c_{11/2}+c_{13/2}t$.

23.4 The universal piece R_e is wall-independent through $t^{13/2}$

For H_s write, as in Theorem 22.1(ii) and Propositions 22.6, 22.7, $H_r=\sqrt{2\pi s}\,[R_r(s)+S_r(s,g)\sqrt{\Delta}]$.

Proposition 23.8 (Wall-independence of the universal piece)**.** *For every $r\in\{-\frac12,\frac12,\dots,\frac{13}{2}\}$, $R_r(s)=S_r^{\mathrm{ax}}(s)$ identically in s. Equivalently, the $\sqrt{2\pi s}\cdot\mathbb{Q}(s)$ part of the radial expansion is the same for the cone of H_s and for the axes cone through $t^{13/2}$, while $R_r\neq R_r^{\mathrm{ax}}$.*

Status (Category A). Exact rational-function identities: $r \le \frac{5}{2}$ by comparison with Theorem 22.1(ii); $r = \frac{7}{2}, \dots, \frac{13}{2}$ by exact comparison (difference 0 at $e = 7, 9, 11, 13$). *Open:* the identity at general order. Comparison at $t^{15/2}$, $t^{17/2}$ needs the axes forms at $K \ge 15$, which were not computed.

Category B. *Interpretation:* the c_1 wall produces the universal piece and the c_2 wall the remainder ($S_r\sqrt{\Delta}$ for the cone of H_s, R_r^{ax} for the axes cone).

24 The structure of C_{third}

24.1 One coset, one cone weight, one kernel

Theorem 24.1 (C_{third}: one coset, one cone weight, one kernel, two chains)**.** *Let $F = [e_1, \gamma e_1)$, Q_1 the open first quadrant, $F' = F \setminus Q_1$, and $E_\mu^X = \sum_{\nu \in \mu + \mathbb{Z}^2,\, \nu \in X} q^{Q(\nu)}$. (i) For every $s \ge 5$, $2C_{\mathrm{third},s} - 1 + \theta_3 = q^{c_0}\Theta_w(a)$ with $\Theta_w(a) = \sum_{\nu \in a + \mathbb{Z}^2} w(\nu) q^{Q(\nu)}$ and $w = 1_{F'} + 1_{-Q_1}$: one coset, one quadratic form, one non-negative piecewise-constant cone weight. (ii)* Reflection lemma. *With $R_s = \begin{pmatrix} 1 & s \\ 0 & -1 \end{pmatrix}$ ($R_s^{\mathsf{T}} A_s R_s = A_s$, $R_s a = -a$, $R_s F$ again a γ-fundamental cone): $E_a^F = E_{-a}^F$ for every $s \ge 5$. Hence $\Theta_w(a) = E_L + \vartheta_{\mathrm{hol}}$: the part of w that is even under $\nu \mapsto -\nu$ gives E_L, the odd part gives Zwegers' ϑ_{hol}, and the would-be weight-1 piece on the unit cone is exactly 0. With $1_K = \frac{1}{4}(\rho_{\mathrm{odd}} + \rho_{\mathrm{even}})$, the parity of the cone weight, the radial parity at $x = 0$ and the automorphy type coincide. (iii)* One kernel, two chains. *On the path of lines $c(t)$ in the signature-$(1,1)$ plane let $\Xi_0 = \sqrt{y}\sum_{a+\mathbb{Z}^2} q^{Q(n)} e^{-\pi y B(n,c(t))^2}$ and $\Xi_1 = \sqrt{y}\sum_{a+\mathbb{Z}^2} 2B(n, c'(t)) q^{Q(n)} e^{-\pi y B(n,c(t))^2}$, and $\Xi = \Xi_0 + \Xi_1$. On the chain Γ_{Q_1}, $\int \Xi_1\, dt$ is exactly Zwegers' 2002 completion, with holomorphic part ϑ_{hol}. On the unit-cone chain $\Gamma_F = [c_1, \gamma c_1]$, $\int \Xi_0\, dt$ is Zwegers' 2012 mock Maass theta function Θ_a, which here has no shadow and has $E_L = \frac{1}{2}\vartheta_a$; and $\int_{\Gamma_F} \Xi_1\, dt = 0$. (iv) The two outputs come from different operations (holomorphic projection at weight 1; the Lewis–Zagier period correspondence at weight 0) and carry different automorphy factors ($(c\tau + d)$ versus $|c\tau + d|$). So this is a single construction, not a single modular object; Theorems 24.5 and 24.4 still apply.*

Status (Category A). (i) proved (elementary, from $2P_s^{\mathrm{orth}} = E_s + 1 + \theta_3 - \Theta_{\mathrm{hol}}$ and $Q_1 \subset F$); checked to q^{3000} at $s = 7$ and to q^{2000} for $s = 5..25, 41$. (ii) proved (symbolic checks of $\gamma a - a$, $R_s a + a$, $R_s^{\mathsf{T}} A_s R_s - A_s$, $R_s \gamma e_1 = \gamma^{-1} e_1$); identically zero to q^{3000} at $s = 7$. The parity/radial dictionary uses Proposition 13.1 and Hecke's theorem. (iii), the weight-1 identity: exact, by the fundamental theorem of calculus with Siegel-majorant domination. (iv): the obstruction of Theorems 24.5 and 24.4.

Category B. The weight-0 identification in (iii) is by citation of Zwegers 2012 [19] through the restatement in [2]; numerical agreement $8 \cdot 10^{-16}$ ($s = 7, 9$), $5 \cdot 10^{-15}$ ($s = 13$). Non-γ control: the shadow does not cancel (≈ 0.1). "No exotic single kernel exists" is open; higher depth is unavailable for the rank-2 lattice.

Proposition 24.2 (Unit cone)**.** *(i) On the unit cone, both the holomorphic weight-1 q-series and its Zwegers completion vanish identically. (ii) $\int_{\Gamma_F} \Xi_0\, dt = \sqrt{y}\sum_{\gamma\text{-orbits } O \subset a+\mathbb{Z}^2} \mathbf{e}(Q(O)x) K_0(2\pi |Q(O)| y)$, which is Zwegers' mock Maass theta Θ_a.*

Status (Category A). (i) is proved by the orbit-count symmetry $N(a, +, n) = N(-a, +, n)$, which comes from the automorph R_s, together with a telescoping argument for the completion; $\int_{\Gamma_F} \Xi_1 = 0$ follows. (ii) is proved by orbit unfolding. Modularity is by Siegel/Vignéras, which is standard. *Category B:* that no depth-2 rank-3 object carries C_{third} (a structural argument, not a classification).

Lemma Prod1. Finite sums of products of weight-0 holomorphic forms with logarithmic cocycle and holomorphic weight-1 forms have only $t^{-2}, t^{-1}\log t, t^{-1}$ radial terms, so they cannot

carry either sector. (Category A.) **Lemma Prod2.** The same holds for products involving the Zwegers pieces; it is conditional on the same inputs as Theorem 24.5. (Category B.)

24.2 Modularity status of C_{third}

The statements of this subsection are proved or exactly checked, conditional where stated; the words "quantum modular form" for the factor $|cx+d|$ are terminology (Category B).
Identity. $2C_{\text{third}} = q^8 f_{2,7,7}(-q^{10}, -q^{14}, q) + 1 + E$, checked to q^{8000}, and

$$C_{\text{third}} = \tfrac{1}{2}(1-\theta_3) + \tfrac{1}{2}\,\mathbf{e}(7\tau/20)\big[\vartheta_{\text{hol}} + E_L\big].$$

Lemma 24.3 (Parity splitting)**.** *At every rational x, along $q = \mathbf{e}(x)e^{-t}$, $t \to 0^+$: $\mathbf{e}(7\tau/20)\vartheta_{\text{hol}}$, θ_3 and $q^8 f$ contain only half-integer powers of t; $E = q^{7/20}E_L$ contains only integer powers $t^{-1}, t^0, t^1, \dots$. Hence the integer-order radial data of C_{third} are those of $\frac{1}{2}(1+E)$ and the half-integer ones those of $\frac{1}{2}q^8 f$.*

Status (Category A). Proved, conditional on two-dimensional Euler–Maclaurin (Proposition 13.1) and Hecke's continuation theorem.

Theorem 24.4 (Phase obstruction)**.** *Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ have finite index, ρ any finite-dimensional representation, $j(\gamma,\tau) = (c\tau+d)^k$ or $|c\tau+d|^k$, and V holomorphic with $V(\gamma\tau) = j(\gamma,\tau)\rho(\gamma)V(\tau) + o(1/y)$ at every rational $x \neq \gamma^{-1}\infty$. Then $C_{\text{third}} \neq \lambda \cdot V$ for every λ; likewise for C, P_s^{orth} and E.*

Status (Category A). Unconditional. The proof uses $G(1/m) = \mathbf{e}(7/(20m))P(m \bmod 140)/m$ for $\gcd(m,70) = 1$ (which also follows from Corollary 14.2(iii)) and a linear-recurrence/essential-singularity argument.

Theorem 24.5 (Sign obstruction)**.** *If V is holomorphic and finite-dimensional on $\Gamma \ni S, T^N$, $\rho'(T^N)$ is diagonalisable with unimodular eigenvalues, and $V(\gamma\tau) = (c\tau+d)^k\rho'(\gamma)V(\tau) + o(1/y)$, then none of E_L, $X = \frac{1}{2}(\vartheta_{\text{hol}} + E_L)$ or the 72-component E-vector is $\lambda \cdot V$. Hence E is not the holomorphic part of a weight-1 harmonic Maass form, nor a holomorphic weight-1 form, nor mixed mock with modular factors of weight < 1.*

Status (Category A). Conditional on the cited analyticity of harmonic-Maass cocycles, which is needed only for the class-membership corollaries; the $|x|$ profile law is Theorem 14.1. Objects whose cocycles are themselves of size $\geq 1/y$ are not excluded (open).

Theorem 24.6 (Structure of C_{third})**.** *(i) The decomposition above, with ϑ_{hol} a signature-$(1,1)$ Zwegers holomorphic part and E_L the $(a,0)$ component of the 72-dimensional vector. (ii) E_L is the Lewis–Zagier f-function of the $\lambda = \frac{1}{4}$ Maass Eisenstein lift; its $|cx+d|$ weight-1 cocycle is real-analytic off $\gamma^{-1}\infty$. (iii) Lemma 24.3. (iv) For $x = h/m$, the renormalised value is $c_C(x) = \frac{1}{2} + \frac{1}{2}\mathbf{e}(7x/20)Z_x(0) - \frac{7}{20}I\,G(x) = \alpha(x) + \beta(x)I$ with α algebraic and $\beta = -\frac{7}{20}G(x)$, which vanishes iff $2 \mid m$ or $5 \mid m$ (Theorem 11.4); the Weil-kernel invariance $G_{a,0}(\gamma x)|cx+1| = G_{a,0}(x)$, $\gamma = \left(\begin{smallmatrix} 1 & 0 \\ 140 & 1 \end{smallmatrix}\right)$, is Corollary 14.2(ii). For example $c_C(0) = \frac{11}{20} - \frac{7}{20}I$, $c_C(1/2) = 0$, $c_C(1/4) = 2 + \frac{3}{2}i$. The cusp-form $L(0)$ values do not enter.*

Status (Category A). For (i), (iii) and the formula in (iv); the general genus formula behind $Z_x(0)$ is Category B.

Verdicts. C_{third} is not modular (Theorem 11.5(i) (Baker), Theorem 11.5(ii) and Theorem 24.4); not mock or mixed mock with a holomorphic factor and an $o(1/y)$ cocycle (Theorems 24.5 and 24.4); has no strict quantum-modular values (radial divergence at all 490 rationals with $m \leq 40$; "every rational" is Category B); and is renormalised quantum modular after removing $q^{7/20}$ and the theta pieces.

25 Dependence on s: the ζ-curve, g_s, and the cases $s = 8$, $s = 11$

25.1 $\zeta_E(0;s)$ and $\zeta_E(-1;s)$ on one rational curve

Theorem 25.1 (Rational ζ-curve). *For every integer $s \geq 3$, with $c_0 = s/(4(s-2))$ and $\zeta_E(w;s) = \sum_m e_m(m-c_0)^{-w}$:*

$$\zeta_E(0;s) = \frac{(s-4)(s-5)}{12(s-2)}, \qquad \zeta_E(-1;s) = \frac{2s^5 - 14s^4 - 45s^3 + 425s^2 - 632s - 96}{1440(s-2)^2},$$

the t^{-1} coefficient of $E_s(e^{-t})e^{c_0 t}$ is $\operatorname{arccosh}(s-1)/\sqrt{s(s-2)}$, and the $t^{-1/2}$ and $t^{1/2}$ coefficients vanish.

Status (Category A). All algebraic steps are symbolic in s: in the γ-adapted basis $Q_2 = S^2 + 2(s-1)ST + T^2$; E_s is the sum over one half-open cone with exactly two lattice classes $(\frac{1}{2}+\delta, 1-\delta)$, $(\delta, \frac{1}{2}-\delta)$, $\delta = 1/(2(s-2))$, via the involution $\varphi(n,k) = (n, -2n-k-1)$ (Remark 25.3(a)). Analytic inputs: Proposition 13.1 and the Mellin dictionary (cf. Zagier, Shintani). Cross-checked against 42 exact cone values ($s = 3, \ldots, 20, 25, 30, 41$); the key steps (shift identity, γ-invariance, the involution, checked by brute force for $s = 3..30$, the per-class t^0 coefficient, both closed forms and the vanishing of the $t^{\pm 1/2}$ sums) were re-derived with independent code.

Normalisation note. The ray-class value $\zeta(-1, \mathcal{K}) = 29/60$ and $\zeta_E(-1;7) = 29/1200$ differ only by normalisation: since $\mathrm{N}\mathfrak{a} = 20Q$, $\sum_{\mathfrak{a}\in\mathcal{K}} \mathrm{N}\mathfrak{a}^{-w} = 20^{-w}\zeta_E(w;7)$, and at $w = -1$ this gives $29/60 = 20 \cdot 29/1200$. Likewise $5/12 = 3 \cdot 5/36$ ($\mathrm{N}\mathfrak{a} = 3Q$, $s = 8$) and $1129/36 = 36 \cdot 1129/1296$ ($\mathrm{N}\mathfrak{a} = 36Q$, $s = 11$).

25.2 Radial expansion of g_s

Let $g_s(q) = \sum_{n\geq 0} q^{n^2}/(-q^s;q^s)_n$.

Theorem 25.2 (Radial expansion of g_s). *For every real $s > 0$, $g_s(e^{-t}) \sim \sum_{k\geq 0} c_k(s)t^k$ as $t \to 0^+$, with $c_k \in \mathbb{Q}[s]$ of exact degree k and no t^{-1} or half-integer terms. In particular $c_0 = 2$, $c_1 = 2(s-3)$, $c_2 = 5s^2 - 44s + 75$, $c_3 = (55s^3 - 930s^2 + 3918s - 4683)/3$, $c_4 = (1073s^4 - 28864s^3 + 214620s^2 - 593128s + 545835)/12$. Consequently $\zeta_g(0) = 1$, $\zeta_g(-1) = 6 - 2s$ and $\zeta_g(-k) = (-1)^k k!\, c_k(s)$.*

Status (Category A). Elementary proof: write $1/(-q^s;q^s)_n = 2^{-n}\exp\sum_{j\leq n}\Lambda(sjt)$, expand with Faulhaber polynomials, sum against 2^{-n}, and bound the remainder by splitting at $N = t^{-1/3}$. Confirmed on held-out s; $c_0, \ldots, c_4$ re-derived with independent code (exact match), and $g_s(e^{-t}) - \sum_{k\leq 4} c_k t^k = O(t^5)$ checked numerically at $s = 3, 7.5$. The Dirichlet-series reading of ζ_g is Category B.

P_s^{orth} and Θ_{hol} for all $s \geq 3$ (Category A, by Proposition 13.1), with Res $= \operatorname{arccosh}(s-1)/\sqrt{s(s-2)}$: P_s^{orth} has t^{-1} coefficient Res$/2$ and $t^{-1/2}$ coefficient $\sqrt{\pi}(s^2 - 2\sqrt{2s})/(4s(s-2))$; Θ_{hol} has $t^{-1/2}$ coefficient $\sqrt{\pi}(s^2/2 - 2s + \sqrt{2s})/(s(s-2))$ and no integer-order terms through t^1. Thus Lemma 24.3 holds at $x = 0$ as an identity in s. g_s is not in $\operatorname{span}_{\mathbb{Q}}\{1, \theta_3, R, M_m\}$ (exact rank to q^{400}, $s = 3..20, 25, 30, 41$). H_s is Zwegers-admissible only for integer $s \geq 5$. The higher radial coefficients of H_s are given in §22.

25.3 The cases $s = 8$ and $s = 11$

- $2P_s^{\mathrm{orth}} = E_s + 1 + \theta_3 - \Theta_{\mathrm{hol},s}$ holds exactly as a q-series to q^{3000} for $s = 7, 8, 11$ (Category A).

- Fields and orders: $s = 8$: $\mathbb{Q}(\sqrt{3})$, order $\mathbb{Z}[2\sqrt{3}]$. The maximal order has fundamental unit $2 + \sqrt{3} \notin \mathbb{Z}[2\sqrt{3}]$; the fundamental unit of the order is $7 + 4\sqrt{3} = (2 + \sqrt{3})^2$, which is the γ-eigenvalue. Multiplicity one (no double counting). $s = 11$: $\mathbb{Q}(\sqrt{11})$, order $\mathbb{Z}[3\sqrt{11}]$, $\varepsilon = 10 + 3\sqrt{11}$. E_s is a single-class ray-class series for each s.

- $\zeta_E(0)$, $\zeta_E(-1)$: $1/10, 29/1200$ ($s = 7$); $1/6, 5/36$ ($s = 8$); $7/18, 1129/1296$ ($s = 11$), by Shintani cones (exact) and by PARI Hecke L-functions. The $t^{\pm 1/2}$ coefficients of E_s vanish; the Baker obstruction carries over.

- S/T family: S unitary, $S^2 = P$, $(ST)^3 = I$ for all three s (exact). $s = 8$: even lattice, whole family Eisenstein, $\psi = \psi_{\mathrm{Eis}}$, Herglotz form to $3 \cdot 10^{-13}$ (Category B). $s = 11$: u_L has a genuine dihedral cusp component (6 Maass forms with $\lambda = \frac{1}{4}$ from 12 non-genus characters of conductor (3) and (9); exact ray-class test, Proposition 21.1); $\psi = \psi_{\mathrm{Eis}} + \psi_{\mathrm{cusp}}$ with $\psi_{\mathrm{cusp}} \neq 0$ at 8 rationals (floating, Category B). Exactly, $\dim V = 110$, $d_{\mathrm{Eis}} = 90$ and $d_{\mathrm{cusp}} = 20$ (Theorem 20.1). The count "≥ 90 dihedral newforms" from conductor norms is not a dimension: as at $s = 7$, the forms occur only in fixed combinations.

- Genus: at $s = 11$ there are 4 genera ($2^{\mu-1}$, $\mu = 3$) and $\mathrm{Cl}^+ \cong \mathbb{Z}/4 \times \mathbb{Z}/2$, so $h^+ = 8 >$ #genera.

Class numbers versus genera (Category A for the counts). With D_s the discriminant of the order, h^+ equals the number of genera for every $s \in \{3, \ldots, 20, 30\}$ except $s = 11$ (8 vs 4) and $s = 19$ (8 vs 4); also $s = 25$ (12 vs 4) and $s = 41$ (24 vs 8).

The rule "cusp part $\Leftrightarrow$ $h^+ \neq$ #genera" is false at both levels. For the family span, $d_{\mathrm{cusp}} > 0$ for every odd s in 5..51, including $s = 5, 7, 13, 15$ where $h^+ =$ #genera (Theorem 20.1). For u_L it fails at $s = 14, 17, 18, 22, 23$ (Proposition 21.1). Both notions are invisible in the radial data at $x = 0$, which are s-uniform (§§25.1–25.2).

25.4 Structural remarks

Remark 25.3.

(a) One improper automorph (Category A).
The involution $\varphi(n, k) = (n, -2n - k - 1)$ used in the proof of Theorem 25.1 equals $-\gamma R_s$, and $R_s a = -a$. So φ, the reflection of Theorem 24.1(ii) and $H_5 \equiv 0$ come from a single symmetry. $R_s c_1 = -c_2$ holds only at $s = 5$, which is why $H_5 \equiv 0$ and $Q(c_1) = Q(c_2)$ there. Adding R_s to $\langle \gamma, -I \rangle$ gives no new label relations for any $s = 5, \ldots, 25$ (exact orbit counts).

(b) The factor 20 (identity Category A; its meaning Category B).
The denominator of c_0 is $4(s-2)/\gcd(s, 8) = 20, 3, 36$ at $s = 7, 8, 11$, which is the ideal-norm factor c_s in $\mathrm{N}\mathfrak{a} = c_s Q$. For odd s it equals $-B(c_1, c_2) = -8Q(c_1) = 4(s-2)$ of the H_s cone (at $s = 8$ the two differ: 3 versus 24); equivalently $Q_s = (\mathrm{N}\mathfrak{a} + s)/(4(s-2))$ for odd s.

(c) Residue and Pell unit (Category A).
$\mathrm{Res} = \operatorname{arccosh}(s-1)/\sqrt{n_s} = \log \varepsilon_s / \sqrt{n_s}$ with $\varepsilon_s = (s-1) + \sqrt{n_s}$, the fundamental solution of $x^2 - n_s y^2 = 1$. For even s this is the fundamental unit of the order, not of the maximal order: at $s = 8$, $\varepsilon_8 = 7 + 4\sqrt{3}$.

(d) Δ and the $s = 8$ scale (Category A).
$\Delta = (s-2)(7s-32) = 4Q(c_1)Q(c_2) = 16(s-2)^2 - 9n_s$; $B(c_1, c_2)^2 - 4Q(c_1)Q(c_2) = 9n_s$; $\det(c_1, c_2) = 3$; $\gcd(7s-32, s-8) = g$ and $\gcd(7s-32, s-2) = \gcd(18, s-2)$. $\mathbb{Q}(\sqrt{\Delta}) \neq \mathbb{Q}(\sqrt{n_s})$ for every $s < 10^6$ (exact search), so the H_s half-integer data live in a different quadratic field

from the E-residue except where Δ is a square. At $s = 8$ all the listed half-integer coefficients lie in $\sqrt{\pi}\,\mathbb{Q}$ (Theorem 22.1(v)).

(e) Three parities are one (algebra Category A; same-ρ identification Category B). The label parity $P = S_{\mathrm{lab}}^2$, the cone-weight parity $\nu \mapsto -\nu$ and the integer versus half-integer radial orders coincide. A weight-k vector-valued form satisfies $\rho(S)^2 f = (-1)^k f$, so the E/Eisenstein sector lies in W_P and the Zwegers/ϑ_{hol} sector in $\mathrm{Fix}(-P)$ (dimension $68 = 4(3 \cdot 3 + 2 \cdot 4)$ at $s = 7$). Hence weight-1 vectors cannot realise any part of $V^{\perp} \subset W_P$. Category B: that Zwegers' completed theta transforms under this same 140-dimensional ρ with the $S_{\mathrm{lab}} = \rho(S)^{-1}$ sign.

(f) The constants a_H, a_C are closed forms (Category A). $a_H = H_{-1/2}(7)$ and $a_C = C_{-1/2}(7)$ symbolically; the $t^{-1/2}$ coefficient of C_{third} at $x = 0$, $-\sqrt{\pi}(49 - 2\sqrt{14})/140$, is consistent with the all-s formulas of Appendix 25.2.

26 K3 lattice realisation of the 140-label representation

This section uses only the label representation of Theorem 11.6 at $s = 7$; nothing in it concerns H_s. *Category A (cited)* means proved conditional only on the published theorems named, which we use as stated in the cited sources and do not re-prove.

Notation. Throughout, $A = \left(\begin{smallmatrix} 2 & 7 \\ 7 & 7 \end{smallmatrix}\right)$ is the Gram matrix of Q_7 (§23.1), $B(u, v) = u^{\mathsf{T}} A v$, and a lattice is written with its *norm* $x \cdot x$ (so $Q_2 = \frac{1}{2} x \cdot x$). We write N_1, N_2 (italic) for two even lattices.

26.1 The even lattice N_2 and its discriminant form

The form $(\mathbb{Z}^2, A)$ is odd ($Q_7 \notin \mathbb{Z}$), so it is not itself a K3 lattice. Its even sublattice of index 2 is

$$N_2 := \{x \in \mathbb{Z}^2 : \ x_2 \equiv 0 \bmod 2\}, \qquad \mathrm{Gram}\big((1,0),(0,2)\big) = \begin{pmatrix} 2 & 14 \\ 14 & 28 \end{pmatrix}, \qquad \det = -140.$$

The other natural even version is $N_1 := (\mathbb{Z}^2, 2A)$, with Gram matrix $\left(\begin{smallmatrix} 4 & 14 \\ 14 & 14 \end{smallmatrix}\right)$, also of determinant -140.

Proposition 26.1 (Discriminant form of N_2). *(i) In the basis $(1,0), (-7,2)$ of N_2 (base change $P = \left(\begin{smallmatrix} 1 & -7 \\ 0 & 1 \end{smallmatrix}\right)$ from $(1,0),(0,2)$), $N_2 \cong \langle 2 \rangle \oplus \langle -70 \rangle$, i.e. $x \cdot x = 2(x_1 + 7y)^2 - 70y^2$. In particular N_2 is even, of signature $(1,1)$.*

(ii) $A_{N_2} = N_2'/N_2 \cong \mathbb{Z}/2 \times \mathbb{Z}/70$, $|A_{N_2}| = 140$, generated by $(\frac{1}{2}, 0)$ and $(0, \frac{1}{70})$ in the $\langle 2 \rangle \oplus \langle -70 \rangle$ basis, with $q_{N_2}(x, y) = \frac{x^2}{4} - \frac{y^2}{140} \bmod 1$.

(iii) The local Jordan components and their Gauss-sum signatures are:

p	*group*	q *on the p-part*	*signature mod* 8
2	$(\mathbb{Z}/2)^2$, *generators* $(\frac{1}{2}, 0), (0, \frac{1}{2})$	$\langle \frac{1}{4} \rangle \oplus \langle \frac{1}{4} \rangle$, $q = (x^2 + y^2)/4$	2
5	$\mathbb{Z}/5$	$\langle \frac{3}{5} \rangle \cong \langle \frac{2}{5} \rangle$	4
7	$\mathbb{Z}/7$	$\langle \frac{2}{7} \rangle \cong \langle \frac{1}{7} \rangle$	2

The local signatures sum to $2 + 4 + 2 \equiv 0 = \mathrm{sign}(N_2) \bmod 8$ (oddity formula). The 5- and 7-parts are exactly the local forms $q_5 = 2x^2/5$ and $q_7 = x^2/7$ of Theorem 19.3.

Proof. (i) $(1,0)\cdot(1,0)=2$, $(-7,2)\cdot(-7,2)=98-196+28=-70$, $(1,0)\cdot(-7,2)=-14+14=0$, and $\det P=1$. (ii) is read off from (i): $\langle 2\rangle'=\frac{1}{2}\mathbb{Z}$ with $q=\frac{1}{2}\cdot 2\cdot\frac{1}{4}$, $\langle -70\rangle'=\frac{1}{70}\mathbb{Z}$ with $q=-\frac{1}{140}$. (iii) The 2-part is generated by $(\frac{1}{2},0)$ and $(0,\frac{35}{70})$, with $q(0,\frac{1}{2})=-35^2/140\equiv\frac{1}{4}$; the 5-part by $(0,\frac{1}{5})$, $q=-14^2/140\equiv\frac{3}{5}$; the 7-part by $(0,\frac{1}{7})$, $q=-10^2/140\equiv\frac{2}{7}$. The isomorphisms hold because $3\cdot 2^{-1}\equiv 4$ is a square mod 5 and $2\equiv 3^2$ mod 7. The signatures are the standard Gauss-sum values. □

Status (Category A). Elementary; exact.

26.2 The 140-label representation is the Weil representation of N_2

Conventions. For an even lattice L we use Borcherds' convention [3]: $\rho_L(T)\mathbf{e}_\gamma=\mathbf{e}(q(\gamma))\mathbf{e}_\gamma$ and $\rho_L(S)\mathbf{e}_\gamma=\mathbf{e}(-\mathrm{sign}(L)/8)\,|A_L|^{-1/2}\sum_\delta\mathbf{e}(-b(\gamma,\delta))\mathbf{e}_\delta$; the dual ρ_L^* is the complex conjugate. Since $\mathrm{sign}(N_2)=0$, ρ_{N_2} is an honest representation of $\mathrm{SL}_2(\mathbb{Z})$ and $\mathbf{e}(\mathrm{sign}/8)=1$. On the label side, the labels are (α,β) with $\alpha\in(L'\cup(a+L'))/\mathbb{Z}^2$ (70 classes), $\beta\in\{0,\beta_1\}$, $a=(\frac{7}{10},-\frac{1}{5})$ and β_1 the stored label of $-a$ mod $\mathbb{Z}^2$; the operators S_{lab}, T_{lab} are built in the convention of Remark 19.2 (with exact rational phases, all denominators dividing 140). Following Remark 19.1, the label representation in standard normalisation is

$$\rho_{\mathrm{lab}}:\quad S\mapsto S_{\mathrm{lab}}^{-1},\qquad T\mapsto T_{\mathrm{lab}}.$$

Theorem 26.2 (Weil realisation of the label representation)**.** *(i) $\rho_{\mathrm{lab}}\cong\rho_{N_2}$ as representations of $\mathrm{SL}_2(\mathbb{Z})$. No character twist is needed and no metaplectic sign arises.*

(ii) An explicit intertwiner is the following. Index the basis of ρ_{N_2} by $z\in N_2'/N_2$, written in $\mathbb{Z}^2$-coordinates, and let $[z]=z$ mod $\mathbb{Z}^2$. Every class α has exactly two lifts z, $z+(0,1)$ in N_2'/N_2. Put

$$X\,\mathbf{e}_z\;=\;u_{([z],0)}\;+\;\mathbf{e}\big(B(z,\beta_1)\big)\,u_{([z],\beta_1)}.$$

Then $X\rho_{N_2}(S)=S_{\mathrm{lab}}^{-1}X$, $X\rho_{N_2}(T)=T_{\mathrm{lab}}X$, and X is invertible.

(iii) $\rho_{\mathrm{lab}}\ncong\rho_{N_2}^*$, $\rho_{\mathrm{lab}}\ncong\rho_{N_1}$ and $\rho_{\mathrm{lab}}\ncong\rho_{N_1}^*$.

Proof. (ii) All entries of $\sqrt{140}\,X\rho_{N_2}(S)$, $\sqrt{140}\,S_{\mathrm{lab}}^{-1}X$ (note $\sqrt{140}=2\sqrt{35}$ and $\sqrt{35}\,S_{\mathrm{lab}}^{-1}$ has entries $\mathbf{e}(\cdot)$), $X\rho_{N_2}(T)$ and $T_{\mathrm{lab}}X$ lie in $\mathbb{Z}[\zeta_{280}]$. Each of the $2\cdot 140^2$ entry identities is the vanishing of an explicit integer combination of powers of ζ_{280}; it was checked by exact reduction modulo the cyclotomic polynomial Φ_{280}, with 0 failures for S and 0 for T. X is block diagonal with respect to α; the block of the two lifts $z,z'=z+(0,1)$ is $\left(\begin{smallmatrix}1 & \\ & \mathbf{e}(B(z,\beta_1))\end{smallmatrix}\ \begin{smallmatrix}1 & \\ & \mathbf{e}(B(z',\beta_1))\end{smallmatrix}\right)$, with determinant $\mathbf{e}(B(z,\beta_1))\big(\mathbf{e}(B((0,1),\beta_1))-1\big)\neq 0$ because $B((0,1),\beta_1)=7(-\frac{7}{10})+7\cdot\frac{1}{5}\equiv\frac{1}{2}$ mod 1. (i) follows from (ii); X intertwines the untwisted ρ_{N_2}, and $\mathrm{sign}(N_2)=0$. (iii) T_{lab} is monomial; its label permutation has 70 fixed points and 35 two-cycles, and its eigenvalue multiset (36 distinct values) equals that of $\rho_{N_2}(T)$ but differs (exactly, as multisets of elements of $\mathbb{Q}/\mathbb{Z}$) from those of $\rho_{N_2}^*(T)$, $\rho_{N_1}(T)$ and $\rho_{N_1}^*(T)$. Conjugate representations have conjugate T-spectra. □

Status (Category A). Exact computation in $\mathbb{Z}[\zeta_{280}]$ (0 failures among the $2\cdot 140^2$ entry identities; $B(v,\beta_1)\equiv\frac{1}{2}$; X invertible) and exact T-spectrum comparison. A second exact solution, with offsets $\{0,\frac{4}{5}\}$, is another element of the Hom space of Proposition 26.3. Further T-spectrum tests: $2\langle 5,-7\rangle$, $2\langle -5,7\rangle$, $2\left(\begin{smallmatrix}3 & 1\\ 1 & -12\end{smallmatrix}\right)$ and its negative do not match; $2\langle 1,-35\rangle$ and $2\left(\begin{smallmatrix}1 & 1\\ 1 & -34\end{smallmatrix}\right)$ match only because both are isometric to N_2, and their negatives match the dual.

Proposition 26.3 (Relations and Hom spaces modulo ℓ)**.** *Let $\ell = 840841 \equiv 1 \bmod 280$. Computing modulo ℓ (with a fixed primitive 280-th root of unity in $\mathbb{F}_\ell$):*

(i) $(S_{\mathrm{lab}}^{-1}, T_{\mathrm{lab}})$ *satisfies* $S^4 = I$ *and* $(ST)^3 = S^2$*, as* ρ_{N_2} *does;* $(S_{\mathrm{lab}}, T_{\mathrm{lab}})$ *satisfies* $S^4 = I$ *but* fails $(ST)^3 = S^2$*, so it is not a representation of* $\mathrm{SL}_2(\mathbb{Z})$ *in this normalisation (confirming Remark 19.1).*

(ii) $\dim_{\mathbb{F}_\ell} \mathrm{Hom}(\rho_{N_2}, (S_{\mathrm{lab}}^{-1}, T_{\mathrm{lab}})) = 8$*, and a random element has rank* 140*.*

(iii) $\dim_{\mathbb{F}_\ell} \mathrm{Hom}(\rho_{N_2}, (S_{\mathrm{lab}}, T_{\mathrm{lab}})) = 4$*, and a random element has rank* 72*: the two normalisations agree only on the* 72*-dimensional subspace* W_P*.*

Status (Category A). Exact computation modulo ℓ. The isomorphism itself is Theorem 26.2. By semicontinuity the characteristic-0 Hom dimensions are at most 8 and 4. *Category B (reading):* the value 8 is consistent with eight pairwise non-isomorphic irreducible sectors $W_5^{\pm} \otimes W_7^{\pm} \otimes \{A_3, B_1\}$ (§19); that the characteristic-0 dimension equals 8 is not separately proved here.

Remark 26.4 (Dual, twist, metaplectic sign)**.** $\rho_{\mathrm{lab}} \not\cong \rho_{N_2}^*$ (Theorem 26.2(iii)); no twist by a character of $\mathrm{SL}_2(\mathbb{Z})$ is needed; $\mathrm{sign}(N_2) = 0$, so there is no metaplectic sign. The Weil representation depends only on the discriminant form, so any even lattice with discriminant form $\cong q_{N_2}$ and signature $\equiv 0 \bmod 8$ (e.g. $N_2 \oplus U$, $N_2 \oplus E_8$, and the lattice T of Proposition 26.5, up to dualisation) gives the same representation.

26.3 The transcendental lattice

Let $\Lambda_{K3} = U^3 \oplus E_8(-1)^2$ (even unimodular, signature $(3, 19)$), and let $U \oplus U \subset \Lambda_{K3}$ have bases e_1, f_1, e_2, f_2 with $e_i \cdot f_i = 1$ and all other products 0.

Proposition 26.5 (Explicit primitive embedding and T)**.** *Put $n_1 = e_1 + f_1$ (norm 2), $n_2 = e_2 - 35 f_2$ (norm -70), $t_1 = e_1 - f_1$ (norm -2), $t_2 = e_2 + 35 f_2$ (norm 70).*

(i) $\mathrm{span}(n_1, n_2) \cong N_2$ *and is primitive in* $U \oplus U$ *(the* 2×2 *minors of* $(n_1\ n_2)$ *have gcd* 1*).*

(ii) t_1, t_2 *are orthogonal to* n_1, n_2 *and span the full orthogonal complement of* $\mathrm{span}(n_1, n_2)$ *in* $U \oplus U$ *(index* 1*; their Gram determinant is* -140*).*

(iii) Hence the orthogonal complement of N_2 *in* Λ_{K3} *is*

$$T := N_2^{\perp} = \langle -2 \rangle \oplus \langle 70 \rangle \oplus U \oplus E_8(-1)^2, \qquad \mathrm{sign}(T) = (2, 18), \qquad A_T \cong \mathbb{Z}/2 \times \mathbb{Z}/70,$$

with $q_T(x, y) = -\frac{x^2}{4} + \frac{y^2}{140} = -q_{N_2}(x, y)$ *under the identity identification of generators. The gluing is the one induced by the unimodular overlattice* $U \oplus U$ *(consistent with Nikulin [13], Prop. 1.6.1).*

Proof. Direct computation: $n_1 \cdot n_1 = 2$, $n_2 \cdot n_2 = -70$, $n_1 \cdot n_2 = 0$, $t_1 \cdot t_1 = -2$, $t_2 \cdot t_2 = 70$, and $n_i \cdot t_j = 0$. The complement of a rank-2 primitive sublattice of the unimodular $U \oplus U$ has rank 2 and $|\det| = 140$; $\mathrm{span}(t_1, t_2)$ lies in it with $|\det| = 140$, so the index is 1. The other summands $U \oplus E_8(-1)^2$ are orthogonal to $U \oplus U$. The discriminant form of $\langle -2 \rangle \oplus \langle 70 \rangle$ is read off as in Proposition 26.1. □

Status (Category A). Elementary; exact. Nikulin's existence theorem is not needed, since the embedding is explicit.

Corollary 26.6 (The residual law in K3 terms)**.** $\rho_{\mathrm{lab}} \cong \rho_{N_2} \cong \rho_T^* = \overline{\rho_T}$ *(Borcherds' convention; honest* $\mathrm{SL}_2(\mathbb{Z})$*-representations).*

Proof. $\mathrm{sign}(T) = -16 \equiv 0 \bmod 8$ and $q_T = -q_{N_2}$ (hence $b_T = -b_{N_2}$) under the identification of Proposition 26.5, so in the formulas for $\rho_T(S)$ and $\rho_T(T)$ every phase is the conjugate of the corresponding phase for ρ_{N_2}. Combine with Theorem 26.2. □

Status (Category A). Purely lattice-theoretic; no cited geometric theorem is used. In words: the 140-label residual law transforms under the dual Weil representation of a signature-$(2,18)$ lattice, which (Theorem 26.7) is the transcendental lattice of an N_2-polarised K3 surface. This is a statement about a shared *representation*; it is not a statement that any residual series is a K3 generating function (see Remark 26.11).

26.4 Geometric realisation

Theorem 26.7 (N_2-polarised K3 surfaces)**.** *(i) The primitive embedding* $N_2 \hookrightarrow \Lambda_{K3}$ *is unique up to* $O(\Lambda_{K3})$*; hence* T *is unique up to isometry.*

(ii) There is a projective K3 surface X *with* $\mathrm{NS}(X) \cong N_2$ *and* $T_X \cong T$*.*

(iii) N_2 *represents neither* -2 *nor* 0 *(non-trivially). Hence such an* X *contains no smooth rational curves and admits no elliptic fibration, and its ample cone is the whole component of the positive cone containing an ample class.*

Proof sketch. (i) Nikulin [13, Thm. 1.14.4] (p. 126 of the translation): Λ_{K3} is even unimodular of signature $(3,19)$, N_2 is even of signature $(1,1)$, $3 > 1$, $19 > 1$, and $\mathrm{rk}\,\Lambda_{K3} - \mathrm{rk}\,N_2 = 20 \geq 2 + l(A_{N_2}) = 4$. (ii) Morrison [12, Cor. 2.9(i)] (p. 112; proved there via Cor. 1.9): for $\rho \leq 10$ every even lattice of signature $(1, \rho - 1)$ is the Néron–Severi lattice of some algebraic K3 surface (equivalently, surjectivity of the period map); with (i), $T_X \cong T$. (iii) $x \cdot x = 2\big((x_1 + 7y)^2 - 35y^2\big)$; -2 would need $u^2 - 35y^2 = -1$, impossible since -1 is not a square mod 7; 0 would need 35 to be a square. Without (-2)-classes the ample cone is a full component of the positive cone, and an elliptic fibration requires a non-zero isotropic class in NS (standard, [9]). □

Status (Category A). Category A (cited): we use Nikulin [13, Thm. 1.14.4] and Morrison [12, Cor. 2.9(i)], and standard facts on ample cones and elliptic fibrations [9]. The arithmetic in (iii) is exact.

26.5 Automorphisms

Let $\gamma = \left(\begin{smallmatrix} -1 & -7 \\ 2 & 13 \end{smallmatrix}\right)$, the Pell automorph of Theorem 11.2: $\gamma^{\mathsf{T}} A \gamma = A$, $\det \gamma = 1$, $\mathrm{tr}\,\gamma = 12$, eigenvalues $6 \pm \sqrt{35}$. It preserves N_2 (the second coordinate of γx is $2x_1 + 13x_2$). Let $r = \mathrm{diag}(1,-1)$ in the $\langle 2 \rangle \oplus \langle -70 \rangle$ basis; there $\gamma = \left(\begin{smallmatrix} 6 & 35 \\ 1 & 6 \end{smallmatrix}\right)$ (multiplication by $\varepsilon = 6 + \sqrt{35}$).

Proposition 26.8 (Action on A_{N_2} and $O(N_2)$)**.** *(i) The induced map* $\bar{\gamma}$ *on* A_{N_2} *acts on the local parts as follows: on the 2-part* $(\mathbb{Z}/2)^2$ *as a non-trivial involution (not the identity, which on this part equals* ± 1*); on the 5-part as* $+1$*; on the 7-part as* -1*. Hence* $\bar{\gamma} \neq \pm 1$ *and* $\bar{\gamma}^2 = \mathrm{id}$*. This matches the action of* γ *in Theorem 19.5(ii) on the 7/5/2 parts.*

(ii) $O(N_2) = \{\pm\gamma^k,\ \pm r\gamma^k : k \in \mathbb{Z}\}$*, and* $r\gamma r = \gamma^{-1}$*.*

(iii) On A_{N_2}*:* γ^{2j} *acts as* $+1$ *and* $r\gamma^{2j}$ *as* -1*, both preserving the positive cone; odd powers* $\pm\gamma^{2j+1}$*,* $\pm r\gamma^{2j+1}$ *act as neither* $+1$ *nor* -1*;* $-\gamma^k$ *and* $-r\gamma^k$ *do not preserve the positive cone.*

Proof. (i), (iii): exact computation on the 140 classes. (ii): in the diagonal basis $x\cdot x = 2(u^2-35y^2)$. Proper automorphs of u^2-35y^2 correspond to solutions of $t^2-140w^2=4$, i.e. to norm-1 units of $\mathbb{Z}[\sqrt{35}]$, which are $\pm\varepsilon^k$ with $\varepsilon = 6+\sqrt{35}$ fundamental; there is no unit of norm -1 ($u^2-35y^2=-1$ is impossible mod 7). Improper automorphs are r times proper ones, and $r\gamma r = \left(\begin{smallmatrix} 6 & -35 \\ -1 & 6 \end{smallmatrix}\right) = \gamma^{-1}$. □

Status (Category A). Exact.

Theorem 26.9 (Automorphisms of N_2-polarised K3 surfaces)**.** *Let X be as in Theorem 26.7, with $\mathrm{NS}(X) = N_2$ glued to $T_X = T$ as in Proposition 26.5.*

(i) *$\gamma^2\oplus\mathrm{id}_T$ extends to a Hodge isometry of $H^2(X,\mathbb{Z})$ preserving the ample cone, and is induced by a unique $f\in\mathrm{Aut}(X)$. f has infinite order; f^* has spectral radius $(6+\sqrt{35})^2 = 71+12\sqrt{35}$ on $H^2(X,\mathbb{Z})$ (a quadratic unit, not a Salem number of higher degree), so its topological entropy is*

$$h(f) = 2\log(6+\sqrt{35}) = 4\sqrt{35}\,I \approx 4.9558,$$

where $I = \log(6+\sqrt{35})/(2\sqrt{35})$ is the constant of Theorem 11.3.

(ii) *$r\oplus(-\mathrm{id}_T)$ is induced by a non-symplectic involution $\iota\in\mathrm{Aut}(X)$.*

(iii) *If $\mathrm{NS}(X)\cong N_2$ (so that the Hodge isometries of T_X are only ±1, Proposition 26.10(i)), then $\mathrm{Aut}(X) = \langle f,\iota\rangle \cong \mathbb{Z}\rtimes\mathbb{Z}/2$ (the infinite dihedral group D_∞), and γ itself does not lift to $\mathrm{Aut}(X)$.*

Proof sketch. By Nikulin's gluing criterion [13], Cor. 1.5.2, an isometry $g_{\mathrm{NS}}\oplus g_T$ of $N_2\oplus T$ extends to $H^2(X,\mathbb{Z})$ iff $\bar g_{\mathrm{NS}}$ and $\bar g_T$ agree under the glue isomorphism $A_{N_2}\cong A_T$. By Proposition 26.8, $\bar\gamma^2 = +1$ and $\bar r = -1$, compatible with id_T and $-\mathrm{id}_T$, which are Hodge isometries. Both preserve the positive cone, hence (Theorem 26.7(iii)) the ample cone. By the strong Torelli theorem (Piatetski-Shapiro–Shafarevich, Burns–Rapoport; e.g. [9], Thm. 7.5.3) they are induced by unique automorphisms f, ι; ι acts by -1 on $H^{2,0}$, so it is non-symplectic. The eigenvalues of f^* are $(6\pm\sqrt{35})^2$ on $\mathrm{NS}\otimes\mathbb{R}$ and 1 on T, and the entropy is the logarithm of the spectral radius (Gromov–Yomdin [6]). For (iii): every automorphism acts on T_X by ±1, so its restriction to N_2 preserves the positive cone and acts on A_{N_2} by ±1; by Proposition 26.8(iii) these restrictions are exactly γ^{2j} and $r\gamma^{2j}$, which form $\langle\gamma^2\rangle\rtimes\langle r\rangle\cong D_\infty$; the action on H^2 is injective by Torelli. Since $\bar\gamma\neq\pm1$, γ does not lift. □

Status (Category A). Category A (cited): we use the strong Torelli theorem, Nikulin's gluing criterion [13, Cor. 1.5.2], the Gromov–Yomdin entropy formula, and Proposition 26.10(i) for (iii). The lattice inputs (Proposition 26.8) are exact. Numerically $2\log(6+\sqrt{35}) = 2\log(11.91608\ldots) = 4.9558\ldots$. The identity $2\log(6+\sqrt{35}) = 4\sqrt{35}\,I$ is definitional; *Category B (reading):* that the regulator driving the Baker obstruction is "the same" as this entropy is an observation about definitions, and no theorem linking C to X is claimed.

Proposition 26.10 (Lifting γ)**.** *(i) If $\mathrm{NS}(X)\cong N_2$, then the Hodge isometries of T_X are ±1, and γ does not lift to $\mathrm{Aut}(X)$.*

(ii) If γ lifts on some X with $N_2\subset\mathrm{NS}(X)$ primitive and $M = N_2^\perp\subset\mathrm{NS}(X)$, then M carries the 2- and 7-parts of A_{N_2} with $\mathrm{rk}\,M\equiv4\pmod 8$, or the 2- and 5-parts with $\mathrm{rk}\,M\equiv6\pmod 8$. Hence $\rho(X)\geq6$.

(iii) Let $M = A_1(-1)^2 \oplus K_7$ with $K_7 = \left(\begin{smallmatrix} -2 & 1 \\ 1 & -4 \end{smallmatrix}\right)$. There is an even rank-6 gluing $N_2 \oplus M \subset L$ and an isometry g_M of M such that $\gamma \oplus g_M \oplus \mathrm{id}_T$ is a Hodge isometry of $H^2(X, \mathbb{Z})$ for every K3 surface X with $\mathrm{NS}(X) \cong L$ (such X exist by [12, Cor. 2.9(i)]). On this rank-6 Noether–Lefschetz locus some $w \circ (\gamma \oplus g_M)$, $w \in W(L)$, is induced by an automorphism, so Aut *grows there.*

Proof sketch. (i) The reduction $\bar{\gamma} \in O(A_{N_2})$ has order 2: it swaps on the 2-part, is $+1$ on the 5-part and -1 on the 7-part, so $\bar{\gamma} \neq \pm 1$. The Hodge isometries of T_X form a cyclic group μ_m, with $T_X \otimes \mathbb{Q}$ a $\mathbb{Q}(\zeta_m)$-vector space. If $m > 2$, a prime $p \mid m$ gives, after passing to the kernel of the action on A_T, an element of order p without fixed vectors acting trivially on A_T. This forces $pA_T = 0$, which is impossible because $A_T \cong \mathbb{Z}/2 \times \mathbb{Z}/70$ is not p-elementary. So every automorphism acts on T_X by ± 1, hence on A_{N_2} by ± 1, and γ cannot lift. (ii) The glue must carry $\bar{\gamma}$ to an isometry of A_M. Milgram's formula, with Gauss-sum signatures $4, 2, 4, 6, 6 \bmod 8$ for the relevant p-parts, gives the two congruences. (iii) An exact lattice computation gives the anti-isometric glue and g_M (the swap on $A_1(-1)^2$ and -1 on K_7) with $\bar{g}_M \circ \phi = \phi \circ \bar{\gamma}$. Because L contains (-2)-vectors, $\gamma \oplus g_M \oplus \mathrm{id}$ need not preserve the ample chamber. The strong Torelli theorem [9, Thm. 7.5.3] supplies a Weyl element w for which $w \circ (\gamma \oplus g_M \oplus \mathrm{id})$ is induced by an automorphism. □

Status (Category A). (i), (ii): written proofs. (iii): exact lattice computation, together with [12, Cor. 2.9(i)] and the strong Torelli theorem (cited), so it is conditional on these citations. *Open:* whether $w = 1$ can be taken, i.e. whether γ itself lifts on this locus.

Remark 26.11 (Category B and open items for this section)**.** None of the following is a result of this paper.

(a) *Noether–Lefschetz / Heegner series* (Category B). The Noether–Lefschetz/Heegner generating series of the N_2-polarised family are vector-valued modular forms of weight $1 + 18/2 = 10$ for ρ_T or ρ_T^* (depending on convention; Borcherds, Kudla–Millson, Maulik–Pandharipande, cited by name only). The residual objects of Theorems 11.6–11.7 are weight-0 Maass-type objects with $\lambda = \frac{1}{4}$. A shared *representation* (Corollary 26.6) is not a shared *form*; no theta-lift or Borcherds-product link has been checked.

(b) *Special members* (open). Whether γ itself lifts on the rank-6 locus of Proposition 26.10(iii), or on any X with $\rho(X) \geq 6$.

(c) *Interpretation* (Category B). Any physical or "unification" reading of X, f or ι.

(d) *Scope.* Every K3 statement in this paper concerns N_2, because the 140-label representation is ρ_{N_2} and not ρ_{N_1} (Theorem 26.2(iii)).

27 Uniqueness and normalisation of the vector completion

Throughout this section $s = 7$, $\rho = \rho_{\mathrm{lab}}$ is the 140-label representation ($\rho_{\mathrm{lab}} \cong \rho_{N_2}$, Theorem 26.2), $P = S_{\mathrm{lab}}^2$, and the laws are those of Theorem 11.6: $u(-1/w) = S_{\mathrm{lab}} u(w)$ and $u(w+1) = Tu(w)$. Floating-point checks are Category B and are labelled where they occur.

27.1 The adapted basis and the eight sectors

Let $g = (14, 6)/70$ (the generator of $L'/\mathbb{Z}^2$ used in Theorem 19.3), $a_2 = (\frac{1}{2}, 0)$, and put $u'_{x,\varepsilon,\delta} := u_{xg+\varepsilon a_2, -\delta a_2}$ for $x \in \mathbb{Z}/35$ and $\varepsilon, \delta \in \{0, 1\}$, with the phases of the labels in the convention of Remark 19.2.

Proposition 27.1 (Adapted-basis splitting at $s = 7$)**.** *In the basis $u'_{x,\varepsilon,\delta}$ the* 140 *labels are hit bijectively, and*

$$S' = \Big[\tfrac{1}{\sqrt{35}}\,\mathbf{e}\big(13xy/35\big)\Big]_{x,y} \otimes S_2, \qquad T' = \mathrm{diag}_x\,\mathbf{e}\big(24x^2/35\big) \otimes T_2,$$

with the same theta-characteristic factors S_2, T_2 as in Theorem 19.5. Since $\frac{1}{35} = \frac{3}{5} - \frac{4}{7}$, under $x \leftrightarrow (x \bmod 5, x \bmod 7)$ one has $\mathbf{e}(13xy/35) = \mathbf{e}(4x_5y_5/5)\,\mathbf{e}(4x_7y_7/7)$ and $\mathbf{e}(24x^2/35) = \mathbf{e}(2x_5^2/5)\,\mathbf{e}(2x_7^2/7)$. This is exactly the factorisation $F_5 \otimes F_7$, $T_5 \otimes T_7$ of Theorem 19.3(i).

Status (Category A). Exact check in rational phases: 0 failures over all 4900 S-entries and all 140 T-entries. The CRT identities are elementary. With the generator $A_7^{-1}e_2$ of Theorem 19.5 one obtains the factors $\mathbf{e}(-2xy/35)$, $\mathbf{e}(-x^2/35)$ instead (Remark 19.6).

Write $W_5^{\pm}$, $W_7^{\pm}$ for the (±1)-eigenspaces of $x \mapsto -x$ on the $\mathbb{Z}/5$ and $\mathbb{Z}/7$ factors ($\dim W_5^+ = 3$, $\dim W_5^- = 2$, $\dim W_7^+ = 4$, $\dim W_7^- = 3$), and A_3, B_1 for the 3- and 1-dimensional summands of the 4-dimensional theta-characteristic factor ρ_2 (Theorem 19.3(iii)), as in Proposition 19.4. We abbreviate $X^{\pm} = W_5^{\pm} \otimes W_7^{\pm}$; thus $A_3X^+ = A_3 \otimes W_5^+ \otimes W_7^+ = V_{\mathrm{Eis}}$ and $B_1X^- = B_1 \otimes W_5^- \otimes W_7^- = V_{\mathrm{cusp}}$.

Proposition 27.2 (Sectors, T-fixed vectors and the support of u)**.** *(i) $\mathbb{C}^{140}$ splits into eight ρ-stable sectors $W_5^{\pm} \otimes W_7^{\pm} \otimes \{A_3, B_1\}$:*

sector	*dim*	P	$\dim \mathrm{Fix}(T)$	*meets u*
A_3X^+	36	$+1$	1	*yes (all* 507 *frequencies)*
B_1X^+	12	$+1$	0	*no*
A_3X^-	18	$+1$	0	*no*
B_1X^-	6	$+1$	0	*yes (*131 *of* 507 *frequencies)*
$W_5^+W_7^-A_3$	27	-1	0	*no*
$W_5^+W_7^-B_1$	9	-1	0	*no*
$W_5^-W_7^+A_3$	24	-1	0	*no*
$W_5^-W_7^+B_1$	8	-1	0	*no*

The dimensions sum to 140*, and the $P = +1$ sectors give $\dim W_P = 72$.*

(ii) $\mathrm{Fix}(T) = \mathbb{C}\,v_0$ *with $v_0 = e_{x=0} \otimes (e_{00} + e_{01}) \in A_3X^+$ (and $Pv_0 = v_0$).*

(iii) ρ has no nonzero $\mathrm{SL}_2(\mathbb{Z})$*-invariant vector: $S'v_0 = 35^{-1/2}\sum_y e_y \otimes (e_{00} + e_{10}) \notin \mathbb{C}v_0$.*

(iv) Projected to the six sectors other than A_3X^+ and B_1X^-, the positive-frequency coefficients of u vanish at all 507 *positive frequencies $n \le 40$. The A_3X^+ component is nonzero at all* 507*, first at $n = \frac{1}{14}$. The B_1X^- component is nonzero at* 131 *of them, first at $n = \frac{19}{140}$.*

Status (Category A). Exact. (i)–(iii): exact linear algebra in the adapted basis. (iv): exact in $\mathbb{Z}[\zeta_{20}]$ (1169 lattice points). That the four P-odd sectors carry nothing follows from $Pu = u$. That u lies in $V = A_3X^+ \oplus B_1X^-$ for *all* frequencies is Proposition 19.4; (iv) is an independent exact check up to frequency 40.

27.2 Three levels of uniqueness

We separate three levels:

(κ) functions on $\mathbb{Q}$ (radial profiles and constant terms);

(u) $\lambda = \frac{1}{4}$ Maass vectors and their positive-frequency q-series;

(Φ) real-analytic weight-1 period functions on $\mathbb{R} \setminus \{0\}$, i.e. the Lewis–Zagier class [10].

In this subsection ρ is the representation in which Corollary 14.2 and Theorem 11.6 are written.

Proposition 27.3 (The κ-level gauge). *Let $\mathcal{G}$ be the space of functions $g : \mathbb{Q} \to \mathbb{C}^{140}$ with $g(\gamma x) = |cx + d|\, \rho(\gamma) g(x)$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and all $x \in \mathbb{Q}$ with $\gamma x \in \mathbb{Q}$. Then $\dim \mathcal{G} = \dim \mathrm{Fix}(T) = 1$. A basis is $g_0(h/m) := m^2\, \mathbf{G}(h/m)$ ($\gcd(h, m) = 1$, $m > 0$), where $\mathbf{G}$ is the profile vector of Corollary 14.2. Moreover $\|g_0(h/m)\| = m\, \|g_0(0)\| = m\sqrt{70}$.*

Proof. Dimension. $\mathrm{SL}_2(\mathbb{Z})$ acts transitively on $\mathbb{P}^1(\mathbb{Q})$. So g is determined by $g(0)$, and $g(0)$ must be fixed by the stabiliser of 0, which is $\pm S\langle T\rangle S^{-1}$. On this stabiliser $|c \cdot 0 + d| = 1$. Hence $g(0) \in \rho(S)\big(\mathrm{Fix}(T) \cap \mathrm{Fix}(P)\big) = \rho(S)\mathbb{C}v_0$ (Proposition 27.2(ii)), so $\dim \mathcal{G} \le 1$. Equality holds because $g_0 \neq 0$ lies in $\mathcal{G}$. *Equivariance of g_0.* Corollary 14.2(i) gives $\mathbf{G}(\gamma x) = |cx + d|^{-1} \rho(\gamma) \mathbf{G}(x)$. If $x = h/m$ in lowest terms, the denominator of γx is $m' = m|cx + d|$, and $m'^2 = m^2 |cx + d|^2$ turns weight -1 into weight $+1$. *Norm.* Choose γ with $\gamma(h/m) = 0$; then $|c\, h/m + d| = 1/m$. Corollary 14.2(i) gives $\mathbf{G}(0) = m\, \rho(\gamma) \mathbf{G}(h/m)$, and unitarity gives $\|\mathbf{G}(h/m)\| = \|\mathbf{G}(0)\|/m$. Finally $\mathbf{G}(0) = \sum_\alpha e_{(\alpha,0)}$ has $\|\mathbf{G}(0)\|^2 = 70$. □

Status (Category A). Proved, given Theorem 14.1 and Corollary 14.2. Consequence: the only ambiguity at level κ is the radial gauge $\mathbb{C}g_0$, which grows like m along h/m. Two κ-level objects with the same law whose difference is $o(m)$ along rationals coincide, and the canonical radial normalisation $q = \mathbf{e}(x)e^{-t}$ removes the gauge. The constant coboundary $\Phi_{v_0}(x) = v_0 - |x| S_{\mathrm{lab}} v_0$ (the analogue of Lewis–Zagier's $1 - z^{-2s}$) solves the homogeneous Φ-level law. It is excluded once the q^0 term is prescribed.

Category B. The S- and T-laws for g_0 hold to $\le 5 \cdot 10^{-9}$ at 11 rationals; $\|g_0(h/m)\|/(m\|g_0(0)\|) = 1$ to 10^{-11}; and $ST^{-1}S^{-1} g_0(0) = g_0(0)$ to $4 \cdot 10^{-15}$.

Let $M(\rho)$ be the space of real-analytic $d : \mathbb{H} \to \mathbb{C}^{140}$ of moderate growth at the cusp with $\Delta d = \frac{1}{4} d$, $d(-1/w) = S_{\mathrm{lab}} d(w)$ and $d(w+1) = T d(w)$; such d satisfy $Pd = d$ automatically. Let $S(\rho) \subset M(\rho)$ be the subspace with zero constant term. Write f_d^+ for the positive-frequency part of d, a_d for the coefficient in its constant term, $u_{\mathrm{lift}} = u$ for the lift of Theorem 11.6, and $c_d(z) = \int_0^{i\infty} [d, R_z^{1/2}]$ with the Green form of Lemma 15.1.

Theorem 27.4 (Uniqueness of the Maass completion). *(a) $M(\rho) = \mathbb{C}\, u_{\mathrm{lift}} \oplus S(\rho)$.*

(b) The map $d \mapsto f_d^+$ is injective on $M(\rho)$.

Status (Category A). Proved below. The proof uses Lemma 15.1 (for every real frequency), Lemma 15.2 and the contour argument of §15.4.

Lemma 27.5 (Estimates for Theorem 27.4(b)). *Let $d \in M(\rho)$. By Theorem 27.4(a) its constant term is $\sqrt{y}\, a_d v_0$. Let $C_d = \sup \|d(w) - \sqrt{\mathrm{Im}\, w}\, a_d v_0\|$ over the standard fundamental domain, which is finite because the nonconstant modes decay. Use the notation of Lemmas 15.1 and 15.2: $R_z(w) = \mathrm{Im}\, w/((w - z)(\bar{w} - z))$ and $\varsigma(z) = \mathrm{sgn}\, \mathrm{Re}\, z$.*

(i) The Fourier coefficients of every component satisfy $|c_j(n)| \le K_0(2\pi)^{-1}\big(|a_d|\,|n| + C_d|n|^{1/2}\big)$ *for* $n \neq 0$.

(ii) Fix $z \in \mathbb{H}$ *and* $X > 0$, *and let* $Y \ge 2(|z|+X)$. *On the horizontal segment* $\{w = x'+iY : |x' - \operatorname{Re} z| \le X\}$, *the coefficient of* dw *and of* $d\bar w$ *in the form* $[\sqrt{\operatorname{Im} w}, R_z^{1/2}]$ *are bounded in absolute value by* $8/Y$. *So the constant-term contribution of this segment is at most* $16X|a_d|/Y$, *which tends to* 0 *as* $Y \to \infty$.

(iii) Near the cusp 0, *the pullback by* $w \mapsto -1/w$ *(Lemma 15.2:* $R_{-1/z}(-1/w)^{1/2} = \varsigma(z)\, z\, R_z(w)^{1/2}$*) carries the constant-term contribution to that of (ii), with* z *replaced by* $-1/z$ *and* v_0 *by* $S_{\mathrm{lab}}v_0$. *It therefore also tends to* 0.

Proof. (i) We have $c_j(n)\sqrt{y}K_0(2\pi|n|y) = \int_0^1 d_j(x+iy)\mathbf{e}(-nx)\,dx$; take $y = 1/|n| \le 1$. For $\operatorname{Im} w = y \le 1$ the reduced point w^* satisfies $\operatorname{Im} w^* \le 1/y$. Since ρ is unitary, $\|d(w)\| = \|d(w^*)\| \le |a_d|\,y^{-1/2} + C_d$, and dividing by $|n|^{-1/2}K_0(2\pi)$ gives the bound. (ii) On the segment, $Y/2 \le |w - z| \le 3Y/2$, so $|R_z| \le 4/Y$. With $u = \sqrt{\operatorname{Im} w}$ we have $|u_w| = \frac14 Y^{-1/2}$. Also $\partial_{\bar w}R_z = \frac{i}{2}((w-z)(\bar w - z))^{-1} - \operatorname{Im} w\,(w-z)^{-1}(\bar w - z)^{-2}$, so $|\partial_{\bar w}R_z| \le 10/Y^2$, and $|R_z|^{-1/2} \le \frac32 Y^{1/2}$. Hence $|u_w R_z^{1/2}| \le \frac{1}{2Y}$ and $|u\,\partial_{\bar w}R_z^{1/2}| \le \frac12 \cdot \frac32 \cdot 10\,Y^{-1} < 8/Y$. The segment has length $2X$. (iii) We have $-1/w + 1/z = (w-z)/(wz)$ and $\operatorname{Im}(-1/w) = \operatorname{Im} w/|w|^2$. The S-law moves the constant term to $\sqrt{\operatorname{Im}(-1/w)}\,a_d S_{\mathrm{lab}}v_0$, so (ii) applies at the point $-1/z$. □

Proof (outline; the estimates are Lemma 27.5). (a) Since $\operatorname{Fix}(T) = \mathbb{C}v_0$, the constant term of $d \in M(\rho)$ is $\sqrt{y}\,(a + b\log y)\,v_0$. Green's theorem on the truncated fundamental domain (Maass–Selberg; the Hermitian pairing is legitimate because ρ is unitary) gives $a_1\bar b_2 - b_1\bar a_2 = 0$ for any $d_1, d_2 \in M(\rho)$. The lift has constant term $\sqrt{y}\,(\log\varepsilon)\,v_0$ and no $\log y$ term (§15 and the lift formula). Pairing d with u_{lift} therefore forces $b_d = 0$. So $d - (a_d/\log\varepsilon)u_{\mathrm{lift}}$ has zero constant term; its nonconstant modes decay exponentially. (b) *Holomorphy.* $c_d(z) = \int_0^{i\infty}[d, R_z^{1/2}]$ is taken along the fixed path $i\mathbb{R}_{>0}$, where $R_z(it) = t/(t^2+z^2)$. It is therefore holomorphic wherever $t^2 + z^2 \neq 0$ for all $t > 0$, i.e. on $\mathbb{C} \setminus i\mathbb{R}$. This needs no contour argument. *Fourier interchange.* By Lemma 27.5(i) the coefficients satisfy $|c(n)| \ll |n|$. Since $\sqrt{y}K_0(2\pi|n|y)$ decays exponentially in $|n|y$, the Fourier series converges absolutely and uniformly on $y \ge y_0 > 0$, so it may be integrated term by term on every segment with $y \ge y_0$. *The S-relation.* Lemma 15.1 holds for every real frequency $n \neq 0$ by translation covariance (and gives $\frac12$ for the constant term). Apply it on $\mathbb{H}$ to the modes of d, close the contour as in the proof of Theorem 11.7, and pull back by $w \mapsto -1/w$ using $d(-1/w) = S_{\mathrm{lab}}d(w)$ and Lemma 15.2. This gives, on $\mathbb{H} \setminus i\mathbb{R}$,

$$c_d(z) = I_d(z) - \varsigma(z)\,z^{-1}S_{\mathrm{lab}}I_d(-1/z), \qquad I_d = -\tfrac{\pi}{2}f_d^+ + \tfrac12 a_d v_0, \qquad \varsigma(z) = \operatorname{sgn}\operatorname{Re} z.$$

The contour step is not verbatim from Theorem 11.7. There u_L has no constant term and decays at the cusp 0, but for general d the constant term $\sqrt{y}\,a_d v_0$ is carried to the cusp 0 by the S-law. Its contribution is controlled exactly as at $i\infty$. On the connecting segments the constant-term integrand is $-\frac{i}{2}z\,a\,t^{-2}$ times a fixed vector, i.e. $O(t^{-2})$, so a segment at height Y (or $1/Y$ after the S-pullback) contributes $O(1/Y) \to 0$ (Lemma 27.5(ii),(iii)). All other terms decay exponentially. *Reflection.* The reflection $\epsilon(w) = -\bar w$ satisfies $\epsilon^*[u,v] = [\tilde v, \tilde u] = \mathrm{d}(\tilde u\tilde v) - [\tilde u, \tilde v]$ (here d is the exterior derivative, not the Maass vector d). Hence $c_d(z) = a_d\big(v_0 - \varsigma(z)z^{-1}S_{\mathrm{lab}}v_0\big) - c_{\tilde d}(-z)$ with $\tilde d(w) = d(-\bar w)$. Here $\tilde d$ obeys the same S-law and the T^{-1}-twisted T-law, and the positive frequencies of $\tilde d$ are the negative frequencies of d. *Parity step.* Suppose $f_d^+ = 0$. Then $I_d = \frac12 a_d v_0$ is constant, and the S-relation shows that c_d is a rational function on each of the two quadrants of $\mathbb{H} \setminus i\mathbb{R}$. By the

reflection identity, the negative-mode series F of d (the I-function of $\tilde{d}$) then satisfies, on the two quadrants,

$$F(w) = \kappa\, w^{-1} S_{\mathrm{lab}} F(-1/w) + L_1(w) \qquad \text{(first quadrant)},$$
$$F(w) = -\kappa\, w^{-1} S_{\mathrm{lab}} F(-1/w) + L_2(w) \qquad \text{(second quadrant)},$$

for a fixed $\kappa \in \{\pm 1\}$ and rational functions L_1, L_2. The relative sign is $\varsigma(z)$. The first-quadrant relation has the holomorphic coefficient w^{-1}, so it continues analytically to the whole half-plane on which F is holomorphic. Subtracting the second-quadrant relation there gives $2\kappa\, w^{-1} S_{\mathrm{lab}} F(-1/w) = L_2(w) - L_1(w)$. So F is rational, and since it decays exponentially, $F = 0$. (Using the relations quadrant by quadrant without the continuation only gives $F = PF$, which is vacuous because $Pd = d$.) Then $d = \sqrt{y}(a + b \log y)v_0$, and the S-law forces $d = 0$ because $S_{\mathrm{lab}} v_0 \notin \mathbb{C} v_0$ (Proposition 27.2(iii)). □

Corollary 27.6 (Cocycle and period function determine u)**.** *For $d \in M(\rho)$: if $c_d \equiv 0$ on $\mathbb{H}$, then $d = 0$. Moreover, the restriction $\Phi|_{(0,\infty)}$ of the period function of Theorem 11.7 determines u, and hence the positive-frequency series f.*

Status (Category A). Proved below; rests on Theorem 27.4.

Proof (outline). Key step. Let $e \in M(\rho)$ and suppose that on the first quadrant $c_e(z) = r(z)$ with r rational. The S-relation of Theorem 27.4(b) with $\varsigma(z) = +1$ gives $I_e(z) = z^{-1} S_{\mathrm{lab}} I_e(-1/z) + r(z)$ there. I_e is holomorphic on $\mathbb{H}$, $z \mapsto -1/z$ preserves $\mathbb{H}$, and z^{-1} is holomorphic there, so this relation continues analytically to all of $\mathbb{H}$. Applying it twice, with $I_e(-1/z) = -z\, S_{\mathrm{lab}} I_e(z) + r(-1/z)$, gives $I_e = -PI_e + (\text{rational})$. Since $Pe = e$, $PI_e = I_e$, so $2I_e$ is rational. f_e^+ decays exponentially as $\operatorname{Im} z \to \infty$, so $f_e^+ = 0$, and $e = 0$ by Theorem 27.4(b). *First claim.* If $c_d \equiv 0$, apply the key step with $r = 0$. *Second claim.* Φ (in the normalisation of §15) and c_u differ by an explicit constant-coboundary term, rational in z with coefficient a_u, and a scalar. $(0, \infty)$ lies in the right half-plane, where c_u is holomorphic (and $\Omega = \Psi$, Theorem 11.7(i)). Suppose $u, u' \in M(\rho)$ have the same Φ on $(0, \infty)$. By the identity theorem, $c_{u-u'}$ is rational on the first quadrant, and the key step gives $u = u'$. □

Theorem 27.7 (Uniqueness and residual ambiguity of the vector completion)**.** *Let $u = u_{\mathrm{lift}}$, with positive-frequency series f and period function Φ (Theorems 11.6, 11.7).*

(i) Uniqueness given strong data. *Among $d \in M(\rho)$, u is the only element with positive-frequency series f and q^0-coefficient a_u. Moreover, among period functions of elements of $M(\rho)$, Φ is determined by its restriction to $(0, \infty)$.*

(ii) Non-uniqueness under weak conditions. *Consider period functions of elements of $M(\rho)$ with the same ρ-law (weight 1, factor $|cx + d|$), the same singular I/t data, and real-analytic on $\mathbb{R} \setminus \{0\}$. They form the affine space $\Phi + \{\Phi_c : c \in S(\rho)\}$, enlarged by $\mathbb{C}\,\Phi_{v_0}$ when the q^0 term is not prescribed. The parametrisation $c \mapsto \Phi_c$ is injective.*

(iii) The cusp space is nonzero. $\dim S(\rho) \geq 1$. *An explicit witness is $u_{\mathrm{cusp}} := \Pi_{B_1 X^-} u_{\mathrm{lift}}$, the projection to the ρ-stable sector $B_1 X^-$. It lies in $M(\rho)$, has zero constant term because $\mathrm{Fix}(T) \cap B_1 X^- = 0$, and is nonzero because its coefficient at $n = \frac{19}{140}$ is nonzero. Hence, by the injectivity in (ii), the class in (ii) contains more than one element.*

(iv) Shape of the ambiguity. *The non-cusp part of $M(\rho)$ is exactly one-dimensional and lies in A_3X^+, and the P-odd sectors contribute nothing. An explicit basis of the residual ambiguity is*

$$\begin{aligned} &\{\Phi_{v_0}\} && (\textit{only if } q^0 \textit{ is free}) \\ \oplus\ &\{\Phi_c :\ c \in \textit{a basis of } S(\rho)\} && (\textit{containing } \Phi_{u_{\mathrm{cusp}}} \neq 0) \\ \oplus\ &\mathbb{C}\, g_0 && (\kappa\textit{-level, Proposition 27.3}). \end{aligned}$$

Status (Category A). (i) and the injectivity in (ii): Theorem 27.4 and Corollary 27.6. (iii): $\dim S(\rho) \geq 1$ with witness u_{cusp} is independent of Theorem 27.4: sector projectors commute with ρ, the vanishing of the constant term is Proposition 27.2(i),(ii), and non-vanishing is an exact $\mathbb{Z}[\zeta_{20}]$ value. That the class in (ii) has more than one element, and $\Phi_{u_{\mathrm{cusp}}} \neq 0$ in (iv), use the injectivity (Corollary 27.6). (iv): given Theorem 27.4(a) and Proposition 27.2. *Not claimed:* that every real-analytic ρ-cocycle with Lewis–Zagier growth comes from $M(\rho)$ (Remark 27.16(a)); for cusp-type period functions with the growth conditions stated there this follows from Deitmar–Hilgert, but the non-cusp part is not covered.

Proposition 27.8 (Sector-by-sector uniqueness and the scalars)**.** *u_{lift} has components only in $\sigma \in \{A_3X^+, B_1X^-\}$. For each such σ:*

(i) the σ-component is the unique element of $\Pi_\sigma M(\rho)$ with positive-frequency series f_σ;

(ii) it is unique up to a scalar given only the line $\mathbb{C}f_\sigma$;

(iii) the scalar is fixed by the first nonzero coefficient: $n = \frac{1}{14}$ for A_3X^+ and $n = \frac{19}{140}$ for B_1X^-, both exact cyclotomic numbers (Appendix C).

Only A_3X^+ has a non-cusp part, and it is one-dimensional. In B_1X^- the completion is unique given f. In the weak class of Theorem 27.7(ii) it is ambiguous by $S(\rho) \cap B_1X^-$, which contains u_{cusp}.

Status (Category A). Support: Propositions 19.4 and 27.2. Uniqueness in (i)–(ii): Theorem 27.4 applied to $\Pi_\sigma d$. (iii): exact values.

Proposition 27.9 (Sector bounds for $\dim S(\rho)$)**.** *Let σ run over the eight ρ-stable sectors of Proposition 27.2, and write $M(\rho)_\sigma$, $S(\rho)_\sigma$ for the σ-components.*

(i) For the four P-odd sectors, $M(\rho)_\sigma = S(\rho)_\sigma = 0$.

(ii) $M(\rho)_{A_3X^+} = \mathbb{C}\,\Pi_{A_3X^+}u_{\mathrm{lift}} \oplus S(\rho)_{A_3X^+}$, and $M(\rho)_\sigma = S(\rho)_\sigma$ for $\sigma \in \{B_1X^+, A_3X^-, B_1X^-\}$.

(iii) $\dim S(\rho)_{B_1X^-} \geq 1$, with witness $u_{\mathrm{cusp}} = \Pi_{B_1X^-}u_{\mathrm{lift}}$.

sector	$\dim\sigma$	P	$\dim S(\rho)_\sigma$	*status*
A_3X^+	36	$+1$	0	*A (computer-assisted, Theorem 27.13)*
B_1X^+	12	$+1$	0	*A (computer-assisted, Theorem 27.13)*
A_3X^-	18	$+1$	0	*A (computer-assisted, Theorem 27.13)*
B_1X^-	6	$+1$	1	*A (≥ 1: exact witness at $n = \frac{19}{140}$; ≤ 1: Theorem 27.13)*
$W_5^+W_7^-A_3$	27	-1	0	*A*
$W_5^+W_7^-B_1$	9	-1	0	*A*
$W_5^-W_7^+A_3$	24	-1	0	*A*
$W_5^-W_7^+B_1$	8	-1	0	*A*

The upper bounds in the P-even sectors are Theorem 27.13.

Proof. (i) Every $d \in M(\rho)$ satisfies $Pd = d$, because $S_{\mathrm{lab}}^2 = P$ and $d(-1/(-1/w)) = d(w)$. So d has no component in $\mathrm{Fix}(-P)$. (ii) Theorem 27.4(a), together with $\mathrm{Fix}(T) = \mathbb{C}v_0 \subset A_3X^+$, so constant terms vanish outside A_3X^+. (iii) Theorem 27.7(iii). The coefficient at $n = \frac{19}{140}$ was also recomputed exactly: 8 lattice points, all phases ± 1, rational projectors, 4 nonzero B_1X^- entries. □

Status. (i) and (iii): Category A. (ii): Category A given Theorem 27.4(a). *Irreducibility* of the eight sectors: each has a simple T-spectrum with exact exponents in $\frac{1}{140}\mathbb{Z}$ (Category A), and the S-graph between T-eigenvectors is connected, a zero/nonzero test done in floating point (Category B). *Category B (consistency):* theta lifts of all even binary lattices $ax^2 - by^2$ with discriminant group of order ≤ 2000 give, inside ρ, only multiples of the components of u_{lift}, in agreement with Theorem 27.13.

27.3 The dimension of $S(\rho)$

Fix a P-even sector σ of dimension d. Its T-spectrum is simple, with exponents $r_1, \dots, r_d \in \{0, \dots, 139\}$ (Proposition 27.9). Let $v_1, \dots, v_d$ be the unit T-eigenvectors, $Tv_k = \mathbf{e}(r_k/140)v_k$. They are orthonormal, and we write $S_\sigma = (\langle Sv_l, v_k\rangle)_{k,l}$, a unitary matrix with $S_\sigma^2 = I$. For $d \in S(\rho)_\sigma$ put $g_k = \langle d, v_k\rangle$. Then $g(w+1) = \mathrm{diag}(\mathbf{e}(r_k/140))\, g(w)$ and $g(-1/w) = S_\sigma g(w)$. Because the constant term vanishes and d has moderate growth,

$$g_k(x+iy) = \sum_{n \in r_k/140+\mathbb{Z},\ n\neq 0} c_k(n)\sqrt{y}\, K_0(2\pi|n|y)\, \mathbf{e}(nx), \tag{27.1}$$

so g decays exponentially at the cusp and $\|d\|_\infty := \sup_{\mathbb{H}} \|d(w)\|_2 < \infty$. Since ρ is unitary, $\|d(w)\|_2$ is Γ-invariant, and the supremum is attained at some point w_0 of the closed standard fundamental domain. In particular $\Im w_0 \ge \frac{\sqrt{3}}{2}$.

Lemma 27.10. *$K_0(x) \le \sqrt{\pi/(2x)}\, e^{-x}$ for $x > 0$.*

Proof. $K_0(x) = \int_0^\infty e^{-x\cosh t}\, dt \le \int_0^\infty e^{-x(1+t^2/2)}\, dt = e^{-x}\sqrt{\pi/(2x)}$, since $\cosh t \ge 1 + t^2/2$. □

Lemma 27.11 (A priori bounds)**.** *Let $d \in S(\rho)_\sigma$ with $\|d\|_\infty = 1$, and let $\mu_0 = \sqrt{1/2}\, K_0(1/2)$. Then:*

(i) $|c_k(n)| \le \sqrt{2\pi|n|}/\mu_0$ *for every k and every n.*

(ii) For $M \ge 1$ and $y > 0$, the part of (27.1) with $|n| > M$ is bounded in absolute value by

$$\tau_M(y) := \frac{\sqrt{\pi/2}}{\mu_0} \cdot \frac{2e^{-2\pi My}}{1 - e^{-2\pi y}}.$$

(iii) For $y \ge \frac{\sqrt{3}}{2}$, $\sqrt{y}\, K_0(2\pi|n|y) \le \omega_n := \min\bigl(\sqrt{\pi/2}/\sqrt{2\pi|n|},\ (4|n|)^{-1/2}e^{-\pi\sqrt{3}|n|}\bigr)$.

Proof. (i) We have $c_k(n)\sqrt{y}K_0(2\pi|n|y) = \int_0^1 g_k(x+iy)\mathbf{e}(-nx)\, dx$, and $|g_k| \le \|d\|_2 \le 1$. Take $y = 1/(4\pi|n|)$. (ii) By (i) and Lemma 27.10, each term is at most $\sqrt{\pi/2}\,\mu_0^{-1}e^{-2\pi|n|y}$. The frequencies $n > M$ in $r/140 + \mathbb{Z}$ are $> M + j$ for $j = 0, 1, \dots$, and likewise for $n < -M$. (iii) The first bound holds because $\sqrt{t}K_0(t) \le \sqrt{\pi/2}$, by Lemma 27.10. The second follows from Lemma 27.10 and monotonicity in y. □

Proposition 27.12 (Certification criterion). *Fix $M \geq 1$ and let $\mathcal{C} = \{(k,n) : n \in r_k/140+\mathbb{Z},\ n \neq 0,\ n = r_k/140 + j,\ |j| \leq M\}$. Fix sample points $w_m = x_m + iY_m$, $m = 1,\dots,R$, and elements $\gamma_m \in \mathrm{SL}_2(\mathbb{Z})$, and let $w_m^* = \gamma_m^{-1} w_m$. Let Γ_m be the matrix with $g(w_m) = \Gamma_m\, g(w_m^*)$; it is a product of S_σ and diagonal phase matrices, and it is unitary. Let G be the $(Rd) \times |\mathcal{C}|$ matrix whose block row m is $E_m - \Gamma_m F_m$. Here E_m and F_m evaluate the basis functions $\sqrt{y}K_0(2\pi|n|y)\mathbf{e}(nx)$ of column (k,n), placed in row k, at w_m and at w_m^*. Let $D = \mathrm{diag}(s_{(k,n)})$ with all $s > 0$, and $H = (GD^{-1})^*(GD^{-1})$. Put*

$$\varrho = \sum_m d\,\big(\tau_M(Y_m) + \tau_M(\Im w_m^*)\big)^2, \quad \Omega = \max_k \sum_{(k,n)\in\mathcal{C}} (\omega_n/s_{(k,n)})^2, \quad t_0 = \sqrt{d}\,\tau_M(\tfrac{\sqrt{3}}{2}).$$

If $t_0 < 1$ and the $(j+1)$-st smallest eigenvalue of H exceeds $\varrho\,\Omega/(1-t_0)^2$, then $\dim S(\rho)_\sigma \leq j$.

Proof. Let $d \in S(\rho)_\sigma$ and write $c = (c_k(n))_{(k,n)\in\mathcal{C}}$, $c'' = Dc$. The map $d \mapsto c''$ is linear.

Step 1: the map is injective. Normalise $\|d\|_\infty = 1$ and evaluate at w_0. By Lemma 27.11(ii),(iii) and Cauchy–Schwarz in each component,

$$1 = \|d(w_0)\|_2 \leq \Big(\sum_k \big(\sum_n \omega_n |c_k(n)|\big)^2\Big)^{1/2} + t_0 \leq \Omega^{1/2}\|c''\|_2 + t_0.$$

Hence $\|c''\|_2 \geq (1-t_0)\Omega^{-1/2} > 0$.

Step 2: the residual is small. The identity $g(w_m) = \Gamma_m g(w_m^*)$ is exact. Splitting each side of (27.1) into the part in $\mathcal{C}$ and the rest gives $GD^{-1}c'' = \varepsilon$. Here $\varepsilon_m = -(\text{tail of } g(w_m)) + \Gamma_m(\text{tail of } g(w_m^*))$, so $\|\varepsilon_m\|_2 \leq \sqrt{d}(\tau_M(Y_m) + \tau_M(\Im w_m^*))$ by Lemma 27.11(ii) and the unitarity of Γ_m. Hence $\|GD^{-1}c''\|_2^2 \leq \varrho$.

Step 3: conclusion. Suppose $\dim S(\rho)_\sigma \geq j+1$. By Step 1 the image of $d \mapsto c''$ has dimension at least $j+1$. By the Courant–Fischer theorem it contains a vector $c'' \neq 0$ with $\|GD^{-1}c''\|^2 \geq \lambda_{j+1}(H)\|c''\|^2$. Rescale the corresponding d to $\|d\|_\infty = 1$. Steps 1 and 2 then give $\lambda_{j+1}(H)(1-t_0)^2/\Omega \leq \varrho$, which contradicts the hypothesis. □

Theorem 27.13 (Dimension of the cusp space). $\dim S(\rho)_{A_3X^+} = \dim S(\rho)_{B_1X^+} = \dim S(\rho)_{A_3X^-} = 0$ *and* $\dim S(\rho)_{B_1X^-} = 1$. *Hence* $\dim S(\rho) = 1$ *and* $S(\rho) = \mathbb{C}\,u_{\mathrm{cusp}}$.

Proof. We apply Proposition 27.12 with $M = 6$, $Y_m \in \{\frac{4}{5}, \frac{7}{10}, \frac{3}{5}, \frac{1}{2}\}$ and $x_m = \frac{2m+1}{80} - \frac{1}{2}$, $m = 0,\dots,39$. Each γ_m is the usual reduction word, chosen in floating point; the identity holds for any word. The law $d(-1/w) = S_{\mathrm{lab}}d(w)$ used here agrees with $\rho(S) = S_{\mathrm{lab}}^{-1}$ (Remark 19.1), since $S_{\mathrm{lab}}^2 = P$ gives $S_{\mathrm{lab}}^{-1} = S_{\mathrm{lab}}$ on every P-even sector. All of the following are computed in ball arithmetic (128-bit precision) from the exact rational phase data of S_{lab}, T_{lab}, the adapted basis and the exact sector projectors: v_k (via the exact eigenprojection $\frac{1}{140}\sum_j \mathbf{e}(-r_kj/140)T^j$), S_σ, Γ_m, w_m^*, the entries of G (with $K_0(x) = \sqrt{\pi}e^{-x}U(\frac{1}{2},1,2x)$), H, ϱ, Ω and t_0. The ball checks $TV = V\,\mathrm{diag}(\mathbf{e}(r_k/140))$, $V^*V = I$, $SV = VS_\sigma$, $S_\sigma^2 = I$ and $S_\sigma S_\sigma^* = I$ all contain 0. Eigenvalue lower bounds for H come from an approximate unitary Q and two estimates. Weyl's inequality applied to $Q^*HQ - \Lambda$ gives $\lambda_k(Q^*HQ) \geq \Lambda_k - \|Q^*HQ - \Lambda\|_F$. Ostrowski's theorem gives $\lambda_k(H) \geq \lambda_k(Q^*HQ)/(1+\|Q^*Q-I\|_F)$. The resulting certified values are:

sector	d	$\lvert\mathcal{C}\rvert$	rows	threshold $\varrho\Omega/(1-t_0)^2 \leq$	certified lower bound	conclusion
A_3X^+	36	467	5760	$1.04\cdot10^{-14}$	$\lambda_1 \geq 0.0251$	$\dim = 0$
B_1X^+	12	156	1920	$3.6\cdot10^{-15}$	$\lambda_1 \geq 0.0373$	$\dim = 0$
A_3X^-	18	234	2880	$5.2\cdot10^{-15}$	$\lambda_1 \geq 0.0154$	$\dim = 0$
B_1X^-	6	78	960	$4.4\cdot10^{-15}$	$\lambda_2 \geq 0.1317$	$\dim \leq 1$

A second run with $M = 8$ and $40 \to 48$ points per horocycle gives thresholds $\le 4 \cdot 10^{-20}$ and the same lower bounds to three digits. For $B_1 X^-$, the lower bound $\dim \ge 1$ is Proposition 27.9(iii). The P-odd sectors contribute 0 by Proposition 27.9(i). □

Status: Category A (computer-assisted). This is a computer-assisted proof in the sense of §10.1. The analytic reduction (Lemmas 27.10, 27.11, Proposition 27.12) is proved in full. It uses no coefficient bound beyond the trivial sup-norm bound and no assumption on λ other than $\lambda = \frac{1}{4}$. The finite verification uses ball arithmetic (Arb via python-flint), and every quantity entering the final inequality is a rigorous enclosure. Floating point only chooses the reduction words, column scalings and the approximate eigenbasis Q; these choices are not part of the proof. The margin is at least 10^{12} in every sector.

Corollary 27.14. $\dim M(\rho) = 2$ *and* $M(\rho) = \mathbb{C}\, u_{\mathrm{lift}} \oplus \mathbb{C}\, u_{\mathrm{cusp}}$. *In Theorem 27.7(ii), the ambiguity of period functions under the weak conditions is exactly the line* $\mathbb{C}\, \Phi_{u_{\mathrm{cusp}}}$.

Proof. Theorem 27.4(a), which gives $M(\rho) = \mathbb{C} u_{\mathrm{lift}} \oplus S(\rho)$, together with Theorem 27.13. □

Status. Category A, given Theorem 27.4(a) and the same conditions as Theorem 27.7(ii).

Remark 27.15 (Dihedral consistency check)**.** Independently of Theorem 27.13: among the ray class characters $\chi \ne \chi^\sigma$ of $\mathbb{Q}(\sqrt{D})$, $D \in \{2, 5, 7, 10, 14, 35, 70\}$, with infinity type $(+,+)$ or $(-,-)$ and $\mathrm{Ind}\, \chi$ of conductor dividing 19600, there are exactly 120, giving 36 even dihedral representations, and none has projective image field $\mathbb{Q}(\sqrt{7}, \sqrt{10})$. So that field contributes no dihedral $\lambda = \frac{1}{4}$ form at this level, as Theorem 27.13 requires.

Status (Category A). Exact enumeration (PARI/GP). Completeness: the sum over primitive characters equals $|\mathrm{Cl}_{\mathfrak{m}}|$ for all seven fields, and split primes ≤ 3000 generate each $\mathrm{Cl}_{\mathfrak{m}}$, so the test $\chi \ne \chi^\sigma$ is exact.

Remark 27.16 (Further remarks on §27)**.** (a) *Period functions of cusp forms.* We use Deitmar–Hilgert [5, Thm. 3.3 and Thm. 4.4] with $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$, the finite-image representation $\eta = \rho$ and spectral parameter $\nu = 0$ (our $\lambda = \frac{1}{4}$, as $\Delta u = (\frac{1}{4} - \nu^2)u$; the hypotheses $\nu \notin \frac{1}{2} + \mathbb{Z}$ and $\mathrm{Re}\, \nu > -\frac{1}{2}$ hold). The dictionary is $\psi(x) = \rho(S)^{-1}\Phi(x)/x$ on $x > 0$, with $\Phi(x) = \kappa(-1/x) - |x| S_{\mathrm{lab}} \kappa(x)$; ψ satisfies the three-term equation $\eta(T)\psi(z) = \psi(z+1) + (z+1)^{-1}\eta(ST^{-1})\psi(z/(z+1))$, and the growth conditions of [5, Thm. 4.4] read $\Phi(x) = o(1)$ as $x \to 0^+$ and $\Phi(x) = o(x)$ as $x \to \infty$. Under these conditions period functions correspond bijectively to elements of $S(\rho)$, so by Theorem 27.13 they form the line spanned by $\Phi_{u_{\mathrm{cusp}}}$. Lewis–Zagier [10] treats the scalar case for $\mathrm{SL}_2(\mathbb{Z})$ only. The non-cusp form u_{lift} is not covered by either reference. Status: Category A (cited; hypotheses checked against [5]); the numerical check of the dictionary (residual $7 \cdot 10^{-15}$) is Category B.

(b) *Numerical support for Proposition 27.3*: the S- and T-laws for g_0 hold to $\le 5 \cdot 10^{-9}$ at 11 rationals; $\|g_0(h/m)\|/(m\|g_0(0)\|) = 1$ to 10^{-11}; and $ST^{-1}S^{-1}g_0(0) = g_0(0)$ to $4 \cdot 10^{-15}$. Floating point: Category B.

28 Obstructions to coefficient-emitting maps for g_s

One may ask whether the coefficients of g_s (§25.2), or the cone data, can be produced ("emitted") by a natural construction from E_8 data or from the integer pattern $\mathrm{PW}(7, 11; 3^5)$: $B' = 7^2 \cdot 11 = 539$, window 18, fibres $68 = 3 \cdot 11 + 5 \cdot 7$ and 67, towers $243 = 3^5$, body 521, auxiliary 8 (the rank of E_8). We prove obstructions for several natural classes of such maps (Propositions 28.1–28.4) and

record what is not excluded (Remark 28.5). Data of this kind can select the parameter s; once s is fixed, g_s is defined independently of E_8 and of PW, so selecting s does not emit coefficients.

Proposition 28.1 (The t^1 obstruction for modular-type sums)**.** *Let $s > 0$ with $s \neq 3$.*

(i) $g_s(e^{-t}) = 2 + 2(s-3)\,t + (5s^2 - 44s + 75)\,t^2 + O(t^3)$ *as* $t \to 0^+$ *(Theorem 25.2). For* $s = 1, 7, 8, 11$*:* $(c_1, c_2) = (-4, 36), (8, 12), (10, 43), (16, 196)$.

(ii) Let F be any finite sum of holomorphic modular or quasimodular forms of nonnegative real weights (any level, any multiplier) that are holomorphic at the cusp 0. Then $F(e^{-t}) = \sum_a C_a t^{-a}$ with all $a \geq 0$, up to an exponentially small error. Hence g_s is not such a sum.

(iii) Nor is any single shift $q^r g_s$ ($r \in \mathbb{R}$) such a sum; equivalently, $g_s \neq q^{-r}F$. Cancelling the t-term forces $r = s - 3$, and the t^2 coefficient of $q^{s-3}g_s$ is then $2(2s^2 - 19s + 33)$, which is nonzero for $s = 7, 8, 11$ (values -4, 18, 132) and for $s = 1$ (value 32).

The class in (ii) contains $\theta_{E_8} = E_4$, every $\theta_{E_8}(q^m)$, the Eisenstein series built from σ_3 and their derivatives, and every lattice theta series, of E_8 or of any other positive-definite lattice.

Proof. (i) is Theorem 25.2 (c_0, c_1, c_2 were also re-derived exactly). (ii) For a form of weight k, $F(e^{-t})$ is $(2\pi/t)^k$ times the expansion of $F|_k S$ at $i\infty$, i.e. $C\,t^{-k}$ plus exponentially small terms. A quasimodular form of depth $p \leq k/2$ contributes t^{-k+j} with $0 \leq j \leq p$, so all exponents are ≤ 0. Finitely many such terms give only t^{-a} with $a \geq 0$, whereas g_s has the coefficient $c_1 = 2(s-3) \neq 0$ of t^1. (iii) With $q = e^{-t}$, $q^r g_s = e^{-rt} g_s(e^{-t})$ has t-coefficient $2(s-3) - 2r$ and, at $r = s - 3$, t^2-coefficient $c_2 - rc_1 + r^2 = 2(2s^2 - 19s + 33)$. □

Status (Category A). Proved, given Theorem 25.2. Category B support: $(g_s - c_0 - c_1 t - c_2 t^2)/t^3$ at $t = 10^{-3}, \ldots, 1.25 \cdot 10^{-4}$ is $-1355, -1338, -1329, -1325$ for $s = 7$. This is consistent with the value $c_3(7) = -3962/3 \approx -1320.7$ of Theorem 25.2, and similarly for $s = 1, 8, 11$ ($c_3 = -546.7, -1566.3, -303.3$). *Sanity:* the same test excludes $f_0 = g_1$, consistent with f_0 being mock.

Proposition 28.2 (E_8 is invisible to the vector and cone data)**.** *E_8 is even unimodular, so $A_{E_8} = 0$ and $\rho_{L \oplus E_8} = \rho_L$ for every even lattice L. Consequently the Weil representation of Theorem 26.2, the cone/vector transformation law, the eight sectors of Proposition 27.2, the Hom dimension 8 (Proposition 26.3) and the q-exponents modulo 1 of the 140-vector are unchanged by adjoining E_8. No construction that factors through discriminant-form data can carry E_8 information into them.*

Status (Category A). Standard (discriminant forms of orthogonal sums are orthogonal sums). *Arithmetic aside, not a result:* the primes involved are $\{2, 5, 7\}$ at $s = 7$, $\{2, 3\}$ at $s = 8$ and $\{2, 3, 11\}$ at $s = 11$. Only 7 and 11 are also E_8 exponents.

Proposition 28.3 (Exact linear-span test to q^{1499})**.** *Let $\mathcal{F}$ be the following 1036 distinct integer sequences, truncated to $q^0, \ldots, q^{1499}$:*

F1 $q^r \theta_{E_8}(q^m)$ *for* $1 \leq m \leq 12$ *and* $0 \leq r \leq 20$ *(252 series);*

F2 $W(E_8)$ *dominant-weight counts per norm, rescaled by* $1 \leq m \leq 6$ *and shifted by* $0 \leq r \leq 20$ *(126 series);*

F3 the $W(E_8)$ Molien series $\prod_{d \in \{2,8,12,14,18,20,24,30\}} (1 - q^d)^{-1}$, with the same rescalings and shifts (126 series);

F4 partition counts into parts from each nonempty subset of $\{7, 8, 11, 18, 67, 68, 243, 521, 539\}$ *(*511 *series);*

δ $\delta_{n,r}$ *for* $0 \le r \le 20$ *(*21 *series). No two of the* 1036 *series coincide.*

None of g_7, g_8, g_{11}, g_1, $g_7 - g_8$, $g_{11} - g_7$ *lies in the* $\mathbb{Q}$*-span of* $\mathcal{F}$*. Moreover* g_1, g_7, g_8, g_{11} *are* $\mathbb{Q}$*-linearly independent.*

Proof. For $p = 2^{61} - 1$, the mod-p rank of $\mathcal{F}$ is 880. The exact $\mathbb{Q}$-rank of F4 $\cup\, \delta$ is 376 (532 columns; exact integer rank). F1–F3 have 504 columns, so $\operatorname{rank}_{\mathbb{Q}} \mathcal{F} \le 504 + 376 = 880$. For an integer matrix $\operatorname{rank}_p \le \operatorname{rank}_{\mathbb{Q}}$, so $\operatorname{rank}_{\mathbb{Q}} \mathcal{F} = 880$. Adjoining any one target raises the mod-p rank to 881, so its $\mathbb{Q}$-rank is at least $881 > 880$, and the target is not in the $\mathbb{Q}$-span of the truncated sequences, a fortiori not of the full ones. The independence statement is an exact rank-4 computation. □

Status (Category A). Exact. Sanity checks of F2: $d(1..4) = 1, 1, 1, 2$ and $r_{E_8}(4) = 17520 = 17280 + 240$.

Proposition 28.4 (No low-degree rational generating function)**.** *For* $s \in \{7, 8, 11\}$ *and* $(R, n_0) \in \{(1000, 0), (1000, 999), (600, 1200)\}$*, the* $(R+1) \times (R+1)$ *Hankel matrix* $(a_{n_0+i+j})_{0 \le i,j \le R}$ *of the coefficients of* g_s *is nonsingular modulo* $p = 2^{61} - 1$*, hence over* $\mathbb{Q}$*. Consequently* $g_s \ne P/Q$ *with* $P, Q \in \mathbb{Q}[q]$*,* $\deg Q \le 1000$ *and* $\deg P \le 999$*. The same holds with* $\deg Q \le 600$ *and* $\deg P \le 1200$*.*

Proof. Suppose $g_s = P/Q$ with $0 \ne Q = \sum_{k \le D} b_k q^k$. Then $\sum_k b_k a_{n-k} = [q^n]P = 0$ for all $n > \deg P$. Let $R \ge D$ and put $b_k = 0$ for $D < k \le R$ and $c_j = b_{R-j}$ $(0 \le j \le R)$. Row i of the Hankel matrix applied to c gives $\sum_k b_k a_{n_0+i+R-k}$, which vanishes when $n_0 + R > \deg P$; all indices are $\ge n_0 \ge 0$. So the matrix is singular whenever $D \le R$ and $\deg P \le n_0 + R - 1$. Nonsingularity at $(R, n_0) = (1000, 0)$ therefore excludes $\deg Q \le 1000$, $\deg P \le 999$, and at $(600, 1200)$ it excludes $\deg Q \le 600$, $\deg P \le 1799$, which contains the stated range. □

Status (Category A). Exact modular arithmetic. This excludes, for example, any finite partition-count or Molien-type map with total part sum ≤ 1000, and any linear combination of such maps with common denominator of degree ≤ 1000. The $W(E_8)$ Molien degree sum is 128.

Remark 28.5 (Scope: what is *not* excluded)**.** Propositions 28.1–28.4 exclude only the stated classes:

- modular and quasimodular sums of nonnegative weight holomorphic at 0, with at most one q^r shift;
- constructions factoring through discriminant-form data;
- $\mathbb{Q}$-linear combinations of the 1036 tested sequences;
- rational generating functions in the stated degree ranges.

They do **not** exclude:

- nonlinear or non-modular maps;
- mock or false-theta constructions with E_8 input;
- sums with several different q^r prefactors;

- weakly holomorphic forms with poles at the cusp 0;
- rational functions of higher degree;
- linear combinations of untested sequences.

Category B (reading): absence of a map in the tested classes is evidence, not proof, that no natural map exists.

29 Consequences

29.1 No scalar residual law; the law that exists

Proposition 29.1 (No scalar residual law)**.** *(i) There is no identity $C = F + \mathcal{E}$ with F an algebraic combination of holomorphic modular forms, signature-$(1,1)$ mock (Zwegers) holomorphic parts, unary, partial or false theta series, and $\mathcal{E}(e^{-t}) = o(1/t)$.*

(ii) There is no quotient law $C \cdot g = F$ with holomorphic modular $g \not\equiv 0$, F.

(iii) There is no scalar W_7 law: for no weight k and no constant λ is $C(\tau) - \lambda(7\tau)^{-k} C(-1/(7\tau))$ radially bounded at all rationals.

Status (Category A). This restates Theorem 11.5 and §13.5: (i), (ii) are conditional on Baker's theorem and the q-expansion principle, using the leading term $C(e^{-t}) = I/t + a_C t^{-1/2} + \cdots$ and $a_C \neq 0$; (iii) is an exact finite computation: at 178 of the 270 test rationals (denominator < 30), including $x = 5/7$, exactly one of $G(x)$, $G(-1/(7x))$ vanishes (Theorem 11.4).

Proposition 29.2 (The law that exists has weight 1 and is vector-valued)**.** *(i) Where both Gauss means are nonzero, $|G|$ obeys an exact weight-1 relation with $|\lambda| = 7^{-1/2}$ (§13.5); the leading $1/t$ radial profiles obey the S/T law with factor $1/|x|$ (Theorem 14.1); and the vector cocycle has automorphy factor $|cx + d|$ (Theorem 11.7).*

(ii) The vector law consists of the 140-label Siegel lift to $\lambda = \frac{1}{4}$ Maass forms (Theorem 11.6), the Lewis–Zagier period formula (Theorem 11.7), the 42-dimensional function span $V = V_{\mathrm{Eis}}(36) \oplus V_{\mathrm{cusp}}(6)$ (Theorem 11.8), the Herglotz closed form (Theorems 11.9, 11.10), the identification $\rho_{\mathrm{lab}} \cong \rho_{N_2}$ (Theorem 26.2) and the uniqueness statements of §27. It is a law for the vector of cone sums, not for scalar C.

(iii) The characteristic $a = A^{-1}(0, \frac{s}{2})^{\mathsf{T}} = \left(\frac{s}{2(s-2)}, -\frac{1}{s-2}\right)$ completes the square: $Q_+(v) = Q_2(v + a) + c_0$ with $c_0 = \frac{s}{4(s-2)}$.

Status (Category A). (i), (ii) with the status of the cited theorems as stated where they appear; Theorem 11.11 is not part of the law and stays Category B. (iii): $A^{-1} = \frac{-1}{s(s-2)} \left(\begin{smallmatrix} s & -s \\ -s & 2 \end{smallmatrix}\right)$ gives a, $B(v,a) = \frac{s}{2} k$ and $-Q_2(a) = \frac{s}{4(s-2)}$.

Remark 29.3 (Weight $\frac{1}{2}$ cannot carry the residue)**.** Any weight-$\frac{1}{2}$ harmonic Maass form has radial growth $O(t^{-1/2})$, so no weight-$\frac{1}{2}$ object can carry the I/t term of C (Category A, conditional on the standard, cited $t^{-1/2}$ radial bound).

Remark 29.4 (Sample values)**.** Four values 19.80, 1.82, 9.24, 20.90 at $x = 1/7, 3/7, 5/7, 1$ that have been proposed as "residual samples" are not radial values of C: $|C(0.995\,\mathbf{e}(x))| \approx 19.5, 18.5, 19.9, 54.0$ (floating point, Category B). The meaningful radial targets are $\lim_{t \to 0} t\, C(\mathbf{e}(x) e^{-t}) = I \cdot G(x)$ at every rational x (Theorem 11.4; Category A).

29.2 Uniformity in s

Proposition 29.5 (What depends on $s = 7$)**.** *$7s - 32 = -2Q(c_2)$ for the wall $c_2 = (-4, 1)$ (Proposition 23.1); only the factor $s - 2$ of Δ is intrinsic given c_1 (Remark 23.5). The results are otherwise uniform in s: the tensor theorem holds for all odd s (Theorem 19.5), the radial closed forms are rational functions of s, and Theorem 25.2 holds for all real $s > 0$. The structurally special values are $s = 5$ (the radial coefficients of §22 vanish; $H_5 \equiv 0$) and $s = 8$ (the cone of H_s is the axes cone, $\Delta = 144 = 12^2$).*

Status (Category A). As proved in §§22–23.

29.3 The case $s = 1$

$g_1 = f_0$, the fifth-order mock theta function (Hickerson–Mortenson [8]). At $s = 1$ the lattice $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has $\det = -s(s-2) = 1 > 0$ and is positive definite, so the indefinite part of the construction $(H_s, C, \mathbb{Q}(\sqrt{s(s-2)}))$ does not specialise; H_s is Zwegers-admissible only for integer $s \geq 5$. Theorem 25.2 gives a proved radial expansion of g_s uniform in s, including $s = 1$. Quantum modularity of g_s is not claimed.

29.4 A formal contact with A_5

$\mathrm{PSL}_2(\mathbb{F}_5) \cong A_5$. At $s = 7$ the label representation has the tensor factor $\rho_{2x^2/5}$ (Theorem 19.3); $V_{\mathrm{Eis}} = A_3 \otimes W_5^+ \otimes W_7^+$ contains W_5^+ (dimension 3, factoring projectively through A_5) and $V_{\mathrm{cusp}} = B_1 \otimes W_5^- \otimes W_7^-$ contains W_5^- (dimension 2, a faithful representation of $\mathrm{SL}_2(\mathbb{F}_5) = 2 \cdot A_5$). This is standard theory together with Theorem 19.3. Numerically (Category B) the S, T images on $W_5^{\pm}$ generate groups of order 120 with scalar subgroup $\{\pm 1\}$ and quotient of order 60. Any further reading of this contact is Category B.

29.5 Status of three questions

For reference we record the status of three questions about C: (N1) a map from E_8 or PW data that emits the coefficients; (N2) a unique completed period; (N3) a tail. (These labels are unrelated to the lattices N_1, N_2 of §26.)

question	status	category
(N1) emitting map	No coefficient-emitting map in the classes of Propositions 28.1–28.4; such data select s but do not emit coefficients (§28). Not excluded: Remark 28.5.	A for the stated obstructions
(N2) completed period	Scalar version excluded (Theorem 11.5); existence as the 140-component vector cocycle (Theorems 11.6, 11.7, 11.9, 11.10). Unique given the positive-frequency q-series (with q^0) or Φ on $(0, \infty)$ (Theorem 27.7(i)); not unique under the weak conditions: residual ambiguity $\mathbb{C}\Phi_{v_0} \oplus \{\Phi_c : c \in S(\rho)\}$ with $\dim S(\rho) = 1$ (Theorem 27.13), and the κ-level gauge $\mathbb{C}g_0$.	A (computer-assisted for $\dim S(\rho) = 1$); a correspondence covering the non-cusp part open

question	status	category
(N3) tail	Exact closed forms at $x = 0$ through $t^{17/2}$, at $x = \frac{1}{2}, \frac{1}{3}$ through $t^{13/2}$, at $x = \frac{1}{4}, \frac{1}{6}$ for $t^{11/2}, t^{13/2}$, the period-72 theorem at $x = \frac{1}{3}$, and the Gauss evaluation and CFP at general $x = a/N$ (§22). Not given in closed form at all rationals; $t^{19/2}$ onward not computed.	A

30 Numerical verification

"Exact" means rational/integer arithmetic. An entry that is exact supports a Category A statement; a floating-point entry is supporting evidence for a proved statement where one exists, and is otherwise Category B.

check	range / points	result
Theorem 11.1 identities	through q^{3000}	exact
$C_{\text{mixed}} = 0$	through q^{293}	exact; $-\frac{1}{2}q^{294}$ first term
cone sum (12.1), four cones; $E_{-a} = E_a$	through q^{3000}	exact
H_s cone decomposition	through q^{2000}	exact
Gauss pattern (Thm. 11.4)	3374 pairs, $m \le 105$	0 mismatches
$\zeta_E(0)$, $\zeta_E(-1)$	$1 + 5$ Shintani cones	exact $\frac{1}{10}$, $\frac{29}{1200}$
Radial fit, P_s^{orth}	$t \in [0.00125, 0.08]$	0.209419 (I), 0.52569 (a_P)
EM remainder (Table 13)	$t \ge 2.5 \cdot 10^{-5}$	scales as $\sqrt{t}$ (t for E)
region $\leftrightarrow$ ray class; u_L Eisenstein	$n \le 60\,000$	exact (max diff 0)

Table 19: Exact and finite checks.

check	points	max deviation
Maass S-, T-law, P-invariance, all 140 rows	3 points, $y \approx 1$	$1.2 \cdot 10^{-15}$, $7 \cdot 10^{-16}$, $6 \cdot 10^{-16}$
$S^2 = I$, $(ST)^3 = I$ on 72-dim space	—	$1.3 \cdot 10^{-15}$
Green constant $-\pi/2$ (cross-check)	$n = 1, 2, 3$, two z	30 digits
period formula vs. Bernoulli ψ	84 rationals	$1.1 \cdot 10^{-13}$
all-row closed form; three-term relation	7 pts; 4 pts	$2.3 \cdot 10^{-14}$; $7 \cdot 10^{-15}$
component fits (140×2)	$n \le 54\,880$	$4.4 \cdot 10^{-14}$
Thm. 11.9, 34 digits	44 rationals in $[\frac{1}{8}, 2]$	$5.7 \cdot 10^{-33}$
Thm. 11.9, random rationals	40, den. ≤ 60	$2.5 \cdot 10^{-12}$
Thm. 11.9 on $\mathbb{H}$ (36 digits)	6 points	$5.5 \cdot 10^{-35}$
Thm. 11.9 off $\mathbb{H}$ (double precision)	$5 + 3$ points	$2.5 \cdot 10^{-16}$; $9.2 \cdot 10^{-16}$
Thm. 11.9 at rationals (36 digits)	8 rationals	$5.7 \cdot 10^{-35}$
split $G_\chi = 2R_\chi + T_\chi$	8 chars, 5 points	$\le 8 \cdot 10^{-34}$
Thm. 11.10	$N \le 14\,000$	$1.84 \cdot 10^{-15}$; 627 coeffs equal in $\mathbb{Z}[\phi]$
$\Omega = \Psi$ ($\operatorname{Re} z > 0$), $\Omega = -\Psi^-$ ($\operatorname{Re} z < 0$)	$3 + 3$ points	$\approx 10^{-16}$
cusp rows = $L(0)$ combinations	24 rows, 7 points	$4 \cdot 10^{-14}$

Table 20: Numerical checks. For the rational comparisons at rationals an independent from-scratch period formula at double precision agrees to $5.7 \cdot 10^{-17}$.

x	Φ_c dev. (24 rows)	$\Phi_{73}(x)$
1	$3.8 \cdot 10^{-15}$	$0.120446470003 + 0.176437422094i$
-1	$4.5 \cdot 10^{-15}$	$0.006264073515 + 0.213537531676i$
2	$9.6 \cdot 10^{-15}$	$0.206217773055 + 0.263200146342i$
$7/4$	$3.4 \cdot 10^{-13}$	$0.188336862451 + 0.245368413466i$
$1/3$	$6.7 \cdot 10^{-15}$	$0.038631276713 + 0.077116144808i$
$3/5$	$4.0 \cdot 10^{-14}$	$0.073437646967 + 0.123627119436i$
$-2/3$	$1.2 \cdot 10^{-14}$	$0.012354928128 + 0.156183157173i$

Table 21: Cusp-row cocycle from $L(0)$ values vs. exact Bernoulli sums.

31 External dependencies

The following published results are used. Each is cited, not re-proved; "we use [X]" in the text refers to this list.

result	where used
Baker's theorem [1]	Theorem 11.5(i); the obstruction at the cusp pair (§14.1); Proposition 29.1
q-expansion principle (Shimura)	Theorem 11.5
Hecke's continuation theorem [7]	Theorem 22.1(vi); Lemma 24.3; §24
Siegel–Klingen [15]	rationality of the partial zeta values in Theorem 11.3
Congruence theorem for Weil representations [16]	Theorems 19.3, 19.5, 20.1(iii); Proposition 19.4(iii)
Nobs–Wolfart / Cliff–McNeilly–Szechtman / Scheithauer, decomposition of Weil representations of $\mathbb{Z}/p^2$	(C3) of Theorem 20.11 for primes not covered by Propositions 20.14, 20.15
Zwegers, thesis Cor. 2.9 [18]	modular completion of H_s (Theorem 22.1)
Zwegers 2012 [19] via [2]	identification of $\int \Xi_0$ in Theorem 24.1(iii) (also proved directly, Proposition 24.2)
Chevalley's ambiguous class number formula; Gauss genus theory; Tate's $\ker \mathrm{N} = C_K^{1-\sigma}$	Theorem 21.3; Proposition 21.1(i)
Maass's dihedral construction [11]	the identification of V_{cusp} with dihedral forms (§16.2; Category B parts)
Nikulin [13], Thm. 1.14.4, Cor. 1.5.2	Theorems 26.7, 26.9
Morrison [12], Cor. 2.9(i)	Theorem 26.7; Proposition 26.10(iii)
Deitmar–Hilgert [5], Thms. 3.3, 4.4	Remark 27.16(a)
Arb ball arithmetic (python-flint)	Theorem 27.13 (computer-assisted)
Strong Torelli theorem [9]	Theorem 26.9
Gromov–Yomdin [6]	entropy in Theorem 26.9
Standard Bessel asymptotics (DLMF 10.40.2)	Lemma 15.1

PARI dependencies. The following use PARI/GP and are otherwise unverified: the ray-class data at $\mathfrak{p}_2^3\mathfrak{p}_5\mathfrak{p}_7\infty_1\infty_2$ used for the 140-component fits and the Eisenstein/cusp classification of

the components; `lfun` values for $L(\frac{1}{2},\chi)$, $L(1,\chi)$ and root numbers; the numerical side of Theorem 11.11; the dihedral enumeration of Remark 27.15. Not dependent on PARI: u_L Eisenstein, $\mathrm{Cl}(K)$, $|\mathrm{Cl}_{\mathfrak{f}}^{+}| = 16$, the structure of $\mathrm{Cl}^{+}_{(2\sqrt{35})\infty_1\infty_2}$ and the 12 values $L(0,\chi)$ (§18.1).

32 Open problems

1. Prove the cusp-row identities of Theorem 11.11 (currently Category B).
2. Prove the symmetry $\psi(-x) = \overline{\psi(x)}$ (observed numerically to $8 \cdot 10^{-14}$; not used).
3. An arithmetic (Gauss-sum or ideal-theoretic) proof of Theorem 11.10.
4. Whether Zwegers' completed theta transforms under the same 140-dimensional ρ (Remark 25.3(e)); whether an exotic single kernel produces both sectors of C_{third}.
5. Objects whose cocycles are of size $\geq 1/y$ are not excluded by Theorems 24.5, 24.4.
6. $\dim V = s(s-1)$ for all odd s (the lower bound); equality $d_{\text{Eis}} = 3\prod(p^e+1)/2$ in general; $d_2(k,v)$ for $k \geq 7$.
7. Sharpness of the Proposition K1 bounds for $m = 2k^2 - 1$ with $k \geq 6$ and on the full space ($\dim V(578) \leq 83532$); composite m and g-consistency for all odd m in Proposition 20.4.
8. A uniform proof of the prime-square decompositions; (C3) for general p without citation; irreducibility of the complements at $s = 121$ and $s = 169$; the label-space decomposition at $s = 3481$ (whose p-local invariants are out of reach in label space).
9. The ambiguous-class count $|\bar{G}^{\sigma}| = 2^{\omega(s)}\varphi(s-2)$ in general; sufficiency of the modulus $M = 2dc$ for all norms.
10. The pattern of S_e for all odd e; the identity $R_r = S_r^{\text{ax}}$ at general order; closed forms at general rationals; the $t^{19/2}$ coefficient and beyond.
11. A period-function correspondence for ρ at $\lambda = \frac{1}{4}$ that also covers the non-cusp element u_{lift} (Remark 27.16(a)).
12. Whether γ itself lifts on the rank-6 Noether–Lefschetz locus of Proposition 26.10(iii) (Remark 26.11(b)); any link between Noether–Lefschetz series and the residual objects beyond the shared representation.
13. Coefficient-emitting maps outside the classes of §28 (Remark 28.5).
14. Quantum modularity of scalar C and of g_s (not claimed).

Part III
The 35-component law vector and the normalized Bridge Series

33 The 35-component law vector

Notation is that of §2. In Part III, x denotes a rational number h/m with $m \geq 1$, $\gcd(h,m)=1$ (and $m=1$ for $x=0$).

33.1 Gauss means

Definition 33.1. For $\alpha,\beta \in \mathbb{Q}^2$ and $x \in \mathbb{Q}$ the function $\nu \mapsto \mathbf{e}(xQ(\nu)+B(\nu,\beta))$ on $\alpha+\mathbb{Z}^2$ is invariant under $\nu \mapsto \nu+Mu$ ($u \in \mathbb{Z}^2$) for every positive integer M divisible by a suitable $M_0(\alpha,\beta,x)$, because $xQ(\nu+Mu)+B(\nu+Mu,\beta)-xQ(\nu)-B(\nu,\beta) = xMB(\nu,u)+xM^2Q(u)+MB(u,\beta)$ and $B(\nu,u)$, $Q(u)$, $B(u,\beta)$ have bounded denominators. The *Gauss mean* is

$$G_{\alpha,\beta}(x) = M^{-2} \sum_{n \in \mathbb{Z}^2/M\mathbb{Z}^2} \mathbf{e}\big(xQ(\alpha+n)+B(\alpha+n,\beta)\big),$$

which does not depend on the admissible M.

Let $h_A = (\tfrac{1}{2},0)$. For $\mu \in D$ we use the representative of μ in $[0,1)^2$.

Definition 33.2 (The 35-component law vector)**.** For $\mu \in D$ put

$$K1_\mu = G_{a+\mu,a}, \quad K2_\mu = G_{a+\mu,0}, \quad K3_\mu = G_{\mu,a}, \quad K4_\mu = G_{\mu,0}.$$

The vector $(K2_\mu)_{\mu\in D}$ is the *35-component law vector*; $K3$ and $K4$ are its S- and T-companions. $\mathcal{V}_{35}$ denotes the complex span of the 105 functions $K2_\mu, K3_\mu, K4_\mu$ on $\mathbb{Q}$.

Lemma 33.3 (Elementary rules)**.** *For $\alpha,\beta \in \mathbb{Q}^2$ and $x \in \mathbb{Q}$:*

(i) $G_{\alpha+n,\beta} = G_{\alpha,\beta}$ *for* $n \in \mathbb{Z}^2$*;*

(ii) $G_{\alpha,\beta+\lambda} = \mathbf{e}(B(\alpha,\lambda))\, G_{\alpha,\beta}$ *whenever* $A\lambda \in \mathbb{Z}^2$*;*

(iii) $G_{g\alpha,g\beta} = G_{\alpha,\beta}$ *for* $g \in \mathrm{GL}_2(\mathbb{Z})$ *with* $g^{\mathsf{T}}Ag = A$.

Proof. (i) $\alpha+n+\mathbb{Z}^2 = \alpha+\mathbb{Z}^2$. (ii) For $\nu = \alpha+n$, $B(\nu,\lambda) = B(\alpha,\lambda)+n^{\mathsf{T}}A\lambda \in B(\alpha,\lambda)+\mathbb{Z}$. (iii) $\nu \mapsto g\nu$ maps $\alpha+\mathbb{Z}^2$ bijectively onto $g\alpha+\mathbb{Z}^2$, commutes with translation by $M\mathbb{Z}^2$ up to the bijection g of $\mathbb{Z}^2/M\mathbb{Z}^2$, and $Q(g\nu)=Q(\nu)$, $B(g\nu,g\beta)=B(\nu,\beta)$. □

Theorem 33.4 (T-law and S-law)**.** *For $\alpha,\beta \in \mathbb{Q}^2$ and $x \in \mathbb{Q}$,*

$$G_{\alpha,\beta}(x+1) = \mathbf{e}\big(-Q(\alpha)-B(\alpha,h_A)\big)\, G_{\alpha,\beta+\alpha+h_A}(x),$$
$$G_{\alpha,\beta}(-1/x) = \frac{\mathbf{e}(B(\alpha,\beta))}{|x|\sqrt{35}} \sum_{p\in D} G_{\beta+p,-\alpha}(x) \qquad (x \neq 0).$$

Proof. T-law. For $n \in \mathbb{Z}^2$, $Q(n) = n_1^2 + 7n_1n_2 + \frac{7}{2}n_2^2 \equiv \frac{7}{2}n_2 \equiv B(n, h_A) \pmod 1$, since $n_2^2 \equiv n_2 \pmod 2$ and $B(n, h_A) = n_1 + \frac{7}{2}n_2$. For $\nu = \alpha + n$, $Q(\nu) = Q(\alpha) + B(n, \alpha) + Q(n) \equiv Q(\alpha) + B(n, \alpha) + B(n, h_A)$, while $B(\nu, \alpha + h_A) = 2Q(\alpha) + B(\alpha, h_A) + B(n, \alpha) + B(n, h_A)$. Hence $Q(\nu) \equiv B(\nu, \alpha + h_A) - Q(\alpha) - B(\alpha, h_A) \pmod 1$, and inserting this into the mean defining $G_{\alpha,\beta}(x + 1)$ gives the law.

S-law. Fix $y \in \mathbb{Q} \setminus \{0\}$ and $t > 0$, and let $Z = yA + itI$. Then Z is symmetric with $\operatorname{Im} Z = tI$ positive definite, so $f(v) = \exp(\pi i v^{\mathsf{T}} Z v + 2\pi i v^{\mathsf{T}} A\beta)$ is a Schwartz function on $\mathbb{R}^2$. Its Fourier transform is $\hat{f}(w) = \det(-iZ)^{-1/2} \exp(-\pi i (w - A\beta)^{\mathsf{T}} Z^{-1} (w - A\beta))$, where $\det(-iZ)^{1/2}$ is the branch that is positive for $y = 0$; diagonalising A by an orthogonal matrix, with eigenvalues $\lambda_1 > 0 > \lambda_2$, it equals $\prod_j (t - iy\lambda_j)^{1/2}$ with principal square roots. Poisson summation for $v \mapsto f(\alpha + v)$ gives

$$\sum_{\nu \in \alpha + \mathbb{Z}^2} f(\nu) = \det(-iZ)^{-1/2} \sum_{w \in \mathbb{Z}^2} \mathbf{e}(w^{\mathsf{T}} \alpha) \exp\bigl(-\pi i (w - A\beta)^{\mathsf{T}} Z^{-1} (w - A\beta)\bigr). \tag{33.1}$$

Left side. $f(\nu) = \mathbf{e}(yQ(\nu) + B(\nu, \beta)) e^{-\pi t |\nu|^2}$. The phase is $M\mathbb{Z}^2$-periodic on $\alpha + \mathbb{Z}^2$; for each residue class $r + M\mathbb{Z}^2$, Poisson summation gives $\sum_{n \in r + M\mathbb{Z}^2} e^{-\pi t |n|^2} = (M^2 t)^{-1} (1 + O(e^{-c/t}))$ with $c = \pi / M^2$. Hence the left side is $t^{-1} G_{\alpha,\beta}(y) + O(t^{-1} e^{-c/t})$.

Right side. Put $w - A\beta = -A\gamma$; as w runs over $\mathbb{Z}^2$, $\gamma = \beta - A^{-1}w$ runs over $\beta + A^{-1}\mathbb{Z}^2$, and $w^{\mathsf{T}}\alpha = (\beta - \gamma)^{\mathsf{T}} A \alpha = B(\alpha, \beta) - B(\gamma, \alpha)$. The exponent is $-\pi i \gamma^{\mathsf{T}} A Z^{-1} A \gamma$. Since $Z^{-1} = (yA)^{-1}(I + it(yA)^{-1})^{-1}$, one has $AZ^{-1}A = y^{-1}A - ity^{-2}I + t^2 R_t$ with $\|R_t\|$ bounded for $t \le t_0$. Hence the summand is

$$\mathbf{e}\bigl(B(\alpha, \beta) - B(\gamma, \alpha) - Q(\gamma)/y\bigr)\, e^{-\pi t |\gamma|^2 / y^2} \exp(-\pi i t^2 \gamma^{\mathsf{T}} R_t \gamma).$$

On the support of the Gaussian the last factor is $1 + O(t^2 |\gamma|^2)$, and $\sum_\gamma t^2 |\gamma|^2 e^{-\pi t |\gamma|^2 / y^2} = O(1)$. The phase $\gamma \mapsto \mathbf{e}(-B(\gamma, \alpha) - Q(\gamma)/y)$ is periodic on $\beta + A^{-1}\mathbb{Z}^2$, a coset of a lattice of covolume $1/35$; its mean over that coset is $\frac{1}{35} \sum_{p \in D} G_{\beta + p, -\alpha}(-1/y)$, because $\beta + A^{-1}\mathbb{Z}^2 = \bigsqcup_{p \in D} (\beta + p + \mathbb{Z}^2)$. The same residue-class argument as on the left gives the right side as $\det(-iZ)^{-1/2}\, \mathbf{e}(B(\alpha, \beta))\, 35 \cdot \frac{y^2}{t} \cdot \frac{1}{35} \sum_p G_{\beta + p, -\alpha}(-1/y) + O(1)$. Finally $\det(-iZ)^{1/2} \to (-iy\lambda_1)^{1/2} (-iy\lambda_2)^{1/2} = |y| \sqrt{|\lambda_1 \lambda_2|} = |y| \sqrt{35}$, since the two arguments are $\mp \pi/2$ and $\pm \pi/2$.

Multiplying (33.1) by t and letting $t \to 0^+$ gives $G_{\alpha,\beta}(y) = \frac{|y|}{\sqrt{35}} \mathbf{e}(B(\alpha, \beta)) \sum_{p \in D} G_{\beta + p, -\alpha}(-1/y)$. Substituting $y = -1/x$ gives the S-law. □

Proposition 33.5 (Closure)**.** *The laws of Theorem 33.4, followed by the reductions of Lemma 33.3(i),(ii), map each of the* 105 *functions $K2_\mu, K3_\mu, K4_\mu$ to a linear combination of them with coefficients in $\mathbb{Q}(\zeta_{9800}, \sqrt{35})$: T maps $K2$ to $K2$ and exchanges $K3$ and $K4$; S maps $K2$ into the span of $K3$, $K3$ into the span of $K2$, and $K4$ into the span of $K4$. Hence $\mathcal{V}_{35}$ is stable under $x \mapsto x + 1$ and under $x \mapsto -1/x$ with the factor $|x|^{-1}$.*

Proof. $a - h_A = (\frac{1}{5}, -\frac{1}{5})$ and $A(a - h_A) = (-1, 0)$, so $a \equiv h_A$ modulo $A^{-1}\mathbb{Z}^2$; $A(2a) = (0, 7)$, so $2a \in A^{-1}\mathbb{Z}^2$; and $D \subset A^{-1}\mathbb{Z}^2 / \mathbb{Z}^2$. *$T$.* For $K2_\mu$ the new second argument is $a + \mu + h_A \equiv 2a + \mu \equiv 0$; for $K3_\mu = G_{\mu, a}$ it is $a + \mu + h_A \equiv 0$ (giving $K4_\mu$); for $K4_\mu$ it is $\mu + h_A \equiv a$ (giving $K3_\mu$); all congruences modulo $A^{-1}\mathbb{Z}^2$, where Lemma 33.3(ii) applies. *S.* $K2_\mu \mapsto \sum_p G_{p, -a-\mu}$ and $-a - \mu \equiv a$; $K3_\mu \mapsto \sum_p G_{a+p, -\mu}$ and $-\mu \equiv 0$; $K4_\mu \mapsto \sum_p G_{p, -\mu}$ and $-\mu \equiv 0$. The phases introduced are $\mathbf{e}(B(\alpha, \lambda))$ with $\alpha \in \frac{1}{70}\mathbb{Z}^2$ and $\lambda \in A^{-1}\mathbb{Z}^2$, roots of unity of order dividing 9800. □

Proposition 33.6 (Dimension; computer-assisted)**.** $\dim \mathcal{V}_{35} = 36$.

Proof. Upper bound. Lemma 33.3(iii), applied to twelve automorphs g of A (including $\pm I$ and $\gamma = \left(\begin{smallmatrix} -1 & -7 \\ 2 & 13 \end{smallmatrix}\right)$), followed by (i) and (ii), shows that each of the 105 functions equals a root of unity times one of 36 orbit representatives, 12 in each family (exact rational computation, `A35_lawspace_exact/orbits.py`). *Lower bound.* At the 216 rationals $X_{216} = \{h/m : 1 \le m \le 15,\ 1 \le h \le 3m,\ \gcd(h,m) = 1\}$ every value $G_{\alpha,\beta}(h/m)$ is a finite mean over a $4m \times 4m$ box and lies in $R = \mathbb{Z}[\zeta_L][1/(4m)^2]$ with $L = 3\,531\,528\,000$. For a prime $p \equiv 1 \pmod{L}$, sending ζ_L to a primitive L-th root of unity in $\mathbb{F}_p$ defines a ring homomorphism $R \to \mathbb{F}_p$, under which the rank of a matrix cannot increase. The 216×105 value matrix has rank 36 modulo each of $p = 17\,657\,640\,001$, $21\,189\,168\,001$, $35\,315\,280\,001$, $42\,378\,336\,001$ (`A51_bridgeC_exact/rank216.py`). So the rank in characteristic 0 is at least 36. □

Proposition 33.7 (Exact SL_2-relations and irreducibility; computer-assisted)**.** *Let M_S, M_T be the matrices of S and T on $\mathcal{V}_{35}$ in the basis of orbit representatives.*

(i) $M_S^4 = I$ *and* $(M_S M_T)^3 = M_S^2$ *exactly over* $\mathbb{Q}(\zeta_{9800})$*, with* $\sqrt{35} = -i\sum_{k \bmod 35} \left(\frac{k}{35}\right)\zeta_{35}^k$.

(ii) $\dim \mathrm{End}_{\mathrm{SL}_2(\mathbb{Z})}(\mathcal{V}_{35}) = 1$*, and $\mathcal{V}_{35}$ is isomorphic to the 36-dimensional summand $\mathcal{E}_{36}$ of the decomposition $\mathcal{E}_{36} \oplus \mathcal{C}_6$ of the 42-dimensional lift representation of Part II at $s = 7$; $\mathcal{E}_{36}$ and $\mathcal{C}_6$ are irreducible.*

In particular every non-zero vector of $\mathcal{V}_{35}$ generates $\mathcal{V}_{35}$ under S and T.

Proof. (i) Exact reduction modulo the cyclotomic polynomial Φ_{9800} of all 1296 entries (`A51_bridgeC_exact/sl2_exact.py`). (ii) The intertwining equations were solved over $\mathbb{F}_p$ at two primes; a dimension over $\mathbb{F}_p$ bounds the dimension in characteristic 0 from above, and the identity gives $\dim \mathrm{End} \ge 1$. An explicit isomorphism $X : \mathcal{V}_{35} \to \mathcal{E}_{36}$ was verified exactly modulo Φ_{58800} ($X\,S_V = S_E\,X$ in all 1296 entries, X a product of invertible matrices; `A35_lawspace_exact/exact_intertwiner.py`). The lift representation is unitary, hence semisimple, and its commutant has dimension 2 (exact upper bound, `A32_35_to_rho7/exact_hom.py`); with two non-zero invariant summands $\mathcal{E}_{36}$, $\mathcal{C}_6$ this forces both to be irreducible and non-isomorphic. The final statement holds for any irreducible representation. □

34 The Bridge Series and the normalized Bridge Series at leading order

34.1 Companion series

Definition 34.1. Let $K_+ = \{(n,k) \in \mathbb{R}^2 : n \ge 0,\ n + 21k \ge 0\}$ and $K_- = -K_+$. For $v \in \mathbb{Z}^2$ let $\mathrm{wt}(v) = \mathbf{1}[v_1 \ge 0, v_2 \ge 0] - w(v)$. For each of the 105 pairs (α,β) of Definition 33.2 ($\alpha \in \{a+\mu, \mu\}$, $\beta \in \{0, a\}$, excluding the $K1$ pairs) the *companion series* is

$$S_{\alpha,\beta}(\tau) = \sum_{v \in \mathbb{Z}^2} \mathrm{wt}(v)\,\mathbf{e}\big(B(\alpha+v,\beta)\big)\,\mathbf{e}\big(\tau Q(\alpha+v)\big).$$

Since $Q(a+v) = Q_7(v) - \frac{7}{20}$, $S_{a,0} = q^{-7/20}(P_7^{\mathrm{orth}} - H_7) = q^{-7/20}C$: the *normalized Bridge Series*.

Lemma 34.2 (Weights)**.** $B(c_1,v) = 5v_1$, $B(c_2,v) = -v_1 - 21v_2$, *so* $w(v) = \frac{1}{2}(\mathrm{sgn}\,v_1 + \mathrm{sgn}(v_1 + 21v_2))$. *Hence $w = 1$ on the interior of K_+, $w = -1$ on the interior of K_-, $w = \pm\frac{1}{2}$ on the boundary rays of $K_\pm$ other than 0, and $w = 0$ elsewhere. The series $S_{\alpha,\beta}$ converges absolutely for $\tau \in \mathbb{H}$.*

Proof. The formulas for B are direct. If $w(v) \neq 0$ then $v \in K_+ \cup K_-$, and $Q(\xi u_1 + \eta u_2) = \frac{595}{2}\xi^2 + 140\xi\eta + \frac{7}{2}\eta^2$ with $u_1 = (21, -1)$, $u_2 = (0, 1)$ is positive definite on $\xi, \eta \geq 0$; on the orthant $Q(n, k) \geq n^2 + \frac{7}{2}k^2$. So $\operatorname{Re}(-2\pi i\tau Q) \leq -c|v|^2$ on the support of wt, for some $c > 0$ depending on $\operatorname{Im}\tau$. □

34.2 Decomposition into pieces

A *piece* is a triple (c, U, σ) with $c \in \mathbb{Z}^2$, $U = (u_1\, u_2) \in M_2(\mathbb{Z})$ and a sign σ; it stands for the sum over $c + U\mathbb{Z}^2_{\geq 0}$. A *ray* is a pair (c, u) with weight σ, standing for the sum over $c + u\mathbb{N}$.

Proposition 34.3 (Exact decomposition)**.** *Let $u_1 = (21, -1)$, $u_2 = (0, 1)$, $e_1 = (1, 0)$. For every function Φ on $\mathbb{Z}^2$ with $\sum |\operatorname{wt}(v)\Phi(v)| < \infty$ and $\sum_{K_\pm \cap \mathbb{Z}^2} |\Phi| < \infty$,*

$$\sum_v \operatorname{wt}(v)\Phi(v) = \sum_{z \in \mathbb{Z}^2_{\geq 0}} \Phi(z) - \sum_{r=0}^{20}\sum_{z \geq 0} \Phi(re_1 + Uz) + \sum_{r=0}^{20}\sum_{z \geq 0} \Phi(-re_1 - Uz) + \tfrac{1}{2}\sum_{b \geq 0}\big(\Phi(bu_2) + \Phi(bu_1) - \Phi(-bu_2) - \Phi(-bu_1)\big),$$

with $U = (u_1\, u_2)$.

Proof. For $(n, k) \in \mathbb{Z}^2$ write $n = 21j + r$ with $0 \leq r < 21$. Then $n \geq 0 \iff j \geq 0$, and $n + 21k \geq 0 \iff j + k \geq 0$ because $0 \leq r < 21$. As $(n, k) = re_1 + ju_1 + (j + k)u_2$, the map $(r, j, i) \mapsto re_1 + ju_1 + iu_2$ is a bijection from $\{0, \dots, 20\} \times \mathbb{Z}^2_{\geq 0}$ onto $K_+ \cap \mathbb{Z}^2$. The two boundary rays of K_+ are $\mathbb{N}u_2$ ($n = 0$) and $\mathbb{N}u_1$ ($n + 21k = 0$); they meet only at 0. By Lemma 34.2, $\sum_v w(v)\Phi(v) = \big[\sum_{K_+} \Phi - \frac{1}{2}\sum_{\mathbb{N}u_1} \Phi - \frac{1}{2}\sum_{\mathbb{N}u_2} \Phi\big] - \big[\sum_{K_-} \Phi - \frac{1}{2}\sum_{-\mathbb{N}u_1} \Phi - \frac{1}{2}\sum_{-\mathbb{N}u_2} \Phi\big]$: interior points get ± 1, non-zero ray points $\pm\frac{1}{2}$, and 0 gets $(1 - \frac{1}{2} - \frac{1}{2}) - (1 - \frac{1}{2} - \frac{1}{2}) = 0$. Subtracting from the orthant sum gives the identity. □

For a piece (c, U) and a pair (α, β) put

$$p = Q(u_1), \quad r = Q(u_2), \quad \mathsf{q} = B(u_1, u_2), \quad \tilde{Q}(z) = Q(Uz) = pz_1^2 + \mathsf{q}z_1z_2 + rz_2^2,$$
$$l_i = B(\alpha + c, u_i), \quad Q_0 = Q(\alpha + c),$$

so that $Q(\alpha + c + Uz) = E(z) := \tilde{Q}(z) + l_1z_1 + l_2z_2 + Q_0$. For the orthant $(p, \mathsf{q}, r) = (1, 7, \frac{7}{2})$; for the pieces of $K_\pm$, $(p, \mathsf{q}, r) = (\frac{595}{2}, 140, \frac{7}{2})$. In both cases $p, r > 0$, $\mathsf{q} \geq 0$ and $\Delta := \mathsf{q}^2 - 4pr > 0$ ($\Delta = 35$ and $\Delta = 15435 = 35 \cdot 441$).

Lemma 34.4 (Periodicity)**.** *Let $x = h/m$, $N = 2m$, and $\varphi(\nu) = \mathbf{e}(xQ(\nu) + B(\nu, \beta))$ for $\nu \in \alpha + \mathbb{Z}^2$, with (α, β) one of the 105 pairs or $(\alpha, \beta) = (a, 0)$. Then $\varphi(\nu + Nu) = \varphi(\nu)$ for all $u \in \mathbb{Z}^2$. Consequently, for every piece, $f(z) := \varphi(\alpha + c + Uz)$ is N-periodic in each coordinate, and for every ray $b \mapsto \varphi(\alpha + c + bu)$ is N-periodic.*

Proof. $\varphi(\nu + Nu)/\varphi(\nu) = \mathbf{e}(xNB(\nu, u) + xN^2Q(u) + NB(u, \beta))$. Here $A\nu \in A\alpha + \mathbb{Z}^2$, and $A\alpha \in \mathbb{Z}^2$ (if $\alpha = \mu$) or $A\alpha \in (0, \frac{7}{2}) + \mathbb{Z}^2$ (if $\alpha = a + \mu$), since $A\mu \in \mathbb{Z}^2$. So $B(\nu, u) = u^{\mathsf{T}}A\nu \in \frac{1}{2}\mathbb{Z}$ and $xNB(\nu, u) \in h\mathbb{Z}$. $Q(u) \in \frac{1}{2}\mathbb{Z}$ gives $xN^2Q(u) \in 2mh\mathbb{Z}$. $B(u, 0) = 0$ and $B(u, a) = \frac{7}{2}u_2$ give $NB(u, \beta) \in m\mathbb{Z}$. □

34.3 Leading order

Theorem 34.5 (Profile of C and of $q^{-7/20}C$)**.** *Let $G(x) = (2m)^{-2}\sum_{v \bmod 2m} \mathbf{e}(xQ_7(v))$. Then $G(x) = \mathbf{e}(7x/20)\,K2_0(x)$ for every rational x, and $G \notin \mathcal{V}_{35}$ (computer-assisted).*

Proof. $Q_7(v) = Q(v+a) + \frac{7}{20}$ and $v \mapsto v+a$ maps $\mathbb{Z}^2$ onto $a+\mathbb{Z}^2$, so $G(x) = \mathbf{e}(7x/20)G_{a,0}(x) = \mathbf{e}(7x/20)K2_0(x)$. If $G \in \mathcal{V}_{35}$, then on X_{216} the column of values of G lies in the complex span of the 105 columns, and the 216×106 matrix has rank at most 36 over $\mathbb{Q}(\zeta_L)$, with the notation of the proof of Proposition 33.6 (the values of G also lie in R). Reduction modulo p cannot raise this rank. The computed rank modulo each of the four primes of Proposition 33.6 is 37, and each run verifies the identity $G = \mathbf{e}(7x/20)K2_0$ exactly modulo p at all 216 points (`BRIDGE_C_35/scripts/testA.py`). □

Theorem 34.6 (Radial asymptotics at leading order)**.** *Let $I = \log(6+\sqrt{35})/(2\sqrt{35})$. For every rational x, as $t \to 0^+$,*

$$t\,P_7^{\mathrm{orth}}(\mathbf{e}(x)e^{-t}) = I\,G(x) + O(t^{1/2}), \quad t\,H_7(\mathbf{e}(x)e^{-t}) = O(t^{1/2}), \quad t\,C(\mathbf{e}(x)e^{-t}) = I\,G(x) + O(t^{1/2}),$$

and $t\,S_{\alpha,\beta}(\mathbf{e}(x)e^{-t}) = I\,G_{\alpha,\beta}(x) + O(t^{1/2})$ for every companion series. In particular the leading profile of $q^{-7/20}C$ is $I\,K2_0 \in \mathcal{V}_{35}$, and that of C is $I\,G \notin \mathcal{V}_{35}$.

Proof. This is the t^{-1} part of Theorem 35.6 below, together with Corollary 35.7 and Theorem 34.5; the statements for P_7^{orth} and H_7 are the orthant term and the $K_\pm$ terms of the same computation. □

Theorem 34.7 (Vector-valued S-law of C at leading order)**.** *Let $\Lambda_C(y) = \lim_{t\to 0^+} t\,C(\mathbf{e}(y)e^{-t})$. For every rational $x > 0$,*

$$\Lambda_C(-1/x) = I\,\mathbf{e}\Big(-\frac{7}{20x}\Big)\frac{1}{x\sqrt{35}}\sum_{p\in D}\mathbf{e}(-B(p,2a))\,K3_p(x).$$

The leading profile $\Lambda_{a,0} = I\,K2_0$ of $q^{-7/20}C$, together with its images under $x \mapsto x+1$ and under $x \mapsto -1/x$ with the factor $|x|^{-1}$, spans $\mathcal{V}_{35}$; S and T act on it by the matrices of Proposition 33.7.

Proof. By Theorems 34.5 and 34.6, $\Lambda_C(y) = I\mathbf{e}(7y/20)K2_0(y)$. Theorem 33.4 with $(\alpha,\beta) = (a,0)$ gives $K2_0(-1/x) = (x\sqrt{35})^{-1}\sum_p G_{p,-a}(x)$. Since $A(2a) \in \mathbb{Z}^2$, Lemma 33.3(ii) with $\lambda = -2a$ gives $G_{p,-a} = G_{p,a-2a} = \mathbf{e}(-B(p,2a))G_{p,a} = \mathbf{e}(-B(p,2a))K3_p$. Multiply by $I\mathbf{e}(7(-1/x)/20)$. The spanning statement follows from $K2_0 \neq 0$ ($|K2_0(1)| = |G(1)| = 1$ by Theorem 11.4) and Proposition 33.7. □

Remark 34.8. By Theorem 11.4, $|\Lambda_C(x)|/I$ equals $\sqrt{7}/7$ at $x = \frac{1}{7}, \frac{3}{7}, \frac{5}{7}$, equals 1 at $x = 1$ and at $x = -1$, equals $\frac{1}{3}$ at $-\frac{1}{3}$, 0 at $-\frac{1}{5}$ and $\sqrt{7}/7$ at $-\frac{1}{7}$. Hence $|\Lambda_C(-1/(7x))|/(|7x|^{-1/2}|\Lambda_C(x)|)$ takes the values $\sqrt{7}, \sqrt{7/3}, 0, 1$ at $x = \frac{1}{7}, \frac{3}{7}, \frac{5}{7}, 1$: no scalar factor $|7x|^{-1/2}$ relates Λ_C to its W_7-image, in accordance with Theorem 11.5(iii).

35 The radial expansion through order t^0

35.1 One-variable lemmas

Lemma 35.1 (Euler–Maclaurin with offset)**.** *Let $N \ge 1$, $\rho \in \{0, \dots, N-1\}$, $y_0 = \rho/N$, and let $h \in C^3[0,\infty)$ satisfy $|h^{(i)}(u)| \le c_1 e^{-c_2 u^2}$ for $i \le 3$, with constants $c_1, c_2 > 0$. Then*

$$\sum_{j\ge 0} h(\rho + Nj) = N^{-1}\int_0^\infty h + R_1, \qquad |R_1| \le \tfrac12 |h(0)| + \tfrac12 \int_0^\infty |h'|,$$

$$\sum_{j\ge 0} h(\rho + Nj) = N^{-1}\int_0^\infty h - B_1(y_0)h(0) + R_2, \qquad |R_2| \le \tfrac{N}{6}\int_0^\infty |h''|,$$

$$\sum_{j\ge 0} h(\rho + Nj) = N^{-1}\int_0^\infty h - B_1(y_0)h(0) - \tfrac{N}{2}B_2(y_0)h'(0) + R_3, \qquad |R_3| \le \tfrac{N^2}{9}\int_0^\infty |h'''|.$$

Proof. Let $\phi(u) = h(Nu)$, so the sum is $\sum_{j\ge0}\phi(y_0+j)$. For $u \ge 0$ the counting function $S(u) = \#\{j \ge 0 : y_0 + j \le u\}$ equals $u + \frac12 - y_0 - \bar B_1(u - y_0) = u - B_1(y_0) - \bar B_1(u-y_0)$, and $S(0^-) = 0$. Stieltjes integration by parts on $[0,\infty)$ gives

$$\sum_{j\ge0}\phi(y_0+j) = -\int_0^\infty \phi'(u)S(u)\,du = \int_0^\infty \phi - B_1(y_0)\phi(0) + \int_0^\infty \phi'(u)\bar B_1(u-y_0)\,du.$$

Since $|\bar B_1| \le \frac12$ and $|B_1(y_0)| \le \frac12$, and $\int\phi = N^{-1}\int h$, $\int|\phi'| = \int|h'|$, the first statement follows. Using $\bar B_1 = \frac12 \bar B_2'$ and continuity of $\bar B_2$, $\int_0^\infty \phi'\bar B_1(u-y_0)du = -\frac12\phi'(0)B_2(y_0) - \frac12\int_0^\infty \phi''\bar B_2(u-y_0)du$ (note $\bar B_2(-y_0) = B_2(y_0)$). Since $|\phi'(0)| \le \int|\phi''|$ and $|\bar B_2| \le \frac16$, the remainder after $-B_1(y_0)\phi(0)$ is at most $\frac16\int|\phi''| = \frac{N}{6}\int|h''|$. One further integration by parts, with $\bar B_2 = \frac13 \bar B_3'$, gives $-\frac12\int\phi''\bar B_2 = \frac16\phi''(0)\bar B_3(-y_0) + \frac16\int\phi'''\bar B_3(u-y_0)du$; with $|\bar B_3| \le \sqrt3/36$ and $|\phi''(0)| \le \int|\phi'''|$ this is at most $\frac{\sqrt3}{108}\int|\phi'''| \le \frac19 N^2\int|h'''|$. Finally $\phi'(0) = Nh'(0)$. □

Lemma 35.2 (Gaussian bounds)**.** *Let $E(z) = \tilde Q(z) + l_1 z_1 + l_2 z_2 + Q_0$ with $\tilde Q(z) = pz_1^2 + \mathsf{q} z_1 z_2 + r z_2^2$, $p, r > 0$, $\mathsf{q} \ge 0$, and put $\kappa = \min(p, r)$ and $F_t = e^{-tE}$. For every multi-index a with $|a| \le 3$ there is C_a such that, for $0 < t \le 1$ and $z \in \mathbb{R}^2_{\ge0}$,*

$$|\partial^a F_t(z)| \le C_a\, t^{|a|/2}\, e^{-t\kappa|z|^2/4}.$$

Consequently, uniformly in the remaining variable, $\int_0^\infty |\partial^a F_t|\,dz_i = O(t^{(|a|-1)/2})$, $\iint_{\mathbb{R}^2_{\ge0}}|\partial^a F_t| = O(t^{|a|/2-1})$, and $\sum_{j\ge0} e^{-t\kappa(\rho+Nj)^2/4} = O(t^{-1/2})$.

Proof. On $z \ge 0$, $\tilde Q(z) \ge \kappa|z|^2$, so $tE(z) \ge t\kappa|z|^2 - t|l||z| - t|Q_0| \ge \frac{t\kappa}{2}|z|^2 - c_1$ with $c_1 = |l|^2/(2\kappa) + |Q_0|$. A derivative of order k of F_t is F_t times a sum of products of $k-m$ factors $-t\partial_j E$ (each $\le Ct(1+|z|)$) and m factors $-t\partial_{ij}E$ (each $\le Ct$), with $0 \le m \le k/2$. Such a product is at most $Ct^{k-m}(1+|z|)^{k-2m} = Ct^{k/2}(\sqrt t(1+|z|))^{k-2m}$. Since $\sqrt t(1+|z|) \le 1 + \sqrt t|z|$ and $(1+s)^j e^{-\kappa s^2/4}$ is bounded on $s \ge 0$, the bound follows. The consequences follow from $\int_0^\infty e^{-t\kappa s^2/4}ds = O(t^{-1/2})$ and from the first statement of Lemma 35.1. □

Lemma 35.3 (Integrals)**.** *With the notation of Lemma 35.2 and $\Delta = \mathsf{q}^2 - 4pr > 0$, as $t \to 0^+$,*

$$\iint_{\mathbb{R}^2_{\ge 0}} F_t = t^{-1}\mathrm{Ar} - t^{-1/2}(l_1 J_a + l_2 J_b) + \big(\tfrac{1}{2}K(l) - Q_0\mathrm{Ar}\big) + O(t^{1/2}),$$

$$\int_0^\infty F_t(s,0)\,ds = \tfrac{1}{2}\sqrt{\pi/(tp)} - \tfrac{l_1}{2p} + O(t^{1/2}), \qquad \int_0^\infty F_t(0,s)\,ds = \tfrac{1}{2}\sqrt{\pi/(tr)} - \tfrac{l_2}{2r} + O(t^{1/2}),$$

$$\int_0^\infty \partial_2 F_t(s,0)\,ds = -\tfrac{\mathsf{q}}{2p} + O(t^{1/2}), \qquad \int_0^\infty \partial_1 F_t(0,s)\,ds = -\tfrac{\mathsf{q}}{2r} + O(t^{1/2}), \qquad F_t(0,0) = 1 + O(t),$$

where $J_a = \iint u_1 e^{-\tilde{Q}(u)}du$, $J_b = \iint u_2 e^{-\tilde{Q}(u)}du$, $K(l) = \iint (l_1u_1 + l_2u_2)^2 e^{-\tilde{Q}(u)}du$ (over $\mathbb{R}^2_{\ge 0}$), and

$$\mathrm{Ar} = \iint_{\mathbb{R}^2_{\ge 0}} e^{-\tilde{Q}} = \frac{\log\frac{\mathsf{q}+\sqrt{\Delta}}{\mathsf{q}-\sqrt{\Delta}}}{2\sqrt{\Delta}}, \quad J_a = \frac{\sqrt{\pi}}{4}\cdot\frac{2}{4pr-\mathsf{q}^2}\Big(2\sqrt{r} - \frac{\mathsf{q}}{\sqrt{p}}\Big), \quad J_b = \frac{\sqrt{\pi}}{4}\cdot\frac{2}{4pr-\mathsf{q}^2}\Big(2\sqrt{p} - \frac{\mathsf{q}}{\sqrt{r}}\Big),$$

$$K(l) = \int_0^\infty \frac{(l_1 + l_2\lambda)^2}{2\tilde{Q}(1,\lambda)^2}\,d\lambda.$$

Proof. Substituting $z = u/\sqrt{t}$, $\iint F_t = t^{-1}\iint e^{-\tilde{Q}(u)-w(u)}du$ with $w = \sqrt{t}(l_1u_1 + l_2u_2) + tQ_0$. By Taylor's theorem $|e^{-w} - 1 + w - \frac{1}{2}w^2| \le \frac{1}{6}|w|^3 e^{|w|}$, and $e^{-\tilde{Q}(u)+|w(u)|} \le e^{-\kappa|u|^2+|l||u|+|Q_0|}$ is integrable, so the expansion follows with the t^0 term $\iint(\frac{1}{2}(l_1u_1 + l_2u_2)^2 - Q_0)e^{-\tilde{Q}}$. The one-variable integrals are treated in the same way with $\int_0^\infty e^{-pu^2}du = \frac{1}{2}\sqrt{\pi/p}$, $\int_0^\infty ue^{-pu^2}du = \frac{1}{2p}$. For the derivative integrals, $\partial_2 F_t(s,0) = -t(\mathsf{q}s + l_2)F_t(s,0)$, and the substitution $s = u/\sqrt{t}$ gives $-t\cdot t^{-1}\mathsf{q}\int ue^{-pu^2}du + O(t^{1/2})$. $F_t(0,0) = e^{-tQ_0}$. For the closed forms write $u = \rho(1,\lambda)$, $du = \rho\,d\rho\,d\lambda$. Then $\mathrm{Ar} = \int_0^\infty d\lambda/(2\tilde{Q}(1,\lambda))$; the roots of $r\lambda^2 + \mathsf{q}\lambda + p$ are $\lambda_\pm = (-\mathsf{q} \pm \sqrt{\Delta})/(2r) < 0$ since $\mathsf{q} > \sqrt{\Delta}$, and partial fractions give $\int_0^\infty d\lambda/(r(\lambda-\lambda_+)(\lambda-\lambda_-)) = \Delta^{-1/2}\log(\lambda_-/\lambda_+)$. Next $J_a = \int_0^\infty\int_0^\infty \rho^2 e^{-\rho^2\tilde{Q}(1,\lambda)}d\rho\,d\lambda = \frac{\sqrt{\pi}}{4}\int_0^\infty \tilde{Q}(1,\lambda)^{-3/2}d\lambda$, and $\frac{d}{d\lambda}\frac{2(2r\lambda+\mathsf{q})}{(4pr-\mathsf{q}^2)\sqrt{r\lambda^2+\mathsf{q}\lambda+p}} = (r\lambda^2+\mathsf{q}\lambda+p)^{-3/2}$; evaluating at 0 and ∞ gives J_a. J_b follows by the parametrisation $u = \rho(\lambda, 1)$, which exchanges p and r. $K(l)$ follows from $\int_0^\infty \rho^3 e^{-\rho^2 c}d\rho = 1/(2c^2)$. □

35.2 The expansion

Theorem 35.4 (Expansion of a piece)**.** *Let (p,q,r), l_1, l_2, Q_0 be as above with $p, r > 0$, $\mathsf{q} \ge 0$, $\Delta > 0$, and let $f : \mathbb{Z}^2_{\ge 0} \to \mathbb{C}$ be N-periodic in each coordinate with $|f| \le 1$. Put, with z running over $\{0,\dots,N-1\}^2$,*

$$\mathrm{mean}(f) = N^{-2}\sum_z f(z), \qquad W_i = N^{-1}\sum_z f(z)B_1(z_i/N),$$

$$V_i = \sum_z f(z)B_2(z_i/N), \qquad \mathsf{C} = \sum_z f(z)B_1(z_1/N)B_1(z_2/N).$$

Then, as $t \to 0^+$,

$$\sum_{z\in\mathbb{Z}^2_{\ge 0}} f(z)e^{-tE(z)} = t^{-1}A_0^P + t^{-1/2}A_1^P + A_2^P + O(t^{1/2}),$$

with

$$A_0^P = \mathrm{mean}(f)\,\mathrm{Ar},$$

$$A_1^P = -\,\mathrm{mean}(f)(l_1J_a + l_2J_b) - \tfrac{\sqrt{\pi}}{2}\big(r^{-1/2}W_1 + p^{-1/2}W_2\big),$$

$$A_2^P = \mathrm{mean}(f)\big(\tfrac{1}{2}K(l) - Q_0\mathrm{Ar}\big) + \frac{l_2W_1}{2r} + \frac{l_1W_2}{2p} + \frac{\mathsf{q}}{4}\Big(\frac{V_1}{r} + \frac{V_2}{p}\Big) + \mathsf{C}.$$

Proof. Write the sum as $\sum_\rho f(\rho)\Sigma_\rho$ with $\Sigma_\rho = \sum_{j\in\mathbb{Z}^2_{\ge 0}} F_t(\rho+Nj)$, $\rho\in\{0,\dots,N-1\}^2$, and $y_i=\rho_i/N$. Fix ρ. For each $z_1=\rho_1+Nj_1$ apply the third statement of Lemma 35.1 in z_2:

$$\Sigma_\rho = \sum_{j_1\ge 0}\Big[N^{-1}\int_0^\infty F_t(z_1,s)ds - B_1(y_2)F_t(z_1,0) - \tfrac{N}{2}B_2(y_2)\partial_2F_t(z_1,0) + R_3(z_1)\Big].$$

By Lemma 35.2, $\sum_{j_1}|R_3(z_1)| \le \frac{N^2}{9}\sum_{j_1}\int_0^\infty|\partial_2^3F_t(z_1,s)|ds \le Ct^{3/2}\cdot t^{-1/2}\cdot t^{-1/2} = O(t^{1/2})$. For the first term apply the third statement of Lemma 35.1 in z_1 to $g(z_1)=N^{-1}\int_0^\infty F_t(z_1,s)ds$ (differentiation under the integral is justified by Lemma 35.2); its remainder is at most $\frac{N^2}{9}\int|g'''| \le CN\iint|\partial_1^3F_t| = O(t^{1/2})$. This gives $N^{-2}\iint F_t - N^{-1}B_1(y_1)\int_0^\infty F_t(0,s)ds - \frac12 B_2(y_1)\int_0^\infty\partial_1F_t(0,s)ds + O(t^{1/2})$. For the second term apply the second statement of Lemma 35.1 to $F_t(\cdot,0)$; the remainder is at most $\frac{N}{6}\int|\partial_1^2F_t(s,0)|ds = O(t^{1/2})$. This gives $-B_1(y_2)[N^{-1}\int_0^\infty F_t(s,0)ds - B_1(y_1)F_t(0,0)] + O(t^{1/2})$. For the third term apply the first statement of Lemma 35.1 to $\partial_2F_t(\cdot,0)$; the remainder is at most $\frac12|\partial_2F_t(0,0)| + \frac12\int|\partial_1\partial_2F_t(s,0)|ds = O(t^{1/2})$. This gives $-\frac12 B_2(y_2)\int_0^\infty\partial_2F_t(s,0)ds + O(t^{1/2})$. Altogether

$$\begin{aligned}\Sigma_\rho =& N^{-2}\iint F_t - N^{-1}B_1(y_1)\int_0^\infty F_t(0,s)ds - N^{-1}B_1(y_2)\int_0^\infty F_t(s,0)ds + B_1(y_1)B_1(y_2)F_t(0,0)\\ &- \tfrac12 B_2(y_1)\int_0^\infty\partial_1F_t(0,s)ds - \tfrac12 B_2(y_2)\int_0^\infty\partial_2F_t(s,0)ds + O(t^{1/2}).\end{aligned}$$

Insert Lemma 35.3, multiply by $f(\rho)$ and sum over ρ. The coefficient of t^{-1} is $N^{-2}\sum f\cdot \mathrm{Ar}$. The coefficient of $t^{-1/2}$ is $-\operatorname{mean}(f)(l_1J_a+l_2J_b) - N^{-1}\sum_\rho f(\rho)B_1(y_1)\frac{\sqrt\pi}{2\sqrt r} - N^{-1}\sum_\rho f(\rho)B_1(y_2)\frac{\sqrt\pi}{2\sqrt p}$. The t^0 coefficient is $\operatorname{mean}(f)(\frac12 K - Q_0\mathrm{Ar}) + W_1\frac{l_2}{2r} + W_2\frac{l_1}{2p} + \mathsf{C} + \frac12 V_1\frac{\mathfrak{q}}{2r} + \frac12 V_2\frac{\mathfrak{q}}{2p}$. □

Lemma 35.5 (Expansion of a ray)**.** *Let $Q(u)>0$, $l_u = B(u,\alpha+c)$, and let $b\mapsto f(b)$ be N-periodic with $|f|\le 1$. Then*

$$\sum_{b\ge 0} f(b)e^{-tQ(\alpha+c+bu)} = t^{-1/2}\frac{\sqrt\pi}{2}Q(u)^{-1/2}\operatorname{mean}(f) - \operatorname{mean}(f)\frac{l_u}{2Q(u)} - \sum_{b=0}^{N-1} f(b)B_1(b/N) + O(t^{1/2}),$$

with $\operatorname{mean}(f) = N^{-1}\sum_{b<N} f(b)$.

Proof. Here $h(b) = e^{-t(Q(u)b^2+l_ub+Q(\alpha+c))}$. Apply the second statement of Lemma 35.1 for each residue ρ; the remainder is $O(\int|h''|) = O(t^{1/2})$. Then $\int_0^\infty h = \frac12\sqrt{\pi/(tQ(u))} - \frac{l_u}{2Q(u)} + O(t^{1/2})$ and $h(0) = 1+O(t)$, as in Lemma 35.3. □

Theorem 35.6 (Radial expansion of the companion series)**.** *For each of the* 105 *companion series and for $q^{-7/20}C = S_{a,0}$, and every rational x,*

$$S_{\alpha,\beta}(\mathbf{e}(x)e^{-t}) = t^{-1}A_0(x) + t^{-1/2}A_1(x) + A_2(x) + O(t^{1/2}) \qquad (t\to 0^+),$$

where $A_j = \sum_{\text{pieces}}\sigma A_j^P + \sum_{\text{rays}}\sigma A_j^{\text{ray}}$ over the signed pieces and rays of Proposition 34.3, with $f(z) = \mathbf{e}(xQ(\alpha+c+Uz) + B(\alpha+c+Uz,\beta))$ and $N=2m$, A_j^P as in Theorem 35.4, and $A_1^{\text{ray}} = \frac{\sqrt\pi}{2}Q(u)^{-1/2}\operatorname{mean}(f)$, $A_2^{\text{ray}} = -\operatorname{mean}(f)\frac{l_u}{2Q(u)} - \sum_{b<N}f(b)B_1(b/N)$, $A_0^{\text{ray}} = 0$.

Proof. With $\Phi(v) = \mathbf{e}(B(\alpha+v,\beta))\mathbf{e}(xQ(\alpha+v))e^{-tQ(\alpha+v)}$, $S_{\alpha,\beta}(\mathbf{e}(x)e^{-t}) = \sum_v \mathrm{wt}(v)\Phi(v)$, and Proposition 34.3 applies by Lemma 34.2. Each piece is a sum as in Theorem 35.4 with f N-periodic by Lemma 34.4, and each ray is as in Lemma 35.5. There are finitely many pieces and rays, so the $O(t^{1/2})$ remainders add up. □

Corollary 35.7 (Order t^{-1})**.** *$A_0(x) = I\,G_{\alpha,\beta}(x)$ for every companion. In particular $A_0 = I\,K2_0$ for $q^{-7/20}C$, and $t\,H_7(\mathbf{e}(x)e^{-t}) = O(t^{1/2})$.*

Proof. The orthant has $U = I$ and $\mathrm{Ar} = \log((7+\sqrt{35})/(7-\sqrt{35}))/(2\sqrt{35}) = I$, because $(7+\sqrt{35})/(7-\sqrt{35}) = 6+\sqrt{35}$; its mean is $G_{\alpha,\beta}(x)$. The 21 pieces of K_+ have a common Ar_+. As (r,j) runs over $\{0,\dots,20\}\times\{0,\dots,N-1\}$, $21j+r$ runs over $\{0,\dots,21N-1\}$, which covers each residue modulo N exactly 21 times; for each (r,j), $i-j$ covers each residue once as i does. Hence $\sum_r \mathrm{mean}(f_r) = 21\,G_{\alpha,\beta}(x)$. For K_- the same count applies to $\nu = \alpha - v$, and $\alpha - v$ runs over $\alpha+\mathbb{Z}^2$ modulo N as v does, so $\sum_r \mathrm{mean}(f'_r) = 21\,G_{\alpha,\beta}(x)$. The signs are -1 and $+1$, so the cone pieces cancel at order t^{-1}. The statement for H_7 is the $K_\pm$ part of the same computation for $(\alpha,\beta) = (a,0)$, multiplied by $q^{7/20}$, which changes the expansion by a factor $1+O(t)$ and the phase $\mathbf{e}(7x/20)$. □

Corollary 35.8 (Order $t^{-1/2}$: algebraicity)**.** *For every companion and every rational x, $A_1(x)/\sqrt{\pi}$ is an algebraic number. The number $\log(6+\sqrt{35})$ does not enter A_1.*

Proof. By Theorems 35.4 and 35.6, $A_1/\sqrt{\pi}$ is a finite sum of products of roots of unity (values of f), rational numbers (B_1-weights, l_i, signs, $\frac{1}{2}$), and the algebraic numbers $J_a/\sqrt{\pi}$, $J_b/\sqrt{\pi}$, $r^{-1/2}$, $p^{-1/2}$, $Q(u)^{-1/2}$ with $p, \mathtt{q}, r, Q(u)$ rational. □

Remark 35.9. For the orthant piece alone at $x = 0$ and $(\alpha,\beta) = (a,0)$ the formula gives $0.5256172043\ldots$, the $t^{-1/2}$ coefficient $a_P = \sqrt{\pi}(49-2\sqrt{14})/140$ of P_7^{orth} in Part II (Table 13).

Proposition 35.10 (Order $t^{-1/2}$ on the zero set of the leading term; computer-assisted)**.** *Let $Z_0 = \{x \in X_{216} : 2 \mid m \text{ or } 5 \mid m\}$ (105 points). For $q^{-7/20}C$, $A_1(x) \neq 0$ for every $x \in Z_0$ and for $x = -\frac{1}{5}$. Since $A_0(x) = 0$ there (Theorem 11.4), $t\,|q^{-7/20}C(\mathbf{e}(x)e^{-t})| = |A_1(x)|\,t^{1/2} + O(t)$ at these points. The minimum of $|A_1|$ over these points is $0.1229\ldots$, and $|A_1(-\frac{1}{5})| = 0.4285\ldots$.*

Proof. The formula of Theorem 35.6 is a finite expression. The phase histograms were computed in exact rational arithmetic, and the trigonometric and algebraic values were enclosed in interval arithmetic at 60 digits (`BRIDGE_C_35/order2/data/map2_interval.json`); the enclosure of $|A_1|^2$ excludes 0 at all 106 points, with maximal width $6\cdot 10^{-59}$. □

35.3 Decomposition of A_1 by quadratic class

Each term of A_1 carries a label: the drift terms $-\mathrm{mean}(f)(l_1J_a + l_2J_b)$ of the pieces; the sawtooth terms $-\frac{\sqrt{\pi}}{2}Q(u_i)^{-1/2}W_i$ of the pieces, labelled by $Q(u_i)$ for the edge direction u_i whose B_1-weight they carry (the W_1 term carries $r^{-1/2} = Q(u_2)^{-1/2}$, the W_2 term $p^{-1/2} = Q(u_1)^{-1/2}$); and the ray terms $\pm\frac{1}{2}\cdot\frac{\sqrt{\pi}}{2}Q(u)^{-1/2}\,\mathrm{mean}(f)$, labelled by $Q(u)$.

Proposition 35.11 (Exact decomposition of A_1)**.** *$Q(e_1) = 1$, $Q(e_2) = Q(0,\pm1) = \frac{7}{2}$ and $Q(\pm(21,-1)) = \frac{595}{2}$. Consequently, for every companion and every rational x,*

$$A_1 = D + R + W[1] + W[\tfrac{7}{2}] + W[\tfrac{595}{2}], \qquad R = R[\tfrac{7}{2}] + R[\tfrac{595}{2}],$$

where D is the sum of the drift terms, $W[\lambda]$ the sum of the sawtooth terms with label λ, and $R[\lambda]$ the sum of the ray terms with label λ; $R[\frac{7}{2}]$ collects the rays $\pm\mathbb{N}(0,1)$ and $R[\frac{595}{2}]$ the rays $\pm\mathbb{N}(21,-1)$.

Proof. $Q(21,-1) = \frac{1}{2}(2\cdot 441 - 2\cdot 7\cdot 21 + 7) = \frac{595}{2}$. The edge directions of the pieces are e_1, e_2 (orthant) and $\pm u_1, \pm u_2$ (cones), and the rays of Proposition 34.3 are $\pm\mathbb{N}u_1$, $\pm\mathbb{N}u_2$. Every term of A_1 in Theorem 35.6 is therefore of exactly one of the listed kinds with one of the listed labels, and the identities are regroupings of a finite sum. □

Remark 35.12 (Structure of A_2)**.** By Theorem 35.6, A_2 consists of the moments $\frac{1}{2}K(l) - Q_0\mathrm{Ar}$ of the drift, edge terms $l_i W_j$, terms $\frac{\mathfrak{q}}{4}V_i$ weighted by the cross coefficient, corner terms C (finite sums of Dedekind-sum type), and the ray constants. Only $K(l)$ and Ar are transcendental integrals.

36 Order four: the shadow candidate and the law at order t^0

The next two questions are (a) whether A_1 becomes S-covariant after subtracting an explicit shadow contribution, and (b) whether the t^0 coefficient A_2 of Theorem 35.6, or A_2 minus a completion term, satisfies a weight-0 vector-valued law.

Both questions were tested numerically; neither produced a Category A statement, and the results are recorded only in the following remarks and in Appendix D.

Remark 36.1 (Category B: the sawtooth shadow candidate)**.** The candidate for (a), modelled on an erf-completion, removes all sawtooth groups: the completed coefficient is $A_1 - \Xi_\varepsilon$ with $\Xi_\varepsilon = \sum_\lambda \varepsilon_\lambda W[\lambda]$, $\varepsilon = (1,1,1)$, that is $A_1 - \Xi_\varepsilon = D + R$ (Proposition 35.11). Its S-image is not in its span, so this candidate does not close; nor does any of the six secondary subsets $\varepsilon \neq (0,0,0), (1,1,1)$. The sign control (closure of A_0 at weight 1, failure at the wrong weight) passed (item D.8). The candidate is not the actual completion: the erf-completion constants of H_7 have not been derived (Problem 1.).

Remark 36.2 (Category B: a weight-0 law for A_2)**.** For (b), A_2 does not satisfy a pure weight-0 vector-valued law, nor one of weight $\frac{1}{2}$ or 1; a mixed law $A_2(-1/x) \in \mathrm{span}[A_2(x), |x|^{1/2}A_1(x), |x|A_0(x)]$ is not closed either, but this test has little discriminating power (rank 214 on 216 points). The sign control passed (item D.9).

37 Shadow structure of the companion series

Throughout this section $Q(\nu) = \nu_1^2 + 7\nu_1\nu_2 + \frac{7}{2}\nu_2^2$, $B(\mu,\nu) = Q(\mu+\nu) - Q(\mu) - Q(\nu)$, so that $B(\nu,\nu) = 2Q(\nu)$ and det of the Gram matrix $A = \left(\begin{smallmatrix} 2 & 7 \\ 7 & 7 \end{smallmatrix}\right)$ is -35. The index set J consists of the 105 companion pairs (α_j, β_j) of Definition 34.1 (families $K2, K3, K4$). For a function f on $\mathbb{R}^2$ we write

$$S^f_{\alpha,\beta}(\tau) = \sum_{\nu \in \alpha + \mathbb{Z}^2} f(\nu)\, \mathbf{e}(B(\nu,\beta))\, q^{Q(\nu)}, \qquad q = \mathbf{e}(\tau),$$

whenever the series converges absolutely and locally uniformly on $\mathbb{H}$, and $S^f = (S^f_{\alpha_j,\beta_j})_{j\in J}$.

37.1 The matrices M, T, P

The reduction rules of Definition 34.1 express every pair $(\alpha,\beta) \in (\mathbb{Q}/\mathbb{Z})^2 \times \mathbb{Q}^2$ occurring below as a root of unity times a pair indexed by J; they act on the terms of S^f and are therefore valid for every f depending only on ν. Define the formal 105×105 matrices

- $M = (M_{jk})$: the coefficient matrix of the right-hand side of Theorem 33.4 (the S-law) after reduction to J, without the factor $35^{-1/2}$;
- $T = (T_{jk})$: the monomial matrix of the T-law, $T_{jk} = \mathbf{e}(Q(\alpha_j))\,\delta$ after reduction;
- $P = (P_{jk})$: the monomial matrix defined by $S^f_{-\alpha_j,-\beta_j} = \sum_k P_{jk} S^f_{\alpha_k,\beta_k}$.

All entries lie in $\mathbb{Z}[\zeta_{9800}]$, $\zeta_N = \mathbf{e}(1/N)$.

Lemma 37.1. *For every f one has $(PS^f)_j = S^{f^-}_{\alpha_j,\beta_j}$ with $f^-(\nu) = f(-\nu)$. In particular $P^2 = I$ on the image of the series, and $P^2 = I$ holds as a matrix identity.*

Proof. Substituting $\nu \mapsto -\nu$ gives $S^f_{-\alpha,-\beta} = \sum_{\nu\in\alpha+\mathbb{Z}^2} f(-\nu)\mathbf{e}(B(\nu,\beta))q^{Q(\nu)} = S^{f^-}_{\alpha,\beta}$. The matrix identity is a finite check on the exponents of the monomial entries: P is a permutation matrix with root-of-unity weights $\zeta_{9800}^{e_j}$, the permutation is an involution and $e_j + e_{\pi(j)} \equiv 0 \pmod{9800}$ for all j (script `BRIDGE_C_35/order9/cyclo.py`). □

Proposition 37.2 (exact, computer-assisted)**.** *$M^2 = 35P$ and $(MT)^3 = 35\sqrt{35}\,P$, where $\sqrt{35} > 0$ is embedded in $\mathbb{Q}(\zeta_{140}) \subset \mathbb{Q}(\zeta_{9800})$ via the identity embedding $\zeta_N \mapsto \mathbf{e}(1/N)$.*

Proof. Let X be an entry of $M^2 - 35P$. It is an algebraic integer in $K = \mathbb{Q}(\zeta_{9800})$, $[K:\mathbb{Q}] = \varphi(9800) = 3360$. Evaluating X under each of the 3360 embeddings $\zeta_{9800} \mapsto \zeta_{9800}^g$, $g \in (\mathbb{Z}/9800)^\times$, in ball arithmetic [21] with 200 bits gives $|\sigma_g(X)| < \frac{1}{2}$ for every g. Then $|N_{K/\mathbb{Q}}(X)| < 2^{-3360} < 1$; since $N_{K/\mathbb{Q}}(X) \in \mathbb{Z}$ this forces $N_{K/\mathbb{Q}}(X) = 0$, hence $X = 0$. For the second identity put $Y = (MT)^3P$. The same argument applied to all off-diagonal entries of Y, to all differences $Y_{jj} - Y_{11}$ and to $Y_{11}^2 - 35^3$ shows that $Y = sI$ with $s^2 = 35^3$. Under the identity embedding s evaluates to a positive real ball, so $s = 35\sqrt{35}$, and $(MT)^3 = sP^{-1} = 35\sqrt{35}P$ by Lemma 37.1. Scripts: `BRIDGE_C_35/order9/cyclo.py`, `BRIDGE_C_35/order13/w1.py`. □

For $F : \mathbb{H} \to \mathbb{C}^{105}$, $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and a constant matrix X put $(F|_1 g)(\tau) = (c\tau + d)^{-1}F(g\tau)$; then $(F|_1 g)|_1 h = F|_1(gh)$ and $(XF)|_1 g = X(F|_1 g)$. Let $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $T_0 = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$.

Lemma 37.3 (parity of a weight-one law)**.** *Let $F : \mathbb{H} \to \mathbb{C}^{105}$ and $\chi \in \mathbb{C}^\times$ with $F(-1/\tau) = \chi\,\tau\,35^{-1/2}MF(\tau)$ for all $\tau \in \mathbb{H}$. Then $F = -\chi^2 PF$.*

Proof. Apply the law at $-1/\tau$: $F(\tau) = \chi(-1/\tau)35^{-1/2}MF(-1/\tau) = -\chi^2\,35^{-1}M^2F(\tau) = -\chi^2 PF(\tau)$ by Proposition 37.2. □

Lemma 37.4 (no even weight-one law)**.** *Let $F \not\equiv 0$ satisfy the law of Lemma 37.3 and in addition $F(\tau+1) = TF(\tau)$. Then $\chi = 1$ and $PF = -F$.*

Proof. The hypotheses read $F|_1 S = \chi 35^{-1/2}MF$ and $F|_1 T_0 = TF$. Hence $F|_1(ST_0) = (\chi 35^{-1/2}MF)|_1 T_0 = \chi 35^{-1/2}MTF$ and, iterating, $F|_1(ST_0)^3 = \chi^3 35^{-3/2}(MT)^3F = \chi^3 PF$ by Proposition 37.2. Since $(ST_0)^3 = -I$ and $F|_1(-I) = -F$, we get $\chi^3 PF = -F$. Lemma 37.3 gives $\chi^2 PF = -F$. Subtracting, $\chi^2(\chi - 1)PF = 0$; as P is invertible and $F \not\equiv 0$, $\chi = 1$, and then $PF = -F$. □

Corollary 37.5. *If F satisfies the hypotheses of Lemma 37.4 for some χ, then its P-even part $\frac{1}{2}(F + PF)$ vanishes. No nonzero P-even vector satisfies a weight-one law with matrices $(35^{-1/2}\chi M, T)$ for any χ.*

37.2 Sign weights and parity

Let $s_1 = \operatorname{sgn}\nu_1$, $s_2 = \operatorname{sgn}\nu_2$, $s_3 = \operatorname{sgn}(\nu_1 + 21\nu_2)$ (with $\operatorname{sgn} 0 = 0$) and define the ν-sign weight

$$p(\nu) = \tfrac{1}{4}(1+s_1)(1+s_2) - \tfrac{1}{2}(s_1+s_3).$$

Lemma 37.6. *$\frac{1}{2}(p(\nu)+p(-\nu)) = \frac{1}{4}(1+s_1s_2)$ and $\frac{1}{2}(p(\nu)-p(-\nu)) = \frac{1}{4}(s_1+s_2) - \frac{1}{2}(s_1+s_3)$. Consequently, by Lemma 37.1, the P-even part of S^p is $\frac{1}{2}S^E$ with $E = \frac{1}{2}(1+s_1s_2)$, and the P-odd part is $\frac{1}{2}S^{\rho_O} - S^{\rho_H}$ with $\rho_O = \frac{1}{2}(s_1+s_2)$, $\rho_H = \frac{1}{2}(s_1+s_3)$.*

Proof. The signs are odd functions of ν; expand p and separate even and odd monomials in s_1, s_2, s_3. $\square$

The series S^{ρ_H} and S^{ρ_O} are indefinite theta series attached to the cones spanned by $c_1 = (-1, 1)$, c_2 and c_1, $c' = (7, -2)$, respectively, where $c_2 = (4, -1)$; one has $Q(c_1) = -\frac{5}{2}$, $Q(c_2) = -\frac{17}{2}$, $Q(c') = -35$, and $B(c_1, \nu) = 5\nu_1$, $B(c_2, \nu) = \nu_1 + 21\nu_2$, $B(c', \nu) = 35\nu_2$.

37.3 The P-odd shadow

For a negative vector c put $s_c(\nu) = B(c, \nu)/\sqrt{-Q(c)}$ and $s'_c(\nu)$ for the coordinate along a unit vector of $c^{\perp}$, normalised so that $Q(\nu) = \frac{1}{4}(s'^2_c - s^2_c)$. With $E(u) = 2\int_0^u e^{-\pi t^2} dt$ define

$$\widehat{S}^c_{\alpha,\beta}(\tau) = \sum_{\nu \in \alpha + \mathbb{Z}^2} E\big(s_c(\nu)\sqrt{y}\big)\mathbf{e}(B(\nu,\beta)) q^{Q(\nu)}, \qquad X^c_{\alpha,\beta}(\tau) = y^{1/2} \sum_{\nu \in \alpha + \mathbb{Z}^2} s_c(\nu)\, \mathbf{e}\big(\tfrac{s_c^2}{4}\tau\big)\, \overline{\mathbf{e}\big(\tfrac{s_c'^2}{4}\tau\big)}\, \mathbf{e}(-B(\nu,\beta)).$$

Proposition 37.7. *$\xi_1 \widehat{S}^c = X^c$, where $\xi_1 = 2iy\,\overline{\partial_{\bar\tau}}$.*

Proof. $E'(u) = 2e^{-\pi u^2}$, so $\partial_y E(s\sqrt{y}) = s\, y^{-1/2} e^{-\pi s^2 y}$ and $\partial_{\bar\tau} = \frac{i}{2}\partial_y$ on functions of y. Since $q^{Q(\nu)} e^{-\pi s^2 y} = \mathbf{e}(\frac{s'^2}{4}\tau)\,\overline{\mathbf{e}(\frac{s^2}{4}\tau)}$, termwise differentiation (justified by the Gaussian majorant $e^{-\pi y (s^2 + s'^2)/2}$, locally uniform in τ) gives $\partial_{\bar\tau}\widehat{S}^c = \frac{i}{2} y^{-1/2} \sum s\, \mathbf{e}(\frac{s'^2}{4}\tau)\overline{\mathbf{e}(\frac{s^2}{4}\tau)}\mathbf{e}(B(\nu,\beta))$; apply $2iy\,\bar{\cdot}$. $\square$

For c_1, c_2, c' one has $s_c^2/4 = \frac{5}{2}\nu_1^2$, $\frac{1}{34}(\nu_1 + 21\nu_2)^2$, $\frac{35}{4}\nu_2^2$; the corresponding weight-$\frac{3}{2}$ unary series are $\sum u\, \mathbf{e}(\frac{5}{2}u^2\tau)$, $\sum R\, \mathbf{e}(\frac{R^2}{34}\tau)$, $\sum m\, \mathbf{e}(\frac{35}{4}m^2\tau)$, and the weight-$\frac{1}{2}$ partners in s'_c have quadratic forms $\frac{7}{2}$, $\frac{35}{34}$ and 1 in suitable coordinates. Thus exactly three distinct unary theta series of weight $\frac{3}{2}$ occur; the one in $\frac{5}{2}u^2$ is shared between the cones of ρ_H and ρ_O. The completions $\widehat{H} = \frac{1}{2}(\widehat{S}^{c_1} + \widehat{S}^{c_2})$ and $\widehat{G}_O = \frac{1}{2}(\widehat{S}^{c_1} + \widehat{S}^{c'})$ transform as vector-valued forms of weight one by [18, Cor. 2.9], and their ξ_1-images are $\frac{1}{2}(X^{c_1} + X^{c_2})$ and $\frac{1}{2}(X^{c_1} + X^{c'})$.

37.4 Boundary series and their vanishing

For a unit negative vector z ($Q(z) = -1$) with unit positive orthogonal z' let $s = B(\nu, z)$, $s' = B(\nu, z')$ and

$$\mathcal{B}_{z,z'}(\tau) = \sum_{\nu \in \alpha + \mathbb{Z}^2} \mathbf{e}(B(\nu,\beta))\, s\, s'\, \mathbf{e}\big(\tfrac{s'^2}{4}\tau\big)\, \mathbf{e}\big(-\tfrac{s^2}{4}\bar\tau\big).$$

Let $\hat c' = c'/\sqrt{35}$ and $\hat c_1 = c_1/\sqrt{5/2}$ (both with $Q = -1$), $e_1 = (1, 0)$, $\hat e_2 = e_2/\sqrt{7/2}$.

Proposition 37.8. *$\mathcal{B}_{\hat c', e_1} \equiv 0$ for every companion.*

Proof. Here $s = \sqrt{35}\nu_2$ and $s' = B(\nu, e_1) = 2\nu_1 + 7\nu_2$. The map $\rho(\nu) = (-\nu_1 - 7\nu_2, \nu_2)$ preserves Q, fixes s, sends $s' \mapsto -s'$, and maps $\alpha + \mathbb{Z}^2$ to $\rho(\alpha) + \mathbb{Z}^2$. Writing $\beta = (b_0, b_1)$ one checks from Definition 34.1 that every companion satisfies $\rho(\alpha) \equiv \alpha \pmod{\mathbb{Z}^2}$ and $B(\rho\nu, \beta) \equiv B(\nu, \beta) \pmod 1$ for $\nu \in \alpha + \mathbb{Z}^2$ (a finite check on the 105 pairs, `BRIDGE_C_35/order13/cprime_all.py`, exact in $\mathbb{Q}$). Hence the summand is odd under the bijection $\nu \mapsto \rho\nu$ of $\alpha + \mathbb{Z}^2$, and the sum vanishes. $\square$

For $\hat c_1$ take $z' = \hat e_2$. With $u = \nu_1$ and $m = \nu_1 + \nu_2$ one has $s = \sqrt{10}\, u$ up to sign, $s' = \pm\sqrt{14}\, m$, $Q = \frac{7}{2}m^2 - \frac{5}{2}u^2$, and $B(\nu, \beta) = 7(b_0 + b_1)m - 5b_0 u$. Define

$$\vartheta^+(\tau) = \sum_{m \in \alpha_1 + \alpha_2 + \mathbb{Z}} m\, \mathbf{e}\big(\tfrac{7}{2}m^2\tau + 7(b_0 + b_1)m\big), \qquad \vartheta^-(\tau) = \sum_{u \in \alpha_1 + \mathbb{Z}} u\, \mathbf{e}\big(\tfrac{5}{2}u^2\tau + 5b_0 u\big).$$

Lemma 37.9. *Let $A>0$, $\delta,c\in\mathbb{R}$. Then $\vartheta(\tau)=\sum_{m\in\delta+\mathbb{Z}} m\,\mathbf{e}(Am^2\tau+cm)\equiv 0$ on $\mathbb{H}$ if and only if $2\delta\in\mathbb{Z}$, $2c\in\mathbb{Z}$ and $2c\delta\in\mathbb{Z}$.*

Proof. The exponents Am^2, $m\in\delta+\mathbb{Z}$, take each value at most at m and $-m$, and $-m\in\delta+\mathbb{Z}$ iff $2\delta\in\mathbb{Z}$. If $2\delta\notin\mathbb{Z}$, the coefficient of q^{Am^2} for any $m\neq 0$ is $m\,\mathbf{e}(cm)\neq 0$. If $2\delta\in\mathbb{Z}$, the coefficient of q^{Am^2} ($m>0$, $m\in\delta+\mathbb{Z}$) is $m(\mathbf{e}(cm)-\mathbf{e}(-cm))=2im\sin(2\pi cm)$, which vanishes for all such m iff $2cm\in\mathbb{Z}$ for all $m\in\delta+\mathbb{Z}$, i.e. iff $2c\delta\in\mathbb{Z}$ and $2c\in\mathbb{Z}$. The term $m=0$ contributes 0. Distinct exponents give linearly independent functions, which proves both directions. □

Theorem 37.10. *$\mathcal{B}_{\hat c_1,\hat e_2}=\varepsilon\sqrt{140}\,\vartheta^+\,\overline{\vartheta^-}$ with $\varepsilon\in\{\pm1\}$ fixed by the orientation of $\hat e_2$. It vanishes identically if and only if $2(\alpha_1+\alpha_2)$, $14(b_0+b_1)$, $14(b_0+b_1)(\alpha_1+\alpha_2)\in\mathbb{Z}$, or $2\alpha_1$, $10b_0$, $10b_0\alpha_1\in\mathbb{Z}$.*

Proof. (u,m) is a $\mathbb{Z}$-basis change, so $\alpha+\mathbb{Z}^2\to(\alpha_1+\mathbb{Z})\times(\alpha_1+\alpha_2+\mathbb{Z})$ is a bijection and the sum factors: $ss'=\varepsilon\sqrt{140}\,um$, $\mathbf{e}(\frac{s'^2}{4}\tau)=\mathbf{e}(\frac{7}{2}m^2\tau)$, $\mathbf{e}(-\frac{s^2}{4}\bar\tau)=\overline{\mathbf{e}(\frac{5}{2}u^2\tau)}$ and $\mathbf{e}(B(\nu,\beta))=\mathbf{e}(7(b_0+b_1)m)\,\overline{\mathbf{e}(5b_0u)}$. The product of the holomorphic ϑ^+ and the antiholomorphic $\overline{\vartheta^-}$ vanishes identically iff one factor does: if both are $\not\equiv 0$, pick τ_0 with $\vartheta^-(\tau_0)\neq 0$; then $\vartheta^+(\tau)\overline{\vartheta^-(\tau_0)}$ is the restriction to $\bar\tau=\bar\tau_0$ of the real-analytic function $(\tau,w)\mapsto\vartheta^+(\tau)\overline{\vartheta^-(\bar w)}$, which vanishes on the totally real set $\{w=\bar\tau\}$ and hence identically, contradicting $\vartheta^+\not\equiv 0$. Apply Lemma 37.9 with $(A,\delta,c)=(\frac{7}{2},\alpha_1+\alpha_2,7(b_0+b_1))$ and $(\frac{5}{2},\alpha_1,5b_0)$. □

The two conditions are exactly the conditions that the involutions $\rho_m:(u,m)\mapsto(u,-m)$, resp. $\rho_u:(u,m)\mapsto(-u,m)$, preserve the companion; this is the statement that the endpoint term vanishes if and only if one of the two reflections fixing the boundary ray preserves the companion.

37.5 The Siegel lift over the orthant arc and the P-even defect

For a unit negative z put $\Theta_z(\tau)=y^{1/2}\sum_{\nu\in\alpha+\mathbb{Z}^2}\mathbf{e}(B(\nu,\beta))\,\mathbf{e}(Q(\nu)x)\,e^{-\pi y(s^2+s'^2)/2}$, with s,s' as above. Let $z(\mu)=\cosh\mu\,\hat c'-\sinh\mu\,e_1$ and $L=\frac12\log(6+\sqrt{35})$. Then $Q(z(\mu))=-1$, $z'(\mu)=\frac{d}{d\mu}z(\mu)$ is unit positive and orthogonal to $z(\mu)$, $\cosh L=\sqrt{7/2}$, $\sinh L=\sqrt{5/2}$, and a direct computation gives $z(L)=-\hat c_1$, $z'(L)=-\hat e_2$, $z(0)=\hat c'$, $z'(0)=-e_1$. Define

$$\Phi_{\mathrm{orth}}(\tau)=\int_0^L\Theta_{z(\mu)}(\tau)\,d\mu,\qquad \Delta_0=-y^2(\partial_x^2+\partial_y^2).$$

Theorem 37.11. *$(\Delta_0-\frac14)\Phi_{\mathrm{orth}}=\frac{\pi}{2}y^{3/2}\big[\mathcal{B}_{z(\mu),z'(\mu)}\big]_{\mu=0}^{\mu=L}=\frac{\pi}{2}\sqrt{140}\,y^{3/2}\,\vartheta^+\,\overline{\vartheta^-}$ (the sign being that of Theorem 37.10 for $z'=-\hat e_2$, $z=-\hat c_1$, for which $ss'=+\sqrt{140}\,um$).*

Proof. Fix ν and write the summand of $\Theta_{z(\mu)}$ as $t=\mathbf{e}(B(\nu,\beta))\,y^{1/2}\mathbf{e}(Qx)e^{-\pi yK}$ with $K=\frac12(s^2+s'^2)$, where $s=s(\mu)$, $s'=s'(\mu)$ satisfy $\partial_\mu s=s'$, $\partial_\mu s'=s$ (because $z''=z$), so $\partial_\mu K=2ss'$ and $\partial_\mu(ss')=s^2+s'^2=2K$. Using $Q=\frac14(s'^2-s^2)$, a direct differentiation gives

$$(\Delta_0-\tfrac14)t=\partial_\mu\Big(\tfrac{\pi}{2}y^{3/2}ss'\,\mathbf{e}(Qx)e^{-\pi yK}\Big);$$

explicitly, both sides equal $\frac{\pi}{2}y^{3/2}\mathbf{e}(Qx)e^{-\pi yK}\big(2K-2\pi y\,s^2s'^2\big)$, as one checks from $\partial_x^2t=-4\pi^2Q^2t$, $\partial_y^2t=\big(-\frac14y^{-2}-\pi Ky^{-1}+\pi^2K^2\big)t$, and $K^2-4Q^2=s^2s'^2$. Summing over ν (the Gaussian majorant is uniform for $\mu\in[0,L]$ and τ in compact sets, justifying termwise differentiation and the exchange with $\int d\mu$) and integrating in μ gives the first equality, since $y^{3/2}ss'\mathbf{e}(Qx)e^{-\pi yK}=y^{3/2}ss'\mathbf{e}(\frac{s'^2}{4}\tau)\mathbf{e}(-\frac{s^2}{4}\bar\tau)$. At $\mu=0$ the boundary term vanishes by Proposition 37.8 (the sign of z' is immaterial for vanishing); at $\mu=L$ Theorem 37.10 applies with $ss'=+\sqrt{140}\,um$. □

Since $E=\frac12(1+s_1s_2)$, the P-even part of S^p is $\frac12 S^E$; for Φ_{orth} the corresponding defect is $\frac12(\Delta_0-\frac14)\Phi_{\mathrm{orth}}=\frac{\pi}{4}\sqrt{140}\,y^{3/2}\vartheta^+\overline{\vartheta^-}$, which by Theorem 37.10 vanishes exactly on the companions preserved by ρ_m or ρ_u.

37.6 Line terms of the literal series

Write $\nu=\alpha+v$ with α the representative of Definition 34.1 and $v=(n,k)\in\mathbb{Z}^2$. The literal weight is $W(v)=\mathbf{1}[n\ge0]\mathbf{1}[k\ge0]-\frac12(\operatorname{sgn}n+\operatorname{sgn}(n+21k))$. Put $\sigma=\alpha_1+21\alpha_2$, $m=n+21k$ and

$$h_1=\mathbf{1}[n\ge0]-\tfrac12(1+s_1),\quad h_2=\mathbf{1}[k\ge0]-\tfrac12(1+s_2),\quad g_1=\operatorname{sgn}n-s_1,\quad g_3=\operatorname{sgn}m-\operatorname{sgn}(m+\sigma).$$

Proposition 37.12. *$W(v)-p(\nu)=\frac12(1+s_1)h_2+\frac12(1+s_2)h_1+h_1h_2-\frac12g_1-\frac12g_3$. Each of h_1,g_1 (resp. h_2; g_3) is supported on the integers n (resp. k; m) in the closed interval between 0 and $-\alpha_1$ (resp. $-\alpha_2$; $-\sigma$).*

Proof. By definition $\mathbf{1}[n\ge0]=\frac12(1+s_1)+h_1$ and $\mathbf{1}[k\ge0]=\frac12(1+s_2)+h_2$; multiplying, $\mathbf{1}[n\ge0]\mathbf{1}[k\ge0]=\frac14(1+s_1)(1+s_2)+\frac12(1+s_1)h_2+\frac12(1+s_2)h_1+h_1h_2$. Also $\operatorname{sgn}n=s_1+g_1$ and $\operatorname{sgn}(n+21k)=s_3+g_3$ since $\nu_1=n+\alpha_1$ and $\nu_1+21\nu_2=m+\sigma$. Subtract p. The functions $n\mapsto\mathbf{1}[n\ge0]$ and $n\mapsto\frac12(1+\operatorname{sgn}(n+\alpha_1))$ agree unless n and $n+\alpha_1$ lie on different sides of 0 or one of them is 0, which confines n to the stated interval; likewise for the others. □

Hence S^W-S^p is a finite sum of unary series, by type:

(a) half-lines $\nu_2=c$ fixed (weight $h_2\cdot\frac12(1+s_1)$): $Q=w^2-\frac{35}{4}c^2$, $w=\nu_1+\frac72c$, a partial theta in $w\ge\frac72c$ with Eichler partner $\sum w\,\mathbf{e}(w^2\tau)$;

(b) half-lines $\nu_1=c$ fixed (weight $h_1\cdot\frac12(1+s_2)$): $Q=\frac72m'^2-\frac52c^2$, $m'=c+\nu_2\ge c$, a partial theta with partner $\sum m'\,\mathbf{e}(\frac72m'^2\tau)$;

(c) the finite set where $h_1h_2\neq0$;

(d) full lines $\nu_1=c$ (weight $-\frac12g_1$): $q^{-5c^2/2}$ times a weight-$\frac12$ unary theta in $\frac72m'^2$;

(e) full lines $n+21k=M$ (weight $-\frac12g_3$): with $\nu_0=(\alpha_1+M,\alpha_2)$, $u=(-21,1)$, $Q(\nu_0+ku)=\frac{595}{2}\big(k+\frac{B(\nu_0,u)}{595}\big)^2+Q(\nu_0)-\frac{B(\nu_0,u)^2}{1190}$, a weight-$\frac12$ unary theta of norm $\frac{595}{2}$.

No line term introduces a weight-$\frac32$ unary theta beyond the three of Proposition 37.7.

37.7 Summary

Theorem 37.13. *For the 105 companions:*

(i) the P-odd part of S^p equals the holomorphic part of $\frac12\widehat{G}_O-\widehat{H}$, whose ξ_1-image is $\frac14(X^{c'}-X^{c_1})-\frac12X^{c_2}$ and involves exactly three distinct weight-$\frac32$ unary thetas;

(ii) the P-even part admits no weight-one law with the matrices $(\chi35^{-1/2}M,T)$ (Corollary 37.5);

(iii) the orthant Siegel lift Φ_{orth} has the Laplacian defect of Theorem 37.11, a single product of the weight-$\frac32$ unary series ϑ^+ (in $\frac72m^2$) and the conjugate of ϑ^- (in $\frac52u^2$), vanishing iff ρ_m or ρ_u preserves the companion;

(iv) the literal series differs from S^p by the line terms of Proposition 37.12.

Remark (Category B) 37.14. The identification of the holomorphic projection of Φ_{orth} with S^E, the identification of the transformation law of $\widehat{H}, \widehat{G}_O$ given by [18, Cor. 2.9] with the pair $(35^{-1/2}M, T)$ on the companions, and the independent tests of the P-even law and of the vanishing set are numerical; see Appendix D, items D.10–D.13.

Remark 37.15. Descriptively, the P-odd part is a local (cone-type) mock modular object of weight one and the P-even part is a non-local (arc-integral) object whose defect lives on the endpoint ray; these words describe the structure above and are not further statements.

Part IV
Foundation

38 The foundation attempt: what ties RTA to the framework

This section lists the written RTA objects and, for each, what is proved about its relation to the structures of Parts I–III. References to "RTA" are to the RTA manuscript (Residual Ternary Algebra: Packaging Identities, Residual Series g_s, and Bridge Series C, draft of 24 August 2026), which has no algebraic axioms: its six numbered rules are process rules.

38.1 Proved links

Proposition 38.1 (Identification of objects)**.** *The following hold.*

(i) The RTA residual series g_s is the series g_k of Part I with $k = s$.

(ii) The RTA double-sum identity for $(-q^s;q^s)_\infty g_s$ holds for every $s \geq 1$ as an identity in $\mathbb{Z}[[q]]$ (Proposition 2.2).

(iii) The RTA path quadratic Q_s, cones c_1, c_2, weight w, cone series H_s, orthant series and Bridge Series $C = P_7^{\mathrm{orth}} - H_7$ are those of §2 and of Part II; C_{axis} and C_{third} are those of Part II, §10; the RTA natural characteristic $\chi_{\mathrm{nat}}(7) = (\frac{7}{10}, -\frac{1}{5})$ is the vector a.

Proof. (i) and (iii) are comparisons of definitions: RTA Definition 6.1 and §§9–10 against Definition 2.1, §2 and Part II, §10; $\chi_{\mathrm{nat}}(s) = (s/(2(s-2)), -1/(s-2))$ at $s = 7$ equals a. (ii) is Proposition 2.2. □

Corollary 38.2 (Residue law of g_7 and its converse)**.** $g_7(d) = 0$ *whenever* $d \equiv 3, 5, 6 \pmod 7$. *Conversely,* $g_7(d) \neq 0$ *for every* $d \equiv 0, 1, 2, 4 \pmod 7$ *outside the finite set* Z_7 *of* 24 *elements listed in Appendix B (*$\max Z_7 = 247$*), and the sign of* $g_7(d)$ *is* $(-1)^{d+1}$ *for all such* $d \geq D_0(7)$.

Proof. By Proposition 3.2 (Part I), $g_7(d)$ is a sum over $n \geq 0$ with $n^2 \equiv d \pmod 7$; the squares modulo 7 are $0, 1, 2, 4$, so the sum is empty for $d \equiv 3, 5, 6$. The converse is the case $k = 7$ of Theorem 8.1 (Part I, §8.1), parts 1 and 2. □

Theorem 38.3 (The normalized Bridge Series and the 35-component law vector)**.** *The following hold.*

(i) The leading radial profile of the RTA Bridge Series C is $I\,\mathbf{e}(7x/20)K2_0(x)$, and it does not lie in $\mathcal{V}_{35}$.

(ii) The leading radial profile of $q^{-7/20}C$ is $I\,K2_0$, a non-zero element of $\mathcal{V}_{35}$; together with its S- and T-images it spans $\mathcal{V}_{35}$, and it obeys the vector-valued S-law of Theorem 34.7.

(iii) Up to order t^0, the radial expansion of $q^{-7/20}C$ at every rational point is given in closed form by Theorem 35.6; at order $t^{-1/2}$ the coefficient divided by $\sqrt{\pi}$ is algebraic, and at every point of $Z_0 \cup \{-\frac{1}{5}\}$ it is non-zero.

Proof. Theorems 34.5, 34.6, 34.7, 35.6, Corollary 35.8 and Proposition 35.10. □

38.2 Negative results

Proposition 38.4 (No scalar residual law)**.** *There is no scalar modular, mock-modular (signature $(1,1)$) or unary/partial/false theta law for C of the kinds listed in Theorem 11.5; in particular for no weight k and constant λ is $C(\tau) - \lambda(7\tau)^{-k}C(-1/(7\tau))$ radially bounded at all rationals.*

Proof. Theorem 11.5, with the stated dependence on Baker's theorem for part (i). □

Proposition 38.5 (No vector-valued Fricke law at leading order; computer-assisted)**.** *Let $D_7 f(x) = f(7x)$. Then $\mathcal{V}_{35} \cap D_7\mathcal{V}_{35} = 0$ as spaces of functions on $\mathbb{Q}$. In particular $D_7\mathcal{V}_{35} \not\subset \mathcal{V}_{35}$.*

Proof. Both spaces have dimension at most 36 (Proposition 33.6). The 216×210 matrix of values of the 105 functions and of their dilations by 7 at X_{216} has rank 72 modulo each of the four primes of Proposition 33.6 (`A53_RTA_maps`); by the reduction argument of that proof its rank in characteristic 0 is at least 72. Hence $\dim(\mathcal{V}_{35} + D_7\mathcal{V}_{35}) = 72$ and the sum is direct. □

Proposition 38.6 (Facts about T_3 and 539)**.** *Let $T_3(3k) = k$, $T_3(3k+1) = 4k+2$, $T_3(3k+2) = 2k+1$.*

(i) If $539 = a_1 + \cdots + a_8$ with positive integers and $\max a_i - \min a_i \le 1$, then $\{a_i\} = \{68, 68, 68, 67, 67, 67, 67, 67\}$.

(ii) For M coprime to 6 and $L \ge 1$, as n runs over $\mathbb{Z}/3^L M$, $T_3^L(n) \bmod M$ is well defined and takes each value exactly 3^L times.

(iii) Extended to $\mathbb{Z}_3$, T_3 preserves Haar measure, and the itinerary map $n \mapsto (T_3^j(n) \bmod 3)_{j\ge 0}$ is a homeomorphism $\mathbb{Z}_3 \to \{0,1,2\}^{\mathbb{N}}$ conjugating T_3 to the shift.

Proof. (i) Let $q = \min a_i$ and let b be the number of parts equal to $q+1$. If $b = 8$ then $8 \mid 539$, false. So $539 = 8q+b$ with $0 \le b \le 7$, and $q = 67$, $b = 3$. (ii) For $N = 3^i M$ ($i \ge 1$) and $n = 3k+r$, replacing n by $n + N$ changes $T_3(n)$ by $N/3$, $4N/3$ or $2N/3$, all $\equiv 0 \pmod{N/3}$; and for fixed r the maps $k \mapsto k, 4k+2, 2k+1$ are bijections of $\mathbb{Z}/(N/3)$ because $N/3$ is odd. So $n \bmod N \mapsto T_3(n) \bmod N/3$ is well defined and 3-to-1; iterate L times. (iii) Each branch $k \mapsto k, 4k+2, 2k+1$ is an affine bijection of $\mathbb{Z}_3$ with unit linear coefficient, and $k \mapsto 3k+r$ maps $\mathbb{Z}_3$ onto $r + 3\mathbb{Z}_3$ scaling Haar measure by $\frac{1}{3}$; hence $\mu(T_3^{-1}B) = \sum_r \mu(B)/3 = \mu(B)$. By induction on j, the points whose itinerary begins with a given word of length j form one class modulo 3^j; so the itinerary map is a continuous bijection of compact Hausdorff spaces, hence a homeomorphism, and it conjugates T_3 to the shift by definition. □

Remark 38.7 (No written link)**.** Proposition 38.6 is a statement about T_3 and the integer 539 alone. No written object (in RTA, in Parts I–III, or in any source used here) defines a map from T_3, the carrier $\mathcal{O}_{\mathrm{res}} = \{21 + 9j : 0 \le j \le 538\}$, the packaging identities ($539 = \lfloor 4859/9 \rfloor$, $18 + 521$, $3 \cdot 68 + 5 \cdot 67$) or the linking series Λ_B to g_s, C, $\mathcal{V}_{35}$ or the lift representation. RTA states that

its packaging identities "are independent of any dynamical map" and of the modular status of the series. The choice $s \in \{7, 8, 11\}$ is attributed in RTA to the packaging arithmetic, but no derivation of these values from it is written. The statement that the $R = ST$ grading of the cusp space is a foundation of RTA has no proof; the only proved statement of that kind is the grading lemma of Part II's framework (the $\mathbb{Z}/3$ grading of equivariant maps by R-degree), which concerns representations and not T_3.

38.3 Where the shadow Ξ enters

RTA describes a weight-$\frac{3}{2}$ shadow $\Xi = \Xi_{\text{even}} + \Xi_{\text{odd}}$ as "the residual-weighted family of shifted partial thetas of squares whose outer coefficients are the residual disc-35 series", and a weight-$\frac{1}{2}$ period candidate as its Eichler integral. Neither is written as an explicit series with stated coefficients, so no statement about Ξ can be proved from the written definitions. What is proved here bounds the place where Ξ must enter:

(i) at order t^{-1} the normalized series is governed by $\mathcal{V}_{35}$ with no correction term (Theorem 34.7);

(ii) at order $t^{-1/2}$ the coefficient A_1 is an explicit finite sum of drift, sawtooth and ray terms (Proposition 35.11) with $A_1/\sqrt{\pi}$ algebraic (Corollary 35.8); any S-covariant completion of A_1 must therefore come from a term that is not visible at order t^{-1};

(iii) at order t^0 the coefficient A_2 is explicit (Theorem 35.6).

For the 105 companion series, Section 37 determines the shadow structure: the P-odd part has the ξ_1-image of Theorem 37.13(i), and the P-even part has no weight-one law and the Laplacian defect of Theorem 37.11. This does not identify Ξ, since Ξ is not written as an explicit series. Whether a completion term of Zwegers type, with coefficients derived from the error-function completion of H_7, makes A_1 or A_2 S-covariant is open (Problem 1.). The numerical tests carried out so far (Appendix D) found no closure for any regrouping of the finite sum or for the simplest sawtooth candidate.

39 Open problems

1. Derive the actual completion constants of H_7 (erf-completion along the walls $(0, 1)$ and $(21, -1)$), form the completed coefficient $A_1 - \Xi^{\text{compl}}$, and decide whether it satisfies a vector-valued S-law of weight $\frac{1}{2}$. This would identify the shadow Ξ. The completions of the cone parts and their ξ_1-images are now explicit (Proposition 37.7); what remains is their effect on A_1. The sawtooth candidate, which removes all $W[\lambda]$, does not close (Appendix D).

2. Decide whether A_2 minus a completion term satisfies a vector-valued law; this is the period-function level. Numerically, A_2 itself satisfies no pure law of weight 0, $\frac{1}{2}$ or 1, and the mixed test with A_1, A_0 is inconclusive (Appendix D).

3. Prove or disprove $|A_1(\frac{1}{2})| = \sqrt{\pi}/4$ (observed numerically) and find the zero set of A_1 on the zero set of A_0.

4. Find a written object linking T_3, $\mathcal{O}_{\text{res}}$ or 539 to g_s, C or $\mathcal{V}_{35}$, or prove that a natural class of such maps does not exist.

5. Extend the parity-dominance theorem of Part I beyond $k = 33$ and to the classes of even k that are not one-sign.

6. The open problems of Part II, §32.

A Computational reproduction for Part I

All files are in the repository `ArvinHampton/539-Labs-Master` under `reports/s_family_paths/PAPER_gk_v1/` (abbreviated `P/`). Environment: Python 3.13, python-flint 0.9.0, mpmath, gcc, Linux; every command was run with `OMP_NUM_THREADS=1` under `nice`. SHA-256 sums of all files are in `P/SHA256SUMS`.

A.1 Algebraic checks (Sections 3–7)

`P/code/gklib.py` evaluates $g_k(d)$ exactly (Python integers). `P/code/verify_algebra.py` (log: `P/data/verify_algebra_log.txt`) checks:

- stabilisation, sign rule, $c_m = (-1)^m Q(m)$, the product form (4.1) and the truncation formula (Lemmas 4.1–4.5), for $n \le 80$, $m \le 400$;
- the expansion (3.1) against the direct series ($k \le 20$, $k = 27, 33$, $d \le 1500$);
- Theorem 5.1 for all $k \le 1001$ (608 squarefree k: all vanish; 392 non-squarefree k: all nonzero);
- formula (5.1) in 5121 cases and the exact failures of Corollary 5.4 for all odd prime powers $k \le 1001$;
- the pair reduction (6.1) for $7 \le k \le 130$ and the values $g_{15}(286) = g_{21}(526) = 0$;
- Theorem 7.7 for even $k \le 130$ (values of g_{2^a});
- the sign law and the zero set of g_2 for $d \le 6000$; zero sets for $k = 2, 6, 10, 14$ and the bounds B_c.

The checks are consistency tests of the proved statements, not parts of proofs, except the exact term lists quoted in proofs (Corollary 5.4, Sections 6.6, 7.2, 7.3), which are exact integer evaluations. `P/code/even_zero_sets.py` produces Table B.2 (`P/data/even_zero_sets_k2mod4.txt`).

Run: `cd P/code && python3 verify_algebra.py` (about 5 seconds).

A.2 The composite-zero theorem (Section 6)

Directory `P/code/composite/`. From `P/code/composite/code`:

1. `python3 make_Q.py` builds the exact table $Q(0..35200)$ (`../data/Q_exact_35200.pkl`; sha256 of the decimal list `6a2037fb2c51b525ae73b77b8593bcfaa6ad2868315b7b01be4bb62a24e06a78`).
2. `python3 pairs_certify.py 2500 300` (`../data/pairs_certify.log`): for the 15 616 pairs $3 \le s \le 2500$, k odd, $k > 2s$, $k \mid s^2 - 1$, an integer lower bound $LB \le A_1$ and the exact A_2; 15 587 pairs have $LB > A_2$; the remaining 29 have $s \le 32$ and are settled by the exact dynamic programme, which was run for all $s \le 300$ (995 pairs). Only $(4, 15)$ and $(8, 21)$ have $A_1 \le A_2$ (equality).
3. `python3 chern_margins.py` (`../data/chern_margins.log`): interval evaluation (mpmath `iv`, 40 digits) of C_E, K, $\bar\beta(n)$ at the thresholds quoted in Lemma 6.5, the monotonicity ratios, the comparison $|Q - M| \le \bar B$ on $n \in [300, 2999]$ and every 17th $n \le 35200$ (no violations; this comparison is an extra check, not used in the proof), and the margins of Proposition 6.8 on $2501 \le s \le 60000$ (minimum margins 0.44805 and 0.99364).

A.3 Parity dominance (Section 8)

Directory `P/code/pd/` holds the programs `arb_compat.py`, `iv56.py` (interval type over Arb), `clt56.py` (Algorithms E and C), `blocks56.py` (Algorithm R), `tails56.py` (Algorithm T), `params56.py`, `make_cells56.py`, `run_pd56.py`, and the data `params56.json`, `cells56_sharp.json`, `consts56_sharp.json`, `pd56_results.json` (hashes in `P/data/pd/sha256_copied.txt`; the files `*_c15.json` belong to a variant not used in the text).

- Window parameters: `python3 params56.py` (Table 8.3).
- Cells and constants: `python3 make_cells56.py sharp` (Theorem 8.12; Table 8.4).
- Block and tail: `python3 run_pd56.py` evaluates three variants; the text uses `SHARP` (Table 8.2). Output `pd56_results.json`.
- Reproduction: `P/data/pd/regen/regen_all.sh` re-ran all of the above from scratch in a clean directory; `params56.json`, `cells56_sharp.json`, `consts56_sharp.json` and `pd56_results.json` are identical to the stored files except the timing field `secs` of the latter (`regen/compare.log`). `P/code/pd/spot_rerun.py` re-runs cells $0, 100, 200, 311$ and $k = 3, 33$ (`P/data/pd/spot_rerun_log.txt`).
- Tables 8.1–8.4 of the text are produced from the JSON files by `P/code/make_tables.py` (`P/data/tables_pd.md`).

Run time on one core: cells about 20 s, `run_pd56.py` about 75 s for all 16 values of k and three variants.

A.4 The modular certificate and zero sets

`P/cert/modcert_copy.c` (sha256 `a228eaff1ac42094a28ec7b32ce495e233d3d78007d5ebd33dce4e35482caf08`); compile `gcc -O2 -o modcert modcert_copy.c`; `P/cert/run_cert.sh` runs `modcert k 60000000 p` for all odd $k \le 33$ and $p \in \{2305843009213693951, 4611686018427387847\}$ (outputs `z_k<k>_p<p>.txt`; log `cert_jobs.log`; about 3 min per job, memory below 900 MB). `python3 P/code/zero_sets.py` intersects the two lists, evaluates every candidate exactly, evaluates all allowed $d \le \max C_k$ exactly, and writes `P/data/zero_sets.json` and `zero_sets.md` (Table 8.5, Appendix B.1).

A.5 Source directories

The programs of A.3 and A.4 are unchanged copies of the following directories of the same repository (`reports/s_family_paths/`): `A56_A49_audit` (block, tail, cell code, modular certificate), `A58_chern_bound_audit` (pair certificate, Q table), `A61_clt_proofs` (written proofs of the lattice-sum, cell, and tail lemmas, `PROOFS.md`), `A59_clt_rederivation` (the independent derivation of Remark 8.9), `A57_gk_31_33` (the cases $k = 31, 33$), `A46_zero_facts_audit`, `A47_composite_zeros`, `A49_gk_runs`, `A50_proofs`, `A54_proofs2`, `A55_small_k_exceptions` (Sections 5–7). The proofs in Part I were written out and checked against these sources.

B Zero sets of g_k

B.1 Odd k, $3 \le k \le 33$

$Z_k = \{d \text{ allowed for } k : \ g_k(d) = 0\}$ (Theorem 8.22). Every element is listed.

$k = 3$ ($\#Z_k = 9$, $\max Z_k = 43$): 3, 4, 6, 7, 15, 28, 31, 40, 43.

$k = 5$ ($\#Z_k = 17$, $\max Z_k = 176$): 5, 9, 10, 15, 16, 19, 20, 21, 24, 31, 35, 41, 46, 71, 86, 161, 176.

$k = 7$ ($\#Z_k = 24$, $\max Z_k = 247$): 2, 7, 14, 16, 18, 21, 28, 32, 35, 36, 42, 43, 44, 57, 60, 63, 78, 79, 85, 100, 107, 135, 191, 247.

$k = 9$ ($\#Z_k = 20$, $\max Z_k = 118$): 7, 22, 25, 27, 31, 36, 49, 52, 63, 64, 67, 70, 73, 76, 85, 88, 91, 97, 112, 118.

$k = 11$ ($\#Z_k = 43$, $\max Z_k = 273$): 3, 5, 11, 14, 22, 26, 31, 33, 36, 37, 38, 44, 49, 55, 60, 64, 66, 69, 70, 71, 75, 77, 80, 82, 88, 92, 93, 99, 100, 102, 104, 110, 111, 126, 133, 135, 136, 143, 147, 148, 166, 196, 273.

$k = 13$ ($\#Z_k = 54$, $\max Z_k = 545$): 3, 10, 12, 13, 23, 26, 30, 35, 39, 42, 43, 49, 51, 52, 64, 65, 77, 78, 82, 88, 91, 94, 95, 101, 104, 114, 116, 117, 120, 121, 130, 143, 144, 146, 147, 156, 157, 159, 160, 173, 179, 183, 195, 205, 212, 222, 224, 256, 269, 324, 334, 480, 519, 545.

$k = 15$ ($\#Z_k = 55$, $\max Z_k = 565$): 6, 10, 15, 16, 21, 30, 31, 34, 39, 45, 55, 60, 61, 66, 75, 76, 81, 90, 96, 100, 105, 106, 109, 120, 126, 135, 150, 154, 165, 180, 184, 186, 195, 199, 201, 210, 211, 216, 231, 241, 246, 255, 259, 276, 286, 291, 324, 369, 376, 384, 414, 436, 460, 520, 565.

$k = 17$ ($\#Z_k = 80$, $\max Z_k = 1137$): 2, 8, 13, 15, 17, 19, 30, 32, 34, 38, 43, 47, 50, 51, 55, 59, 68, 70, 81, 83, 85, 100, 101, 102, 106, 117, 119, 121, 123, 132, 136, 138, 144, 149, 151, 153, 155, 166, 168, 170, 174, 183, 186, 187, 191, 204, 206, 221, 225, 234, 237, 238, 254, 255, 256, 259, 268, 272, 273, 274, 276, 285, 291, 293, 305, 307, 322, 323, 339, 342, 344, 356, 358, 359, 393, 400, 451, 485, 502, 1137.

$k = 19$ ($\#Z_k = 89$, $\max Z_k = 766$): 5, 6, 7, 11, 17, 19, 24, 26, 30, 38, 42, 43, 45, 47, 54, 57, 61, 62, 63, 74, 76, 87, 95, 100, 102, 111, 114, 118, 121, 133, 137, 140, 144, 152, 157, 163, 171, 176, 178, 190, 194, 195, 197, 209, 213, 214, 225, 226, 228, 235, 247, 252, 266, 270, 273, 282, 285, 290, 292, 304, 308, 321, 323, 324, 330, 342, 343, 347, 349, 377, 381, 384, 396, 397, 399, 403, 415, 434, 438, 444, 463, 472, 491, 492, 503, 579, 598, 747, 766.

$k = 21$ ($\#Z_k = 88$, $\max Z_k = 2055$): 7, 15, 18, 21, 25, 28, 39, 42, 51, 58, 60, 63, 64, 67, 78, 84, 85, 88, 91, 105, 109, 121, 123, 126, 127, 142, 144, 147, 148, 165, 168, 172, 183, 189, 196, 207, 210, 211, 228, 231, 232, 246, 252, 273, 274, 289, 294, 295, 310, 312, 315, 316, 336, 337, 357, 373, 378, 394, 399, 420, 436, 442, 457, 463, 483, 498, 499, 520, 526, 561, 582, 603, 610, 624, 639, 645, 660, 681, 702, 723, 862, 1114, 1135, 1141, 1261, 1303, 1534, 2055.

$k = 23$ ($\#Z_k = 105$, $\max Z_k = 837$): 2, 3, 6, 8, 12, 13, 18, 23, 26, 29, 31, 35, 41, 46, 50, 52, 54, 55, 58, 62, 69, 71, 73, 75, 77, 82, 92, 95, 98, 110, 115, 127, 131, 138, 142, 144, 146, 161, 165, 169, 184, 192, 196, 197, 207, 213, 219, 225, 230, 234, 236, 238, 253, 257, 259, 261, 276, 282, 299, 322, 326, 328, 345, 349, 351, 368, 374, 391, 414, 418, 420, 422, 437, 460, 464, 483, 484, 506, 507, 514, 535, 537, 545, 553, 556, 560, 575, 579, 593, 606, 614, 622, 637, 657, 660, 662, 675, 708, 726, 754, 768, 791, 800, 814, 837.

$k = 25$ ($\#Z_k = 76$, $\max Z_k = 936$): 6, 11, 14, 19, 21, 24, 31, 39, 44, 46, 54, 56, 59, 66, 69, 71, 75, 79, 86, 94, 96, 99, 100, 114, 119, 125, 131, 141, 154, 169, 171, 179, 196, 200, 211, 221, 244, 254, 256, 269, 271, 279, 289, 294, 296, 319, 354, 369, 379, 394, 396, 419, 454, 479, 494, 519, 529, 544, 569, 576, 579, 601, 604, 629, 651, 666, 669, 704, 726, 744, 766, 791, 884, 911, 914, 936.

$k = 27$ ($\#Z_k = 68$, $\max Z_k = 1375$): 7, 10, 13, 19, 22, 27, 34, 36, 37, 40, 46, 54, 58, 61, 67, 70, 73, 79, 85, 88, 94, 103, 115, 118, 135, 142, 151, 154, 166, 175, 193, 196, 252, 256, 274, 277, 283, 289, 301, 304, 331, 358, 360, 382, 397, 409, 412, 414, 439, 466, 490, 495, 517, 520, 598, 652, 655, 676, 703, 757, 760, 787, 799, 838, 853, 880, 1143, 1375.

$k = 29$ ($\#Z_k = 121$, $\max Z_k = 1934$): 5, 6, 7, 13, 20, 22, 23, 24, 28, 29, 34, 35, 42, 51, 52, 53, 57, 58, 62, 63, 67, 71, 74, 80, 82, 83, 86, 87, 91, 92, 94, 107, 109, 111, 115, 116, 122, 138, 139, 140, 145, 158, 161, 167, 174, 178, 179, 202, 203, 207, 225, 227, 232, 239, 256, 261, 285, 289, 290, 294, 312, 318, 319, 323, 324, 341, 343, 348, 370, 372, 377, 399, 406, 410, 435, 439, 457, 464, 486, 488,

493, 515, 521, 522, 526, 551, 555, 573, 580, 602, 609, 631, 638, 642, 667, 671, 696, 725, 729, 754, 783, 784, 787, 812, 813, 816, 845, 857, 863, 871, 892, 899, 932, 958, 1002, 1008, 1031, 1060, 1227, 1591, 1934.

$k = 31$ ($\#Z_k = 127$, $\max Z_k = 1712$): 2, 5, 7, 8, 10, 14, 18, 19, 20, 28, 31, 33, 38, 39, 41, 45, 50, 51, 59, 62, 66, 69, 70, 71, 72, 76, 78, 82, 87, 90, 93, 97, 98, 101, 103, 107, 111, 113, 124, 126, 132, 134, 138, 143, 155, 162, 163, 165, 171, 183, 186, 190, 194, 206, 217, 221, 231, 248, 253, 256, 258, 279, 289, 310, 314, 320, 324, 341, 345, 349, 355, 361, 372, 380, 382, 403, 411, 413, 434, 438, 441, 442, 465, 469, 496, 504, 527, 535, 537, 558, 562, 566, 589, 593, 620, 628, 651, 659, 661, 682, 686, 713, 717, 744, 752, 754, 775, 806, 810, 837, 868, 872, 899, 900, 907, 930, 931, 993, 996, 1023, 1027, 1086, 1101, 1163, 1312, 1400, 1712.

$k = 33$ ($\#Z_k = 88$, $\max Z_k = 1587$): 3, 12, 15, 22, 27, 31, 33, 45, 48, 49, 55, 60, 66, 70, 75, 78, 88, 91, 93, 99, 100, 102, 103, 111, 126, 130, 132, 133, 147, 148, 159, 165, 169, 187, 192, 198, 199, 210, 214, 231, 232, 235, 247, 256, 264, 265, 267, 268, 280, 291, 297, 298, 301, 324, 330, 331, 357, 363, 396, 423, 429, 433, 456, 462, 463, 495, 528, 532, 561, 562, 588, 594, 627, 660, 693, 726, 759, 792, 825, 858, 891, 924, 957, 990, 1023, 1056, 1155, 1587.

B.2 Even $k \equiv 2 \pmod 4$, $2 \le k \le 34$

By Theorem 7.4 and Lemma 7.3 every allowed class is one-sign; $B = \max_c B_c$; the zeros are the allowed $d < B$ with $g_k(d) = 0$ (computed exactly, `code/even_zero_sets.py`).

k	classes	B	$\#Z_k$	$\max Z_k$
2	2	42	4	26
6	4	186	10	78
10	6	710	17	120
14	8	1778	30	224
18	8	2899	19	202
22	12	6314	58	528
26	14	10166	73	728
30	12	15330	59	960
34	18	21998	108	1224

C Reproducibility for Part II

All computations were run with Python 3.13.5, python-flint 0.9.0, sympy 1.14.0, mpmath 1.3.0, numpy 2.5.4 and cypari2 2.2.4 (PARI/GP, used only where stated). The scripts, logs and data files are kept in the companion folders below (under `quantum-comp/reports/heavy-daily/`); each folder contains a report and the scripts it describes.

folder	content
`2026-10-17-bridge-c-residual-law`, `2026-10-17-P_s-structure`	Theorems 11.1–11.4; radial fits; Gauss pattern
`2026-10-17-psi-cocycle`, `2026-10-17-psi-conditions`	lift, S/T law, period formula, row closed forms

folder	content
2026-10-17-cusp-part, 2026-10-17-psiEis-Lvalues, 2026-10-18-PARI-free	Eisenstein/cusp split, $L(0)$ identities, PARI-free certification
2026-10-18-ER-cancellation, 2026-10-18-ER-cancellation-audit	Herglotz closed form (Theorems 11.9, 11.10)
2026-10-18-Audit-and-ProfileLaw, 2026-10-18-S-Eigenvalues, 2026-10-18-V42-Structure	profile law; spectra; function span
2026-10-18-Tensor-Pattern-Proof, 2026-10-18-Tensor-Pattern-Closure, 2026-10-18-General-s-Tensor, 2026-10-18-Symmetry-Audit	tensor theorems, dimensions, label operators
2026-10-18-Cusp-Onset, 2026-10-18-Cusp-Onset-Closure, 2026-10-18-uL-Criterion-Closure, 2026-10-18-s8-s11	u_L criterion; $s = 8, 11$
2026-10-18-H_s-Closed-Forms, 2026-10-18-H_s-Closure, 2026-10-18-Open-Analytic, 2026-10-18-Analytic-CatB-Push	Euler–Maclaurin; H_s closed forms; Gauss data; $t^{11/2}$–$t^{17/2}$; data files of the closed forms
2026-10-18-Algebraic-Round3, 2026-10-18-Open-Algebraic	h-relations; Proposition K1; Weil inputs; cone and axes cone
2026-10-19-HighRAM-Items	prime-square splits and commutants (`sec_comm2.py`, transpose convention; $s = 49$ certificate at two primes), p-local invariants at $s = 121, 169$ (`famfast.py`, `pinv2.py`, `pinv3.py`)
2026-10-19-Open-Items-Pursuit	Proposition 20.8 (`item7_k1_sharp.py`); Remark 27.15 (`dihedral_enum.py`, `dihedral_diag.py`, `dihedral_gen.py`); Proposition 26.10(iii) (`item6_candidate.py`); dictionary check in Remark 27.16(a) (`item4_dictionary.py`); Lemma 27.5 checks (`item3_estimates.py`)
2026-10-19-Certified-SRho	Theorem 27.13 (`certify.py`; outputs `cert_*.json`)
2026-10-18-C_third-Unified, 2026-10-18-C_third-Closure, 2026-10-18-C_third-modularity, 2026-10-18-Cocycle-Closure	§24; Theorem 15.3
2026-10-18-Zeta-Rational-Curve-Proof, 2026-10-18-g_s-Closed-Forms	Theorems 25.1, 25.2
2026-10-18-AtkinLehner-Local	Proposition 21.5
2026-10-19-Implications-checks (incl. k3_140)	§26; coefficient values of C; A_5 group closure
2026-10-19-N1-and-Uniqueness	§27 (sectors, q-coefficients to frequency 40, g_0); §28
2026-10-19-Audit-U2-U3-ThmG-SRho	numerical checks of Theorem 11.7 on both half-planes, Lemma 15.1 quadrature, Theorem 27.4, sector bounds

To reproduce, run each script with the stated Python environment from its folder (e.g. `python3 n1_tests.py 1500`); scripts write their JSON or log outputs next to themselves. Exact computations are deterministic; floating-point runs agree to the stated tolerances.

D Category B record for Part III

Everything in this appendix is Category B: floating-point evaluation, least-squares fits, or numerical ranks of real or complex matrices by singular value decomposition with relative threshold 10^{-9}. None of it is used in a proof. Data, scripts and logs are in `reports/bridges/BRIDGE_C_35/` and its subfolders `order2`, `order3`, `order3b`, `order3c`, `order4a`, and `order5`–`order16` for items D.10–D.13; every run was preregistered with a SHA-256 hash recorded before the run.

D.1 *Leading order on the series.* $t\,C(\mathbf{e}(y)e^{-t})$ was fitted at 30 rational points y using up to $3.4\cdot10^6$ lattice points. On a first grid ($t \ge 10^{-3}$ scale) 19 points agree with $I\,G(y)$ to $\le 2\%$ (most $\le 0.1\%$); on $t \in [3.1\cdot10^{-6}, 10^{-4}]$, 10 of the remaining 11 agree to $\le 0.7\%$ and $y = -\frac{5}{12}$ differs by 6.4%. The remaining discrepancies occur at denominators ≥ 9, where the constant in the $O(t^{1/2})$ term of Theorem 34.6 is large.

D.2 *Order $t^{-1/2}$ on the series.* On grids $t_j = 2^{-j}\cdot 0.5/(297.5\,(2m)^2)$, scaled to the denominator and to the long cone edge $Q(21,-1) = 297.5$, the fitted $t^{-1/2}$ coefficient of $q^{-7/20}C$ agrees with A_1 of Theorem 35.6 to relative error $\le 10^{-5}$ at $x = \frac12, \frac13, \frac14, \frac56, \frac38, \frac29, \frac{3}{10}, \frac15, \frac17$, and to $\le 7.5\cdot10^{-6}$ at $x = 1, \frac12, \frac13, \frac23, \frac14, \frac15, \frac16$ in an independent run. Grids scaled to the denominator only ($t(2m)^2 \le 0.08$, so $t\,Q(u_1)(2m)^2$ up to 24) are outside the regime and do not reproduce A_1 at denominators $2, 3, 4, 6, 8, 9$.

D.3 *Order t^0 on the series.* On the same cone-scaled grid ($j = 0, \dots, 7$, fit on $1, t^{1/2}, t, t^{3/2}$), the fitted constant term agrees with A_2 of Theorem 35.6 at $x \in \{1, \frac12, \frac13, \frac23, \frac14, \frac15, \frac16, \frac17, \frac37\}$ with maximal relative error $1.4\cdot10^{-7}$; for example $A_2(1) = -0.3232819 - 0.4449593i$, $A_2(\frac14) = 2.4890282 + 0.2339631i$, $A_2(\frac12) = 0$ to 10^{-15}. With the sign of the corner term C reversed, the formula disagrees by relative errors 0.25 to 2.92 at the seven points where $|\mathsf{C}| > 0.05$.

D.4 *Values.* $|A_1(\frac12)| = |A_1(\frac52)| = 0.44311346272638\ldots = \sqrt{\pi}/4$ to 15 digits. The piece decomposition reproduces $A_0 = I\,G_{\alpha,\beta}$ to $1.4\cdot10^{-17}$ in 63 cases.

D.5 *No single-weight law at order $t^{-1/2}$.* Let $\mathbf{A}_1(x)$ be the 105-vector of A_1 over the companions, at 138 rational points. With $B_w(x) = |x|^w\mathbf{A}_1(-1/x)$: $\operatorname{rank}\mathbf{A}_1 = 105$ and $\operatorname{rank}[\mathbf{A}_1 \mid B_w] = 138$ for $w = \frac12$ and $w = 1$ ($\sigma_{\min}/\sigma_{\max} \approx 0.013$). A law $\mathbf{A}_1(-1/x) = |x|^{-w}M\mathbf{A}_1(x)$ would force rank ≤ 105. The same pipeline applied to $\mathbf{A}_0$ gives rank 36 and joint rank 36 at $w = 1$ (and 72 with the wrong sign of the exponent).

D.6 *Parts of A_1.* With the same test on 216 points (ranks; joint ranks at $w = \frac12$ / $w = 1$): D: 58; 116/98. R: 105; 210/210. $W[1]$: 90; 178/156. $W[\frac72]$: 60; 116/120. $W[\frac{595}{2}]$: 100; 200/200. W: 105; 210/210. None of the parts of Proposition 35.11 closes.

D.7 *Regrouping by quadratic class and by characteristic.* Groups $W[1]$, $W[\frac72] + R[\frac72]$, $W[\frac{595}{2}] + R[\frac{595}{2}]$, D: ranks $90, 60, 100, 58$, joint ranks at $w = \frac12$: $178, 116, 200, 116$. Restricted to the characteristic blocks $\Theta_1 = K2 \cup K3$ (70 columns) and $\Theta_2 = K4$ (35 columns), A_1 has ranks 70 and 35 with joint ranks 140 and 70. No group closes. A_0 restricted to Θ_1 and Θ_2 has ranks 24 and 12, each closed.

D.8 *Sawtooth shadow candidate.* For $A_1 - \sum_\lambda \varepsilon_\lambda W[\lambda]$, $\varepsilon \in \{0,1\}^3$, the ranks are $105, 91, 104, 90, 105, 87, 103, 58$ for $\varepsilon = 000, 001, 010, 011, 100, 101, 110, 111$ (order $W[1], W[\frac72], W[\frac{595}{2}]$), with joint ranks at $w = \frac12$ equal to twice the rank except 170 for 101, and at $w = 1$ equal to twice the rank except 156 for 101 and 98 for 111. No choice closes. The primary candidate $\varepsilon = 111$ (completed $A_1 = D + R$) has rank 58 and is falsified;

the six secondary subsets do not close. Sign control (A_0: rank 36 and closed at $w = 1$; joint rank 72 at the wrong weight) passed. Source: `BRIDGE_C_35/order4a`.

D.9 *Weight-0 law for A_2.* For A_2 on the 105 companions at 216 points: rank 73, joint rank with the S-image 146 at $w = 0$, and likewise at $w = \frac{1}{2}$ and $w = 1$; not closed. Mixed span $[A_2, |x|^{1/2}A_1, |x|A_0]$: rank 214 of 216 rows, not closed, weakly discriminating. Anchor (order3c value of A_2 at $x = 1$) reproduced exactly; sign control passed. Source: `BRIDGE_C_35/order4b`.

D.10 *Holomorphic projection.* The holomorphic projection of Φ_{orth} matches S^E coefficientwise on the companions tested (order8), and in a fresh preregistered test on $K1$ with residual $7.5 \cdot 10^{-32}$ (order15, C1′).

D.11 *Law of the completions.* The completed $\widehat{H}$, $\widehat{G}_O$ satisfy the law with $(35^{-1/2}M, T)$ and $\chi = 1$ at sample points to residual below 10^{-20} (orders 5, 6, 11; control $\widehat{G}_O$ residual $3.6 \cdot 10^{-21}$).

D.12 *P-even sector.* The P-even part obeys a law in $|\tau|$ with factor $|x|^{-1}$ on $x > 0$ to about 10^{-18} and fails for $x < 0$ (order12); a scalar multiplier $\chi = \pm i$ fails (order12), consistent with Lemma 37.4.

D.13 *Vanishing set.* The predicted zero set of Theorem 37.10 agrees with numerically computed endpoint terms on all 105 companions, on a further family of 22 predicted zeros (gap 0.86), and on $K1$ (orders 14–16). The line-term identity of Proposition 37.12 was checked on a box of radius 100 for 140 companions with 0 mismatches (order16).

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Note on citations. The cited statements of Nikulin (Thm. 1.14.4) and Morrison (Thm. 2.8, Cor. 2.9(i)) were checked against the original publications; those of Deitmar–Hilgert (Thms. 3.3, 4.4) against the arXiv version, whose numbering is used here.